EXERCISE SET A Graphing Utility

1–4 Use a graphing utility to generate the graph of f in the given viewing windows, and specify the window that you think gives the best view of the graph.

1. $f(x) = x^4 - x^2$
(a) $[-50, 50] \times [-50, 50]$ (b) $[-5, 5] \times [-5, 5]$
(c) $[-2, 2] \times [-2, 2]$ (d) $[-2, 2] \times [-1, 1]$
(e) $[-1.5, 1.5] \times [-0.5, 0.5]$

2. $f(x) = x^5 - x^3$
(a) $[-50, 50] \times [-50, 50]$ (b) $[-5, 5] \times [-5, 5]$
(c) $[-2, 2] \times [-2, 2]$ (d) $[-2, 2] \times [-1, 1]$
(e) $[-1.5, 1.5] \times [-0.5, 0.5]$

3. $f(x) = x^2 + 12$
(a) $[-1, 1] \times [13, 15]$ (b) $[-2, 2] \times [11, 15]$
(c) $[-4, 4] \times [10, 28]$ (d) A window of your choice

4. $f(x) = -12 - x^2$
(a) $[-1, 1] \times [-15, -13]$ (b) $[-2, 2] \times [-15, -11]$
(c) $[-4, 4] \times [-28, -10]$ (d) A window of your choice

5–6 Use the domain and range of f to determine a viewing window that contains the entire graph, and generate the graph in that window.

5. $f(x) = \sqrt{16 - 2x^2}$ **6.** $f(x) = \sqrt{3 - 2x - x^2}$

7–14 Generate the graph of f in a viewing window that you think is appropriate.

7. $f(x) = x^2 - 9x - 36$ **8.** $f(x) = \dfrac{x+7}{x-9}$

9. $f(x) = 2\cos(80x)$ **10.** $f(x) = 12\sin(x/80)$

11. $f(x) = 300 - 10x^2 + 0.01x^3$

12. $f(x) = x(30 - 2x)(25 - 2x)$

13. $f(x) = x^2 + \dfrac{1}{x}$ **14.** $f(x) = \sqrt{11x - 18}$

15–16 Generate the graph of f and determine whether your graphs contain false line segments. Sketch the actual graph and see if you can make the false line segments disappear by changing the viewing window.

15. $f(x) = \dfrac{x}{x^2 - 1}$ **16.** $f(x) = \dfrac{x^2}{4 - x^2}$

17. The graph of the equation $x^2 + y^2 = 16$ is a circle of radius 4 centered at the origin.
(a) Find a function whose graph is the upper semicircle and graph it.
(b) Find a function whose graph is the lower semicircle and graph it.
(c) Graph the upper and lower semicircles together. If the combined graphs do not appear circular, see if you can adjust the viewing window to eliminate the aspect ratio distortion.
(d) Graph the portion of the circle in the first quadrant.
(e) Is there a function whose graph is the right half of the circle? Explain.

18. In each part, graph the equation by solving for y in terms of x and graphing the resulting functions together.
(a) $x^2/4 + y^2/9 = 1$ (b) $y^2 - x^2 = 1$

19. Read the documentation for your graphing utility to determine how to graph functions involving absolute values, and graph the given equation.
(a) $y = |x|$ (b) $y = |x - 1|$
(c) $y = |x| - 1$ (d) $y = |\sin x|$
(e) $y = \sin|x|$ (f) $y = |x| - |x + 1|$

20. Based on your knowledge of the absolute value function, sketch the graph of $f(x) = |x|/x$. Check your result using a graphing utility.

21–22 Most graphing utilities provide some way of graphing functions that are defined piecewise; read the documentation for your graphing utility to find out how to do this. However, if your goal is just to find the general shape of the graph, you can graph each portion of the function separately and combine the pieces with a hand-drawn sketch. Use this method in these exercises.

21. Draw the graph of

$$f(x) = \begin{cases} \sqrt[3]{x-2}, & x \le 2 \\ x^3 - 2x - 4, & x > 2 \end{cases}$$

22. Draw the graph of

$$f(x) = \begin{cases} x^3 - x^2, & x \le 1 \\ \dfrac{1}{1-x}, & 1 < x < 4 \\ x^2 \cos\sqrt{x}, & 4 \le x \end{cases}$$

23–24 We noted in the text that for functions involving fractional exponents (or radicals), graphing utilities sometimes omit portions of the graph. If $f(x) = x^{p/q}$, where p/q is a positive fraction in *lowest terms*, then you can circumvent this problem as follows:

- If p is even and q is odd, then graph $g(x) = |x|^{p/q}$ instead of $f(x)$.
- If p is odd and q is odd, then graph $g(x) = (|x|/x)|x|^{p/q}$ instead of $f(x)$.

23. (a) Generate the graphs of $f(x) = x^{2/5}$ and $g(x) = |x|^{2/5}$, and determine whether your graphing utility missed part of the graph of f.
(b) Generate the graphs of the functions $f(x) = x^{1/5}$ and $g(x) = (|x|/x)|x|^{1/5}$, and determine whether your graphing utility missed part of the graph of f.
(c) Generate a graph of the function $f(x) = (x - 1)^{4/5}$ that shows all of its important features.
(d) Generate a graph of the function $f(x) = (x + 1)^{3/4}$ that shows all of its important features.

24. The graphs of $y = (x^2 - 4)^{2/3}$ and $y = [(x^2 - 4)^2]^{1/3}$ should be the same. Does your graphing utility produce

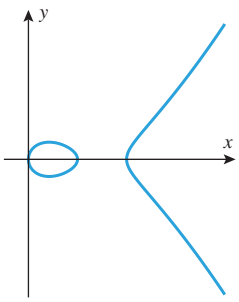
the same graph for both equations? If not, what do you think is happening?

25. In each part, graph the function for various values of c, and write a paragraph or two that describes how changes in c affect the graph in each case.
(a) $y = cx^2$ (b) $y = x^2 + cx$ (c) $y = x^2 + x + c$

26. The graph of an equation of the form $y^2 = x(x-a)(x-b)$ (where $0 < a < b$) is called a ***bipartite cubic***. The accompanying figure shows a typical graph of this type.
(a) Graph the bipartite cubic $y^2 = x(x-1)(x-2)$ by solving for y in terms of x and graphing the two resulting functions.
(b) Find the x-intercepts of the bipartite cubic

$$y^2 = x(x-a)(x-b)$$

and make a conjecture about how changes in the values of a and b would affect the graph. Test your conjecture by graphing the bipartite cubic for various values of a and b.



Bipartite cubic

◀ **Figure Ex-26**

27. Based on your knowledge of the graphs of $y = x$ and $y = \sin x$, make a sketch of the graph of $y = x \sin x$. Check your conclusion using a graphing utility.

28. What do you think the graph of $y = \sin(1/x)$ looks like? Test your conclusion using a graphing utility. [*Suggestion:* Examine the graph on a succession of smaller and smaller intervals centered at $x = 0$.]

29–30 Graph the equation using a graphing utility. ■

29. (a) $x = y^2 + 2y + 1$
(b) $x = \sin y, \ -2\pi \le y \le 2\pi$

30. (a) $x = y + 2y^3 - y^5$
(b) $x = \tan y, \ -\pi/2 < y < \pi/2$

31–34 Use a graphing utility and parametric equations to display the graphs of f and f^{-1} on the same screen. ■

31. $f(x) = x^3 + 0.2x - 1, \quad -1 \le x \le 2$

32. $f(x) = \sqrt{x^2 + 2} + x, \quad -5 \le x \le 5$

33. $f(x) = \cos(\cos 0.5x), \quad 0 \le x \le 3$

34. $f(x) = x + \sin x, \quad 0 \le x \le 6$

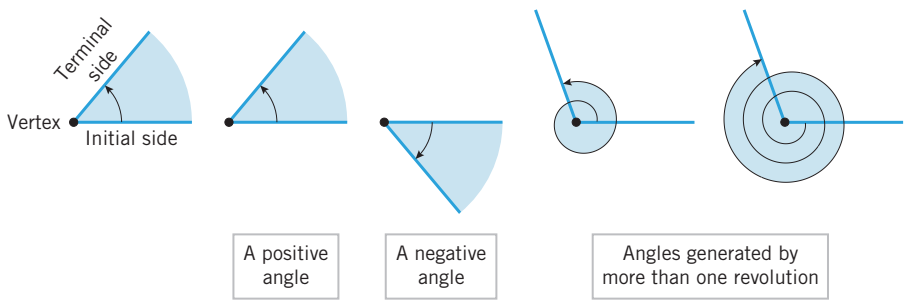
35. (a) Find parametric equations for the ellipse that is centered at the origin and has intercepts $(4, 0)$, $(-4, 0)$, $(0, 3)$, and $(0, -3)$.
(b) Find parametric equations for the ellipse that results by translating the ellipse in part (a) so that its center is at $(-1, 2)$.
(c) Confirm your results in parts (a) and (b) using a graphing utility.

B

TRIGONOMETRY REVIEW

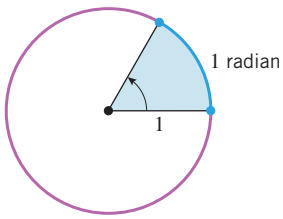
ANGLES

Angles in the plane can be generated by rotating a ray about its endpoint. The starting position of the ray is called the ***initial side*** of the angle, the final position is called the ***terminal side*** of the angle, and the point at which the initial and terminal sides meet is called the ***vertex*** of the angle. We allow for the possibility that the ray may make more than one complete revolution. Angles are considered to be ***positive*** if generated counterclockwise and ***negative*** if generated clockwise (Figure B.1).



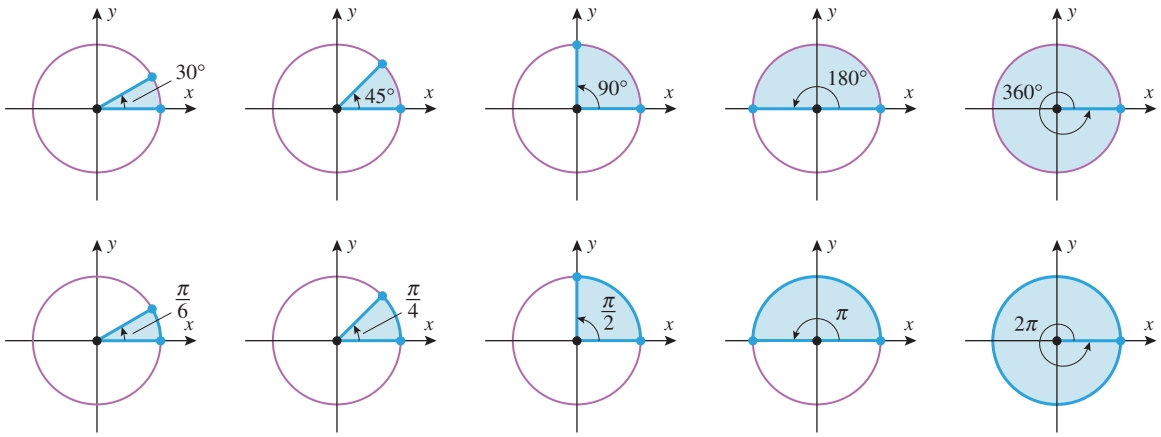
▶ **Figure B.1**

There are two standard measurement systems for describing the size of an angle: ***degree measure*** and ***radian measure***. In degree measure, one degree (written $1°$) is the measure of an angle generated by $1/360$ of one revolution. Thus, there are $360°$ in an angle of one revolution, $180°$ in an angle of one-half revolution, $90°$ in an angle of one-quarter revolution (a *right angle*), and so forth. Degrees are divided into sixty equal parts, called ***minutes***, and minutes are divided into sixty equal parts, called ***seconds***. Thus, one minute (written $1'$) is $1/60$ of a degree, and one second (written $1''$) is $1/60$ of a minute. Smaller subdivisions of a degree are expressed as fractions of a second.



▲ **Figure B.2**

In radian measure, angles are measured by the length of the arc that the angle subtends on a circle of radius 1 when the vertex is at the center. One unit of arc on a circle of radius 1 is called one ***radian*** (written 1 radian or 1 rad) (Figure B.2), and hence the entire circumference of a circle of radius 1 is 2π radians. It follows that an angle of $360°$ subtends an arc of 2π radians, an angle of $180°$ subtends an arc of π radians, an angle of $90°$ subtends an arc of $\pi/2$ radians, and so forth. Figure B.3 and Table B.1 show the relationship between degree measure and radian measure for some important positive angles.



▲ **Figure B.3**

Observe that in Table B.1, angles in degrees are designated by the degree symbol, but angles in radians have no units specified. This is standard practice—when no units are specified for an angle, it is understood that the units are radians.

Table B.1

DEGREES	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
RADIANS	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

From the fact that π radians corresponds to 180°, we obtain the following formulas, which are useful for converting from degrees to radians and conversely.

$$1^\circ = \frac{\pi}{180}\ \text{rad} \approx 0.01745\ \text{rad} \tag{1}$$

$$1\ \text{rad} = \left(\frac{180}{\pi}\right)^\circ \approx 57^\circ\, 17'\, 44.8'' \tag{2}$$

▶ Example 1

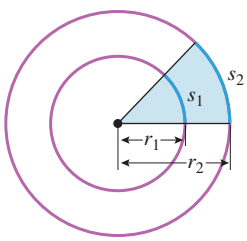
(a) Express 146° in radians. (b) Express 3 radians in degrees.

Solution (a). From (1), degrees can be converted to radians by multiplying by a conversion factor of $\pi/180$. Thus,

$$146^\circ = \left(\frac{\pi}{180} \cdot 146\right)\text{rad} = \frac{73\pi}{90}\ \text{rad} \approx 2.5482\ \text{rad}$$

Solution (b). From (2), radians can be converted to degrees by multiplying by a conversion factor of $180/\pi$. Thus,

$$3\ \text{rad} = \left(3 \cdot \frac{180}{\pi}\right)^\circ = \left(\frac{540}{\pi}\right)^\circ \approx 171.9^\circ \quad ◀$$



$$\frac{s_1}{s_2} = \frac{r_1}{r_2}$$

▲ **Figure B.4**

RELATIONSHIPS BETWEEN ARC LENGTH, ANGLE, RADIUS, AND AREA

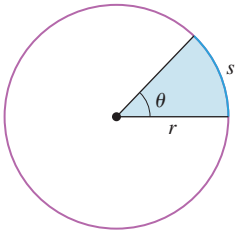
There is a theorem from plane geometry which states that for two concentric circles, the ratio of the arc lengths subtended by a central angle is equal to the ratio of the corresponding radii (Figure B.4). In particular, if s is the arc length subtended on a circle of radius r by a

central angle of θ radians, then by comparison with the arc length subtended by that angle on a circle of radius 1 we obtain

$$\frac{s}{\theta} = \frac{r}{1}$$

from which we obtain the following relationships between the central angle θ, the radius r, and the subtended arc length s when θ is in radians (Figure B.5):

$$\theta = s/r \qquad \text{and} \qquad s = r\theta \tag{3–4}$$



If θ is in radians, then $\theta = s/r$.

▲ **Figure B.5**

The shaded region in Figure B.5 is called a ***sector***. It is a theorem from plane geometry that the ratio of the area A of this sector to the area of the entire circle is the same as the ratio of the central angle of the sector to the central angle of the entire circle; thus, if the angles are in radians, we have

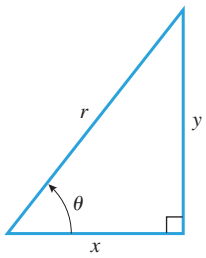
$$\frac{A}{\pi r^2} = \frac{\theta}{2\pi}$$

Solving for A yields the following formula for the area of a sector in terms of the radius r and the angle θ in radians:

$$A = \tfrac{1}{2}r^2\theta \tag{5}$$

TRIGONOMETRIC FUNCTIONS FOR RIGHT TRIANGLES

The ***sine***, ***cosine***, ***tangent***, ***cosecant***, ***secant***, and ***cotangent*** of a positive acute angle θ can be defined as ratios of the sides of a right triangle. Using the notation from Figure B.6, these definitions take the following form:

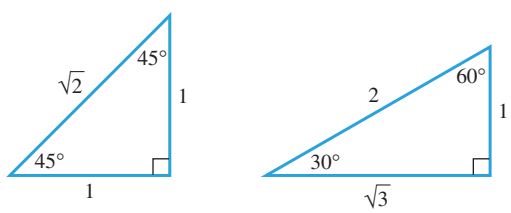


▲ **Figure B.6**

$$\begin{aligned}
\sin\theta &= \frac{\text{side opposite } \theta}{\text{hypotenuse}} = \frac{y}{r}, & \csc\theta &= \frac{\text{hypotenuse}}{\text{side opposite } \theta} = \frac{r}{y} \\
\cos\theta &= \frac{\text{side adjacent to } \theta}{\text{hypotenuse}} = \frac{x}{r}, & \sec\theta &= \frac{\text{hypotenuse}}{\text{side adjacent to } \theta} = \frac{r}{x} \\
\tan\theta &= \frac{\text{side opposite } \theta}{\text{side adjacent to } \theta} = \frac{y}{x}, & \cot\theta &= \frac{\text{side adjacent to } \theta}{\text{side opposite } \theta} = \frac{x}{y}
\end{aligned} \tag{6}$$

We will call sin, cos, tan, csc, sec, and cot the ***trigonometric functions***. Because similar triangles have proportional sides, the values of the trigonometric functions depend only on the size of θ and not on the particular right triangle used to compute the ratios. Moreover, in these definitions it does not matter whether θ is measured in degrees or radians.

► **Example 2** Recall from geometry that the two legs of a $45°–45°–90°$ triangle are of equal size and that the hypotenuse of a $30°–60°–90°$ triangle is twice the shorter leg, where the shorter leg is opposite the $30°$ angle. These facts and the Theorem of Pythagoras yield Figure B.7. From that figure we obtain the results in Table B.2. ◄



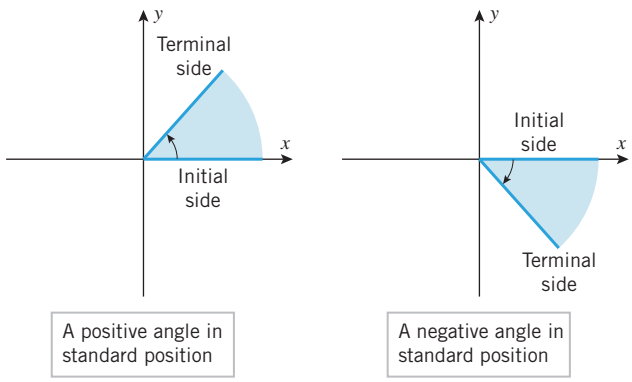
► **Figure B.7**

Table B.2

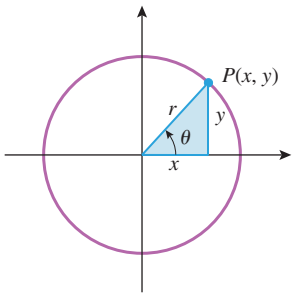
$\sin 45° = 1/\sqrt{2}$,	$\cos 45° = 1/\sqrt{2}$,	$\tan 45° = 1$
$\csc 45° = \sqrt{2}$,	$\sec 45° = \sqrt{2}$,	$\cot 45° = 1$
$\sin 30° = 1/2$,	$\cos 30° = \sqrt{3}/2$,	$\tan 30° = 1/\sqrt{3}$
$\csc 30° = 2$,	$\sec 30° = 2/\sqrt{3}$,	$\cot 30° = \sqrt{3}$
$\sin 60° = \sqrt{3}/2$,	$\cos 60° = 1/2$,	$\tan 60° = \sqrt{3}$
$\csc 60° = 2/\sqrt{3}$,	$\sec 60° = 2$,	$\cot 60° = 1/\sqrt{3}$

ANGLES IN RECTANGULAR COORDINATE SYSTEMS

Because the angles of a right triangle are between $0°$ and $90°$, the formulas in (6) are not directly applicable to negative angles or to angles greater than $90°$. To extend the trigonometric functions to include these cases, it will be convenient to consider angles in rectangular coordinate systems. An angle is said to be in ***standard position*** in an xy-coordinate system if its vertex is at the origin and its initial side is on the positive x-axis (Figure B.8).



▶ **Figure B.8**



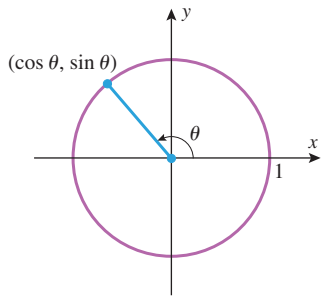
▲ **Figure B.9**

To define the trigonometric functions of an angle θ in standard position, construct a circle of radius r, centered at the origin, and let $P(x, y)$ be the intersection of the terminal side of θ with this circle (Figure B.9). We make the following definition.

B.1 DEFINITION

$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}, \quad \tan\theta = \frac{y}{x}$$
$$\csc\theta = \frac{r}{y}, \quad \sec\theta = \frac{r}{x}, \quad \cot\theta = \frac{x}{y}$$

Note that the formulas in this definition agree with those in (6), so there is no conflict with the earlier definition of the trigonometric functions for triangles. However, this definition applies to all angles (except for cases where a zero denominator occurs).



▲ **Figure B.10**

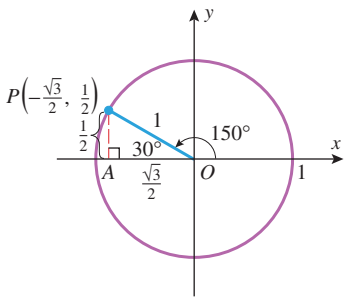
In the special case where $r = 1$, we have $\sin\theta = y$ and $\cos\theta = x$, so the terminal side of the angle θ intersects the unit circle at the point $(\cos\theta, \sin\theta)$ (Figure B.10). It follows from Definition B.1 that the remaining trigonometric functions of θ are expressible as (verify)

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta} \tag{7–10}$$

These observations suggest the following procedure for evaluating the trigonometric functions of common angles:

- Construct the angle θ in standard position in an xy-coordinate system.
- Find the coordinates of the intersection of the terminal side of the angle and the unit circle; the x- and y-coordinates of this intersection are the values of $\cos\theta$ and $\sin\theta$, respectively.
- Use Formulas (7) through (10) to find the values of the remaining trigonometric functions from the values of $\cos\theta$ and $\sin\theta$.

▶ **Example 3** Evaluate the trigonometric functions of $\theta = 150°$.



▲ **Figure B.11**

Solution. Construct a unit circle and place the angle $\theta = 150°$ in standard position (Figure B.11). Since $\angle AOP$ is $30°$ and $\triangle OAP$ is a $30°-60°-90°$ triangle, the leg AP has length $\frac{1}{2}$ (half the hypotenuse) and the leg OA has length $\sqrt{3}/2$ by the Theorem of Pythagoras. Thus, the coordinates of P are $(-\sqrt{3}/2, 1/2)$, from which we obtain

$$\sin 150° = \frac{1}{2}, \quad \cos 150° = -\frac{\sqrt{3}}{2}, \quad \tan 150° = \frac{\sin 150°}{\cos 150°} = \frac{1/2}{-\sqrt{3}/2} = -\frac{1}{\sqrt{3}}$$

$$\csc 150° = \frac{1}{\sin 150°} = 2, \quad \sec 150° = \frac{1}{\cos 150°} = -\frac{2}{\sqrt{3}}$$

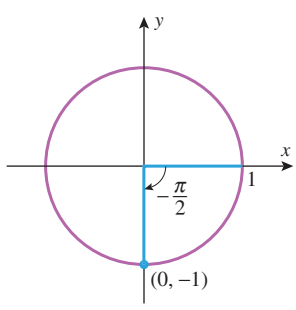
$$\cot 150° = \frac{1}{\tan 150°} = -\sqrt{3} \quad ◀$$

▶ **Example 4** Evaluate the trigonometric functions of $\theta = 5\pi/6$.

Solution. Since $5\pi/6 = 150°$, this problem is equivalent to that of Example 3. From that example we obtain

$$\sin\frac{5\pi}{6} = \frac{1}{2}, \quad \cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \quad \tan\frac{5\pi}{6} = -\frac{1}{\sqrt{3}}$$

$$\csc\frac{5\pi}{6} = 2, \quad \sec\frac{5\pi}{6} = -\frac{2}{\sqrt{3}}, \quad \cot\frac{5\pi}{6} = -\sqrt{3} \quad ◀$$



▲ **Figure B.12**

▶ **Example 5** Evaluate the trigonometric functions of $\theta = -\pi/2$.

Solution. As shown in Figure B.12, the terminal side of $\theta = -\pi/2$ intersects the unit circle at the point $(0, -1)$, so

$$\sin(-\pi/2) = -1, \quad \cos(-\pi/2) = 0$$

and from Formulas (7) through (10),

$$\tan(-\pi/2) = \frac{\sin(-\pi/2)}{\cos(-\pi/2)} = \frac{-1}{0} \quad \text{(undefined)}$$

$$\cot(-\pi/2) = \frac{\cos(-\pi/2)}{\sin(-\pi/2)} = \frac{0}{-1} = 0$$

$$\sec(-\pi/2) = \frac{1}{\cos(-\pi/2)} = \frac{1}{0} \quad \text{(undefined)}$$

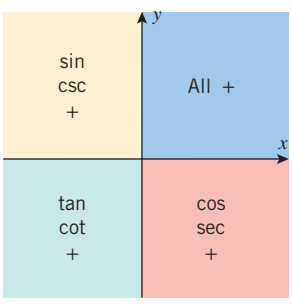
$$\csc(-\pi/2) = \frac{1}{\sin(-\pi/2)} = \frac{1}{-1} = -1 \quad ◀$$

You should be able to obtain all of the results in Table B.3 by the methods illustrated in the last three examples. The dashes indicate quantities that are undefined.

Table B.3

	$\theta = 0$ (0°)	$\pi/6$ (30°)	$\pi/4$ (45°)	$\pi/3$ (60°)	$\pi/2$ (90°)	$2\pi/3$ (120°)	$3\pi/4$ (135°)	$5\pi/6$ (150°)	π (180°)	$3\pi/2$ (270°)	2π (360°)
$\sin\theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0	-1	0
$\cos\theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0	$-1/2$	$-1/\sqrt{2}$	$-\sqrt{3}/2$	-1	0	1
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	—	$-\sqrt{3}$	-1	$-1/\sqrt{3}$	0	—	0
$\csc\theta$	—	2	$\sqrt{2}$	$2/\sqrt{3}$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	—	-1	—
$\sec\theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	—	-2	$-\sqrt{2}$	$-2/\sqrt{3}$	-1	—	1
$\cot\theta$	—	$\sqrt{3}$	1	$1/\sqrt{3}$	0	$-1/\sqrt{3}$	-1	$-\sqrt{3}$	—	0	—

REMARK It is only in special cases that exact values for trigonometric functions can be obtained; usually, a calculating utility or a computer program will be required.



▲ **Figure B.13**

The signs of the trigonometric functions of an angle are determined by the quadrant in which the terminal side of the angle falls. For example, if the terminal side falls in the first quadrant, then x and y are positive in Definition B.1, so all of the trigonometric functions have positive values. If the terminal side falls in the second quadrant, then x is negative and y is positive, so sin and csc are positive, but all other trigonometric functions are negative. The diagram in Figure B.13 shows which trigonometric functions are positive in the various quadrants. You will find it instructive to check that the results in Table B.3 are consistent with Figure B.13.

TRIGONOMETRIC IDENTITIES

A ***trigonometric identity*** is an equation involving trigonometric functions that is true for all angles for which both sides of the equation are defined. One of the most important identities in trigonometry can be derived by applying the Theorem of Pythagoras to the triangle in Figure B.9 to obtain

$$x^2 + y^2 = r^2$$

Dividing both sides by r^2 and using the definitions of $\sin\theta$ and $\cos\theta$ (Definition B.1), we obtain the following fundamental result:

$$\sin^2\theta + \cos^2\theta = 1 \tag{11}$$

The following identities can be obtained from (11) by dividing through by $\cos^2\theta$ and $\sin^2\theta$, respectively, then applying Formulas (7) through (10):

$$\tan^2\theta + 1 = \sec^2\theta \tag{12}$$

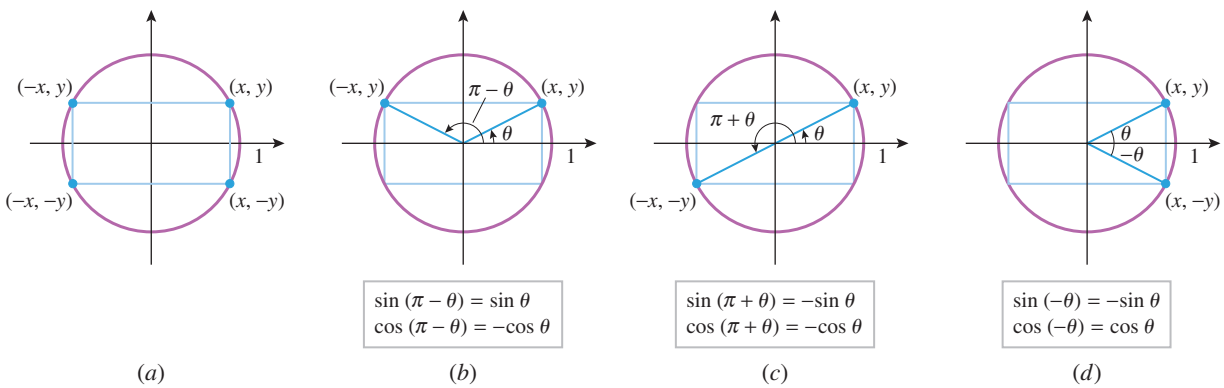
$$1 + \cot^2\theta = \csc^2\theta \tag{13}$$

If (x, y) is a point on the unit circle, then the points $(-x, y)$, $(-x, -y)$, and $(x, -y)$ also lie on the unit circle (why?), and the four points form corners of a rectangle with sides parallel to the coordinate axes (Figure B.14*a*). The x- and y-coordinates of each corner represent the cosine and sine of an angle in standard position whose terminal side passes through the corner; hence we obtain the identities in parts (*b*), (*c*), and (*d*) of Figure B.14 for sine and cosine. Dividing those identities leads to identities for the tangent. In summary:

$$\sin(\pi-\theta) = \sin\theta, \qquad \sin(\pi+\theta) = -\sin\theta, \qquad \sin(-\theta) = -\sin\theta \tag{14–16}$$

$$\cos(\pi-\theta) = -\cos\theta, \qquad \cos(\pi+\theta) = -\cos\theta, \qquad \cos(-\theta) = \cos\theta \tag{17–19}$$

$$\tan(\pi-\theta) = -\tan\theta, \qquad \tan(\pi+\theta) = \tan\theta, \qquad \tan(-\theta) = -\tan\theta \tag{20–22}$$



▲ **Figure B.14**

Two angles in standard position that have the same terminal side must have the same values for their trigonometric functions since their terminal sides intersect the unit circle at the same point. In particular, two angles whose radian measures differ by a multiple of 2π have the same terminal side and hence have the same values for their trigonometric functions. This yields the identities

$$\sin\theta = \sin(\theta + 2\pi) = \sin(\theta - 2\pi) \tag{23}$$

$$\cos\theta = \cos(\theta + 2\pi) = \cos(\theta - 2\pi) \tag{24}$$

and more generally,

$$\sin\theta = \sin(\theta \pm 2n\pi), \quad n = 0, 1, 2, \ldots \tag{25}$$

$$\cos\theta = \cos(\theta \pm 2n\pi), \quad n = 0, 1, 2, \ldots \tag{26}$$

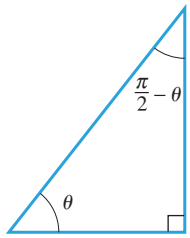
Identity (21) implies that

$$\tan\theta = \tan(\theta + \pi) \qquad \text{and} \qquad \tan\theta = \tan(\theta - \pi) \tag{27–28}$$

Identity (27) is just (21) with the terms in the sum reversed, and identity (28) follows from (21) by substituting $\theta - \pi$ for θ. These two identities state that adding or subtracting π

from an angle does not affect the value of the tangent of the angle. It follows that the same is true for any multiple of π; thus,

$$\tan\theta = \tan(\theta \pm n\pi), \quad n = 0, 1, 2, \ldots \tag{29}$$



▲ **Figure B.15**

Figure B.15 shows complementary angles θ and $(\pi/2) - \theta$ of a right triangle. It follows from (6) that

$$\sin\theta = \frac{\text{side opposite } \theta}{\text{hypotenuse}} = \frac{\text{side adjacent to } (\pi/2) - \theta}{\text{hypotenuse}} = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cos\theta = \frac{\text{side adjacent to } \theta}{\text{hypotenuse}} = \frac{\text{side opposite } (\pi/2) - \theta}{\text{hypotenuse}} = \sin\left(\frac{\pi}{2} - \theta\right)$$

which yields the identities

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta, \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta, \quad \tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta \tag{30–32}$$

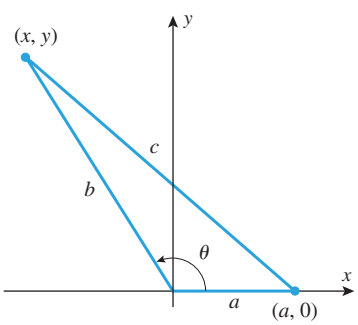
where the third identity results from dividing the first two. These identities are also valid for angles that are not acute and for negative angles as well.

THE LAW OF COSINES

The next theorem, called the ***law of cosines***, generalizes the Theorem of Pythagoras. This result is important in its own right and is also the starting point for some important trigonometric identities.

B.2 THEOREM (*Law of Cosines*) *If the sides of a triangle have lengths a, b, and c, and if θ is the angle between the sides with lengths a and b, then*

$$c^2 = a^2 + b^2 - 2ab\cos\theta$$



▲ **Figure B.16**

PROOF Introduce a coordinate system so that θ is in standard position and the side of length a falls along the positive x-axis. As shown in Figure B.16, the side of length a extends from the origin to $(a, 0)$ and the side of length b extends from the origin to some point (x, y). From the definition of $\sin\theta$ and $\cos\theta$ we have $\sin\theta = y/b$ and $\cos\theta = x/b$, so

$$y = b\sin\theta, \quad x = b\cos\theta \tag{33}$$

From the distance formula in Theorem H.1 of Appendix H, we obtain

$$c^2 = (x - a)^2 + (y - 0)^2$$

so that, from (33),

$$\begin{aligned} c^2 &= (b\cos\theta - a)^2 + b^2\sin^2\theta \\ &= a^2 + b^2(\cos^2\theta + \sin^2\theta) - 2ab\cos\theta \\ &= a^2 + b^2 - 2ab\cos\theta \end{aligned}$$

which completes the proof. ■

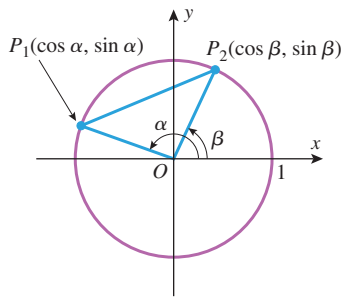
We will now show how the law of cosines can be used to obtain the following identities, called the ***addition formulas*** for sine and cosine:

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta \tag{34}$$

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta \tag{35}$$

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta \tag{36}$$

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta \tag{37}$$



▲ **Figure B.17**

We will derive (37) first. In our derivation we will assume that $0 \le \beta < \alpha < 2\pi$ (Figure B.17). As shown in the figure, the terminal sides of α and β intersect the unit circle at the points $P_1(\cos\alpha, \sin\alpha)$ and $P_2(\cos\beta, \sin\beta)$. If we denote the lengths of the sides of triangle OP_1P_2 by OP_1, P_1P_2, and OP_2, then $OP_1 = OP_2 = 1$ and, from the distance formula in Theorem H.1 of Appendix H,

$$\begin{aligned}(P_1P_2)^2 &= (\cos\beta - \cos\alpha)^2 + (\sin\beta - \sin\alpha)^2\\ &= (\sin^2\alpha + \cos^2\alpha) + (\sin^2\beta + \cos^2\beta) - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta)\\ &= 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta)\end{aligned}$$

But angle $P_2OP_1 = \alpha - \beta$, so that the law of cosines yields

$$\begin{aligned}(P_1P_2)^2 &= (OP_1)^2 + (OP_2)^2 - 2(OP_1)(OP_2)\cos(\alpha - \beta)\\ &= 2 - 2\cos(\alpha - \beta)\end{aligned}$$

Equating the two expressions for $(P_1P_2)^2$ and simplifying, we obtain

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

which completes the derivation of (37).

We can use (31) and (37) to derive (36) as follows:

$$\begin{aligned}\sin(\alpha - \beta) &= \cos\left[\frac{\pi}{2} - (\alpha - \beta)\right] = \cos\left[\left(\frac{\pi}{2} - \alpha\right) - (-\beta)\right]\\ &= \cos\left(\frac{\pi}{2} - \alpha\right)\cos(-\beta) + \sin\left(\frac{\pi}{2} - \alpha\right)\sin(-\beta)\\ &= \cos\left(\frac{\pi}{2} - \alpha\right)\cos\beta - \sin\left(\frac{\pi}{2} - \alpha\right)\sin\beta\\ &= \sin\alpha\cos\beta - \cos\alpha\sin\beta\end{aligned}$$

Identities (34) and (35) can be obtained from (36) and (37) by substituting $-\beta$ for β and using the identities

$$\sin(-\beta) = -\sin\beta, \quad \cos(-\beta) = \cos\beta$$

We leave it for you to derive the identities

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} \qquad \tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} \tag{38–39}$$

Identity (38) can be obtained by dividing (34) by (35) and then simplifying. Identity (39) can be obtained from (38) by substituting $-\beta$ for β and simplifying.

In the special case where $\alpha = \beta$, identities (34), (35), and (38) yield the ***double-angle formulas***

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \tag{40}$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha \tag{41}$$

$$\tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} \tag{42}$$

By using the identity $\sin^2\alpha + \cos^2\alpha = 1$, (41) can be rewritten in the alternative forms

$$\cos 2\alpha = 2\cos^2\alpha - 1 \qquad \text{and} \qquad \cos 2\alpha = 1 - 2\sin^2\alpha \tag{43–44}$$

If we replace α by $\alpha/2$ in (43) and (44) and use some algebra, we obtain the ***half-angle formulas***

$$\cos^2\frac{\alpha}{2} = \frac{1+\cos\alpha}{2} \qquad \text{and} \qquad \sin^2\frac{\alpha}{2} = \frac{1-\cos\alpha}{2} \tag{45–46}$$

We leave it for the exercises to derive the following ***product-to-sum formulas*** from (34) through (37):

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha-\beta)+\sin(\alpha+\beta)] \tag{47}$$

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)] \tag{48}$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)] \tag{49}$$

We also leave it for the exercises to derive the following ***sum-to-product formulas***:

$$\sin\alpha+\sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} \tag{50}$$

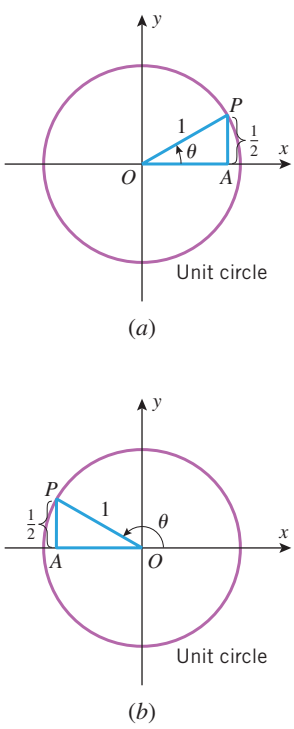
$$\sin\alpha-\sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} \tag{51}$$

$$\cos\alpha+\cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} \tag{52}$$

$$\cos\alpha-\cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} \tag{53}$$

FINDING AN ANGLE FROM THE VALUE OF ITS TRIGONOMETRIC FUNCTIONS

There are numerous situations in which it is necessary to find an unknown angle from a known value of one of its trigonometric functions. The following example illustrates a method for doing this.



▲ **Figure B.18**

▶ **Example 6** Find θ if $\sin\theta = \frac{1}{2}$.

Solution. We begin by looking for positive angles that satisfy the equation. Because $\sin\theta$ is positive, the angle θ must terminate in the first or second quadrant. If it terminates in the first quadrant, then the hypotenuse of $\triangle OAP$ in Figure B.18*a* is double the leg AP, so

$$\theta = 30^\circ = \frac{\pi}{6}\text{ radians}$$

If θ terminates in the second quadrant (Figure B.18*b*), then the hypotenuse of $\triangle OAP$ is double the leg AP, so $\angle AOP = 30^\circ$, which implies that

$$\theta = 180^\circ - 30^\circ = 150^\circ = \frac{5\pi}{6}\text{ radians}$$

Now that we have found these two solutions, all other solutions are obtained by adding or subtracting multiples of 360° (2π radians) to or from them. Thus, the entire set of solutions is given by the formulas

$$\theta = 30^\circ \pm n\cdot 360^\circ, \quad n = 0, 1, 2, \ldots$$

and

$$\theta = 150^\circ \pm n \cdot 360^\circ, \quad n = 0, 1, 2, \ldots$$

or in radian measure,

$$\theta = \frac{\pi}{6} \pm n \cdot 2\pi, \quad n = 0, 1, 2, \ldots$$

and

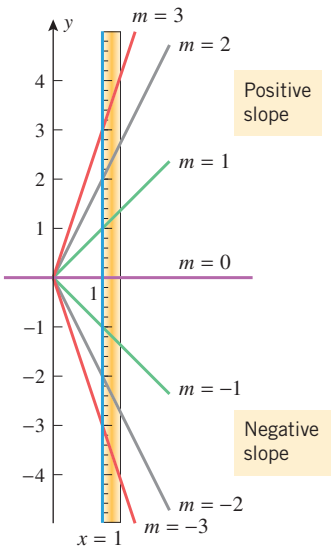
$$\theta = \frac{5\pi}{6} \pm n \cdot 2\pi, \quad n = 0, 1, 2, \ldots \blacktriangleleft$$

ANGLE OF INCLINATION

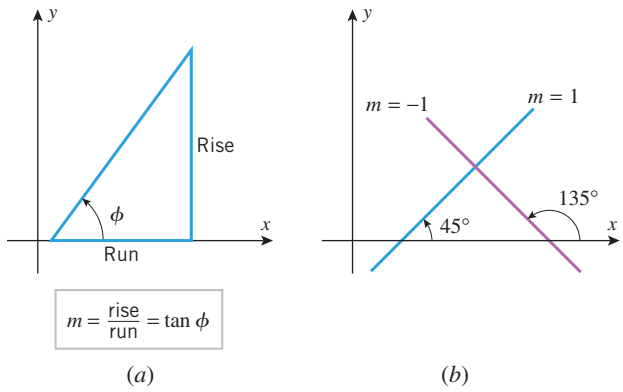
The slope of a nonvertical line L is related to the angle that L makes with the positive x-axis. If ϕ is the smallest positive angle measured counterclockwise from the x-axis to L, then the slope of the line can be expressed as

$$m = \tan\phi \tag{54}$$

(Figure B.19*a*). The angle ϕ, which is called the ***angle of inclination*** of the line, satisfies $0^\circ \le \phi < 180^\circ$ in degree measure (or, equivalently, $0 \le \phi < \pi$ in radian measure). If ϕ is an acute angle, then $m = \tan\phi$ is positive and the line slopes up to the right, and if ϕ is an obtuse angle, then $m = \tan\phi$ is negative and the line slopes down to the right. For example, a line whose angle of inclination is 45° has slope $m = \tan 45^\circ = 1$, and a line whose angle of inclination is 135° has a slope of $m = \tan 135^\circ = -1$ (Figure B.19*b*). Figure B.20 shows a convenient way of using the line $x = 1$ as a "ruler" for visualizing the relationship between lines of various slopes.



▲ **Figure B.20**



▲ **Figure B.19**

EXERCISE SET B

1–2 Express the angles in radians. ■

1. (a) 75° (b) 390° (c) 20° (d) 138°

2. (a) 420° (b) 15° (c) 225° (d) 165°

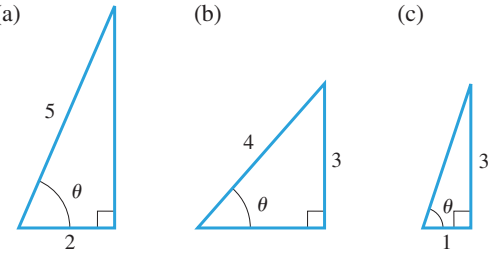
3–4 Express the angles in degrees. ■

3. (a) $\pi/15$ (b) 1.5 (c) $8\pi/5$ (d) 3π

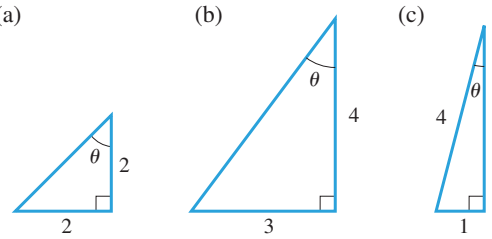
4. (a) $\pi/10$ (b) 2 (c) $2\pi/5$ (d) $7\pi/6$

5–6 Find the exact values of all six trigonometric functions of θ. ■

5. (a) (b) (c)



6. (a) (b) (c)



7–12 The angle θ is an acute angle of a right triangle. Solve the problems by drawing an appropriate right triangle. Do *not* use a calculator.

7. Find $\sin\theta$ and $\cos\theta$ given that $\tan\theta = 3$.

8. Find $\sin\theta$ and $\tan\theta$ given that $\cos\theta = \frac{2}{3}$.

9. Find $\tan\theta$ and $\csc\theta$ given that $\sec\theta = \frac{5}{2}$.

10. Find $\cot\theta$ and $\sec\theta$ given that $\csc\theta = 4$.

11. Find the length of the side adjacent to θ given that the hypotenuse has length 6 and $\cos\theta = 0.3$.

12. Find the length of the hypotenuse given that the side opposite θ has length 2.4 and $\sin\theta = 0.8$.

13–14 The value of an angle θ is given. Find the values of all six trigonometric functions of θ without using a calculator.

13. (a) $225°$ (b) $-210°$ (c) $5\pi/3$ (d) $-3\pi/2$

14. (a) $330°$ (b) $-120°$ (c) $9\pi/4$ (d) -3π

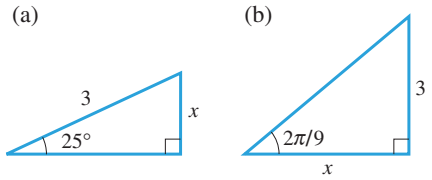
15–16 Use the information to find the exact values of the remaining five trigonometric functions of θ.

15. (a) $\cos\theta = \frac{3}{5}$, $0 < \theta < \pi/2$
(b) $\cos\theta = \frac{3}{5}$, $-\pi/2 < \theta < 0$
(c) $\tan\theta = -1/\sqrt{3}$, $\pi/2 < \theta < \pi$
(d) $\tan\theta = -1/\sqrt{3}$, $-\pi/2 < \theta < 0$
(e) $\csc\theta = \sqrt{2}$, $0 < \theta < \pi/2$
(f) $\csc\theta = \sqrt{2}$, $\pi/2 < \theta < \pi$

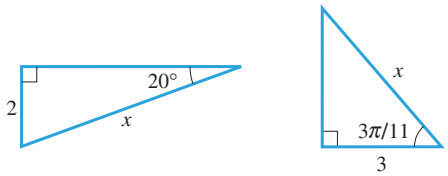
16. (a) $\sin\theta = \frac{1}{4}$, $0 < \theta < \pi/2$
(b) $\sin\theta = \frac{1}{4}$, $\pi/2 < \theta < \pi$
(c) $\cot\theta = \frac{1}{3}$, $0 < \theta < \pi/2$
(d) $\cot\theta = \frac{1}{3}$, $\pi < \theta < 3\pi/2$
(e) $\sec\theta = -\frac{5}{2}$, $\pi/2 < \theta < \pi$
(f) $\sec\theta = -\frac{5}{2}$, $\pi < \theta < 3\pi/2$

17–18 Use a calculating utility to find x to four decimal places.

17. (a) (b)



18. (a) (b)



19. In each part, let θ be an acute angle of a right triangle. Express the remaining five trigonometric functions in terms of a.
(a) $\sin\theta = a/3$ (b) $\tan\theta = a/5$ (c) $\sec\theta = a$

20–27 Find all values of θ (in radians) that satisfy the given equation. Do not use a calculator.

20. (a) $\cos\theta = -1/\sqrt{2}$ (b) $\sin\theta = -1/\sqrt{2}$

21. (a) $\tan\theta = -1$ (b) $\cos\theta = \frac{1}{2}$

22. (a) $\sin\theta = -\frac{1}{2}$ (b) $\tan\theta = \sqrt{3}$

23. (a) $\tan\theta = 1/\sqrt{3}$ (b) $\sin\theta = -\sqrt{3}/2$

24. (a) $\sin\theta = -1$ (b) $\cos\theta = -1$

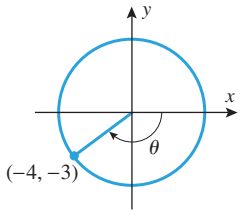
25. (a) $\cot\theta = -1$ (b) $\cot\theta = \sqrt{3}$

26. (a) $\sec\theta = -2$ (b) $\csc\theta = -2$

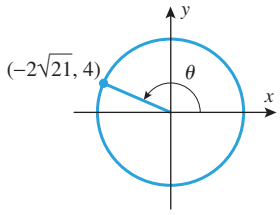
27. (a) $\csc\theta = 2/\sqrt{3}$ (b) $\sec\theta = 2/\sqrt{3}$

28–29 Find the values of all six trigonometric functions of θ.

28.



29.



30. Find all values of θ (in radians) such that
(a) $\sin\theta = 1$ (b) $\cos\theta = 1$ (c) $\tan\theta = 1$
(d) $\csc\theta = 1$ (e) $\sec\theta = 1$ (f) $\cot\theta = 1$.

31. Find all values of θ (in radians) such that
(a) $\sin\theta = 0$ (b) $\cos\theta = 0$ (c) $\tan\theta = 0$
(d) $\csc\theta$ is undefined (e) $\sec\theta$ is undefined
(f) $\cot\theta$ is undefined.

32. How could you use a ruler and protractor to approximate $\sin 17°$ and $\cos 17°$?

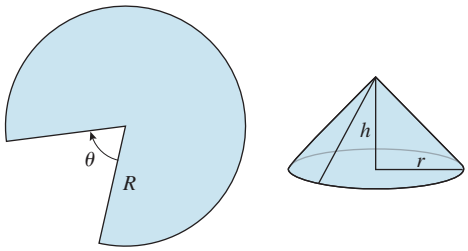
33. Find the length of the circular arc on a circle of radius 4 cm subtended by an angle of
(a) $\pi/6$ (b) $150°$.

34. Find the radius of a circular sector that has an angle of $\pi/3$ and a circular arc length of 7 units.

35. A point P moving counterclockwise on a circle of radius 5 cm traverses an arc length of 2 cm. What is the angle swept out by a radius from the center to P?

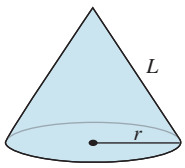
36. Find a formula for the area A of a circular sector in terms of its radius r and arc length s.

37. As shown in the accompanying figure, a right circular cone is made from a circular piece of paper of radius R by cutting out a sector of angle θ radians and gluing the cut edges of the remaining piece together. Find
(a) the radius r of the base of the cone in terms of R and θ
(b) the height h of the cone in terms of R and θ.



▲ **Figure Ex-37**

38. As shown in the accompanying figure, let r and L be the radius of the base and the slant height of a right circular cone. Show that the lateral surface area, S, of the cone is $S = \pi r L$. [*Hint:* As shown in the figure in Exercise 37, the lateral surface of the cone becomes a circular sector when cut along a line from the vertex to the base and flattened.]



◀ **Figure Ex-38**

39. Two sides of a triangle have lengths of 3 cm and 7 cm and meet at an angle of $60°$. Find the area of the triangle.

40. Let ABC be a triangle whose angles at A and B are $30°$ and $45°$. If the side opposite the angle B has length 9, find the lengths of the remaining sides and the size of the angle C.

41. A 10-foot ladder leans against a house and makes an angle of $67°$ with level ground. How far is the top of the ladder above the ground? Express your answer to the nearest tenth of a foot.

42. From a point 120 feet on level ground from a building, the angle of elevation to the top of the building is $76°$. Find the height of the building. Express your answer to the nearest foot.

43. An observer on level ground is at a distance d from a building. The angles of elevation to the bottoms of the windows on the second and third floors are α and β, respectively. Find the distance h between the bottoms of the windows in terms of α, β, and d.

44. From a point on level ground, the angle of elevation to the top of a tower is α. From a point that is d units closer to the tower, the angle of elevation is β. Find the height h of the tower in terms of α, β, and d.

45–46 Do *not* use a calculator in these exercises. ■

45. If $\cos\theta = \frac{2}{3}$ and $0 < \theta < \pi/2$, find
(a) $\sin 2\theta$ (b) $\cos 2\theta$.

46. If $\tan\alpha = \frac{3}{4}$ and $\tan\beta = 2$, where $0 < \alpha < \pi/2$ and $0 < \beta < \pi/2$, find
(a) $\sin(\alpha - \beta)$ (b) $\cos(\alpha + \beta)$.

47. Express $\sin 3\theta$ and $\cos 3\theta$ in terms of $\sin\theta$ and $\cos\theta$.

48–58 Derive the given identities. ■

48. $\dfrac{\cos\theta\sec\theta}{1+\tan^2\theta} = \cos^2\theta$

49. $\dfrac{\cos\theta\tan\theta + \sin\theta}{\tan\theta} = 2\cos\theta$

50. $2\csc 2\theta = \sec\theta\csc\theta$

51. $\tan\theta + \cot\theta = 2\csc 2\theta$

52. $\dfrac{\sin 2\theta}{\sin\theta} - \dfrac{\cos 2\theta}{\cos\theta} = \sec\theta$

53. $\dfrac{\sin\theta + \cos 2\theta - 1}{\cos\theta - \sin 2\theta} = \tan\theta$

54. $\sin 3\theta + \sin\theta = 2\sin 2\theta\cos\theta$

55. $\sin 3\theta - \sin\theta = 2\cos 2\theta\sin\theta$

56. $\tan\dfrac{\theta}{2} = \dfrac{1-\cos\theta}{\sin\theta}$

57. $\tan\dfrac{\theta}{2} = \dfrac{\sin\theta}{1+\cos\theta}$

58. $\cos\left(\dfrac{\pi}{3}+\theta\right) + \cos\left(\dfrac{\pi}{3}-\theta\right) = \cos\theta$

59–60 In these exercises, refer to an arbitrary triangle ABC in which the side of length a is opposite angle A, the side of length b is opposite angle B, and the side of length c is opposite angle C. ■

59. Prove: The area of a triangle ABC can be written as

$$\text{area} = \tfrac{1}{2}bc\sin A$$

Find two other similar formulas for the area.

60. Prove the ***law of sines***: In any triangle, the ratios of the sides to the sines of the opposite angles are equal; that is,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

61. Use identities (34) through (37) to express each of the following in terms of $\sin\theta$ or $\cos\theta$.
(a) $\sin\left(\dfrac{\pi}{2}+\theta\right)$ (b) $\cos\left(\dfrac{\pi}{2}+\theta\right)$
(c) $\sin\left(\dfrac{3\pi}{2}-\theta\right)$ (d) $\cos\left(\dfrac{3\pi}{2}+\theta\right)$

62. Derive identities (38) and (39).

63. Derive identity
(a) (47) (b) (48) (c) (49).

64. If $A = \alpha + \beta$ and $B = \alpha - \beta$, then $\alpha = \frac{1}{2}(A+B)$ and $\beta = \frac{1}{2}(A-B)$ (verify). Use this result and identities (47) through (49) to derive identity
(a) (50) (b) (52) (c) (53).

65. Substitute $-\beta$ for β in identity (50) to derive identity (51).

66. (a) Express $3\sin\alpha + 5\cos\alpha$ in the form

$$C\sin(\alpha + \phi)$$

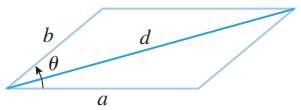
(b) Show that a sum of the form

$$A\sin\alpha + B\cos\alpha$$

can be rewritten in the form $C\sin(\alpha + \phi)$.

67. Show that the length of the diagonal of the parallelogram in the accompanying figure is

$$d = \sqrt{a^2 + b^2 + 2ab\cos\theta}$$



◀ **Figure Ex-67**

68–69 Find the angle of inclination of the line with slope m to the nearest degree. Use a calculating utility, where needed. ■

68. (a) $m = \frac{1}{2}$ (b) $m = -1$
(c) $m = 2$ (d) $m = -57$

69. (a) $m = -\frac{1}{2}$ (b) $m = 1$
(c) $m = -2$ (d) $m = 57$

70–71 Find the angle of inclination of the line to the nearest degree. Use a calculating utility, where needed. ■

70. (a) $3y = 2 - \sqrt{3}x$ (b) $y - 4x + 7 = 0$

71. (a) $y = \sqrt{3}x + 2$ (b) $y + 2x + 5 = 0$

C

SOLVING POLYNOMIAL EQUATIONS

We will assume in this appendix that you know how to divide polynomials using long division and synthetic division. If you need to review those techniques, refer to an algebra book.

A BRIEF REVIEW OF POLYNOMIALS

Recall that if n is a nonnegative integer, then a ***polynomial of degree n*** is a function that can be written in the following forms, depending on whether you want the powers of x in ascending or descending order:

$$c_0 + c_1x + c_2x^2 + \cdots + c_nx^n \quad (c_n \neq 0)$$
$$c_nx^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0 \quad (c_n \neq 0)$$

The numbers $c_0, c_1, \ldots, c_n$ are called the ***coefficients*** of the polynomial. The coefficient c_n (which multiplies the highest power of x) is called the ***leading coefficient***, the term c_nx^n is called the ***leading term***, and the coefficient c_0 is called the ***constant term***. Polynomials of degree 1, 2, 3, 4, and 5 are called ***linear***, ***quadratic***, ***cubic***, ***quartic***, and ***quintic***, respectively. For simplicity, general polynomials of low degree are often written without subscripts on the coefficients:

$$p(x) = a \qquad \text{Constant polynomial}$$
$$p(x) = ax + b \quad (a \neq 0) \qquad \text{Linear polynomial}$$
$$p(x) = ax^2 + bx + c \quad (a \neq 0) \qquad \text{Quadratic polynomial}$$
$$p(x) = ax^3 + bx^2 + cx + d \quad (a \neq 0) \qquad \text{Cubic polynomial}$$

When you attempt to factor a polynomial completely, one of three things can happen:

- You may be able to decompose the polynomial into distinct linear factors using only real numbers; for example,

$$x^3 + x^2 - 2x = x(x^2 + x - 2) = x(x - 1)(x + 2)$$

- You may be able to decompose the polynomial into linear factors using only real numbers, but some of the factors may be repeated; for example,

$$x^6 - 3x^4 + 2x^3 = x^3(x^3 - 3x + 2) = x^3(x - 1)^2(x + 2) \tag{1}$$

- You may be able to decompose the polynomial into linear and quadratic factors using only real numbers, but you may not be able to decompose the quadratic factors into linear factors using only real numbers (such quadratic factors are said to be ***irreducible*** over the real numbers); for example,

$$\begin{aligned} x^4 - 1 = (x^2 - 1)(x^2 + 1) &= (x - 1)(x + 1)(x^2 + 1) \\ &= (x - 1)(x + 1)(x - i)(x + i) \end{aligned}$$

Here, the factor $x^2 + 1$ is irreducible over the real numbers.

In general, if $p(x)$ is a polynomial of degree n with leading coefficient a, and if complex numbers are allowed, then $p(x)$ can be factored as

$$p(x) = a(x - r_1)(x - r_2) \cdots (x - r_n) \tag{2}$$

where $r_1, r_2, \ldots, r_n$ are called the ***zeros*** of $p(x)$ or the ***roots*** of the equation $p(x) = 0$, and (2) is called the ***complete linear factorization*** of $p(x)$. If some of the factors in (2) are repeated, then they can be combined; for example, if the first k factors are distinct and the rest are repetitions of the first k, then (2) can be expressed in the form

$$p(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots (x - r_k)^{m_k} \tag{3}$$

where $r_1, r_2, \ldots, r_k$ are the *distinct* roots of $p(x) = 0$. The exponents $m_1, m_2, \ldots, m_k$ tell us how many times the various factors occur in the complete linear factorization; for example, in (3) the factor $(x - r_1)$ occurs m_1 times, the factor $(x - r_2)$ occurs m_2 times, and so forth. Some techniques for factoring polynomials are discussed later in this appendix. In general, if a factor $(x - r)$ occurs m times in the complete linear factorization of a polynomial, then we say that r is a root or zero of ***multiplicity m***, and if $(x - r)$ has no repetitions (i.e., r has multiplicity 1), then we say that r is a ***simple*** root or zero. For example, it follows from (1) that the equation $x^6 - 3x^4 + 2x^3 = 0$ can be expressed as

$$x^3(x - 1)^2(x + 2) = 0 \tag{4}$$

so this equation has three distinct roots—a root $x = 0$ of multiplicity 3, a root $x = 1$ of multiplicity 2, and a simple root $x = -2$.

Note that in (3) the multiplicities of the roots must add up to n, since $p(x)$ has degree n; that is,

$$m_1 + m_2 + \cdots + m_k = n$$

For example, in (4) the multiplicities add up to 6, which is the same as the degree of the polynomial.

It follows from (2) that a polynomial of degree n can have at most n distinct roots; if all of the roots are simple, then there will be *exactly* n, but if some are repeated, then there will be fewer than n. However, when counting the roots of a polynomial, it is standard practice to count multiplicities, since that convention allows us to say that a polynomial of degree n has n roots. For example, from (1) the six roots of the polynomial $p(x) = x^6 - 3x^4 + 2x^3$ are

$$r = 0, \quad 0, \quad 0, \quad 1, \quad 1, \quad -2$$

In summary, we have the following important theorem.

C.1 THEOREM *If complex roots are allowed, and if roots are counted according to their multiplicities, then a polynomial of degree n has exactly n roots.*

THE REMAINDER THEOREM

When two positive integers are divided, the numerator can be expressed as the quotient plus the remainder over the divisor, where the remainder is less than the divisor. For example,

$$\tfrac{17}{5} = 3 + \tfrac{2}{5}$$

If we multiply this equation through by 5, we obtain

$$17 = 5 \cdot 3 + 2$$

which states that the *numerator is the divisor times the quotient plus the remainder*.

The following theorem, which we state without proof, is an analogous result for division of polynomials.

C.2 THEOREM *If $p(x)$ and $s(x)$ are polynomials, and if $s(x)$ is not the zero polynomial, then $p(x)$ can be expressed as*

$$p(x) = s(x)q(x) + r(x)$$

where $q(x)$ and $r(x)$ are the quotient and remainder that result when $p(x)$ is divided by $s(x)$, and either $r(x)$ is the zero polynomial or the degree of $r(x)$ is less than the degree of $s(x)$.

In the special case where $p(x)$ is divided by a first-degree polynomial of the form $x - c$, the remainder must be some constant r, since it is either zero or has degree less than 1. Thus, Theorem C.2 implies that

$$p(x) = (x - c)q(x) + r$$

and this in turn implies that $p(c) = r$. In summary, we have the following theorem.

C.3 THEOREM (***Remainder Theorem***) *If a polynomial $p(x)$ is divided by $x - c$, then the remainder is $p(c)$.*

▶ **Example 1** According to the Remainder Theorem, the remainder on dividing

$$p(x) = 2x^3 + 3x^2 - 4x - 3$$

by $x + 4$ should be

$$p(-4) = 2(-4)^3 + 3(-4)^2 - 4(-4) - 3 = -67$$

Show that this is so.

Solution. By long division

$$\begin{array}{r|l}
 & \phantom{2x^3 + {}} 2x^2 - 5x + 16 \\ \hline
x + 4 & 2x^3 + 3x^2 - 4x - 3 \\
 & \underline{2x^3 + 8x^2} \\
 & \phantom{2x^3 + {}} -5x^2 - 4x \\
 & \phantom{2x^3 + {}} \underline{-5x^2 - 20x} \\
 & \phantom{2x^3 + {}-5x^2 - {}} 16x - 3 \\
 & \phantom{2x^3 + {}-5x^2 - {}} \underline{16x + 64} \\
 & \phantom{2x^3 + {}-5x^2 - {}16x - {}} -67
\end{array}$$

which shows that the remainder is -67.

Alternative Solution. Because we are dividing by an expression of the form $x - c$ (where $c = -4$), we can use synthetic division rather than long division. The computations are

$$\begin{array}{r|rrrr}
\underline{-4} & 2 & 3 & -4 & -3 \\
 & & -8 & 20 & -64 \\ \hline
 & 2 & -5 & 16 & -67
\end{array}$$

which again shows that the remainder is -67. ◀

THE FACTOR THEOREM

To ***factor*** a polynomial $p(x)$ is to write it as a product of lower-degree polynomials, called ***factors*** of $p(x)$. For $s(x)$ to be a factor of $p(x)$ there must be no remainder when $p(x)$ is divided by $s(x)$. For example, if $p(x)$ can be factored as

$$p(x) = s(x)q(x) \tag{5}$$

then

$$\frac{p(x)}{s(x)} = q(x) \tag{6}$$

so dividing $p(x)$ by $s(x)$ produces a quotient $q(x)$ with no remainder. Conversely, (6) implies (5), so $s(x)$ is a factor of $p(x)$ if there is no remainder when $p(x)$ is divided by $s(x)$.

In the special case where $x - c$ is a factor of $p(x)$, the polynomial $p(x)$ can be expressed as

$$p(x) = (x - c)q(x)$$

which implies that $p(c) = 0$. Conversely, if $p(c) = 0$, then the Remainder Theorem implies that $x - c$ is a factor of $p(x)$, since the remainder is 0 when $p(x)$ is divided by $x - c$. These results are summarized in the following theorem.

C.4 THEOREM (*Factor Theorem*) *A polynomial $p(x)$ has a factor $x - c$ if and only if $p(c) = 0$.*

It follows from this theorem that the statements below say the same thing in different ways:

- $x - c$ is a factor of $p(x)$.
- $p(c) = 0$.
- c is a zero of $p(x)$.
- c is a root of the equation $p(x) = 0$.
- c is a solution of the equation $p(x) = 0$.
- c is an x-intercept of $y = p(x)$.

▶ Example 2 Confirm that $x - 1$ is a factor of

$$p(x) = x^3 - 3x^2 - 13x + 15$$

by dividing $x - 1$ into $p(x)$ and checking that the remainder is zero.

Solution. By long division

$$\begin{array}{r|l}
 & \phantom{x^3 - {}} x^2 - 2x - 15 \\
\hline
x - 1 & x^3 - 3x^2 - 13x + 15 \\
 & \underline{x^3 - x^2} \\
 & \phantom{x^3 - {}} -2x^2 - 13x \\
 & \phantom{x^3 - {}} \underline{-2x^2 + 2x} \\
 & \phantom{x^3 - 3x^2 - {}} -15x + 15 \\
 & \phantom{x^3 - 3x^2 - {}} \underline{-15x + 15} \\
 & \phantom{x^3 - 3x^2 - 13x - {}} 0
\end{array}$$

which shows that the remainder is zero.

Alternative Solution. Because we are dividing by an expression of the form $x - c$, we can use synthetic division rather than long division. The computations are

$$\begin{array}{r|rrrr} \underline{1} & 1 & -3 & -13 & 15 \\ & & 1 & -2 & -15 \\ \hline & 1 & -2 & -15 & 0 \end{array}$$

which again confirms that the remainder is zero. ◀

USING ONE FACTOR TO FIND OTHER FACTORS

If $x - c$ is a factor of $p(x)$, and if $q(x) = p(x)/(x - c)$, then

$$p(x) = (x - c)q(x) \tag{7}$$

so that additional linear factors of $p(x)$ can be obtained by factoring the quotient $q(x)$.

▶ **Example 3** Factor

$$p(x) = x^3 - 3x^2 - 13x + 15 \tag{8}$$

completely into linear factors.

Solution. We showed in Example 2 that $x - 1$ is a factor of $p(x)$ and we also showed that $p(x)/(x - 1) = x^2 - 2x - 15$. Thus,

$$x^3 - 3x^2 - 13x + 15 = (x - 1)(x^2 - 2x - 15)$$

Factoring $x^2 - 2x - 15$ by inspection yields

$$x^3 - 3x^2 - 13x + 15 = (x - 1)(x - 5)(x + 3)$$

which is the complete linear factorization of $p(x)$. ◀

METHODS FOR FINDING ROOTS

A general quadratic equation $ax^2 + bx + c = 0$ can be solved by using the quadratic formula to express the solutions of the equation in terms of the coefficients. Versions of this formula were known since Babylonian times, and by the seventeenth century formulas had been obtained for solving general cubic and quartic equations. However, attempts to find formulas for the solutions of general fifth-degree equations and higher proved fruitless. The reason for this became clear in 1829 when the French mathematician Evariste Galois (1811–1832) proved that it is impossible to express the solutions of a general fifth-degree equation or higher in terms of its coefficients using algebraic operations.

Today, we have powerful computer programs for finding the zeros of specific polynomials. For example, it takes only seconds for a computer algebra system, such as *Mathematica* or *Maple*, to show that the zeros of the polynomial

$$p(x) = 10x^4 - 23x^3 - 10x^2 + 29x + 6 \tag{9}$$

are

$$x = -1, \quad x = -\tfrac{1}{5}, \quad x = \tfrac{3}{2}, \quad \text{and} \quad x = 2 \tag{10}$$

The algorithms that these programs use to find the integer and rational zeros of a polynomial, if any, are based on the following theorem, which is proved in advanced algebra courses.

C.5 THEOREM *Suppose that*

$$p(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$$

is a polynomial with integer coefficients.

(*a*) *If r is an integer zero of $p(x)$, then r must be a divisor of the constant term c_0.*

(*b*) *If $r = a/b$ is a rational zero of $p(x)$ in which all common factors of a and b have been canceled, then a must be a divisor of the constant term c_0, and b must be a divisor of the leading coefficient c_n.*

For example, in (9) the constant term is 6 (which has divisors ± 1, ± 2, ± 3, and ± 6) and the leading coefficient is 10 (which has divisors ± 1, ± 2, ± 5, and ± 10). Thus, the only possible integer zeros of $p(x)$ are

$$\pm 1, \quad \pm 2, \quad \pm 3, \quad \pm 6$$

and the only possible noninteger rational zeros are

$$\pm\tfrac{1}{2}, \quad \pm\tfrac{1}{5}, \quad \pm\tfrac{1}{10}, \quad \pm\tfrac{2}{5}, \quad \pm\tfrac{3}{2}, \quad \pm\tfrac{3}{5}, \quad \pm\tfrac{3}{10}, \quad \pm\tfrac{6}{5}$$

Using a computer, it is a simple matter to evaluate $p(x)$ at each of the numbers in these lists to show that its only rational zeros are the numbers in (10).

▶ **Example 4** Solve the equation $x^3 + 3x^2 - 7x - 21 = 0$.

Solution. The solutions of the equation are the zeros of the polynomial

$$p(x) = x^3 + 3x^2 - 7x - 21$$

We will look for integer zeros first. All such zeros must divide the constant term, so the only possibilities are ± 1, ± 3, ± 7, and ± 21. Substituting these values into $p(x)$ (or using the method of Exercise 6) shows that $x = -3$ is an integer zero. This tells us that $x + 3$ is a factor of $p(x)$ and that $p(x)$ can be written as

$$x^3 + 3x^2 - 7x - 21 = (x + 3)q(x)$$

where $q(x)$ is the quotient that results when $x^3 + 3x^2 - 7x - 21$ is divided by $x + 3$. We leave it for you to perform the division and show that $q(x) = x^2 - 7$; hence,

$$x^3 + 3x^2 - 7x - 21 = (x + 3)(x^2 - 7) = (x + 3)(x + \sqrt{7})(x - \sqrt{7})$$

which tells us that the solutions of the given equation are $x = 3$, $x = \sqrt{7} \approx 2.65$, and $x = -\sqrt{7} \approx -2.65$. ◀

EXERCISE SET C

C CAS

1–2 Find the quotient $q(x)$ and the remainder $r(x)$ that result when $p(x)$ is divided by $s(x)$. ■

1. (a) $p(x) = x^4 + 3x^3 - 5x + 10$; $s(x) = x^2 - x + 2$
(b) $p(x) = 6x^4 + 10x^2 + 5$; $s(x) = 3x^2 - 1$
(c) $p(x) = x^5 + x^3 + 1$; $s(x) = x^2 + x$

2. (a) $p(x) = 2x^4 - 3x^3 + 5x^2 + 2x + 7$; $s(x) = x^2 - x + 1$
(b) $p(x) = 2x^5 + 5x^4 - 4x^3 + 8x^2 + 1$; $s(x) = 2x^2 - x + 1$
(c) $p(x) = 5x^6 + 4x^2 + 5$; $s(x) = x^3 + 1$

3–4 Use synthetic division to find the quotient $q(x)$ and the remainder $r(x)$ that result when $p(x)$ is divided by $s(x)$. ■

3. (a) $p(x) = 3x^3 - 4x - 1$; $s(x) = x - 2$
(b) $p(x) = x^4 - 5x^2 + 4$; $s(x) = x + 5$
(c) $p(x) = x^5 - 1$; $s(x) = x - 1$

4. (a) $p(x) = 2x^3 - x^2 - 2x + 1$; $s(x) = x - 1$
(b) $p(x) = 2x^4 + 3x^3 - 17x^2 - 27x - 9$; $s(x) = x + 4$
(c) $p(x) = x^7 + 1$; $s(x) = x - 1$

5. Let $p(x) = 2x^4 + x^3 - 3x^2 + x - 4$. Use synthetic division and the Remainder Theorem to find $p(0)$, $p(1)$, $p(-3)$, and $p(7)$.

6. Let $p(x)$ be the polynomial in Example 4. Use synthetic division and the Remainder Theorem to evaluate $p(x)$ at $x = \pm 1, \pm 3, \pm 7$, and ± 21.

7. Let $p(x) = x^3 + 4x^2 + x - 6$. Find a polynomial $q(x)$ and a constant r such that
(a) $p(x) = (x - 2)q(x) + r$
(b) $p(x) = (x + 1)q(x) + r$.

8. Let $p(x) = x^5 - 1$. Find a polynomial $q(x)$ and a constant r such that
(a) $p(x) = (x + 1)q(x) + r$
(b) $p(x) = (x - 1)q(x) + r$.

9. In each part, make a list of all possible candidates for the rational zeros of $p(x)$.
(a) $p(x) = x^7 + 3x^3 - x + 24$
(b) $p(x) = 3x^4 - 2x^2 + 7x - 10$
(c) $p(x) = x^{35} - 17$

10. Find all integer zeros of

$$p(x) = x^6 + 5x^5 - 16x^4 - 15x^3 - 12x^2 - 38x - 21$$

11–15 Factor the polynomials completely. ■

11. $p(x) = x^3 - 2x^2 - x + 2$

12. $p(x) = 3x^3 + x^2 - 12x - 4$

13. $p(x) = x^4 + 10x^3 + 36x^2 + 54x + 27$

14. $p(x) = 2x^4 + x^3 + 3x^2 + 3x - 9$

15. $p(x) = x^5 + 4x^4 - 4x^3 - 34x^2 - 45x - 18$

C **16.** For each of the factorizations that you obtained in Exercises 11–15, check your answer using a CAS.

17–21 Find all real solutions of the equations. ■

17. $x^3 + 3x^2 + 4x + 12 = 0$

18. $2x^3 - 5x^2 - 10x + 3 = 0$

19. $3x^4 + 14x^3 + 14x^2 - 8x - 8 = 0$

20. $2x^4 - x^3 - 14x^2 - 5x + 6 = 0$

21. $x^5 - 2x^4 - 6x^3 + 5x^2 + 8x + 12 = 0$

C **22.** For each of the equations you solved in Exercises 17–21, check your answer using a CAS.

23. Find all values of k for which $x - 1$ is a factor of the polynomial $p(x) = k^2x^3 - 7kx + 10$.

24. Is $x + 3$ a factor of $x^7 + 2187$? Justify your answer.

C **25.** A 3 cm thick slice is cut from a cube, leaving a volume of 196 cm^3. Use a CAS to find the length of a side of the original cube.

26. (a) Show that there is no positive rational number that exceeds its cube by 1.
(b) Does there exist a real number that exceeds its cube by 1? Justify your answer.

27. Use the Factor Theorem to show each of the following.
(a) $x - y$ is a factor of $x^n - y^n$ for all positive integer values of n.
(b) $x + y$ is a factor of $x^n - y^n$ for all positive even integer values of n.
(c) $x + y$ is a factor of $x^n + y^n$ for all positive odd integer values of n.

D

SELECTED PROOFS

PROOFS OF BASIC LIMIT THEOREMS

An extensive excursion into proofs of limit theorems would be too time consuming to undertake, so we have selected a few proofs of results from Section 1.2 that illustrate some of the basic ideas.

D.1 THEOREM *Let a be any real number, let k be a constant, and suppose that $\lim_{x \to a} f(x) = L_1$ and that $\lim_{x \to a} g(x) = L_2$. Then:*

(*a*) $\lim_{x \to a} k = k$

(*b*) $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L_1 + L_2$

(*c*) $\lim_{x \to a} [f(x)g(x)] = \left(\lim_{x \to a} f(x)\right)\left(\lim_{x \to a} g(x)\right) = L_1 L_2$

PROOF (*a*) We will apply Definition 1.4.1 with $f(x) = k$ and $L = k$. Thus, given $\epsilon > 0$, we must find a number $\delta > 0$ such that

$$|k - k| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta$$

or, equivalently,

$$0 < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta$$

But the condition on the left side of this statement is *always* true, no matter how δ is chosen. Thus, any positive value for δ will suffice.

PROOF (*b*) We must show that given $\epsilon > 0$ we can find a number $\delta > 0$ such that

$$|(f(x) + g(x)) - (L_1 + L_2)| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta \tag{1}$$

However, from the limits of f and g in the hypothesis of the theorem we can find numbers δ_1 and δ_2 such that

$$|f(x) - L_1| < \epsilon/2 \quad \text{if} \quad 0 < |x - a| < \delta_1$$

$$|g(x) - L_2| < \epsilon/2 \quad \text{if} \quad 0 < |x - a| < \delta_2$$

Moreover, the inequalities on the left sides of these statements *both* hold if we replace δ_1 and δ_2 by any positive number δ that is less than both δ_1 and δ_2. Thus, for any such δ it follows that

$$|f(x) - L_1| + |g(x) - L_2| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta \tag{2}$$

However, it follows from the triangle inequality (Theorem F.5 of Web Appendix F) that

$$\begin{aligned}|(f(x)+g(x))-(L_1+L_2)| &= |(f(x)-L_1)+(g(x)-L_2)| \\ &\le |f(x)-L_1|+|g(x)-L_2|\end{aligned}$$

so that (1) follows from (2).

PROOF (*c*) We must show that given $\epsilon > 0$ we can find a number $\delta > 0$ such that

$$|f(x)g(x)-L_1L_2| < \epsilon \quad \text{if} \quad 0 < |x-a| < \delta \tag{3}$$

To find δ it will be helpful to express (3) in a different form. If we rewrite $f(x)$ and $g(x)$ as

$$f(x) = L_1 + (f(x)-L_1) \quad \text{and} \quad g(x) = L_2 + (g(x)-L_2)$$

then the inequality on the left side of (3) can be expressed as (verify)

$$|L_1(g(x)-L_2)+L_2(f(x)-L_1)+(f(x)-L_1)(g(x)-L_2)| < \epsilon \tag{4}$$

Since

$$\lim_{x\to a} f(x) = L_1 \quad \text{and} \quad \lim_{x\to a} g(x) = L_2$$

we can find positive numbers δ_1, δ_2, δ_3, and δ_4 such that

$$\begin{aligned}
&|f(x)-L_1| < \sqrt{\epsilon/3} && \text{if} \quad 0<|x-a|<\delta_1 \\
&|f(x)-L_1| < \frac{\epsilon}{3(1+|L_2|)} && \text{if} \quad 0<|x-a|<\delta_2 \\
&|g(x)-L_2| < \sqrt{\epsilon/3} && \text{if} \quad 0<|x-a|<\delta_3 \\
&|g(x)-L_2| < \frac{\epsilon}{3(1+|L_1|)} && \text{if} \quad 0<|x-a|<\delta_4
\end{aligned} \tag{5}$$

Moreover, the inequalities on the left sides of these four statements *all* hold if we replace δ_1, δ_2, δ_3, and δ_4 by any positive number δ that is smaller than δ_1, δ_2, δ_3, and δ_4. Thus, for any such δ it follows with the help of the triangle inequality that

$$\begin{aligned}
&|L_1(g(x)-L_2)+L_2(f(x)-L_1)+(f(x)-L_1)(g(x)-L_2)| \\
&\quad \le |L_1(g(x)-L_2)|+|L_2(f(x)-L_1)|+|(f(x)-L_1)(g(x)-L_2)| \\
&\quad = |L_1||g(x)-L_2|+|L_2||f(x)-L_1|+|f(x)-L_1||g(x)-L_2| \\
&\quad < |L_1|\frac{\epsilon}{3(1+|L_1|)}+|L_2|\frac{\epsilon}{3(1+|L_2|)}+\sqrt{\epsilon/3}\sqrt{\epsilon/3} && \boxed{\text{From (5)}} \\
&\quad = \frac{\epsilon}{3}\frac{|L_1|}{1+|L_1|}+\frac{\epsilon}{3}\frac{|L_2|}{1+|L_2|}+\frac{\epsilon}{3} \\
&\quad < \frac{\epsilon}{3}+\frac{\epsilon}{3}+\frac{\epsilon}{3} = \epsilon && \boxed{\text{Since } \frac{|L_1|}{1+|L_1|} < 1 \text{ and } \frac{|L_2|}{1+|L_2|} < 1}
\end{aligned}$$

which shows that (4) holds for the δ selected. ■

Do not be alarmed if the proof of part (*c*) seems difficult; it takes some experience with proofs of this type to develop a feel for choosing a valid δ. Your initial goal should be to understand the ideas and the computations.

PROOF OF A BASIC CONTINUITY PROPERTY

Next we will prove Theorem 1.5.5 for two-sided limits.

D.2 THEOREM (*Theorem 1.5.5*) *If* $\lim_{x\to c} g(x) = L$ *and if the function* f *is continuous at* L, *then* $\lim_{x\to c} f(g(x)) = f(L)$. *That is,*

$$\lim_{x\to c} f(g(x)) = f\left(\lim_{x\to c} g(x)\right)$$

PROOF We must show that given $\epsilon > 0$, we can find a number $\delta > 0$ such that

$$|f(g(x)) - f(L)| < \epsilon \quad \text{if} \quad 0 < |x - c| < \delta \tag{6}$$

Since f is continuous at L, we have

$$\lim_{u \to L} f(u) = f(L)$$

and hence we can find a number $\delta_1 > 0$ such that

$$|f(u) - f(L)| < \epsilon \quad \text{if} \quad |u - L| < \delta_1$$

In particular, if $u = g(x)$, then

$$|f(g(x)) - f(L)| < \epsilon \quad \text{if} \quad |g(x) - L| < \delta_1 \tag{7}$$

But $\lim_{x \to c} g(x) = L$, and hence there is a number $\delta > 0$ such that

$$|g(x) - L| < \delta_1 \quad \text{if} \quad 0 < |x - c| < \delta \tag{8}$$

Thus, if x satisfies the condition on the right side of statement (8), then it follows that $g(x)$ satisfies the condition on the right side of statement (7), and this implies that the condition on the left side of statement (6) is satisfied, completing the proof. ■

PROOF OF THE CHAIN RULE

Next we will prove the chain rule (Theorem 2.6.1), but first we need a preliminary result.

D.3 THEOREM *If f is differentiable at x and if $y = f(x)$, then*

$$\Delta y = f'(x)\Delta x + \epsilon \Delta x$$

where $\epsilon \to 0$ as $\Delta x \to 0$ and $\epsilon = 0$ if $\Delta x = 0$.

PROOF Define

$$\epsilon = \begin{cases} \dfrac{f(x + \Delta x) - f(x)}{\Delta x} - f'(x) & \text{if } \Delta x \neq 0 \\ 0 & \text{if } \Delta x = 0 \end{cases} \tag{9}$$

If $\Delta x \neq 0$, it follows from (9) that

$$\epsilon \Delta x = [f(x + \Delta x) - f(x)] - f'(x)\Delta x \tag{10}$$

But

$$\Delta y = f(x + \Delta x) - f(x) \tag{11}$$

so (10) can be written as

$$\epsilon \Delta x = \Delta y - f'(x)\Delta x$$

or

$$\Delta y = f'(x)\Delta x + \epsilon \Delta x \tag{12}$$

If $\Delta x = 0$, then (12) still holds (why?), so (12) is valid for all values of Δx. It remains to show that $\epsilon \to 0$ as $\Delta x \to 0$. But this follows from the assumption that f is differentiable at x, since

$$\lim_{\Delta x \to 0} \epsilon = \lim_{\Delta x \to 0} \left[\frac{f(x + \Delta x) - f(x)}{\Delta x} - f'(x) \right] = f'(x) - f'(x) = 0 \quad ■$$

We are now ready to prove the chain rule.

D.4 THEOREM (*Theorem 2.6.1*) *If g is differentiable at the point x and f is differentiable at the point $g(x)$, then the composition $f \circ g$ is differentiable at the point x. Moreover, if $y = f(g(x))$ and $u = g(x)$, then*

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

PROOF Since g is differentiable at x and $u = g(x)$, it follows from Theorem D.3 that

$$\Delta u = g'(x)\Delta x + \epsilon_1 \Delta x \tag{13}$$

where $\epsilon_1 \to 0$ as $\Delta x \to 0$. And since $y = f(u)$ is differentiable at $u = g(x)$, it follows from Theorem D.3 that

$$\Delta y = f'(u)\Delta u + \epsilon_2 \Delta u \tag{14}$$

where $\epsilon_2 \to 0$ as $\Delta u \to 0$.

Factoring out the Δu in (14) and then substituting (13) yields

$$\Delta y = [f'(u) + \epsilon_2][g'(x)\Delta x + \epsilon_1 \Delta x]$$

or

$$\Delta y = [f'(u) + \epsilon_2][g'(x) + \epsilon_1]\Delta x$$

or if $\Delta x \neq 0$,

$$\frac{\Delta y}{\Delta x} = [f'(u) + \epsilon_2][g'(x) + \epsilon_1] \tag{15}$$

But (13) implies that $\Delta u \to 0$ as $\Delta x \to 0$, and hence $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $\Delta x \to 0$. Thus, from (15)

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(u)g'(x)$$

or

$$\frac{dy}{dx} = f'(u)g'(x) = \frac{dy}{du} \cdot \frac{du}{dx} \quad \blacksquare$$

PROOF THAT RELATIVE EXTREMA OCCUR AT CRITICAL POINTS

In this subsection we will prove Theorem 4.2.2, which states that the relative extrema of a function occur at critical points.

D.5 THEOREM (*Theorem 4.2.2*) *Suppose that f is a function defined on an open interval containing the point x_0. If f has a relative extremum at $x = x_0$, then $x = x_0$ is a critical point of f; that is, either $f'(x_0) = 0$ or f is not differentiable at x_0.*

PROOF Suppose that f has a relative maximum at x_0. There are two possibilities—either f is differentiable at x_0 or it is not. If it is not, then x_0 is a critical point for f and we are done. If f is differentiable at x_0, then we must show that $f'(x_0) = 0$. We will do this by showing that $f'(x_0) \geq 0$ and $f'(x_0) \leq 0$, from which it follows that $f'(x_0) = 0$. From the definition of a derivative we have

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

so that

$$f'(x_0) = \lim_{h \to 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \tag{16}$$

and

$$f'(x_0) = \lim_{h \to 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \tag{17}$$

Because f has a relative maximum at x_0, there is an open interval (a, b) containing x_0 in which $f(x) \le f(x_0)$ for all x in (a, b).

Assume that h is sufficiently small so that $x_0 + h$ lies in the interval (a, b). Thus,

$$f(x_0 + h) \le f(x_0) \quad \text{or equivalently} \quad f(x_0 + h) - f(x_0) \le 0$$

Thus, if h is negative,

$$\frac{f(x_0 + h) - f(x_0)}{h} \ge 0 \tag{18}$$

and if h is positive,

$$\frac{f(x_0 + h) - f(x_0)}{h} \le 0 \tag{19}$$

But an expression that never assumes negative values cannot approach a negative limit and an expression that never assumes positive values cannot approach a positive limit, so that

$$f'(x_0) = \lim_{h \to 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \ge 0 \qquad \boxed{\text{From (17) and (18)}}$$

and

$$f'(x_0) = \lim_{h \to 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \le 0 \qquad \boxed{\text{From (16) and (19)}}$$

Since $f'(x_0) \ge 0$ and $f'(x_0) \le 0$, it must be that $f'(x_0) = 0$.

A similar argument applies if f has a relative minimum at x_0. ■

PROOFS OF TWO SUMMATION FORMULAS

We will prove parts (a) and (b) of Theorem 5.4.2. The proof of part (c) is similar to that of part (b) and is omitted.

D.6 THEOREM ***(Theorem 5.4.2)***

(a) $$\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

(b) $$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

(c) $$\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

PROOF (a) Writing

$$\sum_{k=1}^{n} k$$

two ways, with summands in increasing order and in decreasing order, and then adding, we obtain

$$\begin{aligned}
\sum_{k=1}^{n} k &= 1 + 2 + 3 + \cdots + (n-2) + (n-1) + n \\
\sum_{k=1}^{n} k &= n + (n-1) + (n-2) + \cdots + 3 + 2 + 1 \\
\hline
2\sum_{k=1}^{n} k &= (n+1) + (n+1) + (n+1) + \cdots + (n+1) + (n+1) + (n+1) \\
&= n(n+1)
\end{aligned}$$

Thus,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

PROOF (*b*) Note that

$$(k+1)^3 - k^3 = k^3 + 3k^2 + 3k + 1 - k^3 = 3k^2 + 3k + 1$$

So,

$$\sum_{k=1}^{n} [(k+1)^3 - k^3] = \sum_{k=1}^{n} (3k^2 + 3k + 1) \tag{20}$$

Writing out the left side of (20) with the index running *down* from $k = n$ to $k = 1$, we have

The sum in (21) is an example of a ***telescoping sum***, since the cancellation of each of the two parts of an interior summand with parts of its neighboring summands allows the entire sum to collapse like a telescope.

$$\begin{aligned}\sum_{k=1}^{n} [(k+1)^3 - k^3] &= [(n+1)^3 - n^3] + \cdots + [4^3 - 3^3] + [3^3 - 2^3] + [2^3 - 1^3] \\ &= (n+1)^3 - 1 \end{aligned} \tag{21}$$

Combining (21) and (20), and expanding the right side of (20) by using Theorem 5.4.1 and part (*a*) of this theorem yields

$$\begin{aligned}(n+1)^3 - 1 &= 3\sum_{k=1}^{n} k^2 + 3\sum_{k=1}^{n} k + \sum_{k=1}^{n} 1 \\ &= 3\sum_{k=1}^{n} k^2 + 3\frac{n(n+1)}{2} + n\end{aligned}$$

So,

$$\begin{aligned}3\sum_{k=1}^{n} k^2 &= [(n+1)^3 - 1] - 3\frac{n(n+1)}{2} - n \\ &= (n+1)^3 - 3(n+1)\left(\frac{n}{2}\right) - (n+1) \\ &= \frac{n+1}{2}[2(n+1)^2 - 3n - 2] \\ &= \frac{n+1}{2}[2n^2 + n] = \frac{n(n+1)(2n+1)}{2}\end{aligned}$$

Thus,

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} \quad \blacksquare$$

PROOF OF THE LIMIT COMPARISON TEST

D.7 THEOREM (*Theorem 9.5.4*) *Let $\sum a_k$ and $\sum b_k$ be series with positive terms and suppose that*

$$\rho = \lim_{k \to +\infty} \frac{a_k}{b_k}$$

If ρ is finite and $\rho > 0$, then the series both converge or both diverge.

PROOF We need only show that $\sum b_k$ converges when $\sum a_k$ converges and that $\sum b_k$ diverges when $\sum a_k$ diverges, since the remaining cases are logical implications of these (why?). The idea of the proof is to apply the comparison test to $\sum a_k$ and suitable multiples of $\sum b_k$. For this purpose let ϵ be any positive number. Since

$$\rho = \lim_{k \to +\infty} \frac{a_k}{b_k}$$

it follows that eventually the terms in the sequence $\{a_k/b_k\}$ must be within ϵ units of ρ; that is, there is a positive integer K such that for $k \geq K$ we have

$$\rho - \epsilon < \frac{a_k}{b_k} < \rho + \epsilon$$

In particular, if we take $\epsilon = \rho/2$, then for $k \geq K$ we have

$$\frac{1}{2}\rho < \frac{a_k}{b_k} < \frac{3}{2}\rho \quad \text{or} \quad \frac{1}{2}\rho b_k < a_k < \frac{3}{2}\rho b_k$$

Thus, by the comparison test we can conclude that

$$\sum_{k=K}^{\infty} \frac{1}{2}\rho b_k \quad \text{converges if} \quad \sum_{k=K}^{\infty} a_k \quad \text{converges} \tag{22}$$

$$\sum_{k=K}^{\infty} \frac{3}{2}\rho b_k \quad \text{diverges if} \quad \sum_{k=K}^{\infty} a_k \quad \text{diverges} \tag{23}$$

But the convergence or divergence of a series is not affected by deleting finitely many terms or by multiplying the general term by a nonzero constant, so (22) and (23) imply that

$$\sum_{k=1}^{\infty} b_k \quad \text{converges if} \quad \sum_{k=1}^{\infty} a_k \quad \text{converges}$$

$$\sum_{k=1}^{\infty} b_k \quad \text{diverges if} \quad \sum_{k=1}^{\infty} a_k \quad \text{diverges} \quad \blacksquare$$

PROOF OF THE RATIO TEST

D.8 THEOREM (*Theorem 9.5.5*) *Let $\sum u_k$ be a series with positive terms and suppose that*

$$\rho = \lim_{k \to +\infty} \frac{u_{k+1}}{u_k}$$

(*a*) *If $\rho < 1$, the series converges.*

(*b*) *If $\rho > 1$ or $\rho = +\infty$, the series diverges.*

(*c*) *If $\rho = 1$, the series may converge or diverge, so that another test must be tried.*

PROOF (*a*) The number ρ must be nonnegative since it is the limit of u_{k+1}/u_k, which is positive for all k. In this part of the proof we assume that $\rho < 1$, so that $0 \leq \rho < 1$.

We will prove convergence by showing that the terms of the given series are eventually less than the terms of a convergent geometric series. For this purpose, choose any real number r such that $0 < \rho < r < 1$. Since the limit of u_{k+1}/u_k is ρ, and $\rho < r$, the terms of the sequence $\{u_{k+1}/u_k\}$ must eventually be less than r. Thus, there is a positive integer K such that for $k \geq K$ we have

$$\frac{u_{k+1}}{u_k} < r \quad \text{or} \quad u_{k+1} < r u_k$$

This yields the inequalities

$$\begin{aligned} u_{K+1} &< ru_K \\ u_{K+2} &< ru_{K+1} < r^2u_K \\ u_{K+3} &< ru_{K+2} < r^3u_K \\ u_{K+4} &< ru_{K+3} < r^4u_K \\ &\vdots \end{aligned} \tag{24}$$

But $0 < r < 1$, so

$$ru_K + r^2u_K + r^3u_K + \cdots$$

is a convergent geometric series. From the inequalities in (24) and the comparison test it follows that

$$u_{K+1} + u_{K+2} + u_{K+3} + \cdots$$

must also be a convergent series. Thus, $u_1 + u_2 + u_3 + \cdots + u_k + \cdots$ converges by Theorem 9.4.3(c).

PROOF (b) In this part we will prove divergence by showing that the limit of the general term is not zero. Since the limit of u_{k+1}/u_k is ρ and $\rho > 1$, the terms in the sequence $\{u_{k+1}/u_k\}$ must eventually be greater than 1. Thus, there is a positive integer K such that for $k \geq K$ we have

$$\frac{u_{k+1}}{u_k} > 1 \quad \text{or} \quad u_{k+1} > u_k$$

This yields the inequalities

$$\begin{aligned} u_{K+1} &> u_K \\ u_{K+2} &> u_{K+1} > u_K \\ u_{K+3} &> u_{K+2} > u_K \\ u_{K+4} &> u_{K+3} > u_K \\ &\vdots \end{aligned} \tag{25}$$

Since $u_K > 0$, it follows from the inequalities in (25) that $\lim_{k\to+\infty} u_k \neq 0$, and thus the series $u_1 + u_2 + \cdots + u_k + \cdots$ diverges by part (a) of Theorem 9.4.1. The proof in the case where $\rho = +\infty$ is omitted.

PROOF (c) The divergent harmonic series and the convergent p-series with $p = 2$ both have $\rho = 1$ (verify), so the ratio test does not distinguish between convergence and divergence when $\rho = 1$. ■

PROOF OF THE REMAINDER ESTIMATION THEOREM

D.9 THEOREM (*Theorem 9.7.4*) *If the function f can be differentiated $n+1$ times on an interval containing the number x_0, and if M is an upper bound for $|f^{(n+1)}(x)|$ on the interval, that is, $|f^{(n+1)}(x)| \leq M$ for all x in the interval, then*

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x - x_0|^{n+1}$$

for all x in the interval.

PROOF We are assuming that f can be differentiated $n+1$ times on an interval containing the number x_0 and that

$$|f^{(n+1)}(x)| \leq M \tag{26}$$

for all x in the interval. We want to show that

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x - x_0|^{n+1} \tag{27}$$

for all x in the interval, where

$$R_n(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \tag{28}$$

In our proof we will need the following two properties of $R_n(x)$:

$$R_n(x_0) = R_n'(x_0) = \cdots = R_n^{(n)}(x_0) = 0 \tag{29}$$

$$R_n^{(n+1)}(x) = f^{(n+1)}(x) \quad \text{for all } x \text{ in the interval} \tag{30}$$

These properties can be obtained by analyzing what happens if the expression for $R_n(x)$ in Formula (28) is differentiated j times and x_0 is then substituted in that derivative. If $j < n$, then the jth derivative of the summation in Formula (28) consists of a constant term $f^{(j)}(x_0)$ plus terms involving powers of $x - x_0$ (verify). Thus, $R_n^{(j)}(x_0) = 0$ for $j < n$, which proves all but the last equation in (29). For the last equation, observe that the nth derivative of the summation in (28) is the constant $f^{(n)}(x_0)$, so $R_n^{(n)}(x_0) = 0$. Formula (30) follows from the observation that the $(n + 1)$-st derivative of the summation in (28) is zero (why?).

Now to the main part of the proof. For simplicity we will give the proof for the case where $x \geq x_0$ and leave the case where $x < x_0$ for you. It follows from (26) and (30) that $|R_n^{(n+1)}(x)| \leq M$, and hence

$$-M \leq R_n^{(n+1)}(x) \leq M$$

Thus,

$$\int_{x_0}^{x} -M\,dt \leq \int_{x_0}^{x} R_n^{(n+1)}(t)\,dt \leq \int_{x_0}^{x} M\,dt \tag{31}$$

However, it follows from (29) that $R_n^{(n)}(x_0) = 0$, so

$$\int_{x_0}^{x} R_n^{(n+1)}(t)\,dt = R_n^{(n)}(t)\Big]_{x_0}^{x} = R_n^{(n)}(x)$$

Thus, performing the integrations in (31) we obtain the inequalities

$$-M(x - x_0) \leq R_n^{(n)}(x) \leq M(x - x_0)$$

Now we will integrate again. Replacing x by t in these inequalities, integrating from x_0 to x, and using $R_n^{(n-1)}(x_0) = 0$ yields

$$-\frac{M}{2}(x - x_0)^2 \leq R_n^{(n-1)}(x) \leq \frac{M}{2}(x - x_0)^2$$

If we keep repeating this process, then after $n + 1$ integrations we will obtain

$$-\frac{M}{(n+1)!}(x - x_0)^{n+1} \leq R_n(x) \leq \frac{M}{(n+1)!}(x - x_0)^{n+1}$$

which we can rewrite as

$$|R_n(x)| \leq \frac{M}{(n+1)!}(x - x_0)^{n+1}$$

This completes the proof of (27), since the absolute value signs can be omitted in that formula when $x \geq x_0$ (which is the case we are considering). ■

PROOF OF THE EQUALITY OF MIXED PARTIALS

D.10 THEOREM (*Theorem 13.3.2*) *Let f be a function of two variables. If f_{xy} and f_{yx} are continuous on some open disk, then $f_{xy} = f_{yx}$ on that disk.*

PROOF Suppose that f is a function of two variables with f_{xy} and f_{yx} both continuous on some open disk. Let (x, y) be a point in that disk and define the function

$$w(\Delta x, \Delta y) = f(x + \Delta x, y + \Delta y) - f(x + \Delta x, y) - f(x, y + \Delta y) + f(x, y)$$

Now fix y and Δy and let

$$g(x) = f(x, y + \Delta y) - f(x, y)$$

so that

$$w(\Delta x, \Delta y) = g(x + \Delta x) - g(x) \tag{32}$$

Since f is differentiable on an open disk containing (x, y), the function g will be differentiable on some interval containing x and $x + \Delta x$ for Δx small enough. The Mean-Value Theorem then applies to g on this interval, and thus there is a c between x and Δx with

$$g(x + \Delta x) - g(x) = g'(c)\Delta x$$

But

$$g'(c) = f_x(c, y + \Delta y) - f_x(c, y)$$

so from Equation (32)

$$w(\Delta x, \Delta y) = g(x + \Delta x) - g(x) = g'(c)\Delta x = (f_x(c, y + \Delta y) - f_x(c, y))\Delta x \tag{33}$$

Now let $h(y) = f_x(c, y)$. Since f_x is differentiable on an open disk containing (x, y), h will be differentiable on some interval containing y and $y + \Delta y$ for Δy small enough. Applying the Mean-Value Theorem to h on this interval gives a d between y and $y + \Delta y$ with

$$h(y + \Delta y) - h(y) = h'(d)\Delta y$$

But $h'(d) = f_{xy}(c, d)$, so by (33) and the definition of h we have

$$\begin{aligned} w(\Delta x, \Delta y) &= (f_x(c, y + \Delta y) - f_x(c, y))\Delta x \\ &= (h(y + \Delta y) - h(y))\Delta x = h'(d)\Delta y \Delta x \\ &= f_{xy}(c, d)\Delta y \Delta x \end{aligned}$$

and

$$f_{xy}(c, d) = \frac{w(\Delta x, \Delta y)}{\Delta y \Delta x} \tag{34}$$

Since c lies between x and Δx and d lies between y and Δy, (c, d) approaches (x, y) as $(\Delta x, \Delta y)$ approaches $(0, 0)$. It then follows from the continuity of f_{xy} and (34) that

$$f_{xy}(x, y) = \lim_{(\Delta x, \Delta y) \to (0,0)} f_{xy}(c, d) = \lim_{(\Delta x, \Delta y) \to (0,0)} \frac{w(\Delta x, \Delta y)}{\Delta y \Delta x}$$

In similar fashion to the above argument, it can be shown that

$$f_{yx}(x, y) = \lim_{(\Delta x, \Delta y) \to (0,0)} \frac{w(\Delta x, \Delta y)}{\Delta y \Delta x}$$

and the result follows. ■

PROOF OF THE TWO-VARIABLE CHAIN RULE FOR DERIVATIVES

D.11 THEOREM (*Theorem 13.5.1*) *If $x = x(t)$ and $y = y(t)$ are differentiable at t, and if $z = f(x, y)$ is differentiable at the point $(x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t and*

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

PROOF Let Δx, Δy, and Δz denote the changes in x, y, and z, respectively, that correspond to a change of Δt in t. Then

$$\frac{dz}{dt} = \lim_{\Delta t \to 0} \frac{\Delta z}{\Delta t}, \quad \frac{dx}{dt} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}, \quad \frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{\Delta y}{\Delta t}$$

Since $f(x, y)$ is differentiable at $(x(t), y(t))$, it follows from (5) in Section 13.4 that

$$\Delta z = \frac{\partial z}{\partial x}\Delta x + \frac{\partial z}{\partial y}\Delta y + \epsilon(\Delta x, \Delta y)\sqrt{(\Delta x)^2 + (\Delta y)^2} \tag{35}$$

where the partial derivatives are evaluated at $(x(t), y(t))$ and where $\epsilon(\Delta x, \Delta y)$ satisfies $\epsilon(\Delta x, \Delta y) \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$ and $\epsilon(0, 0) = 0$. Dividing both sides of (35) by Δt yields

$$\frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial z}{\partial y}\frac{\Delta y}{\Delta t} + \frac{\epsilon(\Delta x, \Delta y)\sqrt{(\Delta x)^2 + (\Delta y)^2}}{\Delta t} \tag{36}$$

Since

$$\lim_{\Delta t \to 0} \frac{\sqrt{(\Delta x)^2 + (\Delta y)^2}}{|\Delta t|} = \lim_{\Delta t \to 0} \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2} = \sqrt{\left(\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}\right)^2 + \left(\lim_{\Delta t \to 0} \frac{\Delta y}{\Delta t}\right)^2}$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

we have

$$\begin{aligned}
\lim_{\Delta t \to 0} \left| \frac{\epsilon(\Delta x, \Delta y)\sqrt{(\Delta x)^2 + (\Delta y)^2}}{\Delta t} \right| &= \lim_{\Delta t \to 0} \frac{|\epsilon(\Delta x, \Delta y)|\sqrt{(\Delta x)^2 + (\Delta y)^2}}{|\Delta t|} \\
&= \lim_{\Delta t \to 0} |\epsilon(\Delta x, \Delta y)| \cdot \lim_{\Delta t \to 0} \frac{\sqrt{(\Delta x)^2 + (\Delta y)^2}}{|\Delta t|} \\
&= 0 \cdot \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 0
\end{aligned}$$

Therefore,

$$\lim_{\Delta t \to 0} \frac{\epsilon(\Delta x, \Delta y)\sqrt{(\Delta x)^2 + (\Delta y)^2}}{\Delta t} = 0$$

Taking the limit as $\Delta t \to 0$ of both sides of (36) then yields the equation

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} \quad \blacksquare$$

ANSWERS TO ODD-NUMBERED EXERCISES

Exercise Set 0.1 (Page 12)

1. **(a)** $-2.9, -2.0, 2.35, 2.9$ **(b)** none **(c)** $y = 0$ **(d)** $-1.75 \le x \le 2.15$ **(e)** $y_{\max} = 2.8$ at $x = -2.6$; $y_{\min} = -2.2$ at $x = 1.2$

3. **(a)** yes **(b)** yes **(c)** no **(d)** no

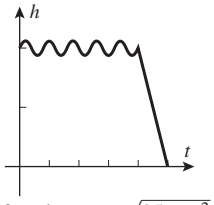
5. **(a)** 1999, about \$47,700 **(b)** 1993, \$41,600 **(c)** first year

7. **(a)** -2; 10; 10; 25; 4; $27t^2 - 2$ **(b)** 0; 4; -4; 6; $2\sqrt{2}$; $f(3t) = 1/(3t)$ for $t > 1$ and $f(3t) = 6t$ for $t \le 1$

9. **(a)** domain: $x \ne 3$; range: $y \ne 0$ **(b)** domain: $x \ne 0$; range: $\{-1, 1\}$ **(c)** domain: $x \le -\sqrt{3}$ or $x \ge \sqrt{3}$; range: $y \ge 0$
(d) domain: $-\infty < x < +\infty$; range: $y \ge 2$
(e) domain: $x \ne \left(2n + \frac{1}{2}\right)\pi, n = 0, \pm 1, \pm 2, \ldots$; range: $y \ge \frac{1}{2}$
(f) domain: $-2 \le x < 2$ or $x > 2$; range: $0 \le y < 2$ or $y > 2$

11. **(a)** no; births and deaths **(b)** decreases for 8 hours, takes a jump upward, and repeats

13.



15. function; $y = \sqrt{25 - x^2}$

17. function; $y = \begin{cases} \sqrt{25 - x^2}, & -5 \le x \le 0 \\ -\sqrt{25 - x^2}, & 0 < x \le 5 \end{cases}$

19. False; for example, the graph of the function $f(x) = x^2 - 1$ crosses the x-axis at $x = \pm 1$.

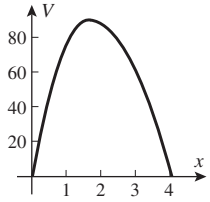
21. False; the range also includes 0.

23. **(a)** 2, 4 **(b)** none **(c)** $x \le 2$; $4 \le x$ **(d)** $y_{\min} = -1$; no maximum

25. $h = L(1 - \cos\theta)$

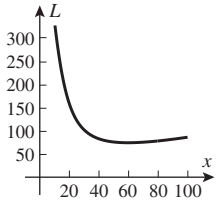
27. **(a)** $f(x) = \begin{cases} 2x + 1, & x < 0 \\ 4x + 1, & x \ge 0 \end{cases}$ **(b)** $g(x) = \begin{cases} 1 - 2x, & x < 0 \\ 1, & 0 \le x < 1 \\ 2x - 1, & x \ge 1 \end{cases}$

29. **(a)** $V = (8 - 2x)(15 - 2x)x$
(b) $0 < x < 4$
(c) $0 < V \le 90$, approximately



(d) V appears to be maximal for $x \approx 1.7$.

31. **(a)** $L = x + 2y$
(b) $L = x + 2000/x$
(c) $0 < x \le 100$
(d) $x \approx 45$ ft, $y \approx 22$ ft



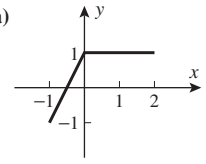
33. **(a)** $r \approx 3.4$, $h \approx 13.7$ **(b)** taller
(c) $r \approx 3.1$ cm, $h \approx 16.0$ cm, $C \approx 4.76$ cents

35. **(i)** $x = 1, -2$ **(ii)** $g(x) = x + 1$, all x

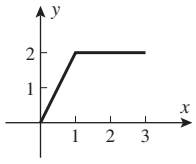
37. **(a)** 25°F **(b)** 13°F **(c)** 5°F **39.** 15°F

Exercise Set 0.2 (Page 24)

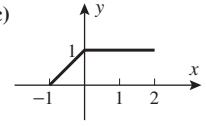
1. **(a)**



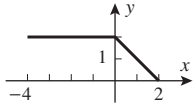
(b)



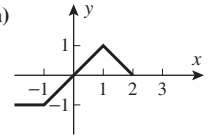
(c)



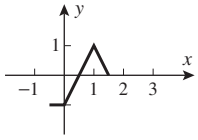
(d)



3. **(a)**



(b)



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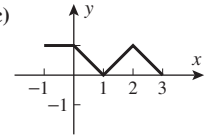
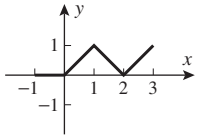


(d)



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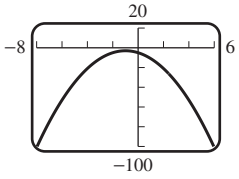

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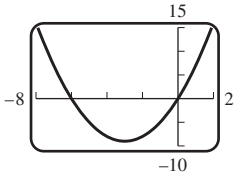

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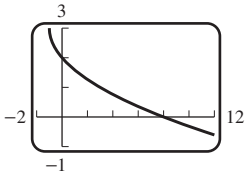

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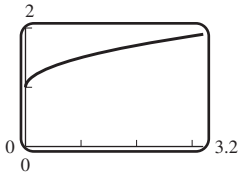

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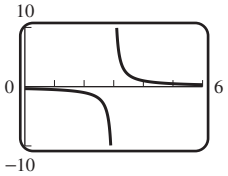

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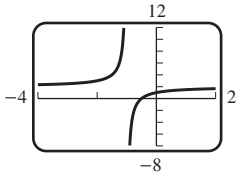

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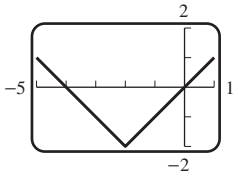

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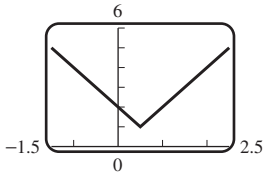

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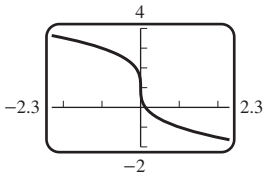

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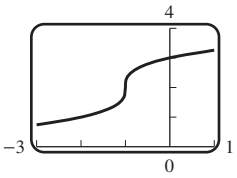
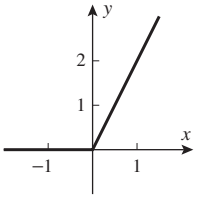


25. **(a)**



(b) $y = \begin{cases} 0, & x \le 0 \\ 2x, & x > 0 \end{cases}$

27. $3\sqrt{x-1}, x \ge 1$; $\sqrt{x-1}, x \ge 1$; $2x-2, x \ge 1$; $2, x > 1$

29. **(a)** 3 **(b)** 9 **(c)** 2 **(d)** 2 **(e)** $\sqrt{2+h}$ **(f)** $(3+h)^3+1$

31. $1-x, x \le 1$; $\sqrt{1-x^2}, |x| \le 1$

33. $\dfrac{1}{1-2x}, x \ne \dfrac{1}{2}$, 1; $-\dfrac{1}{2x}-\dfrac{1}{2}, x \ne 0, 1$ **35.** $x^{-6}+1$

37. **(a)** $g(x)=\sqrt{x}, h(x)=x+2$ **(b)** $g(x)=|x|, h(x)=x^2-3x+5$

39. **(a)** $g(x)=x^2, h(x)=\sin x$ **(b)** $g(x)=3/x, h(x)=5+\cos x$

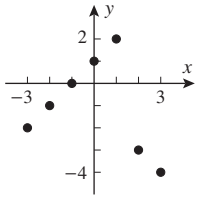
41. **(a)** $g(x)=x^3, h(x)=1+\sin(x^2)$
(b) $g(x)=\sqrt{x}, h(x)=1-\sqrt[3]{x}$

Responses to True–False questions may be abridged to save space.

43. True; see Definition 0.2.1.

45. True; see Theorem 0.2.3 and the definition of even function that follows.

47.



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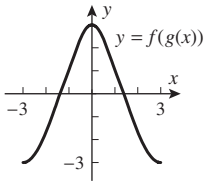
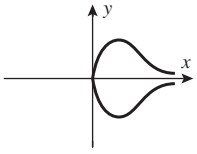


51. $\pm 1.5, \pm 2$ **53.** $6x+3h, 3w+3x$ **55.** $-\dfrac{1}{x(x+h)}, -\dfrac{1}{xw}$

57. f: neither, g: odd, h: even

59. **(a)**



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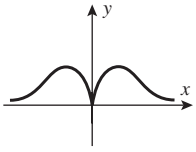

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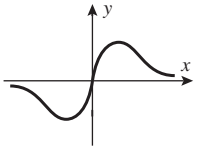
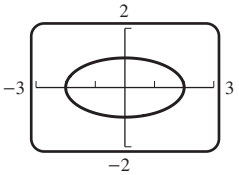


61. **(a)** even **(b)** odd

63. **(a)** even **(b)** odd **(c)** even **(d)** neither **(e)** odd **(f)** even

67. **(a)** y-axis
(b) origin
(c) x-axis, y-axis, origin

69.



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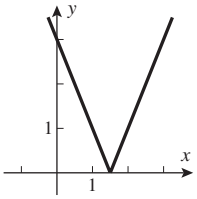
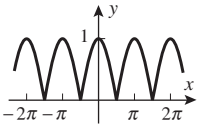
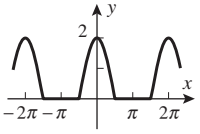


73. **(a)**



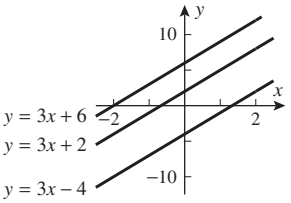
(b)



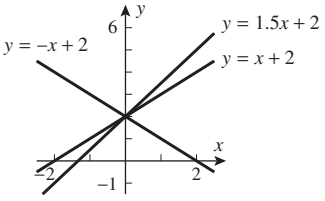
75. yes; $f(x)=x^k, g(x)=x^n$

Exercise Set 0.3 (Page 35)

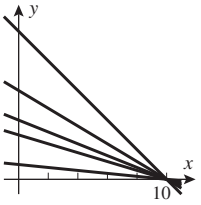
1. **(a)** $y=3x+b$ **(c)**
(b) $y=3x+6$



3. **(a)** $y=mx+2$ **(c)**
(b) $y=-x+2$

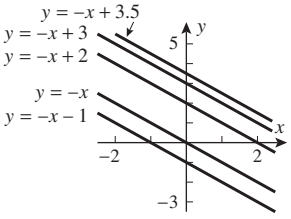


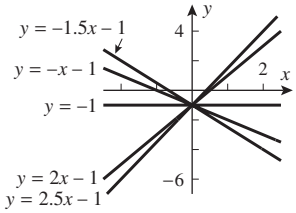
5. $y=\pm\dfrac{9-x_0x}{\sqrt{9-x_0^2}}$ **7.**



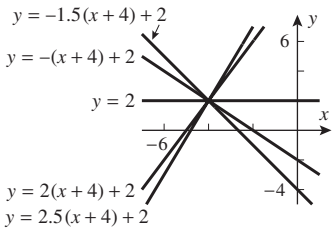
y-intercepts represent current value of item being depreciated.

9. **(a)** slope: -1 **(b)** y-intercept: $y=-1$

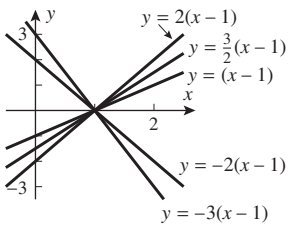




(c) pass through $(-4, 2)$

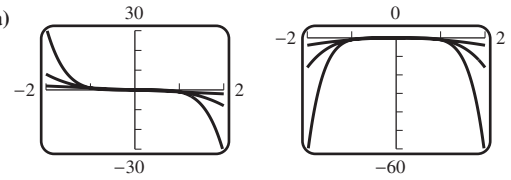


(d) x-intercept: $x = 1$

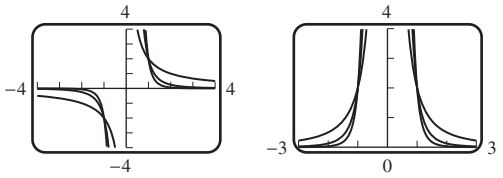


11. **(a)** VI
(b) IV
(c) III
(d) V
(e) I
(f) II

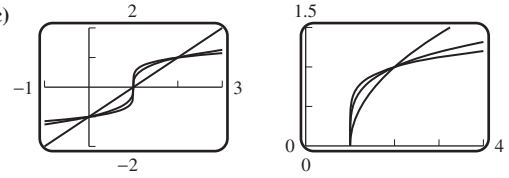
13. **(a)**



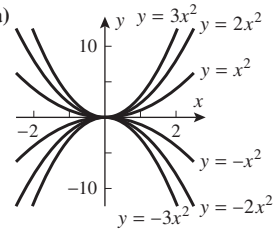
(b)



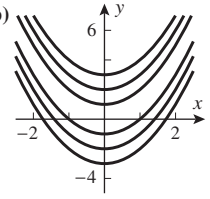
(c)



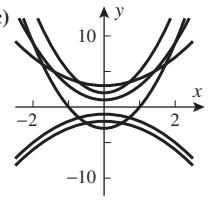
15. **(a)**



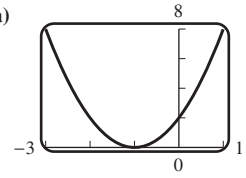
(b)



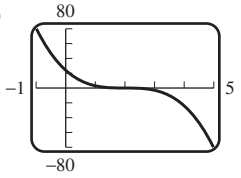
(c)



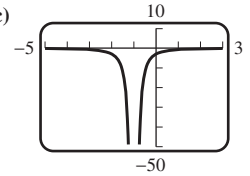
17. **(a)**



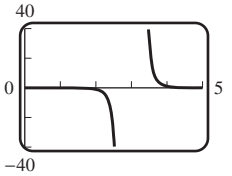
(b)



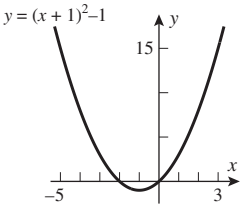
(c)



(d)



19.

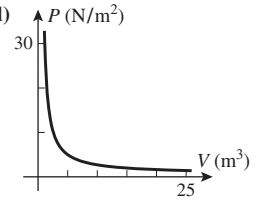


21. **(a)** newton-meters (N·m) **(b)** 20 N·m

(c)

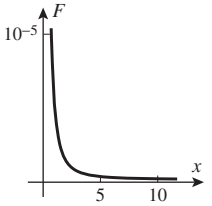
V (L)	0.25	0.5	1.0	1.5	2.0
P (N/m^2)	80×10^3	40×10^3	20×10^3	13.3×10^3	10×10^3

(d)



23. **(a)** $k = 0.000045$ N·m^2
(b) 0.000005 N
(d) The force becomes infinite; the force tends to zero.

(c)



Responses to True–False questions may be abridged to save space.

25. True; see Figure 0.3.2(b).

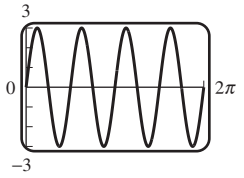
27. False; the constant of proportionality is $2 \cdot 6 = 12$.

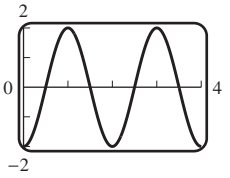
29. **(a)** II; $y = 1, x = -1, 2$ **(b)** I; $y = 0, x = -2, 3$ **(c)** IV; $y = 2$
(d) III; $y = 0, x = -2$

31. **(a)** $y = 3\sin(x/2)$ **(b)** $y = 4\cos 2x$ **(c)** $y = -5\sin 4x$

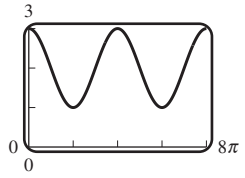
33. **(a)** $y = \sin[x + (\pi/2)]$ **(b)** $y = 3 + 3\sin(2x/9)$
(c) $y = 1 + 2\sin\left(2x - \dfrac{\pi}{2}\right)$

35. **(a)** amplitude $= 3$, period $= \pi/2$ **(b)** amplitude $= 2$, period $= 2$





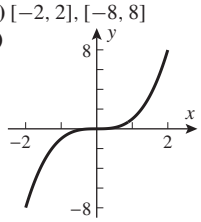
(c) amplitude $= 1$, period $= 4\pi$



37. $x = \frac{5\sqrt{13}}{2}\sin\left(2\pi t + \tan^{-1}\frac{1}{2\sqrt{3}}\right)$

Exercise Set 0.4 (Page 48)

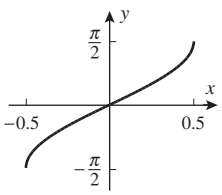
1. (a) yes (b) no (c) yes (d) no
3. (a) yes (b) yes (c) no (d) yes (e) no (f) no
5. (a) yes (b) no
7. (a) $8, -1, 0$
   (b) $[-2, 2], [-8, 8]$
   (c)



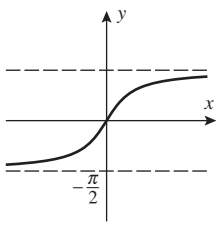
9. $\frac{1}{7}(x+6)$
11. $\sqrt[3]{(x+5)/3}$
13. $-\sqrt{3/x}$
15. $\begin{cases}(5/2)-x, & x > 1/2\\ 1/x, & 0 < x \le 1/2\end{cases}$
17. $x^{1/4} - 2$ for $x \ge 16$
19. $\frac{1}{2}(3 - x^2)$ for $x \le 0$
21. (a) $f^{-1}(x) = \dfrac{-b + \sqrt{b^2 - 4a(c-x)}}{2a}$
    (b) $f^{-1}(x) = \dfrac{-b - \sqrt{b^2 - 4a(c-x)}}{2a}$
23. (a) $y = (6.214 \times 10^{-4})x$ (b) $x = \dfrac{10^4}{6.214}y$
    (c) how many meters in y miles
25. (b) symmetric about the line $y = x$ 27. 10

Responses to True–False questions may be abridged to save space.

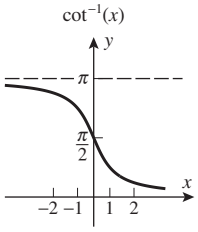
31. False; $f^{-1}(2) = 2$
33. True; see Theorem 0.4.3.
35. $\frac{4}{5}, \frac{3}{5}, \frac{3}{4}, \frac{5}{3}, \frac{5}{4}$
37. (a) $0 \le x \le \pi$ (b) $-1 \le x \le 1$ (c) $-\pi/2 < x < \pi/2$
    (d) $-\infty < x < +\infty$ 39. $\frac{24}{25}$
41. (a) $\dfrac{1}{\sqrt{1+x^2}}$ (b) $\dfrac{\sqrt{1-x^2}}{x}$ (c) $\dfrac{\sqrt{x^2-1}}{x}$ (d) $\dfrac{1}{\sqrt{x^2-1}}$
43. (a)



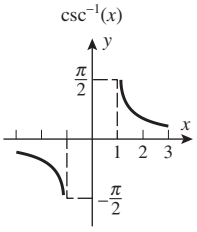
(b)



45. (a) $x = 2.76258$ rad (b) $\theta = 217.59°$
47. (a) 0.25545, error (b) $|x| \le \sin 1$
49. (a) $\cot^{-1}(x)$



$\csc^{-1}(x)$

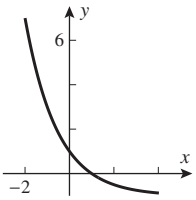


(b) $\cot^{-1} x$: all x, $0 < y < \pi$
$\csc^{-1} x$: $|x| \ge 1$, $0 < |y| < \pi/2$

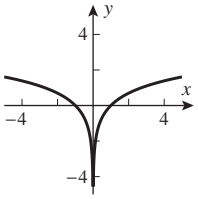
51. (a) 55.0° (b) 33.6° (c) 25.8° 53. (a) 21.1 hours (b) 2.9 hours
55. 29°

Exercise Set 0.5 (Page 61)

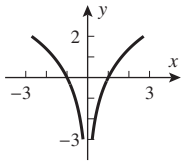
1. (a) -4 (b) 4 (c) $\frac{1}{4}$ 3. (a) 2.9690 (b) 0.0341
5. (a) 4 (b) -5 (c) 1 (d) $\frac{1}{2}$ 7. (a) 1.3655 (b) -0.3011
9. (a) $2r + \dfrac{s}{2} + \dfrac{t}{2}$ (b) $s - 3r - t$
11. (a) $1 + \log x + \frac{1}{2}\log(x-3)$ (b) $2\ln|x| + 3\ln\sin x - \frac{1}{2}\ln(x^2+1)$
13. $\log \frac{256}{3}$ 15. $\ln \dfrac{\sqrt[3]{x}(x+1)^2}{\cos x}$ 17. 0.01 19. e^2 21. 4
23. $\sqrt{3/2}$ 25. $-\dfrac{\ln 3}{2\ln 5}$ 27. $\frac{1}{3}\ln\frac{7}{2}$ 29. -2
31. (a) domain: $(-\infty, +\infty)$; range: $(-1, +\infty)$



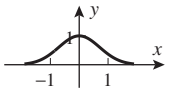
(b) domain: $x \ne 0$; range: $(-\infty, +\infty)$



33. (a) domain: $x \ne 0$; range: $(-\infty, +\infty)$

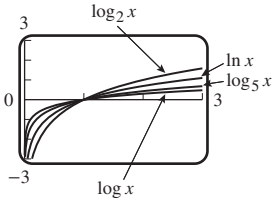


(b) domain: $(-\infty, +\infty)$; range: $(0, 1]$

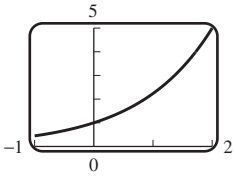


Responses to True–False questions may be abridged to save space.

35. False; exponential functions have constant base and variable exponent.
37. True; $\ln x = \log_e x$ 39. 2.8777, -0.3174
41.



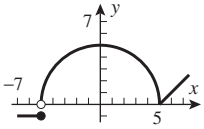
43. $x \approx 1.471, 7.857$
45. (a) no (d) $y = (\sqrt{5})^x$
    (b) $y = 2^{x/4}$
    (c) $y = 2^{-x}$



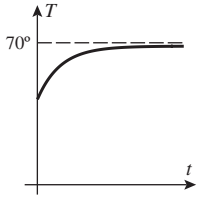
47. $\log\frac{1}{2} < 0$, so $3\log\frac{1}{2} < 2\log\frac{1}{2}$ 49. 201 days
51. (a) 7.4, basic (b) 4.2, acidic (c) 6.4, acidic (d) 5.9, acidic
53. (a) 140 dB, damage (b) 120 dB, damage (c) 80 dB, no damage
    (d) 75 dB, no damage
55. ≈ 200 57. (a) $\approx 5 \times 10^{16}$ J (b) ≈ 0.67

Chapter 0 Review Exercises (Page 63)

1.



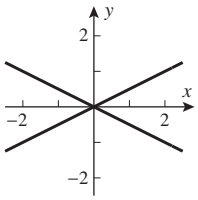
3.



5. **(a)** $C = 5x^2 + (64/x)$ **(b)** $x > 0$

7. **(a)** $V = (6 - 2x)(5 - x)x$ ft^3
(b) $0 < x < 3$
(c) 3.57 ft × 3.79 ft × 1.21 ft

9.



11.

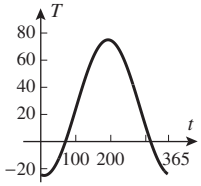
x	−4	−3	−2	−1	0	1	2	3	4
$f(x)$	0	−1	2	1	3	−2	−3	4	−4
$g(x)$	3	2	1	−3	−1	−4	4	−2	0
$(f \circ g)(x)$	4	−3	−2	−1	1	0	−4	2	3
$(g \circ f)(x)$	−1	−3	4	−4	−2	1	2	0	3

13. $0, -2$ 15. $1/(2 - x^2), x \neq \pm 1, \pm\sqrt{2}$

17. **(a)** odd **(b)** even **(c)** neither **(d)** even

19. **(a)** circles of radius 1 centered on the parabola $y = x^2$
(b) parabolas congruent to $y = x^2$ that open up with vertices on the line $y = x/2$

21. **(a)**



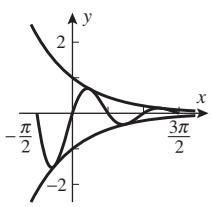
(b) January 11 **(c)** 122 days

23. A: $\left(-\frac{2}{3}\pi, 1 - \sqrt{3}\right)$; B: $\left(\frac{1}{3}\pi, 1 + \sqrt{3}\right)$; C: $\left(\frac{2}{3}\pi, 1 + \sqrt{3}\right)$; D: $\left(\frac{5}{3}\pi, 1 - \sqrt{3}\right)$

27. **(a)** $\frac{1}{2}(x + 1)^{1/3}$ **(b)** none **(c)** $\frac{1}{2}\ln(x - 1)$ **(d)** $\dfrac{x + 2}{x - 1}$
(e) $\dfrac{1}{2 + \sin^{-1} x}$ **(f)** $\tan\left(\dfrac{1}{3x} - \dfrac{1}{3}\right), x < -\dfrac{2}{3\pi - 2}$ or $x > \dfrac{2}{3\pi + 2}$

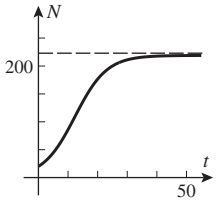
29. **(a)** $\frac{33}{65}$ **(b)** $\frac{56}{65}$ 31. $\frac{10^{60}}{63360} \approx 1.6 \times 10^{55}$ miles 33. $15x + 2$

35. **(a)**



(b) $-\dfrac{\pi}{2}, 0, \dfrac{\pi}{2}, \pi, \dfrac{3\pi}{2}$; $-\dfrac{\pi}{4}, \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}$

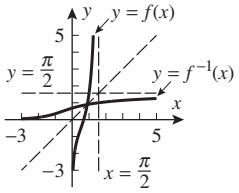
37. **(a)**



(b) about 10 years **(c)** 220 sheep

39. **(b)** 3.654, 332105.108

41. **(a)** f is increasing **(b)** asymptotes for f: $x = 0$ and $x = \pi/2$; asymptotes for f^{-1}: $y = 0$ (as $x \to -\infty$) and $y = \pi/2$ (as $x \to +\infty$)



Exercise Set 1.1 (Page 77)

1. **(a)** 3 **(b)** 3 **(c)** 3 **(d)** 3

3. **(a)** −1 **(b)** 3 **(c)** does not exist **(d)** 1

5. **(a)** 0 **(b)** 0 **(c)** 0 **(d)** 3 7. **(a)** $-\infty$ **(b)** $-\infty$ **(c)** $-\infty$ **(d)** 1

9. **(a)** $+\infty$ **(b)** $+\infty$ **(c)** 2 **(d)** 2 **(e)** $-\infty$ **(f)** the lines $x = -2$, $x = 0$, $x = 2$

11.

x	−0.01	−0.001	−0.0001	0.0001	0.001	0.01
$f(x)$	0.99502	0.99950	0.99995	1.00005	1.00050	1.00502

The limit appears to be 1.

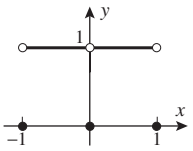
13. **(a)** $\frac{1}{3}$ **(b)** $+\infty$ **(c)** $-\infty$ 15. **(a)** 3 **(b)** does not exist

Responses to True–False questions may be abridged to save space.

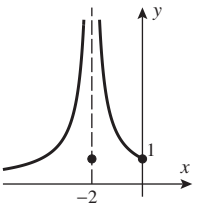
17. False; see Example 6.

19. False; the one-sided limits must also be equal.

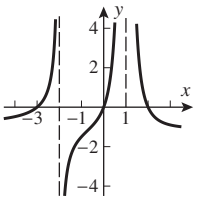
21.



23.



25.



27. $y = -2x - 1$

29. $y = 4x - 3$

31. **(a)** rest length **(b)** 0. As speed approaches c, length shrinks to zero.

Exercise Set 1.2 (Page 87)

1. **(a)** −6 **(b)** 13 **(c)** −8 **(d)** 16 **(e)** 2 **(f)** $-\frac{1}{2}$

3. 6 5. $\frac{3}{4}$ 7. 4 9. $-\frac{4}{5}$ 11. −3 13. $\frac{3}{2}$ 15. $+\infty$

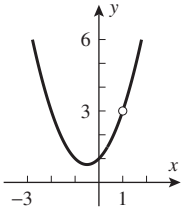
17. does not exist 19. $-\infty$ 21. $+\infty$ 23. does not exist 25. $+\infty$

27. $+\infty$ 29. 6 31. **(a)** 2 **(b)** 2 **(c)** 2

Responses to True–False questions may be abridged to save space.

33. True; this is Theorem 1.2.2(a). 35. False; see Example 9. 37. $\frac{1}{4}$

39. **(a)** 3 **(b)**



41. **(a)** Theorem 1.2.2(a) does not apply.
(b) $\displaystyle\lim_{x \to 0^+}\left(\frac{1}{x} - \frac{1}{x^2}\right) = \lim_{x \to 0^+}\left(\frac{x - 1}{x^2}\right) = -\infty$ 43. $a = 2$

45. The left and/or right limits could be $\pm\infty$; or the limit could exist and equal any preassigned real number.

Exercise Set 1.3 (Page 96)

1. **(a)** $-\infty$ **(b)** $+\infty$ 3. **(a)** 0 **(b)** −1

5. **(a)** −12 **(b)** 21 **(c)** −15 **(d)** 25 **(e)** 2 **(f)** $-\frac{3}{5}$ **(g)** 0
(h) does not exist

7. **(a)**

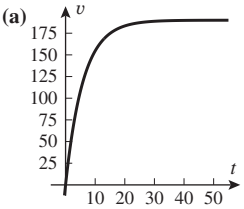
x	0.1	0.01	0.001	0.0001	0.00001	0.000001
$f(x)$	1.471128	1.560797	1.569796	1.570696	1.570786	1.570795

The limit appears to be $\pi/2$. **(b)** $\pi/2$

9. $-\infty$ **11.** $+\infty$ **13.** $\frac{3}{2}$ **15.** 0 **17.** 0 **19.** $-\infty$ **21.** $-\frac{1}{7}$
23. $\dfrac{-\sqrt[3]{5}}{2}$ **25.** $-\sqrt{5}$ **27.** $1/\sqrt{6}$ **29.** $\sqrt{3}$ **31.** 0 **33.** 1
35. 1 **37.** $-\infty$ **39.** e

Responses to True–False questions may be abridged to save space.

41. False; 1^∞ is an indeterminate form. The limit is e^2.
43. True; consider $f(x) = (\sin x)/x$.
45. $\lim_{t\to+\infty} n(t) = +\infty$; $\lim_{t\to+\infty} e(t) = c$ **47.** **(a)** $+\infty$ **(b)** -5
51. **(a)** no **(b)** yes; $\tan x$ and $\sec x$ at $x = n\pi + \pi/2$, and $\cot x$ and $\csc x$ at $x = n\pi$, $n = 0, \pm1, \pm2, \ldots$
55. $+\infty$ **57.** $+\infty$ **59.** 1 **61.** e
65. **(a)**

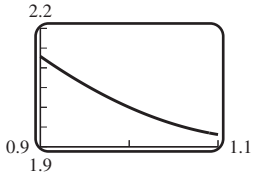


(b) $c = 190$
(c) It is the terminal velocity of the skydiver.

67. **(a)** e **(c)** e^a **69.** $x + 2$ **71.** $1 - x^2$ **73.** $\sin x$

Exercise Set 1.4 (Page 106)

1. **(a)** $|x| < 0.1$ **(b)** $|x - 3| < 0.0025$ **(c)** $|x - 4| < 0.000125$
3. **(a)** $x_0 = 3.8025$, $x_1 = 4.2025$ **(b)** $\delta = 0.1975$
5. $\delta = 0.0442$ **7.** $\delta = 0.13$
9. $\delta = 0.05$
11. $\delta = 0.05$

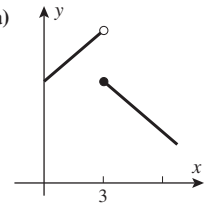


13. $\delta = \sqrt[3]{8.001} - 2 \approx 8.332986 \cdot 10^{-5}$ **15.** $\delta = 1/505 \approx 0.000198$
17. $\delta = 1$ **19.** $\delta = \frac{1}{3}\epsilon$ **21.** $\delta = \epsilon/2$ **23.** $\delta = \epsilon$ **25.** $\delta = \epsilon$
29. **(b)** 65 **(c)** $\epsilon/65$; 65; 65; $\epsilon/65$ **31.** $\delta = \min\left(1, \frac{1}{6}\epsilon\right)$
33. $\delta = \min\left(\frac{1}{2}, \frac{\epsilon}{2}\right)$ **35.** $\delta = 2\epsilon$
39. **(a)** $\sqrt{10}$ **(b)** 99 **(c)** -10 **(d)** -101
41. **(a)** $-\sqrt{\dfrac{1-\epsilon}{\epsilon}}$; $\sqrt{\dfrac{1-\epsilon}{\epsilon}}$ **(b)** $\sqrt{\dfrac{1-\epsilon}{\epsilon}}$ **(c)** $-\sqrt{\dfrac{1-\epsilon}{\epsilon}}$
43. 10 **45.** 999 **47.** -202 **49.** -57.5 **51.** $N = \dfrac{1}{\sqrt{\epsilon}}$
53. $N = -\dfrac{5}{2} - \dfrac{11}{2\epsilon}$ **55.** $N = (1 + 2/\epsilon)^2$
57. **(a)** $|x| < \frac{1}{10}$ **(b)** $|x - 1| < \frac{1}{1000}$
(c) $|x - 3| < \frac{1}{10\sqrt{10}}$ **(d)** $|x| < \frac{1}{10}$
59. $\delta = 1/\sqrt{M}$ **61.** $\delta = 1/M$ **63.** $\delta = 1/(-M)^{1/4}$ **65.** $\delta = \epsilon$
67. $\delta = \epsilon^2$ **69.** $\delta = \epsilon$ **71.** **(a)** $\delta = -1/M$ **(b)** $\delta = 1/M$
73. **(a)** $N = M - 1$ **(b)** $N = M - 1$
75. **(a)** 0.4 amps **(b)** about 0.39474 to 0.40541 amps **(c)** $3/(7.5 + \delta)$ to $3/(7.5 - \delta)$ **(d)** $\delta \approx 0.01870$ **(e)** current approaches $+\infty$

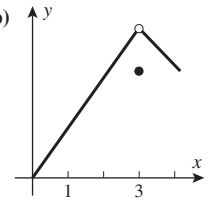
Exercise Set 1.5 (Page 118)

1. **(a)** not continuous, $x = 2$ **(b)** not continuous, $x = 2$
(c) not continuous, $x = 2$ **(d)** continuous **(e)** continuous
(f) continuous
3. **(a)** not continuous, $x = 1, 3$ **(b)** continuous
(c) not continuous, $x = 1$ **(d)** continuous
(e) not continuous, $x = 3$ **(f)** continuous
5. **(a)** no **(b)** no **(c)** no **(d)** yes **(e)** yes **(f)** no **(g)** yes

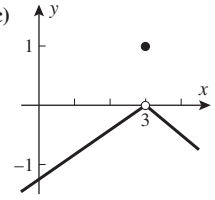
7. **(a)**



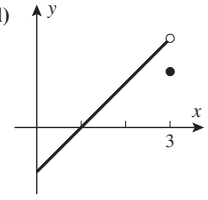
(b)



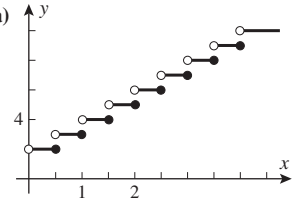
(c)



(d)



9. **(a)**

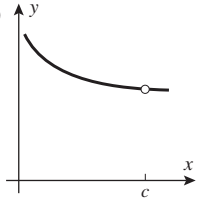


(b) One second could cost you one dollar.

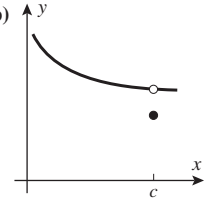
11. none **13.** none **15.** $-1/2, 0$ **17.** $-1, 0, 1$ **19.** none
21. none

Responses to True–False questions may be abridged to save space.

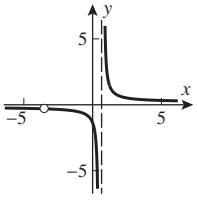
23. True; the composition of continuous functions is continuous.
25. False; let f and g be the functions in Exercise 6.
27. True; $f(x) = \sqrt{f(x)} \cdot \sqrt{f(x)}$
29. **(a)** $k = 5$ **(b)** $k = \frac{4}{3}$ **31.** $k = 4$, $m = 5/3$
33. **(a)**



(b)



35. **(a)** $x = 0$, not removable **(b)** $x = -3$, removable
(c) $x = 2$, removable; $x = -2$, not removable
37. **(a)** $x = \frac{1}{2}$, not removable; at $x = -3$, removable
(b) $(2x - 1)(x + 3)$



45. $f(x) = 1$ for $0 \le x < 1$, $f(x) = -1$ for $1 \le x \le 2$
49. $x = -1.25$, $x = 0.75$ **51.** $x = 2.24$

Exercise Set 1.6 (Page 125)

1. none **3.** $x = n\pi$, $n = 0, \pm1, \pm2, \ldots$
5. $x = n\pi$, $n = 0, \pm1, \pm2, \ldots$
7. $2n\pi + (\pi/6)$, $2n\pi + (5\pi/6)$, $n = 0, \pm1, \pm2, \ldots$
9. $\left[-\frac{1}{2}, \frac{1}{2}\right]$ **11.** $(0, 3)$ and $(3, +\infty)$ **13.** $(-\infty, -1]$ and $[1, +\infty)$
15. **(a)** $\sin x$, $x^3 + 7x + 1$ **(b)** $|x|$, $\sin x$ **(c)** x^3, $\cos x$, $x + 1$
17. 1 **19.** $-\pi/6$ **21.** 1 **23.** 3 **25.** $+\infty$ **27.** $\frac{7}{3}$
29. 0 **31.** 0 **33.** 1 **35.** 2 **37.** does not exist **39.** 0

41. **(a)**

x	4	4.5	4.9	5.1	5.5	6
$f(x)$	0.093497	0.100932	0.100842	0.098845	0.091319	0.076497

The limit appears to be $\frac{1}{10}$. **(b)** $\frac{1}{10}$

Responses to True–False questions may be abridged to save space.

43. True; use the Squeezing Theorem.

45. False; consider $f(x) = \tan^{-1} x$.

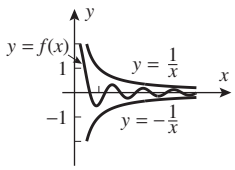
47. **(a)** Using degrees instead of radians **(b)** $\pi/180$

49. $k = \frac{1}{2}$ **51.** **(a)** 1 **(b)** 0 **(c)** 1

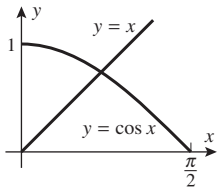
53. $-\pi$ **55.** $-\sqrt{2}$ **57.** 1 **59.** 5 **61.** $-|x| \le x\cos(50\pi/x) \le |x|$

63. $\lim_{x\to 0} \sin(1/x)$ does not exist

65. The limit is 0.

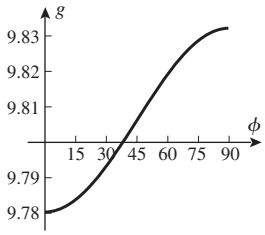


67. **(b)**



(c) 0.739

69. **(a)** Gravity is strongest at the poles and weakest at the equator.



Chapter 1 Review Exercises (Page 128)

1. **(a)** 1 **(b)** does not exist **(c)** does not exist **(d)** 1 **(e)** 3 **(f)** 0 **(g)** 0 **(h)** 2 **(i)** $\frac{1}{2}$

3. **(a)** 0.405 **5.** 1 **7.** $-3/2$ **9.** $32/3$

11. **(a)** $y = 0$ **(b)** none **(c)** $y = 2$ **13.** 1 **15.** $3 - k$ **17.** 0

19. e^{-3} **21.** \$2001.60, \$2009.66, \$2013.62, \$2013.75

23. **(a)** $2x/(x-1)$ is one example.

25. **(a)** $\lim_{x\to 2} f(x) = 5$ **(b)** $\delta = 0.0045$

27. **(a)** $\delta = 0.0025$ **(b)** $\delta = 0.0025$ **(c)** $\delta = 1/9000$ (Some larger values also work.)

31. **(a)** $-1, 1$ **(b)** none **(c)** $-3, 0$ **33.** no; not continuous at $x = 2$

35. Consider $f(x) = x$ for $x \ne 0$, $f(0) = 1$, $a = -1$, $b = 1$, $k = 0$.

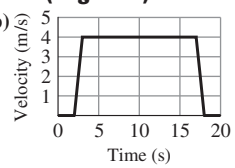
Chapter 1 Making Connections (Page 130)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

4. **(a)** The circle through the origin with center $\left(0, \frac{1}{8}\right)$
(b) The circle through the origin with center $\left(0, \frac{1}{2}\right)$
(c) The circle does not exist.
(d) The circle through the origin with center $\left(0, \frac{1}{2}\right)$
(e) The circle through $(0, 1)$ with center at the origin.
(f) The circle through the origin with center $\left(0, \dfrac{1}{2g(0)}\right)$
(g) The circle does not exist.

Exercise Set 2.1 (Page 140)

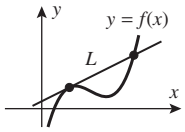
1. **(a)** 4 m/s **(b)**

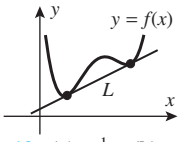


3. **(a)** 0 cm/s **(b)** $t = 0$, $t = 2$, and $t = 4.2$ **(c)** maximum: $t = 1$; minimum: $t = 3$ **(d)** -7.5 cm/s

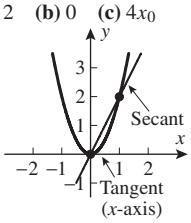
5. straight line with slope equal to the velocity

7. Answers may vary. **9.** Answers may vary.

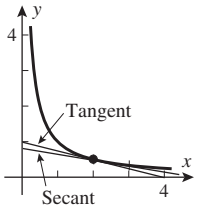




11. **(a)** 2 **(b)** 0 **(c)** $4x_0$ **(d)**



13. **(a)** $-\frac{1}{6}$ **(b)** $-\frac{1}{4}$ **(c)** $-1/x_0^2$ **(d)**



15. **(a)** $2x_0$ **(b)** -2 **17.** **(a)** $1 + \dfrac{1}{2\sqrt{x_0}}$ **(b)** $\frac{3}{2}$

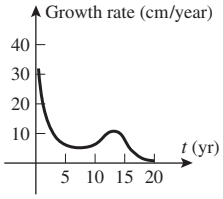
Responses to True–False questions may be abridged to save space.

19. True; set $h = x - 1$, so $x = 1 + h$ and $h \to 0$ is equivalent to $x \to 1$.

21. False; velocity is a ratio of change in position to change in time.

23. **(a)** 72° F at about 4:30 P.M. **(b)** 4° F/h **(c)** -7° F/h at about 9 P.M.

25. **(a)** first year
(b) 6 cm/year
(c) 10 cm/year at about age 14
(d)



27. **(a)** 19,200 ft **(b)** 480 ft/s **(c)** 66.94 ft/s **(d)** 1440 ft/s

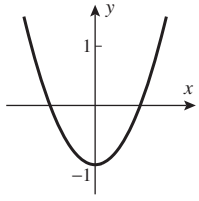
29. **(a)** 720 ft/min **(b)** 192 ft/min

Exercise Set 2.2 (Page 152)

1. $2, 0, -2, -1$

3. **(b)** 3 **(c)** 3

5.



7. $y = 5x - 16$

9. $4x$, $y = 4x - 2$

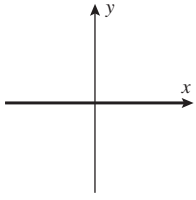
11. $3x^2$; $y = 0$

13. $\dfrac{1}{2\sqrt{x+1}}$; $y = \frac{1}{6}x + \frac{5}{3}$ **15.** $-1/x^2$ **17.** $2x - 1$

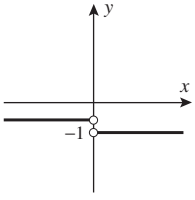
19. $-1/(2x^{3/2})$ **21.** $8t + 1$

23. **(a)** D **(b)** F **(c)** B **(d)** C **(e)** A **(f)** E

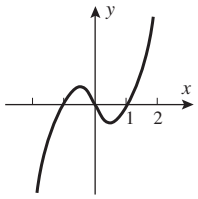
25. **(a)**



(b)



(c)

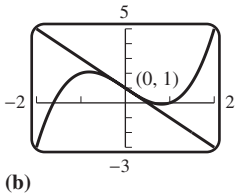


Responses to True–False questions may be abridged to save space.

27. False; $f'(a) = 0$ **29.** False; for example, $f(x) = |x|$

31. **(a)** $\sqrt{x}$, 1 **(b)** x^2, 3 **33.** -2

35. $y = -2x + 1$



37. **(b)**

w	1.5	1.1	1.01	1.001	1.0001	1.00001
$[f(w)-f(1)]/(w-1)$	1.6569	1.4355	1.3911	1.3868	1.3863	1.3863

39. **(a)** 0.04, 0.22, 0.88 **(b)** best: $\dfrac{f(2)-f(0)}{2-0}$; worst: $\dfrac{f(3)-f(1)}{3-1}$

41. **(a)** dollars per foot **(b)** the price per additional foot **(c)** positive **(d)** \$1000

43. **(a)** $F \approx 200$ lb, $dF/d\theta \approx 50$ lb/rad **(b)** $\mu = 0.25$

45. **(a)** $T \approx 115°$F, $dT/dt \approx -3.35°$F/min **(b)** $k = -0.084$

Exercise Set 2.3 (Page 161)

1. $28x^6$ **3.** $24x^7 + 2$ **5.** 0 **7.** $-\frac{1}{3}(7x^6 + 2)$

9. $-3x^{-4} - 7x^{-8}$ **11.** $24x^{-9} + (1/\sqrt{x})$

13. $f'(x) = ex^{e-1} - \dfrac{\sqrt{10}}{x^{(1+\sqrt{10})}}$

15. $12x(3x^2 + 1)$ **17.** 7 **19.** $2t - 1$ **21.** 15 **23.** -8 **25.** 0

27. 0 **29.** $32t$ **31.** $3\pi r^2$

Responses to True–False questions may be abridged to save space.

33. True; apply the difference and constant multiple rules.

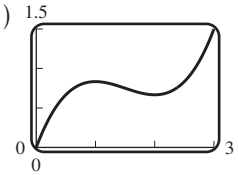
35. False; $\dfrac{d}{dx}[4f(x) + x^3]\Big|_{x=2} = [4f'(x) + 3x^2]\big|_{x=2} = 32$

37. **(a)** $4\pi r^2$ **(b)** 100π **39.** $y = 5x + 17$

41. **(a)** $42x - 10$ **(b)** 24 **(c)** $2/x^3$ **(d)** $700x^3 - 96x$

43. **(a)** $-210x^{-8} + 60x^2$ **(b)** $-6x^{-4}$ **(c)** $6a$

45. **(a)** 0 **49.** $(1, \frac{5}{6})$ $(2, \frac{2}{3})$
(b) 112
(c) 360

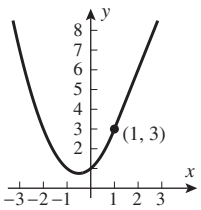


51. $y = 3x^2 - x - 2$ **53.** $x = \frac{1}{2}$

55. $(2 + \sqrt{3}, -6 - 4\sqrt{3})$, $(2 - \sqrt{3}, -6 + 4\sqrt{3})$

57. $-2x_0$ **61.** $-\dfrac{2GmM}{r^3}$ **63.** $f'(x) > 0$ for all $x \neq 0$

65. yes, 3



67. not differentiable at $x = 1$ **69.** **(a)** $x = \frac{2}{3}$ **(b)** $x = \pm 2$ **71.** **(b)** yes

73. **(a)** $n(n-1)(n-2)\cdots 1$ **(b)** 0 **(c)** $a_n n(n-1)(n-2)\cdots 1$

79. $-12/(2x+1)^3$ **81.** $-2/(x+1)^3$

Exercise Set 2.4 (Page 168)

1. $4x + 1$ **3.** $4x^3$ **5.** $18x^2 - \frac{3}{2}x + 12$

7. $-15x^{-2} - 14x^{-3} + 48x^{-4} + 32x^{-5}$ **9.** $3x^2$

11. $\dfrac{-3x^2 - 8x + 3}{(x^2+1)^2}$ **13.** $\dfrac{3x^2 - 8x}{(3x-4)^2}$ **15.** $\dfrac{x^{3/2} + 10x^{1/2} + 4 - 3x^{-1/2}}{(x+3)^2}$

17. $2(1 + x^{-1})(x^{-3} + 7) + (2x + 1)(-x^{-2})(x^{-3} + 7) + (2x + 1)(1 + x^{-1})(-3x^{-4})$ **19.** $3(7x^6 + 2)(x^7 + 2x - 3)^2$ **21.** $\frac{7}{16}$

23. -29 **25.** 0 **27.** **(a)** $-\frac{37}{4}$ **(b)** $-\frac{23}{16}$

29. **(a)** 10 **(b)** 19 **(c)** 9 **(d)** -1 **31.** $-2 \pm \sqrt{3}$ **33.** none **35.** -2

39. $F''(x) = xf''(x) + 2f'(x)$

41. $R'(120) = 1800$; increasing the price by Δp dollars increases revenue by approximately $1800\Delta p$ dollars.

43. $f'(x) = -nx^{-n-1}$

Exercise Set 2.5 (Page 172)

1. $-4\sin x + 2\cos x$ **3.** $4x^2 \sin x - 8x\cos x$

5. $(1 + 5\sin x - 5\cos x)/(5 + \sin x)^2$ **7.** $\sec x \tan x - \sqrt{2}\sec^2 x$

9. $-4\csc x \cot x + \csc^2 x$ **11.** $\sec^3 x + \sec x \tan^2 x$ **13.** $-\dfrac{\csc x}{1 + \csc x}$

15. 0 **17.** $\dfrac{1}{(1 + x\tan x)^2}$ **19.** $-x\cos x - 2\sin x$

21. $-x\sin x + 5\cos x$ **23.** $-4\sin x \cos x$

25. **(a)** $y = x$ **(b)** $y = 2x - (\pi/2) + 1$ **(c)** $y = 2x + (\pi/2) - 1$

29. **(a)** $x = \pm\pi/2, \pm 3\pi/2$ **(b)** $x = -3\pi/2, \pi/2$
(c) no horizontal tangent line **(d)** $x = \pm 2\pi, \pm\pi, 0$

31. 0.087 ft/deg **33.** 1.75 m/deg

Responses to True–False questions may be abridged to save space.

35. False; by the product rule, $g'(x) = f(x)\cos x + f'(x)\sin x$.

37. True; $f(x) = (\sin x)/(\cos x) = \tan x$, so $f'(x) = \sec^2 x$.

39. $-\cos x$ **41.** 3, 7, 11, ...

43. **(a)** all x **(b)** all x **(c)** $x \neq (\pi/2) + n\pi, n = 0, \pm 1, \pm 2, \ldots$
(d) $x \neq n\pi, n = 0, \pm 1, \pm 2, \ldots$ **(e)** $x \neq (\pi/2) + n\pi, n = 0, \pm 1, \pm 2, \ldots$ **(f)** $x \neq n\pi, n = 0, \pm 1, \pm 2, \ldots$ **(g)** $x \neq (2n+1)\pi, n = 0, \pm 1, \pm 2, \ldots$ **(h)** $x \neq n\pi/2, n = 0, \pm 1, \pm 2, \ldots$ **(i)** all x

Exercise Set 2.6 (Page 178)

1. 6 **3.** **(a)** $(2x-3)^5$, $10(2x-3)^4$ **(b)** $2x^5 - 3$, $10x^4$

5. **(a)** -7 **(b)** -8 **7.** $37(x^3 + 2x)^{36}(3x^2 + 2)$

9. $-2\left(x^3 - \dfrac{7}{x}\right)^{-3}\left(3x^2 + \dfrac{7}{x^2}\right)$ **11.** $\dfrac{24(1-3x)}{(3x^2 - 2x + 1)^4}$

13. $\dfrac{3}{4\sqrt{3x}\sqrt{4 + \sqrt{3x}}}$ **15.** $-\dfrac{2}{x^3}\cos\left(\dfrac{1}{x^2}\right)$ **17.** $-20\cos^4 x \sin x$

19. $-\dfrac{3}{\sqrt{x}}\cos(3\sqrt{x})\sin(3\sqrt{x})$ **21.** $28x^6\sec^2(x^7)\tan(x^7)$

23. $-\dfrac{5\sin(5x)}{2\sqrt{\cos(5x)}}$

25. $-3[x + \csc(x^3 + 3)]^{-4}[1 - 3x^2\csc(x^3 + 3)\cot(x^3 + 3)]$

27. $10x^3 \sin 5x \cos 5x + 3x^2 \sin^2 5x$

29. $-x^3 \sec\left(\dfrac{1}{x}\right)\tan\left(\dfrac{1}{x}\right) + 5x^4 \sec\left(\dfrac{1}{x}\right)$

31. $\sin(\cos x)\sin x$ **33.** $-6\cos^2(\sin 2x)\sin(\sin 2x)\cos 2x$

35. $35(5x + 8)^6(1 - \sqrt{x})^6 - \dfrac{3}{\sqrt{x}}(5x + 8)^7(1 - \sqrt{x})^5$

37. $\dfrac{33(x-5)^2}{(2x+1)^4}$ **39.** $-\dfrac{2(2x+3)^2(52x^2 + 96x + 3)}{(4x^2 - 1)^9}$

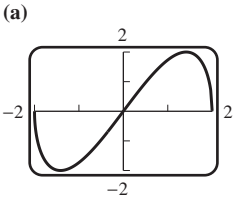
41. $5[x\sin 2x + \tan^4(x^7)]^4[2x\cos 2x + \sin 2x + 28x^6\tan^3(x^7)\sec^2(x^7)]$

43. $y = -x$ **45.** $y = -1$ **47.** $y = 8\sqrt{\pi}x - 8\pi$ **49.** $y = \frac{7}{2}x - \frac{3}{2}$

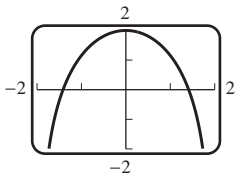
51. $-25x\cos(5x) - 10\sin(5x) - 2\cos(2x)$ **53.** $4(1-x)^{-3}$

55. $3\cot^2\theta\csc^2\theta$ **57.** $\pi(b - a)\sin 2\pi\omega$

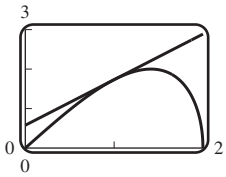
59. **(a)**



(c) $\dfrac{4-2x^2}{\sqrt{4-x^2}}$



(d) $y-\sqrt{3}=\frac{2}{\sqrt{3}}(x-1)$



Responses to True–False questions may be abridged to save space.

61. False; by the chain rule, $\dfrac{d}{dx}[\sqrt{y}]=\dfrac{1}{2\sqrt{y}}\cdot\dfrac{dy}{dx}=\dfrac{f'(x)}{2\sqrt{f(x)}}$.

63. False; by the chain rule, $dy/dx=(-\sin[g(x)])\cdot g'(x)$.

65. **(c)** $f=1/T$ **(d)** amplitude $=0.6$ cm, $T=2\pi/15$ seconds per oscillation, $f=15/(2\pi)$ oscillations per second

67. $\frac{7}{24}\sqrt{6}$ **69.** **(a)** $10\,\text{lb/in}^2$, $-2\,\text{lb/in}^2/\text{mi}$ **(b)** $-0.6\,\text{lb/in}^2/\text{s}$

71. $\begin{cases}\cos x, & 0<x<\pi\\ -\cos x, & -\pi<x<0\end{cases}$

73. **(c)** $-\dfrac{1}{x}\cos\dfrac{1}{x}+\sin\dfrac{1}{x}$ **(d)** limit as x goes to 0 does not exist

75. **(a)** 21 **(b)** -36 **77.** $1/(2x)$ **79.** $\frac{2}{3}x$ **83.** $f'(g(h(x)))g'(h(x))h'(x)$

▶ Chapter 2 Review Exercises (Page 181)

3. **(a)** $2x$ **(b)** 4 **5.** 58.75 ft/s **7.** **(a)** 13 mi/h **(b)** 7 mi/h

9. **(a)** $-2/\sqrt{9-4x}$ **(b)** $1/(x+1)^2$

11. **(a)** $x=-2,-1,1,3$ **(b)** $(-\infty,-2),(-1,1),(3,+\infty)$ **(c)** $(-2,-1),(1,3)$ **(d)** 4

13. **(a)** 78 million people per year **(b)** 1.3% per year

15. **(a)** $x^2\cos x+2x\sin x$ **(c)** $4x\cos x+(2-x^2)\sin x$

17. **(a)** $(6x^2+8x-17)/(3x+2)^2$ **(c)** $118/(3x+2)^3$

19. **(a)** 2000 gal/min **(b)** 2500 gal/min **21.** **(a)** 3.6 **(b)** -0.777778

23. $f(1)=0$, $f'(1)=5$ **25.** $y=-16x$, $y=-145x/4$

29. **(a)** $8x^7-\dfrac{3}{2\sqrt{x}}-15x^{-4}$ **(b)** $(2x+1)^{100}(1030x^2+10x-1414)$

31. **(a)** $\dfrac{(x-1)(15x+1)}{2\sqrt{3x+1}}$ **(b)** $-3(3x+1)^2(3x+2)/x^7$

33. $x=-\frac{7}{2},-\frac{1}{2},2$ **35.** $y=\pm 2x$

37. $x=n\pi\pm(\pi/4)$, $n=0,\pm1,\pm2,\ldots$ **39.** $y=-3x+(1+9\pi/4)$

41. **(a)** $40\sqrt{3}$ **(b)** 7500

▶ Chapter 2 Making Connections (Page 184)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

2. **(c)** $k=2$ **(d)** $h'(x)=0$

3. **(b)** $f'\cdot g\cdot h\cdot k+f\cdot g'\cdot h\cdot k+f\cdot g\cdot h'\cdot k+f\cdot g\cdot h\cdot k'$

4. **(c)** $\dfrac{f'\cdot g\cdot h-f\cdot g'\cdot h+f\cdot g\cdot h'}{g^2}$

▶ Exercise Set 3.1 (Page 190)

1. **(a)** $(6x^2-y-1)/x$ **(b)** $4x-2/x^2$ **3.** $-\dfrac{x}{y}$ **5.** $\dfrac{1-2xy-3y^3}{x^2+9xy^2}$

7. $\dfrac{-y^{3/2}}{x^{3/2}}$ **9.** $\dfrac{1-2xy^2\cos(x^2y^2)}{2x^2y\cos(x^2y^2)}$

11. $\dfrac{1-3y^2\tan^2(xy^2+y)\sec^2(xy^2+y)}{3(2xy+1)\tan^2(xy^2+y)\sec^2(xy^2+y)}$ **13.** $-\dfrac{8}{9y^3}$ **15.** $\dfrac{2y}{x^2}$

17. $\dfrac{\sin y}{(1+\cos y)^3}$ **19.** $-1/\sqrt{3},1/\sqrt{3}$

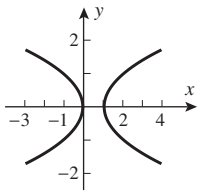
Responses to True–False questions may be abridged to save space.

21. False; the graph of f need only coincide with a portion of the graph of the equation in x and y.

23. False; the equation is equivalent to $x^2=y^2$ and $y=|x|$ satisfies this equation.

25. $-15^{-3/4}\approx-0.1312$ **27.** $-\frac{9}{13}$ **29.** $\dfrac{2t^3+3a^2}{2a^3-6at}$ **31.** $-\dfrac{b^2\lambda}{a^2\omega}$

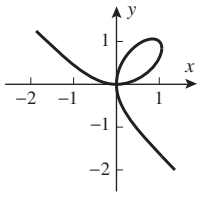
35. **(a)**



(c) $x=-y^2$ or $x=y^2+1$

37. $a=\frac{1}{4}$, $b=\frac{5}{4}$

41. **(a)**



(c) $2\sqrt[3]{2}/3$ **43.** $-\dfrac{2y^3+3t^2y}{(6ty^2+t^3)\cos t}$

▶ Exercise Set 3.2 (Page 195)

1. $1/x$ **3.** $1/(1+x)$ **5.** $2x/(x^2-1)$ **7.** $\dfrac{1-x^2}{x(1+x^2)}$ **9.** $2/x$

11. $\dfrac{1}{2x\sqrt{\ln x}}$ **13.** $1+\ln x$ **15.** $2x\log_2(3-2x)-\dfrac{2x^2}{(\ln 2)(3-2x)}$

17. $\dfrac{2x(1+\log x)-x/(\ln 10)}{(1+\log x)^2}$ **19.** $1/(x\ln x)$ **21.** $2\csc 2x$

23. $-\dfrac{1}{x}\sin(\ln x)$ **25.** $2\cot x/(\ln 10)$ **27.** $\dfrac{3}{x-1}+\dfrac{8x}{x^2+1}$

29. $-\tan x+\dfrac{3x}{4-3x^2}$

Responses to True–False questions may be abridged to save space.

31. True; $\lim\limits_{x\to 0^+}\dfrac{1}{x}=+\infty$ **33.** True; $1/x$ is an odd function.

35. $x\sqrt[3]{1+x^2}\left[\dfrac{1}{x}+\dfrac{2x}{3(1+x^2)}\right]$

37. $\dfrac{(x^2-8)^{1/3}\sqrt{x^3+1}}{x^6-7x+5}\left[\dfrac{2x}{3(x^2-8)}+\dfrac{3x^2}{2(x^3+1)}-\dfrac{6x^5-7}{x^6-7x+5}\right]$

39. **(a)** $-\dfrac{1}{x(\ln x)^2}$ **(b)** $-\dfrac{\ln 2}{x(\ln x)^2}$

41. $y=ex-2$ **43.** $y=-x/e$ **45.** **(a)** $y=x/e$ **47.** $A(w)=w/2$

51. $f(x)=\ln(x+1)$ **53.** **(a)** 3 **(b)** -5 **55.** **(a)** 0 **(b)** $\sqrt{2}$

▶ Exercise Set 3.3 (Page 201)

1. **(b)** $\frac{1}{9}$ **3.** $-2/x^2$ **5.** **(a)** no **(b)** yes **(c)** yes **(d)** yes

7. $\dfrac{1}{15y^2+1}$ **9.** $\dfrac{1}{10y^4+3y^2}$ **13.** $f(x)+g(x)$, $f(g(x))$

15. $7e^{7x}$ **17.** $x^2e^x(x+3)$ **19.** $\dfrac{4}{(e^x+e^{-x})^2}$

21. $(x\sec^2 x+\tan x)e^{x\tan x}$ **23.** $(1-3e^{3x})e^{x-e^{3x}}$

25. $\dfrac{x-1}{e^x-x}$ **27.** $2^x\ln 2$ **29.** $\pi^{\sin x}(\ln\pi)\cos x$

31. $(x^3-2x)^{\ln x}\left[\dfrac{3x^2-2}{x^3-2x}\ln x+\dfrac{1}{x}\ln(x^3-2x)\right]$

33. $(\ln x)^{\tan x}\left[\dfrac{\tan x}{x\ln x}+(\sec^2 x)\ln(\ln x)\right]$

35. $(\ln x)^{\ln x}\left[\dfrac{\ln(\ln x)}{x}+\dfrac{1}{x}\right]$ **37.** $(x^3+x^2-4x+1)e^x$

39. $((\ln 3)x^2+2x+(\ln 3)\sqrt{x}+1/(2\sqrt{x}))3^x$

41. $(\ln 4)(3\cos x-e^x)4^{3\sin x-e^x}$ **43.** $3/\sqrt{1-9x^2}$

45. $-\dfrac{1}{|x|\sqrt{x^2-1}}$ **47.** $3x^2/(1+x^6)$ **49.** $-\dfrac{\sec^2 x}{\tan^2 x}=-\csc^2 x$

51. $\dfrac{e^x}{|x|\sqrt{x^2-1}}+e^x\sec^{-1}x$ **53.** 0 **55.** 0 **57.** $-\dfrac{1}{2\sqrt{x}(1+x)}$

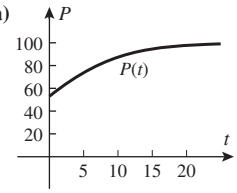
Responses to True–False questions may be abridged to save space.

59. False; consider $y=Ae^x$. **61.** True; use the chain rule.

65. $\dfrac{(3x^2+\tan^{-1}y)(1+y^2)}{(1+y^2)e^y-x}$ **67.** **(b)** $1-(\sqrt{3}/3)$

69. **(b)** $y=(88x-89)/7$

75. **(a)**



(b) $P(t)\to 100\%$
(c) $P'(t)\to 0$

77. 3 **79.** ln 10 **81.** 12π **83.** $9.8t$; if the fluid offers no resistance, then the speed will increase at a constant rate of 9.8 m/s.

Exercise Set 3.4 (Page 208)

1. **(a)** 6 **(b)** $-\frac{1}{3}$ **3.** **(a)** -2 **(b)** $6\sqrt{5}$

5. **(b)** $A=x^2$ **(c)** $\dfrac{dA}{dt}=2x\dfrac{dx}{dt}$ **(d)** 12 ft²/min

7. **(a)** $\dfrac{dV}{dt}=\pi\left(r^2\dfrac{dh}{dt}+2rh\dfrac{dr}{dt}\right)$ **(b)** -20π in³/s; decreasing

9. **(a)** $\dfrac{d\theta}{dt}=\dfrac{\cos^2\theta}{x^2}\left(x\dfrac{dy}{dt}-y\dfrac{dx}{dt}\right)$ **(b)** $-\frac{5}{16}$ rad/s; decreasing

11. $\dfrac{4\pi}{15}$ in²/min **13.** $\dfrac{1}{\sqrt{\pi}}$ mi/h **15.** 4860π cm³/min **17.** $\frac{5}{6}$ ft/s

19. $\dfrac{125}{\sqrt{61}}$ ft/s **21.** 704 ft/s

23. **(a)** 500 mi, 1716 mi **(b)** 1354 mi; 27.7 mi/min

25. $\dfrac{9}{20\pi}$ ft/min **27.** 125π ft³/min **29.** 250 mi/h

31. $\dfrac{36\sqrt{69}}{25}$ ft/min **33.** $\dfrac{8\pi}{5}$ km/s **35.** $600\sqrt{7}$ mi/h

37. **(a)** $-\frac{60}{7}$ units per second **(b)** falling **39.** -4 units per second

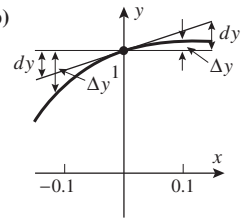
41. $x=\pm\sqrt{\dfrac{-5+\sqrt{33}}{2}}$ **43.** 4.5 cm/s; away **47.** $\dfrac{20}{9\pi}$ cm/s

Exercise Set 3.5 (Page 217)

1. **(a)** $f(x)\approx 1+3(x-1)$ **(b)** $f(1+\Delta x)\approx 1+3\Delta x$ **(c)** 1.06

3. **(a)** $1+\frac{1}{2}x$, 0.95, 1.05 **17.** $|x|<1.692$

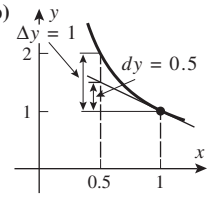
(b)



19. $|x|<0.3158$ **21.** **(a)** 0.0174533 **(b)** $x_0=45°$ **(c)** 0.694765

23. 83.16 **25.** 8.0625 **27.** 8.9944 **29.** 0.1 **31.** 0.8573

33. 0.780398 **37.** **(a)** 0.5, 1 **(b)**



39. $3x^2\,dx$, $3x^2\Delta x+3x(\Delta x)^2+(\Delta x)^3$

41. $(2x-2)\,dx$, $2x\Delta x+(\Delta x)^2-2\Delta x$

43. **(a)** $(12x^2-14x)\,dx$ **(b)** $(-x\sin x+\cos x)\,dx$

45. **(a)** $\dfrac{2-3x}{2\sqrt{1-x}}\,dx$ **(b)** $-17(1+x)^{-18}\,dx$

Responses to True–False questions may be abridged to save space.

47. False; $dy=(dy/dx)\,dx$ **49.** False; consider any linear function.

51. 0.0225 **53.** 0.0048 **55.** **(a)** ± 2 ft² **(b)** side: $\pm 1\%$; area: $\pm 2\%$

57. **(a)** opposite: ± 0.151 in; adjacent: ± 0.087 in
(b) opposite: $\pm 3.0\%$; adjacent: $\pm 1.0\%$

59. $\pm 10\%$ **61.** ± 0.017 cm² **63.** $\pm 6\%$ **65.** $\pm 0.5\%$ **67.** $15\pi/2$ cm³

69. **(a)** $\alpha=1.5\times 10^{-5}/°\text{C}$ **(b)** 180.1 cm long

Exercise Set 3.6 (Page 226)

1. **(a)** $\frac{2}{3}$ **(b)** $\frac{2}{3}$

Responses to True–False questions may be abridged to save space.

3. True; the expression $(\ln x)/x$ is undefined if $x\le 0$.

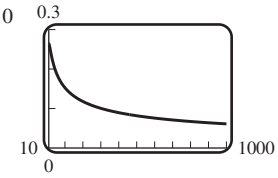
5. False; applying L'Hôpital's rule repeatedly shows that the limit is 0.

7. 1 **9.** 1 **11.** -1 **13.** 0 **15.** $-\infty$ **17.** 0 **19.** 2 **21.** 0

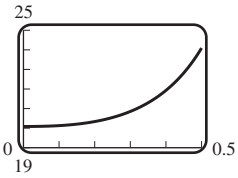
23. π **25.** $-\frac{5}{3}$ **27.** e^{-3} **29.** e^2 **31.** $e^{2/\pi}$ **33.** 0 **35.** $\frac{1}{2}$

37. $+\infty$ **39.** 1 **41.** 1 **43.** 1 **45.** 1 **47.** **(b)** 2

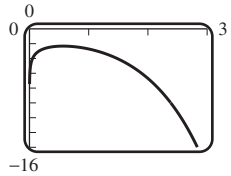
49. 0

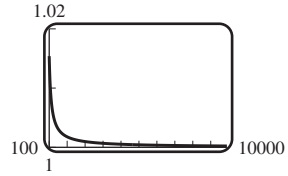


51. e^3



53. no horizontal asymptote **55.** $y=1$





57. **(a)** 0 **(b)** $+\infty$ **(c)** 0 **(d)** $-\infty$ **(e)** $+\infty$ **(f)** $-\infty$ **59.** 1

61. does not exist **63.** Vt/L **67.** **(a)** no **(b)** Both limits equal 0.

69. does not exist

Chapter 3 Review Exercises (Page 228)

1. **(a)** $\dfrac{2-3x^2-y}{x}$ **(b)** $-\dfrac{1}{x^2}-2x$ **3.** $-\dfrac{y^2}{x^2}$

5. $\dfrac{y\sec(xy)\tan(xy)}{1-x\sec(xy)\tan(xy)}$ **7.** $-\dfrac{21}{16y^3}$

9. $2/(2-\pi)$ **13.** $(\sqrt[3]{4}/3,\ \sqrt[3]{2}/3)$

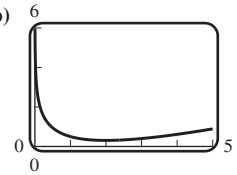
15. $\dfrac{1}{x+1}+\dfrac{2}{x+2}-\dfrac{3}{x+3}-\dfrac{4}{x+4}$ **17.** $\dfrac{1}{x}$ **19.** $\dfrac{1}{3x(\ln x+1)^{2/3}}$

21. $\dfrac{1}{(\ln 10)x\ln x}$ **23.** $\dfrac{3}{2x}+\dfrac{2x^3}{1+x^4}$ **25.** $2x$ **27.** $e^{\sqrt{x}}(2+\sqrt{x})$

29. $\dfrac{2}{\pi(1+4x^2)}$ **31.** $e^x x^{(e^x)}\left(\ln x+\dfrac{1}{x}\right)$ **33.** $\dfrac{1}{|2x+1|\sqrt{x^2+x}}$

35. $\dfrac{x^3}{\sqrt{x^2+1}}\left(\dfrac{3}{x}-\dfrac{x}{x^2+1}\right)$

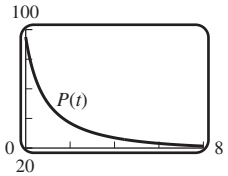
37. **(b)**



(d) curve must have a horizontal tangent line between $x = 1$ and $x = e$ **(e)** $x = 2$

39. e^2 **41.** $e^{1/e}$ **43.** No; for example, $f(x) = x^3$. **45.** $\left(\frac{1}{3}, e\right)$

51. **(a)**



(b) $P(t) \to 19$ **(c)** $P'(t) \to 0$

53. $+\infty, +\infty$: yes; $+\infty, -\infty$: no; $-\infty, +\infty$: no; $-\infty, -\infty$: yes

55. $+\infty$ **57.** $\frac{1}{9}$ **59.** $500\pi \text{ m}^2/\text{min}$

61. **(a)** $-0.5, 1, 0.5$ **(b)** $\pi/4, 1, \pi/2$ **(c)** $3, -1.0$

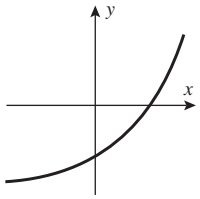
63. **(a)** between 139.48 m and 144.55 m **(c)** $|d\phi| \le 0.98°$

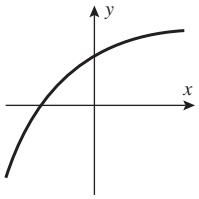
▶ Chapter 3 Making Connections (Page 230)

Answers are provided in the Student Solutions Manual.

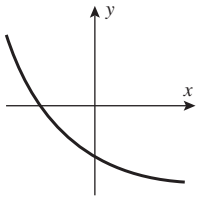
▶ Exercise Set 4.1 (Page 241)

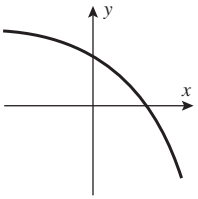
1. **(a)** $f' > 0, f'' > 0$ **(b)** $f' > 0, f'' < 0$





(c) $f' < 0, f'' > 0$ **(d)** $f' < 0, f'' < 0$





3. A: $dy/dx < 0, d^2y/dx^2 > 0$; B: $dy/dx > 0, d^2y/dx^2 < 0$; C: $dy/dx < 0, d^2y/dx^2 < 0$ **5.** $x = -1, 0, 1, 2$

7. **(a)** $[4, 6]$ **(b)** $[1, 4], [6, 7]$ **(c)** $(1, 2), (3, 5)$ **(d)** $(2, 3), (5, 7)$ **(e)** $x = 2, 3, 5$

9. **(a)** $[1, 3]$ **(b)** $(-\infty, 1], [3, +\infty)$ **(c)** $(-\infty, 2), (4, +\infty)$ **(d)** $(2, 4)$ **(e)** $x = 2, 4$

Responses to True–False questions may be abridged to save space.

11. True; see definition of decreasing: $f(x_1) > f(x_2)$ whenever $0 \le x_1 < x_2 \le 2$.

13. False; for example, $f(x) = (x-1)^3$ is increasing on $[0, 2]$ and $f'(1) = 0$.

15. **(a)** $[3/2, +\infty)$ **(b)** $(-\infty, 3/2]$ **(c)** $(-\infty, +\infty)$ **(d)** none **(e)** none

17. **(a)** $(-\infty, +\infty)$ **(b)** none **(c)** $(-1/2, +\infty)$ **(d)** $(-\infty, -1/2)$ **(e)** $-1/2$

19. **(a)** $[1, +\infty)$ **(b)** $(-\infty, 1]$ **(c)** $(-\infty, 0), \left(\frac{2}{3}, +\infty\right)$ **(d)** $\left(0, \frac{2}{3}\right)$ **(e)** $0, \frac{2}{3}$

21. **(a)** $\left[\dfrac{3-\sqrt{5}}{2}, \dfrac{3+\sqrt{5}}{2}\right]$ **(b)** $\left(-\infty, \dfrac{3-\sqrt{5}}{2}\right], \left[\dfrac{3+\sqrt{5}}{2}, +\infty\right)$ **(c)** $\left(0, \dfrac{4-\sqrt{6}}{2}\right), \left(\dfrac{4+\sqrt{6}}{2}, +\infty\right)$ **(d)** $(-\infty, 0), \left(\dfrac{4-\sqrt{6}}{2}, \dfrac{4+\sqrt{6}}{2}\right)$ **(e)** $0, \dfrac{4 \pm \sqrt{6}}{2}$

23. **(a)** $[-1/2, +\infty)$ **(b)** $(-\infty, -1/2]$ **(c)** $(-2, 1)$ **(d)** $(-\infty, -2), (1, +\infty)$ **(e)** $-2, 1$

25. **(a)** $[-1, 0], [1, +\infty)$ **(b)** $(-\infty, -1], [0, 1]$ **(c)** $(-\infty, 0), (0, +\infty)$ **(d)** none **(e)** none

27. **(a)** $(-\infty, 0]$ **(b)** $[0, +\infty)$ **(c)** $(-\infty, -1), (1, +\infty)$ **(d)** $(-1, 1)$ **(e)** $-1, 1$

29. **(a)** $[0, +\infty)$ **(b)** $(-\infty, 0]$ **(c)** $(-2, 2)$ **(d)** $(-\infty, -2), (2, +\infty)$ **(e)** $-2, 2$

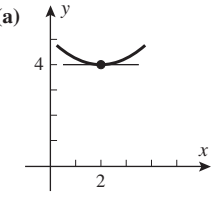
31. **(a)** $[0, +\infty)$ **(b)** $(-\infty, 0]$ **(c)** $\left(-\sqrt{\dfrac{1+\sqrt{7}}{3}}, \sqrt{\dfrac{1+\sqrt{7}}{3}}\right)$ **(d)** $\left(-\infty, -\sqrt{\dfrac{1+\sqrt{7}}{3}}\right), \left(\sqrt{\dfrac{1+\sqrt{7}}{3}}, +\infty\right)$ **(e)** $\pm\sqrt{\dfrac{1+\sqrt{7}}{3}}$

33. increasing: $[-\pi/4, 3\pi/4]$; decreasing: $[-\pi, -\pi/4], [3\pi/4, \pi]$; concave up: $(-3\pi/4, \pi/4)$; concave down: $(-\pi, -3\pi/4), (\pi/4, \pi)$; inflection points: $-3\pi/4, \pi/4$

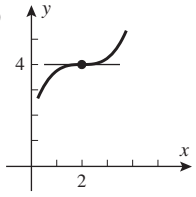
35. increasing: none; decreasing: $(-\pi, \pi)$; concave up: $(-\pi, 0)$; concave down: $(0, \pi)$; inflection point: 0

37. increasing: $[-\pi, -3\pi/4], [-\pi/4, \pi/4], [3\pi/4, \pi]$; decreasing: $[-3\pi/4, -\pi/4], [\pi/4, 3\pi/4]$; concave up: $(-\pi/2, 0), (\pi/2, \pi)$; concave down: $(-\pi, -\pi/2), (0, \pi/2)$; inflection points: $0, \pm\pi/2$

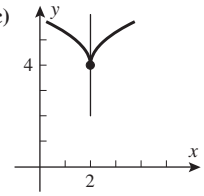
39. **(a)**



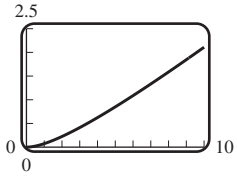
(b)



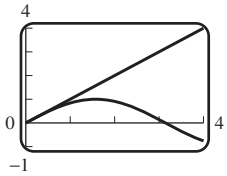
(c)



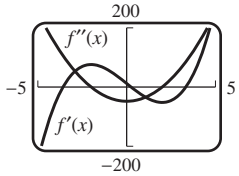
41. $1 + \frac{1}{3}x - \sqrt[3]{1+x} \ge 0$ if $x > 0$



43. $x \ge \sin x$



47.

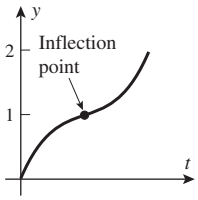


points of inflection at $x = -2, 2$; concave up on $[-5, -2], [2, 5]$; concave down on $[-2, 2]$; increasing on $[-3.5829, 0.2513]$ and $[3.3316, 5]$; decreasing on $[-5, -3.5829]$, $[0.2513, 3.3316]$

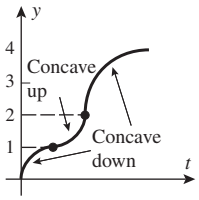
49. $-2.464202, 0.662597, 2.701605$ **53.** **(a)** true **(b)** false

57. **(c)** inflection point $(1, 0)$; concave up on $(1, +\infty)$; concave down on $(-\infty, 1)$

63.

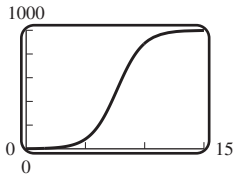


65.



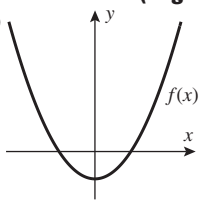
67. **(a)** $\dfrac{LAk}{(1+A)^2}$ **(c)** $\dfrac{1}{k}\ln A$

69. the eighth day

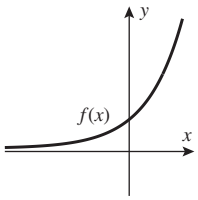


Exercise Set 4.2 (Page 252)

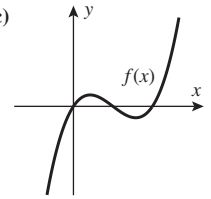
1. **(a)**



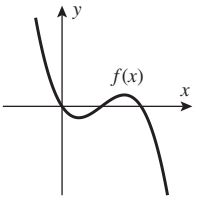
(b)



(c)



(d)



5. **(b)** nothing **(c)** f has a relative minimum at $x = 1$, g has no relative extremum at $x = 1$.

7. critical: $0, \pm\sqrt{2}$; stationary: $0, \pm\sqrt{2}$

9. critical: $-3, 1$; stationary: $-3, 1$ **11.** critical: $0, \pm 5$; stationary: 0

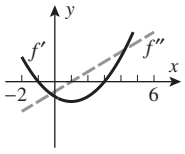
13. critical: $n\pi/2$ for every integer n; stationary: $n\pi + \pi/2$ for every integer n

Responses to True–False questions may be abridged to save space.

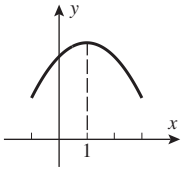
15. False; for example, $f(x) = (x-1)^2(x-1.5)$ has a relative maximum at $x = 1$, but $f(2) = 0.5 > 0 = f(1)$.

17. False; to apply the second derivative test (Theorem 4.2.4) at $x = 1$, $f'(1)$ must equal 0.

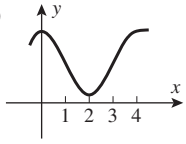
19.



21. **(a)** none **(b)** $x = 1$ **(c)** none **(d)**



23. **(a)** 2 **(b)** 0 **(c)** 1, 3 **(d)**



25. 0 (neither), $\sqrt[3]{5}$ (min) **27.** -2 (min), $2/3$ (max) **29.** 0 (min)

31. -1 (min), 1 (max) **33.** relative maximum at $(4/3, 19/3)$

35. relative maximum at $(\pi/4, 1)$; relative minimum at $(3\pi/4, -1)$

37. relative maximum at $(1, 1)$; relative minima at $(0, 0)$, $(2, 0)$

39. relative maximum at $(-1, 0)$; relative minimum at $(-3/5, -108/3125)$

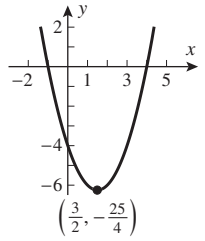
41. relative maximum at $(-1, 1)$; relative minimum at $(0, 0)$

43. no relative extrema **45.** relative minimum at $(0, \ln 2)$

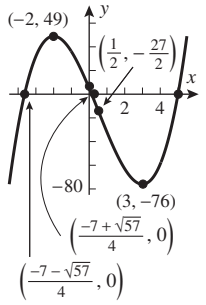
47. relative minimum at $(-\ln 2, -1/4)$

49. relative maximum at $(3/2, 9/4)$; relative minima at $(0, 0)$, $(3, 0)$

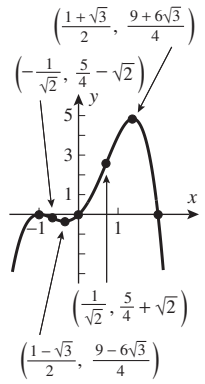
51. intercepts: $(0, -4)$, $(-1, 0)$, $(4, 0)$; stationary point: $(3/2, -25/4)$ (min); inflection points: none



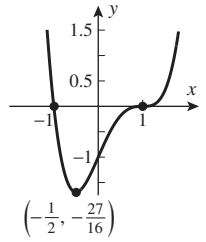
53. intercepts: $(0, 5)$, $\left(\dfrac{-7 \pm \sqrt{57}}{4}, 0\right)$, $(5, 0)$; stationary points: $(-2, 49)$ (max), $(3, -76)$ (min); inflection point: $(1/2, -27/2)$



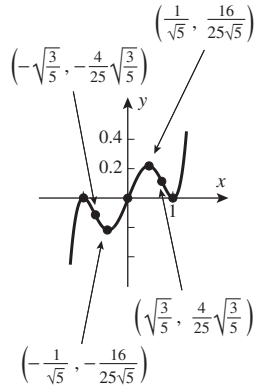
55. intercepts: $(-1, 0)$, $(0, 0)$, $(2, 0)$; stationary points: $(-1, 0)$ (max), $\left(\dfrac{1-\sqrt{3}}{2}, \dfrac{9-6\sqrt{3}}{4}\right)$ (min), $\left(\dfrac{1+\sqrt{3}}{2}, \dfrac{9+6\sqrt{3}}{4}\right)$ (max); inflection points: $\left(-\dfrac{1}{\sqrt{2}}, \dfrac{5}{4} - \sqrt{2}\right)$, $\left(\dfrac{1}{\sqrt{2}}, \dfrac{5}{4} + \sqrt{2}\right)$,



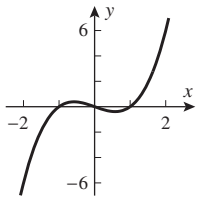
57. intercepts: $(0, -1)$, $(-1, 0)$, $(1, 0)$; stationary points: $(-1/2, -27/16)$ (min), $(1, 0)$ (neither); inflection points: $(0, -1)$, $(1, 0)$



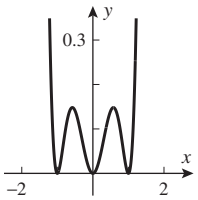
59. intercepts: $(-1, 0)$, $(0, 0)$, $(1, 0)$; stationary points: $(-1, 0)$ (max), $\left(-\dfrac{1}{\sqrt{5}}, -\dfrac{16}{25\sqrt{5}}\right)$ (min), $\left(\dfrac{1}{\sqrt{5}}, \dfrac{16}{25\sqrt{5}}\right)$ (max), $(1, 0)$ (min); inflection points: $\left(-\sqrt{\dfrac{3}{5}}, -\dfrac{4}{25}\sqrt{\dfrac{3}{5}}\right)$, $(0, 0)$, $\left(\sqrt{\dfrac{3}{5}}, \dfrac{4}{25}\sqrt{\dfrac{3}{5}}\right)$



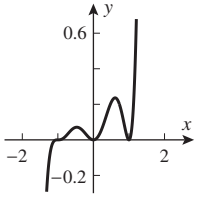
61. **(a)**



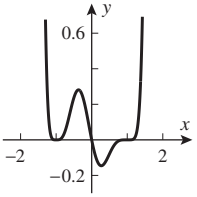
(b)



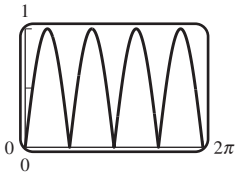
(c)



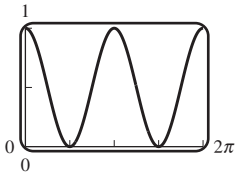
(d)



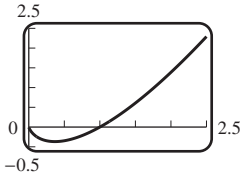
63. relative min of 0 at $x = \pi/2, \pi, 3\pi/2$;
relative max of 1 at $x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$



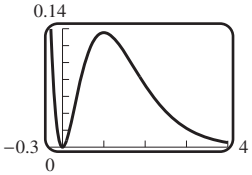
65. relative min of 0 at $x = \pi/2, 3\pi/2$;
relative max of 1 at $x = \pi$



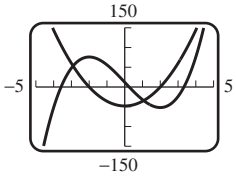
67. relative min of $-1/e$ at $x = 1/e$



69. relative min of 0 at $x = 0$;
relative max of $1/e^2$ at $x = 1$



71. relative minima at $x = -3.58, 3.33$;
relative max at $x = 0.25$

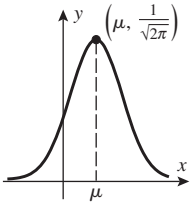


73. relative maximum at $x \approx -0.272$; relative minimum at $x \approx 0.224$

75. relative maximum at $x = 0$; relative minima at $x \approx \pm 0.618$

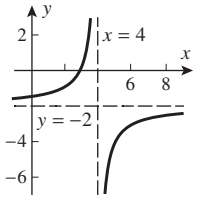
77. **(a)** 54 **(b)** 9

79. **(b)**

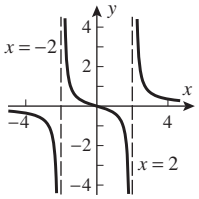


Exercise Set 4.3 (Page 264)

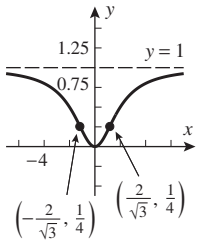
1. stationary points: none;
inflection points: none;
asymptotes: $x = 4$, $y = -2$;
asymptote crossings: none



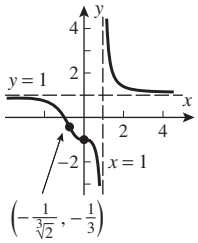
3. stationary points: none;
inflection point: (0, 0);
asymptotes: $x = \pm 2$, $y = 0$;
asymptote crossings: (0, 0)



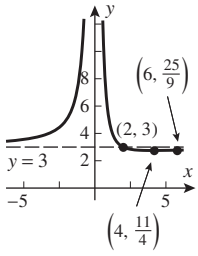
5. stationary point: (0, 0);
inflection points: $\left(\pm\frac{2}{\sqrt{3}}, \frac{1}{4}\right)$;
asymptote: $y = 1$;
asymptote crossings: none



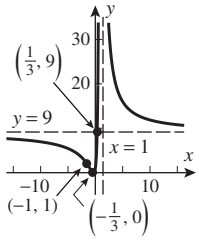
7. stationary point: (0, −1);
inflection points: (0, −1), $\left(-\frac{1}{\sqrt[3]{2}}, -\frac{1}{3}\right)$;
asymptotes: $x = 1$, $y = 1$;
asymptote crossings: none



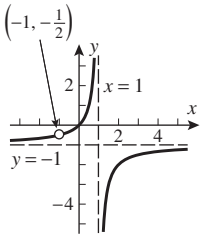
9. stationary point: (4, 11/4);
inflection point: (6, 25/9);
asymptotes: $x = 0$, $y = 3$;
asymptote crossing: (2, 3)



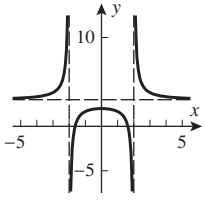
11. stationary point: (−1/3, 0);
inflection point: (−1, 1);
asymptotes: $x = 1$, $y = 9$;
asymptote crossing: (1/3, 9)



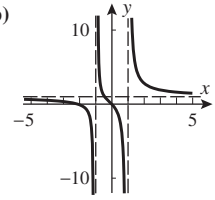
13. stationary points: none;
inflection points: none;
asymptotes: $x = 1$, $y = -1$;
asymptote crossings: none



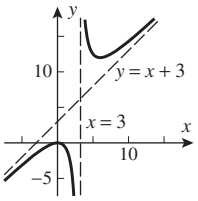
15. **(a)**



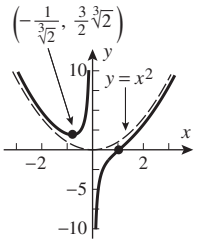
(b)



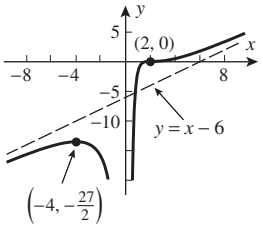
17.



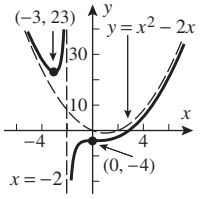
19. stationary point: $\left(-\frac{1}{\sqrt[3]{2}}, \frac{3}{2}\sqrt[3]{2}\right)$;
inflection point: $(1, 0)$;
asymptotes: $y = x^2$, $x = 0$;
asymptote crossings: none



21. stationary points: $(-4, -27/2)$, $(2, 0)$;
inflection point: $(2, 0)$;
asymptotes: $x = 0$, $y = x - 6$;
asymptote crossing: $(2/3, -16/3)$



23. stationary points: $(-3, 23)$, $(0, -4)$;
inflection point: $(0, -4)$;
asymptotes: $x = -2$, $y = x^2 - 2x$;
asymptote crossings: none



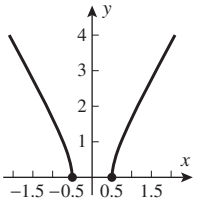
25. **(a)** VI **(b)** I **(c)** III **(d)** V **(e)** IV **(f)** II

Responses to True–False questions may be abridged to save space.

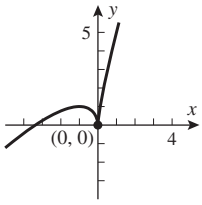
27. True; if deg $P >$ deg Q, then $f(x)$ is unbounded as $x \to \pm\infty$; if deg $P <$ deg Q, then $f(x) \to 0$ as $x \to \pm\infty$.

29. False; for example, $f(x) = (x-1)^{1/3}$ is continuous (with vertical tangent line) at $x = 1$, but $f'(x) = \dfrac{1}{3(x-1)^{2/3}}$ has a vertical asymptote at $x = 1$.

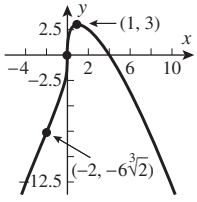
31. critical points: $(\pm 1/2, 0)$;
inflection points: none

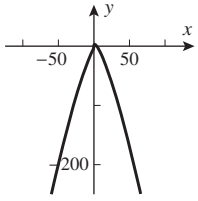


33. critical points: $(-1, 1)$, $(0, 0)$;
inflection points: none

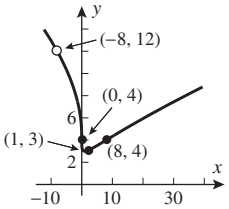


35. critical points: $(0, 0)$, $(1, 3)$; inflection points: $(0, 0)$, $(-2, -6\sqrt[3]{2})$.
It's hard to see all the important features in one graph, so two graphs are shown:

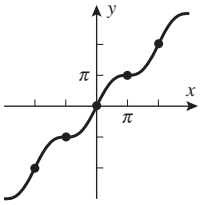




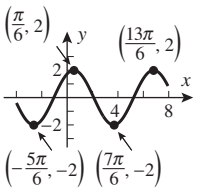
37. critical points: $(0, 4)$, $(1, 3)$;
inflection points: $(0, 4)$, $(8, 4)$



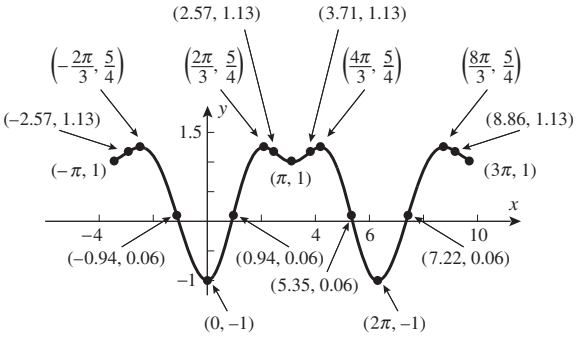
39. extrema: none;
inflection points: $x = \pi n$ for integers n



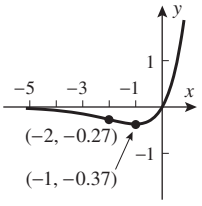
41. minima: $x = 7\pi/6 + 2\pi n$ for integers n;
maxima: $x = \pi/6 + 2\pi n$ for integers n;
inflection points: $x = 2\pi/3 + \pi n$ for integers n



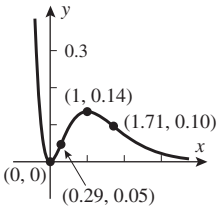
43. relative minima: 1 at $x = \pi$; -1 at $x = 0, 2\pi$;
relative maxima: $5/4$ at $x = -2\pi/3, 2\pi/3, 4\pi/3, 8\pi/3$;
inflection points where $\cos x = \dfrac{-1 \pm \sqrt{33}}{8}$: $(-2.57, 1.13)$, $(-0.94, 0.06)$, $(0.94, 0.06)$, $(2.57, 1.13)$, $(3.71, 1.13)$, $(5.35, 0.06)$, $(7.22, 0.06)$, $(8.86, 1.13)$



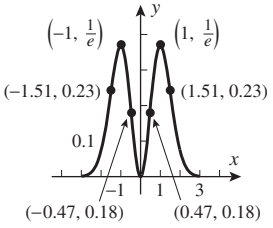
45. **(a)** $+\infty$, 0
(b)



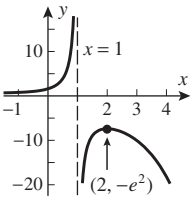
47. **(a)** 0, $+\infty$
(b)



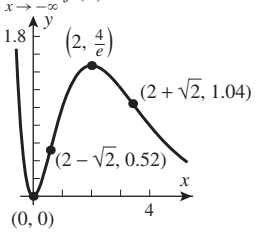
49. **(a)** 0, 0
(b) relative max $= 1/e$ at $x = \pm 1$; relative min $= 0$ at $x = 0$;
inflection points where $x = \pm\sqrt{\dfrac{5 \pm \sqrt{17}}{4}}$:
about $(\pm 0.47, 0.18)$, $(\pm 1.51, 0.23)$; asymptote: $y = 0$



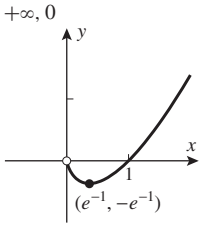
51. **(a)** $-\infty$, 0
(b) relative max $= -e^2$ at $x = 2$;
no relative min;
no inflection points;
asymptotes: $y = 0$, $x = 1$



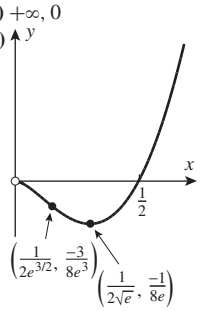
53. **(a)** 0, $+\infty$
(b) critical points at $x = 0.2$;
relative min at $x = 0$,
relative max at $x = 2$;
points of inflection at $x = 2 \pm \sqrt{2}$;
horizontal asymptote $y = 0$ as $x \to +\infty$
$\lim_{x \to -\infty} f(x) = +\infty$



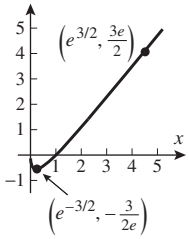
55. **(a)** $+\infty$, 0
(b)

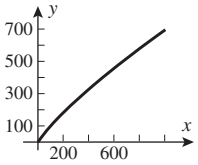


57. **(a)** $+\infty$, 0
(b)

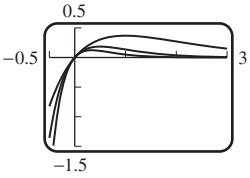


59. **(a)** $+\infty$, 0
(b) no relative max; relative min $= -\dfrac{3}{2e}$ at $x = e^{-3/2}$;
inflection point: $(e^{3/2}, 3e/2)$;
no asymptotes. It's hard to see all the important features in one graph, so two graphs are shown:



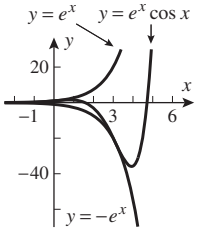


61. **(a)**

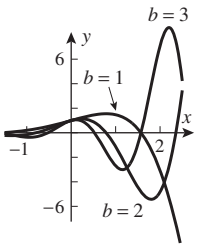


(b) relative max at $x = 1/b$;
inflection point at $x = 2/b$

63. **(a)** does not exist, 0
(b) $y = e^x$ and $y = e^x \cos x$ intersect for $x = 2\pi n$, and $y = -e^x$ and $y = e^x \cos x$ intersect for $x = 2\pi n + \pi$, for all integers n.

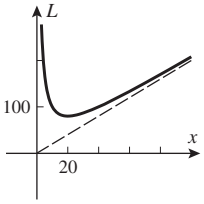


(c) Graphs for $a = 1$:

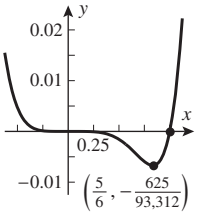


65. **(a)** $x = 1, 2.5, 3, 4$ **(b)** $(-\infty, 1]$, $[2.5, 3]$
(c) relative max at $x = 1, 3$;
relative min at $x = 2.5$ **(d)** $x \approx 0.6, 1.9, 4$

67.



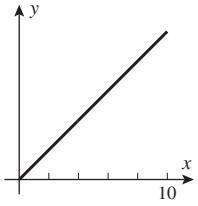
69. Graph misses zeros at $x = 0, 1$ and min at $x = 5/6$.



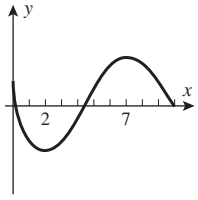
▶ Exercise Set 4.4 (Page 272)

1. relative maxima at $x = 2, 6$; absolute max at $x = 6$;
relative min at $x = 4$; absolute minima at $x = 0, 4$

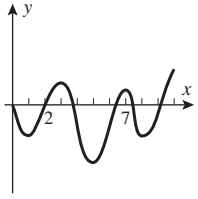
3. **(a)**



(b)



(c)



7. max $= 2$ at $x = 1, 2$;
min $= 1$ at $x = 3/2$

9. max $= 8$ at $x = 4$;
min $= -1$ at $x = 1$

11. maximum value $3/\sqrt{5}$ at $x = 1$;
minimum value $-3/\sqrt{5}$ at $x = -1$

13. max $= \sqrt{2} - \pi/4$ at $x = -\pi/4$;
min $= \pi/3 - \sqrt{3}$ at $x = \pi/3$

15. maximum value 17 at $x = -5$; minimum value 1 at $x = -3$.

Responses to True–False questions may be abridged to save space.

17. True; see the Extreme-Value Theorem (4.4.2).

19. True; see Theorem 4.4.3.

21. no maximum; min $= -9/4$ at $x = 1/2$

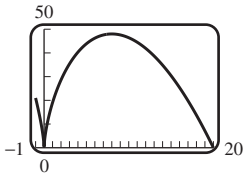
23. maximum value $f(1) = 1$; no minimum

25. no maximum or minimum

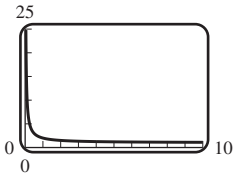
27. max $= -2 - 2\sqrt{2}$ at $x = -1 - \sqrt{2}$; no minimum

29. no maximum; min $= 0$ at $x = 0, 2$

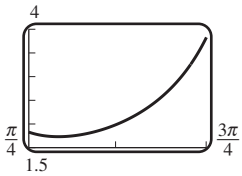
31. maximum value 48 at $x = 8$; minimum value 0 at $x = 0, 20$



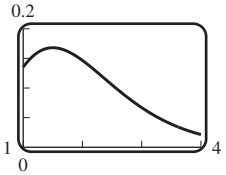
33. no maximum or minimum



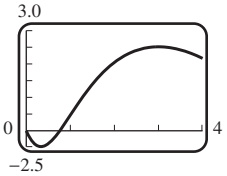
35. max $= 2\sqrt{2} + 1$ at $x = 3\pi/4$; min $= \sqrt{3}$ at $x = \pi/3$



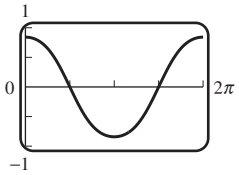
37. maximum value $\frac{27}{8}e^{-3}$ at $x = \frac{3}{2}$; minimum value $64/e^8$ at $x = 4$



39. max $= 5 \ln 10 - 9$ at $x = 3$; min $= 5 \ln(10/9) - 1$ at $x = 1/3$



41. maximum value $\sin(1) \approx 0.84147$; minimum value $-\sin(1) \approx -0.84147$



43. maximum value 2; minimum value $-\frac{1}{4}$

45. max $= 3$ at $x = 2n\pi$; min $= -3/2$ at $x = \pm 2\pi/3 + 2n\pi$ for any integer n **49.** 2, at $x = 1$

53. maximum $y = 4$ at $t = \pi, 3\pi$; minimum $y = 0$ at $t = 0, 2\pi$

Exercise Set 4.5 (Page 283)

1. **(a)** 1 **(b)** $\frac{1}{2}$ **3.** 500 ft parallel to stream, 250 ft perpendicular

5. 500 ft (\$3 fencing) × 750 ft (\$2 fencing) **7.** 5 in × $\frac{12}{5}$ in

9. $10\sqrt{2} \times 10\sqrt{2}$ **11.** 80 ft (\$1 fencing), 40 ft (\$2 fencing)

15. maximum area is 108 when $x = 2$

17. maximum area is 144 when $x = 2$

19. 11,664 in^3 **21.** $\frac{200}{27}$ ft^3 **23.** base 10 cm square, height 20 cm

25. ends $\sqrt[3]{3V/4}$ units square, length $\frac{4}{3}\sqrt[3]{3V/4}$

27. height $= 2R/\sqrt{3}$, radius $= \sqrt{2/3}R$

31. height = radius $= \sqrt[3]{500/\pi}$ cm **33.** $L/12$ by $L/12$ by $L/12$

35. height $= L/\sqrt{3}$, radius $= \sqrt{2/3}L$

37. height $= 2\sqrt[3]{75/\pi}$ cm, radius $= \sqrt{2}\sqrt[3]{75/\pi}$ cm

39. height $= 4R$, radius $= \sqrt{2}R$

41. $R(x) = 225x - 0.25x^2$; $R'(x) = 225 - 0.5x$; 450 tons

43. **(a)** 7000 units **(b)** yes **(c)** \$15 **45.** 13,722 lb **47.** $3\sqrt{3}$

49. height $= r/\sqrt{2}$ **51.** $1/\sqrt{5}$ **53.** $\left(\sqrt{2}, \frac{1}{2}\right)$ **55.** $\left(-1/\sqrt{3}, \frac{3}{4}\right)$

57. **(a)** π mi **(b)** $2\sin^{-1}(1/4)$ mi **59.** $4(1 + 2^{2/3})^{3/2}$ ft

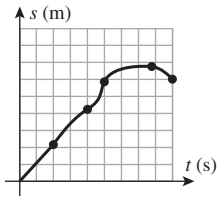
61. 30 cm from the weaker source **63.** $\sqrt{24} = 2\sqrt{6}$ ft

Exercise Set 4.6 (Page 294)

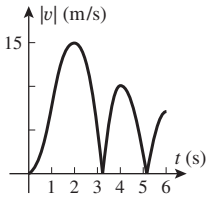
1. **(a)** positive, negative, slowing down
(b) positive, positive, speeding up
(c) negative, positive, slowing down

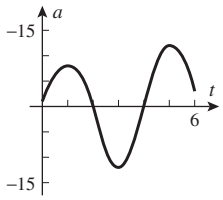
3. **(a)** left
(b) negative
(c) speeding up
(d) slowing down

5.



7.





Responses to True–False questions may be abridged to save space.

9. False; a particle has positive velocity when its position versus time graph is increasing; if that positive velocity is decreasing, the particle would be slowing down.

11. False; acceleration is the derivative of velocity (with respect to time); speed is the absolute value of velocity.

13. **(a)** 6.2 ft/s^2 **(b)** $t = 0$ s

15. **(a)**

t	s	v	a
1	0.71	0.56	−0.44
2	1	0	−0.62
3	0.71	−0.56	−0.44
4	0	−0.79	0
5	−0.71	−0.56	0.44

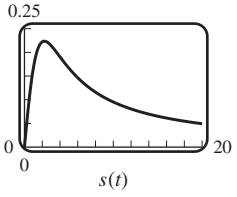
(b) stopped at $t = 2$; moving right at $t = 1$; moving left at $t = 3, 4, 5$
(c) speeding up at $t = 3$; slowing down at $t = 1, 5$; neither at $t = 2, 4$

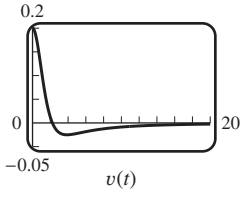
17. **(a)** $v(t) = 3t^2 - 6t$, $a(t) = 6t - 6$
(b) $s(1) = -2$ ft, $v(1) = -3$ ft/s, $|v(1)| = 3$ ft/s, $a(1) = 0$ ft/s^2
(c) $t = 0, 2$ s **(d)** speeding up for $0 < t < 1$ and $2 < t$, slowing down for $1 < t < 2$ **(e)** 58 ft

19. **(a)** $v(t) = 3\pi \sin(\pi t/3)$, $a(t) = \pi^2 \cos(\pi t/3)$ **(b)** $s(1) = 9/2$ ft, $v(1) =$ speed $= 3\sqrt{3}\pi/2$ ft/s, $a(1) = \pi^2/2$ ft/s^2 **(c)** $t = 0$ s, 3 s
(d) speeding up: $0 < t < 1.5, 3 < t < 4.5$;
slowing down: $1.5 < t < 3, 4.5 < t < 5$ **(e)** 31.5 ft

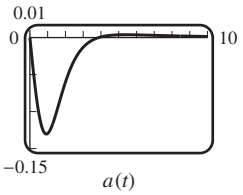
21. **(a)** $v(t) = -\frac{1}{3}(t^2 - 6t + 8)e^{-t/3}$, $a(t) = \frac{1}{9}(t^2 - 12t + 26)e^{-t/3}$
(b) $s(1) = 9e^{-1/3}$ ft, $v(1) = -e^{-1/3}$ ft/s, speed $= e^{-1/3}$ ft/s, $a(1) = \frac{5}{3}e^{-1/3}$ ft/s^2 **(c)** $t = 2$ s, 4 s
(d) speeding up: $2 < t < 6 - \sqrt{10}, 4 < t < 6 + \sqrt{10}$; slowing down: $0 < t < 2, 6 - \sqrt{10} < t < 4, 6 + \sqrt{10} < t$
(e) $8 - 24e^{-2/3} + 48e^{-4/3} - 33e^{-5/3}$

23. **(a)** $\sqrt{5}$ **(b)** $\sqrt{5}/10$





(c) speeding up for $\sqrt{5} < t < \sqrt{15}$; slowing down for $0 < t < \sqrt{5}$ and $\sqrt{15} < t$



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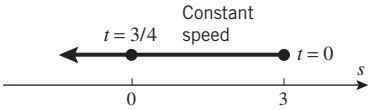

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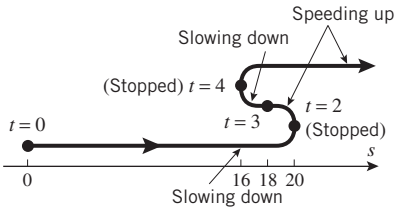

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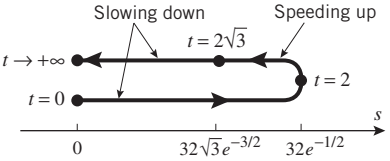
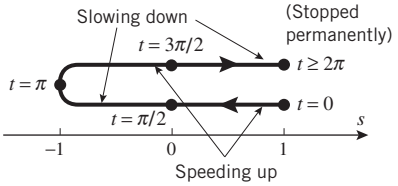


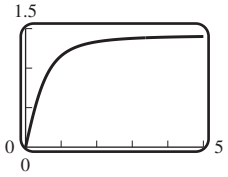
31.



33. **(a)** 12 ft/s **(b)** $t = 2.2$ s, $s = -24.2$ ft

35. **(a)** $t = 2 \pm 1/\sqrt{3}$, $s = \ln 2$, $v = \pm\sqrt{3}$ **(b)** $t = 2$, $s = 0$, $a = 6$

37. **(a)**



(b) $\sqrt{2}$

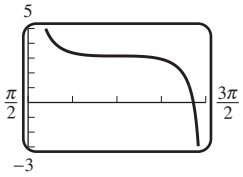
39. **(b)** $\frac{2}{3}$ unit **(c)** $0 \le t < 1$ and $t > 2$

Exercise Set 4.7 (Page 300)

1. 1.414213562 **3.** 1.817120593 **5.** $x \approx 1.76929$

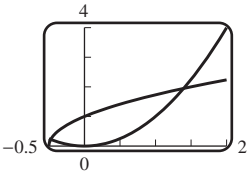
7. $x \approx 1.224439550$ **9.** $x \approx -1.24962$ **11.** $x \approx 1.02987$

13. $x \approx 4.493409458$



15. $x \approx 0.68233$

17. -0.474626618, 1.395336994 **19.** $x \approx 0.58853$ or 3.09636



Responses to True–False questions may be abridged to save space.

21. True; $x = x_{n+1}$ is the x-intercept of the tangent line to $y = f(x)$ at $x = x_n$.

23. False; for example, if $f(x) = x(x-3)^2$, Newton's Method fails (analogous to Figure 4.7.4) with $x_1 = 1$ and approximates the root $x = 3$ for $x_1 > 1$.

25. **(b)** 3.162277660 **27.** -4.098859132

29. $x = -1$ or $x \approx 0.17951$ **31.** (0.589754512, 0.347810385)

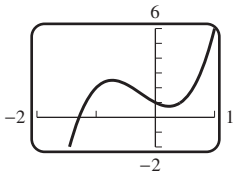
33. **(b)** $\theta \approx 2.99156$ rad or $171°$ **35.** -1.220744085, 0.724491959

37. $i = 0.053362$ or 5.33% **39.** **(a)** The values do not converge.

Exercise Set 4.8 (Page 308)

1. $c = 4$ **3.** $c = \pi$ **5.** $c = 1$ **7.** $c = -\sqrt{5}$

9. **(a)** $[-2, 1]$
(b) $c \approx -1.29$
(c) -1.2885843



Responses to True–False questions may be abridged to save space.

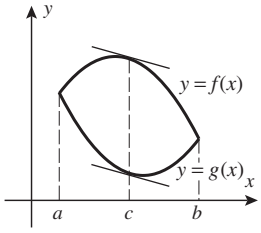
11. False; Rolle's Theorem requires the additional hypothesis that f is differentiable on (a, b) and $f(a) = f(b) = 0$; see Example 2.

13. False; the Constant Difference Theorem applies to two functions with equal derivatives on an interval to conclude that the functions differ by a constant on the interval.

15. **(b)** $\tan x$ is not continuous on $[0, \pi]$. **25.** $f(x) = xe^x - e^x + 2$

35. **(b)** $f(x) = \sin x$, $g(x) = \cos x$

37.



41. $a = 6$, $b = -3$

Chapter 4 Review Exercises (Page 310)

1. **(a)** $f(x_1) < f(x_2)$; $f(x_1) > f(x_2)$; $f(x_1) = f(x_2)$
(b) $f' > 0$; $f' < 0$; $f' = 0$

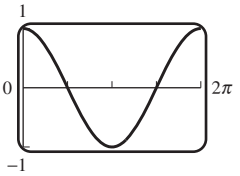
3. **(a)** $\left[\frac{5}{2}, +\infty\right)$ **(b)** $\left(-\infty, \frac{5}{2}\right]$ **(c)** $(-\infty, +\infty)$ **(d)** none **(e)** none

5. **(a)** $[0, +\infty)$ **(b)** $(-\infty, 0]$ **(c)** $(-\sqrt{2/3}, \sqrt{2/3})$
(d) $(-\infty, -\sqrt{2/3})$, $(\sqrt{2/3}, +\infty)$ **(e)** $-\sqrt{2/3}$, $\sqrt{2/3}$

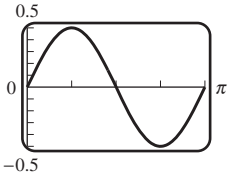
7. **(a)** $[-1, +\infty)$ **(b)** $(-\infty, -1]$ **(c)** $(-\infty, 0)$, $(2, +\infty)$
(d) $(0, 2)$ **(e)** 0, 2

9. **(a)** $(-\infty, 0]$ **(b)** $[0, +\infty)$ **(c)** $(-\infty, -1/\sqrt{2})$, $(1/\sqrt{2}, +\infty)$
(d) $(-1/\sqrt{2}, 1/\sqrt{2})$ **(e)** $\pm 1/\sqrt{2}$

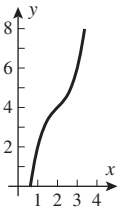
11. increasing on $[\pi, 2\pi]$;
decreasing on $[0, \pi]$;
concave up on $(\pi/2, 3\pi/2)$;
concave down on $(0, \pi/2)$, $(3\pi/2, 2\pi)$;
inflection points: $(\pi/2, 0)$, $(3\pi/2, 0)$



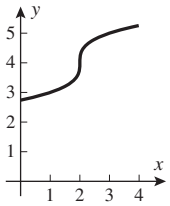
13. increasing on $[0, \pi/4]$, $[3\pi/4, \pi]$;
decreasing on $[\pi/4, 3\pi/4]$;
concave up on $(\pi/2, \pi)$;
concave down on $(0, \pi/2)$;
inflection point: $(\pi/2, 0)$



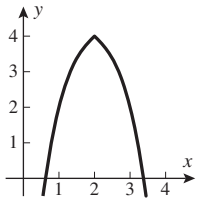
15. **(a)**



(b)



(c)

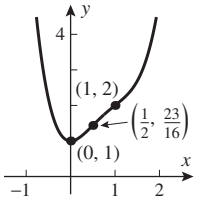


17. $-\dfrac{b}{2a} \le 0$ **19.** $x = -1$ **21.** **(a)** at an inflection point

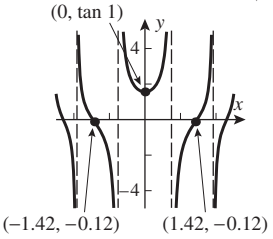
25. **(a)** $x = \pm\sqrt{2}$ (stationary points) **(b)** $x = 0$ (stationary point)

27. **(a)** relative max at $x = 1$, relative min at $x = 7$, neither at $x = 0$
(b) relative max at $x = \pi/2, 3\pi/2$; relative min at $x = 7\pi/6, 11\pi/6$
(c) relative max at $x = 5$

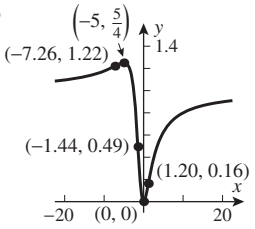
29. $\lim\limits_{x\to-\infty} f(x) = +\infty$, $\lim\limits_{x\to+\infty} f(x) = +\infty$;
relative min at $x = 0$;
points of inflection at $x = \frac{1}{2}, 1$;
no asymptotes



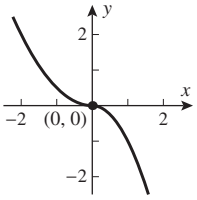
31. $\lim\limits_{x\to\pm\infty} f(x)$ does not exist; critical point at $x = 0$; relative min at $x = 0$; point of inflection when $1 + 4x^2\tan(x^2 + 1) = 0$; vertical asymptotes at $x = \pm\sqrt{\pi\left(n + \frac{1}{2}\right) - 1}$, $n = 0, 1, 2, \ldots$



33. critical points at $x = -5, 0$; relative max at $x = -5$, relative min at $x = 0$; points of inflection at $x \approx -7.26, -1.44, 1.20$; horizontal asymptote $y = 1$ for $x \to \pm\infty$



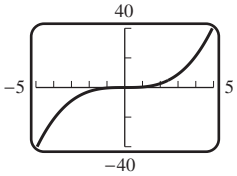
35. $\lim\limits_{x\to-\infty} f(x) = +\infty$, $\lim\limits_{x\to+\infty} f(x) = -\infty$;
critical point at $x = 0$;
no extrema;
inflection point at $x = 0$
(f changes concavity);
no asymptotes



37. no relative extrema **39.** relative min of 0 at $x = 0$

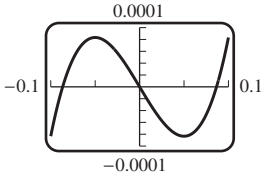
41. relative min of 0 at $x = 0$ **43.** relative min of 0 at $x = 0$

45. **(a)**

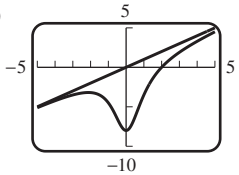


(b) relative max at $x = -\frac{1}{20}$;
relative min at $x = \frac{1}{20}$

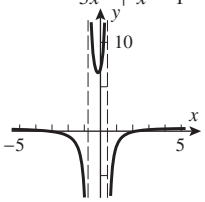
(c) The finer details can be seen when graphing over a much smaller x-window.



47. **(a)**



49. $f(x) = \dfrac{x^2 + x - 7}{3x^2 + x - 1}$, $x \ne \dfrac{1}{2}$

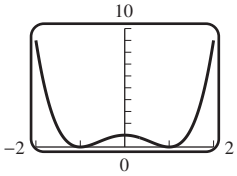


horizontal asymptote $y = 1/3$
vertical asymptotes at
$x = (-1 \pm \sqrt{13})/6$

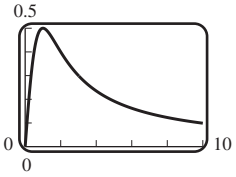
53. **(a)** true **(b)** false

55. **(a)** no max; min $= -13/4$ at $x = 3/2$ **(b)** no max or min
(c) no max; min $m = e^2/4$ at $x = 2$
(d) no max; min $m = e^{-1/e}$ at $x = 1/e$

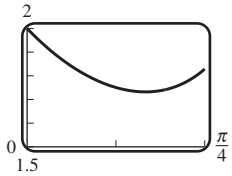
57. **(a)** minimum value 0 for $x = \pm 1$; no maximum



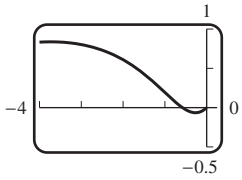
(b) max $= 1/2$ at $x = 1$; min $= 0$ at $x = 0$



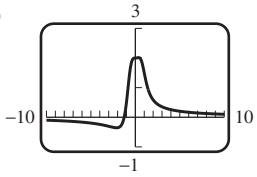
(c) maximum value 2 at $x = 0$; minimum value $\sqrt{3}$ at $x = \pi/6$



(d) maximum value $f(-2 - \sqrt{3}) \approx 0.84$; minimum value $f(-2 + \sqrt{3}) \approx -0.06$



59. **(a)**



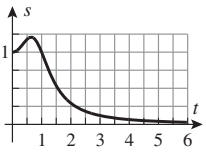
(b) minimum:
$(-2.111985, -0.355116)$;
maximum:
$(0.372591, 2.012931)$

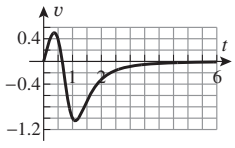
61. width $= 4\sqrt{2}$, height $= 3\sqrt{2}$ **63.** 2 in square

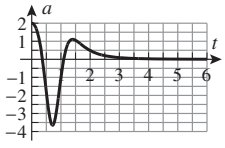
65. **(a)** yes **(b)** yes

67. **(a)** $v = -2\dfrac{t(t^4 + 2t^2 - 1)}{(t^4 + 1)^2}$, $a = 2\dfrac{3t^8 + 10t^6 - 12t^4 - 6t^2 + 1}{(t^4 + 1)^3}$

(b)





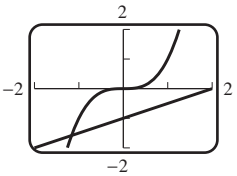


(c) $t \approx 0.64$, $s \approx 1.2$ **(d)** $0 \le t \le 0.64$

(e) speeding up when $0 \le t < 0.36$ and $0.64 < t < 1.1$, otherwise slowing down **(f)** maximum speed ≈ 1.05 when $t \approx 1.10$

69. $x \approx -2.11491, 0.25410, 1.86081$

71. $x \approx -1.165373043$



73. 249×10^6 km

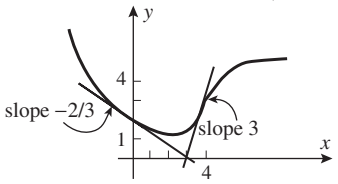
75. **(a)** yes, $c = 0$ **(b)** no **(c)** yes, $c = \sqrt{\pi/2}$

77. use Rolle's Theorem

Chapter 4 Making Connections (Page 314)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** no zeros **(b)** one **(c)** $\lim_{x \to +\infty} g'(x) = 0$



2. **(a)** $(-2.2, 4), (2, 1.2), (4.2, 3)$
(b) critical numbers at $x = -5.1, -2, 0.2, 2$;
local min at $x = -5.1, 2$; local max at $x = -2$;
no extrema at $x = 0.2$; $f''(1) \approx -1.2$

3. $x = -4, 5$ **4.** **(d)** $f(c) = 0$

6. **(a)** route (i): 10 s; route (iv): 10 s
(b) $2 \le x \le 5$; $\dfrac{4\sqrt{10}}{2.1} + \dfrac{5}{0.7} \approx 13.166$ s
(c) $0 \le x \le 2$; 10 s
(d) route (i) or (iv); 10 s

Exercise Set 5.1 (Page 321)

1.

n	2	5	10	50	100
A_n	0.853553	0.749739	0.710509	0.676095	0.671463

3.

n	2	5	10	50	100
A_n	1.57080	1.93376	1.98352	1.99935	1.99984

5.

n	2	5	10	50	100
A_n	0.583333	0.645635	0.668771	0.688172	0.690653

7.

n	2	5	10	50	100
A_n	0.433013	0.659262	0.726130	0.774567	0.780106

9.

n	2	5	10	50	100
A_n	3.71828	2.85174	2.59327	2.39772	2.37398

11.

n	2	5	10	50	100
A_n	1.04720	0.75089	0.65781	0.58730	0.57894

13. $3(x-1)$ **15.** $x(x+2)$ **17.** $(x+3)(x-1)$

Responses to True–False questions may be abridged to save space.

19. False; the limit would be the area of the circle 4π.

21. True; this is the basis of the antiderivative method.

23. area $= A(6) - A(3)$ **27.** $f(x) = 2x$; $a = 2$

Exercise Set 5.2 (Page 330)

1. **(a)** $\displaystyle\int \frac{x}{\sqrt{1+x^2}}\,dx = \sqrt{1+x^2} + C$

(b) $\displaystyle\int (x+1)e^x\,dx = xe^x + C$

5. $\dfrac{d}{dx}\left[\sqrt{x^3+5}\right] = \dfrac{3x^2}{2\sqrt{x^3+5}}$, so $\displaystyle\int \frac{3x^2}{2\sqrt{x^3+5}}\,dx = \sqrt{x^3+5} + C$.

7. $\dfrac{d}{dx}[\sin(2\sqrt{x})] = \dfrac{\cos(2\sqrt{x})}{\sqrt{x}}$, so $\displaystyle\int \frac{\cos(2\sqrt{x})}{\sqrt{x}}\,dx = \sin(2\sqrt{x}) + C$.

9. **(a)** $(x^9/9) + C$ **(b)** $\frac{7}{12}x^{12/7} + C$ **(c)** $\frac{2}{9}x^{9/2} + C$

11. $\dfrac{5}{2}x^2 - \dfrac{1}{6x^4} + C$ **13.** $-\dfrac{1}{2}x^{-2} - \dfrac{12}{5}x^{5/4} + \dfrac{8}{3}x^3 + C$

15. $(x^2/2) + (x^5/5) + C$ **17.** $3x^{4/3} - \frac{12}{7}x^{7/3} + \frac{3}{10}x^{10/3} + C$

19. $\dfrac{x^2}{2} - \dfrac{2}{x} + \dfrac{1}{3x^3} + C$ **21.** $2\ln|x| + 3e^x + C$

23. $-3\cos x - 2\tan x + C$ **25.** $\tan x + \sec x + C$

27. $\tan\theta + C$ **29.** $\sec x + C$ **31.** $\theta - \cos\theta + C$

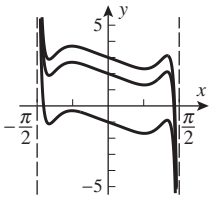
33. $\frac{1}{2}\sin^{-1} x - 3\tan^{-1} x + C$ **35.** $\tan x - \sec x + C$

Responses to True–False questions may be abridged to save space.

37. True; this is Equations (1) and (2).

39. False; the initial condition is not satisfied since $y(0) = 2$.

41.



43. **(a)** $y(x) = \frac{3}{4}x^{4/3} + \frac{5}{4}$
(b) $y = -\cos t + t + 1 - \pi/3$
(c) $y(x) = \frac{2}{3}x^{3/2} + 2x^{1/2} - \frac{8}{3}$

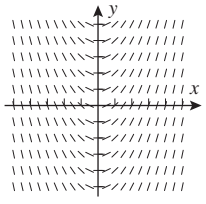
45. **(a)** $y = 4e^x - 3$ **(b)** $y = \ln|t| + 5$

47. $s(t) = 16t^2 + 20$ **49.** $s(t) = 2t^{3/2} - 15$

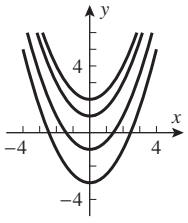
51. $f(x) = \frac{4}{15}x^{5/2} + C_1x + C_2$ **53.** $y = x^2 + x - 6$

55. $f(x) = \cos x + 1$ **57.** $y = x^3 - 6x + 7$

59. **(a)**

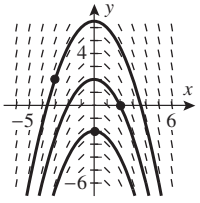


(b)

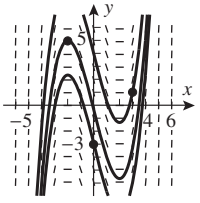


(c) $f(x) = \dfrac{x^2}{2} - 1$

61. (b)



63. (c)



67. **(b)** $\pi/2$ **69.** $\tan x - x + C$

71. **(a)** $\frac{1}{2}(x - \sin x) + C$ **(b)** $\frac{1}{2}(x + \sin x) + C$

73. $v = \dfrac{1087}{\sqrt{273}}T^{1/2}$ ft/s

Exercise Set 5.3 (Page 338)

1. **(a)** $\dfrac{(x^2+1)^{24}}{24} + C$ **(b)** $-\dfrac{\cos^4 x}{4} + C$

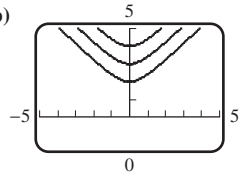
3. **(a)** $\frac{1}{4}\tan(4x+1) + C$ **(b)** $\frac{1}{6}(1+2y^2)^{3/2} + C$

5. **(a)** $-\frac{1}{2}\cot^2 x + C$ **(b)** $\frac{1}{10}(1+\sin t)^{10} + C$

7. **(a)** $\frac{2}{7}(1+x)^{7/2} - \frac{4}{5}(1+x)^{5/2} + \frac{2}{3}(1+x)^{3/2} + C$
(b) $-\cot(\sin x) + C$

9. **(a)** $\ln|\ln x| + C$ **(b)** $-\frac{1}{5}e^{-5x} + C$

11. **(a)** $\frac{1}{3}\tan^{-1}(x^3)+C$ **(b)** $\sin^{-1}(\ln x)+C$
15. $\frac{1}{40}(4x-3)^{10}+C$ **17.** $-\frac{1}{7}\cos 7x+C$ **19.** $\frac{1}{4}\sec 4x+C$
21. $\frac{1}{2}e^{2x}+C$ **23.** $\frac{1}{2}\sin^{-1}(2x)+C$ **25.** $\frac{1}{21}(7t^2+12)^{3/2}+C$
27. $\dfrac{3}{2(1-2x)^2}+C$ **29.** $-\dfrac{1}{40(5x^4+2)^2}+C$ **31.** $e^{\sin x}+C$
33. $-\frac{1}{6}e^{-2x^3}+C$ **35.** $\tan^{-1}e^x+C$ **37.** $\frac{1}{5}\cos(5/x)+C$
39. $-\frac{1}{15}\cos^5 3t+C$ **41.** $\frac{1}{2}\tan(x^2)+C$ **43.** $-\frac{1}{6}(2-\sin 4\theta)^{3/2}+C$
45. $\sin^{-1}(\tan x)+C$ **47.** $\frac{1}{6}\sec^3 2x+C$ **49.** $-e^{-x}+C$
51. $-e^{-2\sqrt{x}}+C$ **53.** $\frac{1}{6}(2y+1)^{3/2}-\frac{1}{2}(2y+1)^{1/2}+C$
55. $-\frac{1}{2}\cos 2\theta+\frac{1}{6}\cos^3 2\theta+C$ **57.** $t+\ln|t|+C$
59. $\int[\ln(e^x)+\ln(e^{-x})]\,dx=C$
61. **(a)** $\sin^{-1}\left(\frac{1}{3}x\right)+C$ **(b)** $\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+C$
(c) $\dfrac{1}{\sqrt{\pi}}\sec^{-1}\left(\dfrac{x}{\sqrt{\pi}}\right)+C$
63. $\dfrac{1}{b}\dfrac{(a+bx)^{n+1}}{n+1}+C$ **65.** $\dfrac{1}{b(n+1)}\sin^{n+1}(a+bx)+C$
67. **(a)** $\frac{1}{2}\sin^2 x+C_1;\ -\frac{1}{2}\cos^2 x+C_2$ **(b)** They differ by a constant.
69. $\frac{2}{15}(5x+1)^{3/2}-\frac{158}{15}$ **71.** $y=-\frac{1}{2}e^{2t}+\frac{13}{2}$
73. **(a)** $\sqrt{x^2+1}+C$ **75.** $f(x)=\frac{2}{9}(3x+1)^{3/2}+\frac{7}{9}$
77. 1280
(b)



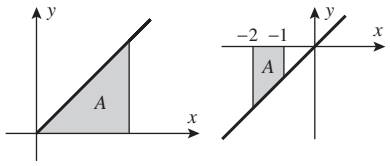
Exercise Set 5.4 (Page 350)

1. **(a)** 36 **(b)** 55 **(c)** 40 **(d)** 6 **(e)** 11 **(f)** 0 **3.** $\sum_{k=1}^{10} k$ **5.** $\sum_{k=1}^{10} 2k$
7. $\sum_{k=1}^{6}(-1)^{k+1}(2k-1)$ **9.** **(a)** $\sum_{k=1}^{50} 2k$ **(b)** $\sum_{k=1}^{50}(2k-1)$ **11.** 5050
13. 2870 **15.** 214,365 **17.** $\frac{3}{2}(n+1)$ **19.** $\frac{1}{4}(n-1)^2$
Responses to True–False questions may be abridged to save space.
21. True; by parts (a) and (c) of Theorem 5.4.2.
23. False; consider $[a,b]=[-1,0]$.
25. **(a)** $\left(2+\frac{3}{n}\right)^4\cdot\frac{3}{n},\ \left(2+\frac{6}{n}\right)^4\cdot\frac{3}{n},\ \left(2+\frac{9}{n}\right)^4\cdot\frac{3}{n},$
$\left(2+\frac{3(n-1)}{n}\right)^4\cdot\frac{3}{n},\ (2+3)^4\cdot\frac{3}{n}$ **(b)** $\sum_{k=0}^{n-1}\left(2+k\cdot\frac{3}{n}\right)^4\frac{3}{n}$
27. **(a)** 46 **(b)** 52 **(c)** 58 **29.** **(a)** $\frac{\pi}{4}$ **(b)** 0 **(c)** $-\frac{\pi}{4}$
31. **(a)** 0.7188, 0.7058, 0.6982 **(b)** 0.6928, 0.6931, 0.6931
(c) 0.6688, 0.6808, 0.6882
33. **(a)** 4.8841, 5.1156, 5.2488 **(b)** 5.3471, 5.3384, 5.3346
(c) 5.6841, 5.5156, 5.4088
35. $\frac{15}{4}$ **37.** 18 **39.** 320 **41.** $\frac{15}{4}$ **43.** 18 **45.** 16 **47.** $\frac{1}{3}$ **49.** 0
51. $\frac{2}{3}$ **53.** **(b)** $\frac{1}{4}(b^4-a^4)$
55. $\dfrac{n^2+2n}{4}$ if n is even; $\dfrac{(n+1)^2}{4}$ if n is odd **57.** $3^{17}-3^4$ **59.** $-\frac{399}{400}$
61. **(b)** $\frac{1}{2}$ **65.** **(a)** yes **(b)** yes

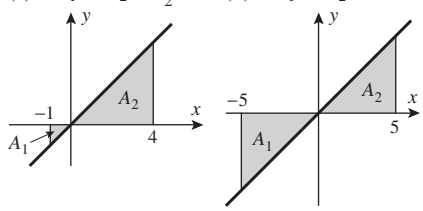
Exercise Set 5.5 (Page 360)

1. **(a)** $\frac{71}{6}$ **(b)** 2 **3.** **(a)** $-\frac{117}{16}$ **(b)** 3 **5.** $\int_{-1}^{2} x^2\,dx$
7. $\int_{-3}^{3} 4x(1-3x)\,dx$
9. **(a)** $\lim_{\max\Delta x_k\to 0}\sum_{k=1}^{n} 2x_k^*\,\Delta x_k;\ a=1,\ b=2$
(b) $\lim_{\max\Delta x_k\to 0}\sum_{k=1}^{n}\dfrac{x_k^*}{x_k^*+1}\Delta x_k;\ a=0,\ b=1$

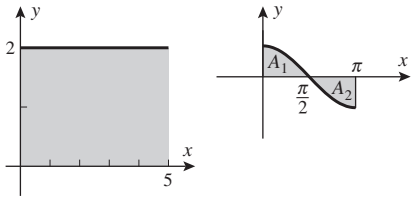
13. **(a)** $A=\frac{9}{2}$ **(b)** $-A=-\frac{3}{2}$



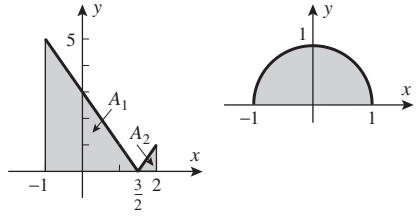
(c) $-A_1+A_2=\frac{15}{2}$ **(d)** $-A_1+A_2=0$



15. **(a)** $A=10$ **(b)** $A_1-A_2=0$ by symmetry



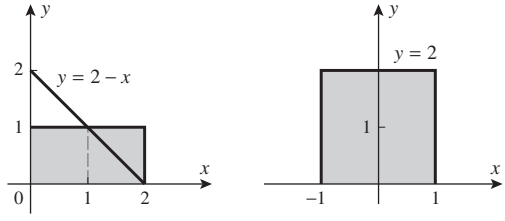
(c) $A_1+A_2=\frac{13}{2}$ **(d)** $\pi/2$



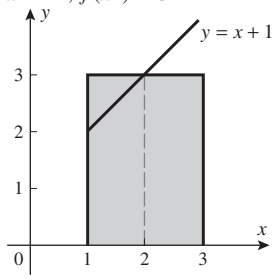
17. **(a)** 2
(b) 4
(c) 10
(d) 10
19. **(a)** 0.8
(b) −2.6
(c) −1.8
(d) −0.3
21. −1 **23.** 3 **25.** −4 **27.** $(1+\pi)/2$
Responses to True–False questions may be abridged to save space.
29. False; see Theorem 5.5.8(a).
31. False; consider $f(x)=x-2$ on $[0,3]$.
33. **(a)** negative **(b)** positive **37.** $\frac{25}{2}\pi$ **39.** $\frac{5}{2}$
45. **(a)** integrable **(b)** integrable **(c)** not integrable **(d)** integrable

Exercise Set 5.6 (Page 373)

1. **(a)** $\int_0^2(2-x)\,dx=2$ **(b)** $\int_{-1}^{1}2\,dx=4$ **(c)** $\int_1^3(x+1)\,dx=6$
3. **(a)** $x^*=f(x^*)=1$ **(b)** x^* is any point in $[-1,1]$, $f(x^*)=2$



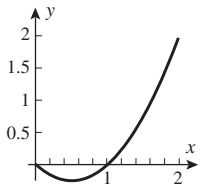
(c) $x^* = 2$, $f(x^*) = 3$

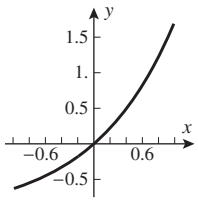


5. $\frac{65}{4}$ **7.** 14 **9.** $\frac{3}{2}$ **11.** **(a)** $\frac{4}{3}$ **(b)** -7 **13.** 48 **15.** 3 **17.** $\frac{844}{5}$
19. 0 **21.** $\sqrt{2}$ **23.** $5e^3 - 10$ **25.** $\pi/4$ **27.** $\pi/12$ **29.** -12
31. **(a)** $5/2$ **(b)** $2 - \frac{\sqrt{2}}{2}$ **33.** **(a)** $e + (1/e) - 2$ **(b)** 1
35. **(a)** $\frac{17}{6}$ **(b)** $F(x) = \begin{cases} \dfrac{x^2}{2}, & x \le 1 \\ \dfrac{x^3}{3} + \dfrac{1}{6}, & x > 1 \end{cases}$

Responses to True–False questions may be abridged to save space.

37. False; since $|x|$ is continuous, it has an antiderivative.
39. True; by the Fundamental Theorem of Calculus.
41. 0.6659; $\frac{2}{3}$ **43.** 3.1060; $2 \tan 1$ **45.** 12 **47.** $\frac{9}{2}$
49. area $= 1$ **51.** area $= e + e^{-1} - 2$



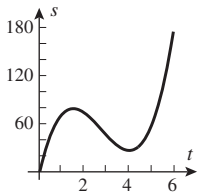


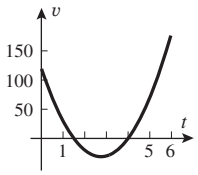
53. **(b)** degree mode, 0.93
55. **(a)** change in height from age 0 to age 10 years; inches
(b) change in radius from time $t = 1$ s to time $t = 2$ s; centimeters
(c) difference between speed of sound at 100°F and at 32°F; feet per second **(d)** change in position from time t_1 to time t_2; centimeters
57. **(a)** $3x^2 - 3$ **59.** **(a)** $\sin(x^2)$ **(b)** $e^{\sqrt{x}}$ **61.** $-x \sec x$
63. **(a)** 0 **(b)** 5 **(c)** $\frac{4}{5}$
65. **(a)** $x = 3$ **(b)** increasing on $[3, +\infty)$, decreasing on $(-\infty, 3]$
(c) concave up on $(-1, 7)$, concave down on $(-\infty, -1)$ and $(7, +\infty)$
67. **(a)** $(0, +\infty)$ **(b)** $x = 1$
69. **(a)** 120 gal **(b)** 420 gal **(c)** 2076.36 gal **71.** 1

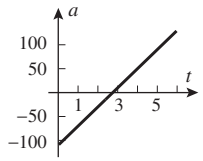
Exercise Set 5.7 (Page 382)

1. **(a)** displacement $= 3$; distance $= 3$
(b) displacement $= -3$; distance $= 3$
(c) displacement $= -\frac{1}{2}$; distance $= \frac{3}{2}$
(d) displacement $= \frac{3}{2}$; distance $= 2$
3. **(a)** 35.3 m/s **(b)** 51.4 m/s **5.** **(a)** $t^3 - t^2 + 1$ **(b)** $4t + 3 - \frac{1}{3}\sin 3t$
7. **(a)** $\frac{3}{2}t^2 + t - 4$ **(b)** $t + 1 - \ln t$
9. **(a)** displacement $= 1$ m; distance $= 1$ m
(b) displacement $= -1$ m; distance $= 3$ m
11. **(a)** displacement $= \frac{9}{4}$ m; distance $= \frac{11}{4}$ m
(b) displacement $= 2\sqrt{3} - 6$ m; distance $= 6 - 2\sqrt{3}$ m
13. 4, 13/3 **15.** 296/27, 296/27
17. **(a)** $s = 2/\pi$, $v = 1$, $|v| = 1$, $a = 0$
(b) $s = \frac{1}{2}$, $v = -\frac{3}{2}$, $|v| = \frac{3}{2}$, $a = -3$ **19.** $t \approx 1.27$ s

21.

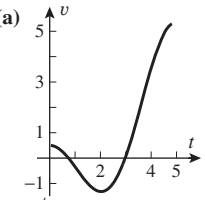






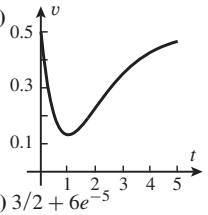
Responses to True–False questions may be abridged to save space.

23. True; if $a(t) = a_0$, then $v(t) = a_0 t + v_0$.
25. False; consider $v(t) = \sin t$ on $[0, 2\pi]$.
27. **(a)**



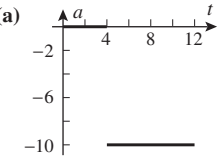
(b) $5/2 - \sin(5) + 5\cos(5)$

29. **(a)**

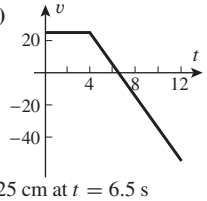


(b) $3/2 + 6e^{-5}$

31. **(a)**



(b)

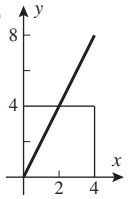


(c) 120 cm, -20 cm **(d)** 131.25 cm at $t = 6.5$ s

33. **(a)** $-\frac{121}{5}$ ft/s^2 **(b)** $\frac{70}{33}$ s **(c)** $\frac{60}{11}$ s **35.** 280 m **37.** 50 s, 5000 ft
39. **(a)** 16 ft/s, -48 ft/s **(b)** 196 ft **(c)** 112 ft/s
41. **(a)** 1 s **(b)** $\frac{1}{2}$ s **43.** **(a)** 6.122 s **(b)** 183.7 m **(c)** 6.122 s **(d)** 60 m/s
45. **(a)** 5 s **(b)** 272.5 m **(c)** 10 s **(d)** -49 m/s
(e) 12.46 s **(f)** 73.1 m/s

Exercise Set 5.8 (Page 388)

1. **(a)** 4 **(b)** 2 **(c)**



3. 6
5. $2/\pi$
7. $\dfrac{1}{e - 1}$
9. $\dfrac{\pi}{12(\sqrt{3} - 1)}$
11. $\dfrac{1 - e^{-8}}{8}$ **13.** **(a)** 5.28 **(b)** 4.305 **(c)** 4 **15.** **(a)** $-\frac{1}{6}$ **(b)** $\frac{1}{2}$

Responses to True–False questions may be abridged to save space.

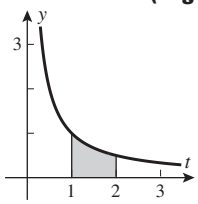
19. False; let $g(x) = \cos x$; $f(x) = 0$ on $[0, 3\pi/2]$.
21. True; see Theorem 5.5.4(*b*).
23. **(a)** $\frac{263}{4}$ **(b)** 31 **25.** 1404π lb **27.** 97 cars/min **31.** 27

Exercise Set 5.9 (Page 393)

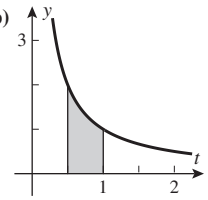
1. **(a)** $\frac{1}{2}\int_1^5 u^3\,du$ **(b)** $\frac{3}{2}\int_9^{25} \sqrt{u}\,du$ **(c)** $\dfrac{1}{\pi}\displaystyle\int_{-\pi/2}^{\pi/2} \cos u\,du$
(d) $\int_1^2 (u+1)u^5\,du$ **3.** **(a)** $\frac{1}{2}\int_{-1}^1 e^u\,du$ **(b)** $\int_1^2 u\,du$
5. 10 **7.** 0 **9.** $\frac{1192}{15}$ **11.** $8 - (4\sqrt{2})$ **13.** $-\frac{1}{48}$ **15.** $\ln\frac{21}{13}$
17. $\pi/6$ **19.** $25\pi/6$ **21.** $\pi/8$ **23.** $2/\pi$ m **25.** 6 **27.** $\pi/18$
29. 1/21 **31.** 2 **33.** $\frac{2}{3}(\sqrt{10} - 2\sqrt{2})$ **35.** $2(\sqrt{7} - \sqrt{3})$ **37.** 1
39. 0 **41.** $(\sqrt{3} - 1)/3$ **43.** $\frac{106}{405}$ **45.** $(\ln 3)/2$ **47.** $\pi/(6\sqrt{3})$
49. $\pi/9$ **51.** **(a)** $\frac{23}{4480}$ **53.** **(a)** $\frac{5}{3}$ **(b)** $\frac{5}{3}$ **(c)** $-\frac{1}{2}$ **57.** Method 2
59. Method 1 **61.** $\approx 48{,}233{,}500{,}000$ **63.** **(a)** 0.45 **(b)** 0.461
65. $(\ln 7)/2$ **67.** **(a)** $2/\pi$ **71.** **(b)** $\frac{3}{2}$ **(c)** $\pi/4$

Exercise Set 5.10 (Page 406)

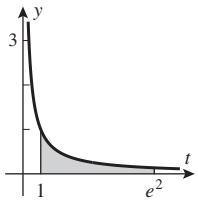
1. (a)



(b)



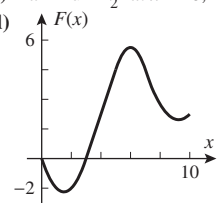
(c)



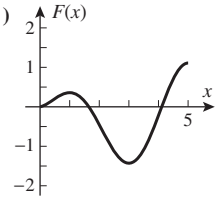
3. (a) 7 (b) -5 (c) -3 (d) 6
5. 1.603210678; magnitude of error is < 0.0063
7. (a) $x^{-1}, x > 0$ (b) $x^2, x \neq 0$ (c) $-x^2, -\infty < x < +\infty$ (d) $-x, -\infty < x < +\infty$ (e) $x^3, x > 0$ (f) $\ln x + x, x > 0$ (g) $x - \sqrt[3]{x}, -\infty < x < +\infty$ (h) $e^x/x, x > 0$
9. (a) $e^{\pi \ln 3}$ (b) $e^{\sqrt{2}\ln 2}$ 11. (a) $\sqrt{e}$ (b) e^2 13. $x^2 - x$
15. (a) $3/x$ (b) 1 17. (a) 0 (b) 0 (c) 1

Responses to True–False questions may be abridged to save space.

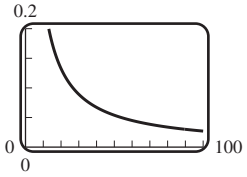
19. True; both equal $-\ln a$.
21. False; the integrand is unbounded on $[-1, e]$ and thus the integrand is undefined.
23. (a) $2x^3\sqrt{1+x^2}$ (b) $-\frac{2}{3}(x^2+1)^{3/2} + \frac{2}{5}(x^2+1)^{5/2} - \frac{4\sqrt{2}}{15}$
25. (a) $-\cos(x^3)$ (b) $-\tan^2 x$ 27. $-3\dfrac{3x-1}{9x^2+1} + 2x\dfrac{x^2-1}{x^4+1}$
29. (a) $3x^2\sin^2(x^3) - 2x\sin^2(x^2)$ (b) $\dfrac{2}{1-x^2}$
31. (a) $F(0) = 0, F(3) = 0, F(5) = 6, F(7) = 6, F(10) = 3$
(b) increasing on $\left[\frac{3}{2}, 6\right]$ and $\left[\frac{37}{4}, 10\right]$, decreasing on $\left[0, \frac{3}{2}\right]$ and $\left[6, \frac{37}{4}\right]$
(c) maximum $\frac{15}{2}$ at $x = 6$, minimum $-\frac{9}{4}$ at $x = \frac{3}{2}$
(d)



33. $F(x) = \begin{cases} (1-x^2)/2, & x < 0 \\ (1+x^2)/2, & x \geq 0 \end{cases}$ 35. $y(x) = x^2 + \ln x + 1$
37. $y(x) = \tan x + \cos x - (\sqrt{2}/2)$
39. $P(x) = P_0 + \int_0^x r(t)\,dt$ individuals 41. I is the derivative of II.
43. (a) $t = 3$ (b) $t = 1, 5$
(c) $t = 5$ (d) $t = 3$
(e) F is concave up on $\left(0, \frac{1}{2}\right)$ and $(2, 4)$, concave down on $\left(\frac{1}{2}, 2\right)$ and $(4, 5)$.
(f)

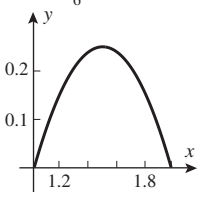


45. (a) relative maxima at $x = \pm\sqrt{4k+1}, k = 0, 1, \ldots$; relative minima at $x = \pm\sqrt{4k-1}, k = 1, 2, \ldots$
(b) $x = \pm\sqrt{2k}, k = 1, 2, \ldots$, and at $x = 0$
47. $f(x) = 2e^{2x}, a = \ln 2$ 49. 0.06



Chapter 5 Review Exercises (Page 408)

1. $-\dfrac{1}{4x^2} + \dfrac{8}{3}x^{3/2} + C$ 3. $-4\cos x + 2\sin x + C$
5. $3x^{1/3} - 5e^x + C$ 7. $\tan^{-1} x + 2\sin^{-1} x + C$
9. (a) $y(x) = 2\sqrt{x} - \frac{2}{3}x^{3/2} - \frac{4}{3}$ (b) $y(x) = \sin x - 5e^x + 5$
(c) $y(x) = \frac{5}{4} + \frac{3}{4}x^{4/3}$ (d) $y(x) = \frac{1}{2}e^{x^2} - \frac{1}{2}$
13. $\frac{1}{2}\sec^{-1}(x^2 - 1) + C$ 15. $\frac{1}{3}\sqrt{5 + 2\sin 3x} + C$
17. $-\dfrac{1}{3a}\dfrac{1}{ax^3+b} + C$
19. (a) $\displaystyle\sum_{k=0}^{14}(k+4)(k+1)$ (b) $\displaystyle\sum_{k=5}^{19}(k-1)(k-4)$
21. $\frac{32}{3}$ 23. 0.35122, 0.42054, 0.38650
27. (a) $\frac{3}{4}$ (b) $-\frac{3}{2}$ (c) $-\frac{35}{4}$ (d) -2 (e) not enough information
(f) not enough information
29. (a) $2 + (\pi/2)$ (b) $\dfrac{1}{3}(10^{3/2} - 1) - \dfrac{9\pi}{4}$ (c) $\pi/8$
31. 48 33. $\frac{2}{3}$ 35. $\frac{3}{2} - \sec 1$ 37. $\frac{5}{2}$ 39. $\frac{52}{3}$ 41. $e^3 - e$
43. area $= \frac{1}{6}$



45. $\frac{22}{3}$
47. (a) $x^3 + 1$
49. e^{x^2}
51. $|x - 1|$
53. $\dfrac{\cos x}{1 + \sin^3 x}$
57. (b) $\dfrac{\pi}{2}$; $\tan^{-1} x + \tan^{-1}\left(\dfrac{1}{x}\right) = \dfrac{\pi}{2}$
59. (a) $F(x)$ is 0 if $x = 1$, positive if $x > 1$, and negative if $x < 1$.
(b) $F(x)$ is 0 if $x = -1$, positive if $-1 < x \leq 2$, and negative if $-2 \leq x < -1$.
61. (a) $\frac{4}{3}$ (b) $e - 1$ 63. $\frac{3}{10}$ 67. $\frac{1}{4}t^4 - \frac{2}{3}t^3 + t + 1$
69. $t^2 - 3t + 7$ 71. 12 m, 20 m 73. $\frac{1}{3}$ m, $\frac{10}{3} - 2\sqrt{2}$ m
75. displacement $= -6$ m; distance $= \frac{13}{2}$ m
77. (a) 2.2 s (b) 387.2 ft 79. $v_0/2$ ft/s 81. $\frac{121}{5}$ 83. $\frac{2}{3}$ 85. 0
87. $2 - 2/\sqrt{e}$ 89. (a) e^2 (b) $e^{1/3}$

Chapter 5 Making Connections (Page 412)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. (b) $b^2 - a^2$ 2. 16/3 3. 12
4. (a) the sum for f is m times that for g
(b) $m\displaystyle\int_0^1 g(x)\,dx = \int_0^m f(x)\,du$
5. (a) they are equal (b) $\displaystyle\int_2^3 g(x)\,dx = \int_4^9 f(u)\,du$

Exercise Set 6.1 (Page 419)

1. 9/2 3. 1 5. (a) 4/3 (b) 4/3 7. 49/192 9. 1/2 11. $\sqrt{2}$
13. $\frac{1}{2}$ 15. $\pi - 1$ 17. 24 19. 37/12 21. $4\sqrt{2}$ 23. $\frac{1}{2}$
25. $\ln 2 - \frac{1}{2}$

Responses to True–False questions may be abridged to save space.

27. True; use area Formula (1) with $f(x) = g(x) + c$.
29. True; the integrand must assume both positive and negative values. By the Intermediate-Value Theorem, the integrand must be equal to 0 somewhere in $[a, b]$.
31. $k \approx 0.9973$ 33. 9152/105 35. $9/\sqrt[3]{4}$
37. (a) 4/3 (b) $m = 2 - \sqrt[3]{4}$ 39. 1.180898334
41. 0.4814, 2.3639, 1.1897 43. 2.54270

45. racer 1's lead over racer 2 at time $t = 0$
47. the increase in population from 1960 to 2010 **49.** $a^2/6$

Exercise Set 6.2 (Page 428)

1. 8π **3.** $13\pi/6$ **5.** $(1-\sqrt{2}/2)\pi$ **7.** 8π **9.** $32/5$ **11.** $256\pi/3$
13. $2048\pi/15$ **15.** 4π **17.** $\pi^2/4$ **19.** $3/5$ **21.** 2π
23. $72\pi/5$ **25.** $\frac{\pi}{2}(e^2-1)$

Responses to True–False questions may be abridged to save space.

27. False; see the solids associated with Exercises 9 and 10.
29. False; see Example 2 where the cross-sectional area is a linear function of x.
31. $4\pi ab^2/3$ **33.** π **35.** $\int_a^b \pi[f(x)-k]^2\,dx$ **37.** **(b)** $40\pi/3$
39. $648\pi/5$ **41.** $\pi/2$ **43.** $\pi/15$ **45.** $40{,}000\pi\ \text{ft}^3$ **47.** $1/30$
49. **(a)** $2\pi/3$ **(b)** $16/3$ **(c)** $4\sqrt{3}/3$ **51.** 0.710172176 **53.** π
57. **(b)** left ≈ 11.157; right ≈ 11.771; $V \approx$ average $= 11.464\ \text{cm}^3$
59. $V = \begin{cases} 3\pi h^2, & 0 \le h < 2 \\ \frac{1}{3}\pi(12h^2 - h^3 - 4), & 2 \le h \le 4 \end{cases}$ **61.** $\frac{2}{3}r^3\tan\theta$ **63.** $16r^3/3$

Exercise Set 6.3 (Page 436)

1. $15\pi/2$ **3.** $\pi/3$ **5.** $2\pi/5$ **7.** 4π **9.** $20\pi/3$ **11.** $\pi\ln 2$ **13.** $\pi/2$
15. $\pi/5$

Responses to True–False questions may be abridged to save space.

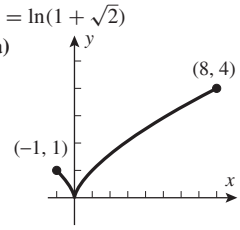
17. True; this is a restatement of Formula (1).
19. True; see Formula (2).
21. $2\pi e^2$ **23.** 1.73680 **25.** **(a)** $7\pi/30$ **(b)** easier
27. **(a)** $\int_0^1 2\pi(1-x)x\,dx$ **(b)** $\int_0^1 2\pi(1+y)(1-y)\,dy$
29. $7\pi/4$ **31.** $\pi r^2 h/3$ **33.** $V = \frac{4}{3}\pi(L/2)^3$ **35.** $b = 1$

Exercise Set 6.4 (Page 441)

1. $L = \sqrt{5}$ **3.** $(85\sqrt{85} - 8)/243$ **5.** $\frac{1}{27}(80\sqrt{10} - 13\sqrt{13})$ **7.** $\frac{17}{6}$

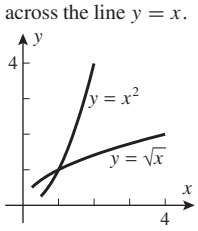
Responses to True–False questions may be abridged to save space.

9. False; f' is undefined at the endpoints ± 1.
11. True; if $f(x) = mx + c$ over $[a, b]$, then $L = \sqrt{1+m^2}(b-a)$, which is equal to the given sum.
13. $L = \ln(1+\sqrt{2})$
15. **(a)**



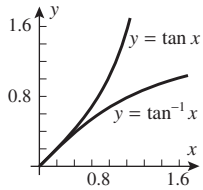
(b) dy/dx does not exist at $x = 0$.
(c) $L = (13\sqrt{13} + 80\sqrt{10} - 16)/27$

17. **(a)** They are mirror images across the line $y = x$.



(b) $\int_{1/2}^{2} \sqrt{1+4x^2}\,dx$, $\int_{1/4}^{4} \sqrt{1+\frac{1}{4x}}\,dx$, $x = \sqrt{u}$ transforms the first integral into the second.
(c) $\int_{1/4}^{4} \sqrt{1+\frac{1}{4y}}\,dy$, $\int_{1/2}^{2} \sqrt{1+4y^2}\,dy$
(d) 4.0724, 4.0716
(e) The first: Both are underestimates of the arc length, so the larger one is more accurate.
(f) 4.0724, 4.0662 **(g)** 4.0729

19. **(a)** They are mirror images across the line $y = x$.



(b) $\int_0^{\pi/3} \sqrt{1+\sec^4 x}\,dx$, $\int_0^{\sqrt{3}} \sqrt{1+\frac{1}{(1+x^2)^2}}\,dx$, $x = \tan^{-1} u$ transforms the first integral into the second.
(c) $\int_0^{\sqrt{3}} \sqrt{1+\frac{1}{(1+y^2)^2}}\,dy$, $\int_0^{\pi/3} \sqrt{1+\sec^4 y}\,dy$
(d) 2.0566, 2.0567
(e) The second: Both are underestimates of the arc length, so the larger one is more accurate. **(f)** 2.0509, 2.0571 **(g)** 2.0570
23. 4354 ft **25.** 196.31 yards **27.** $(2\sqrt{2}-1)/3$
29. π **31.** $L = \sqrt{2}(e^{\pi/2} - 1)$ **33.** **(b)** 9.69 **(c)** 5.16 cm

Exercise Set 6.5 (Page 447)

1. $35\pi\sqrt{2}$ **3.** 8π **5.** $40\pi\sqrt{82}$ **7.** 24π **9.** $16\pi/9$
11. $16,911\pi/1024$ **13.** $S \approx 14.42$ **15.** $S \approx 22.94$

Responses to True–False questions may be abridged to save space.

17. True; use Formula (1) with $r_1 = 0$, $r_2 = r$, $l = \sqrt{r^2+h^2}$.
19. True; the sum telescopes to the surface area of a cylinder.
21. 14.39 **23.** $S = \int_a^b 2\pi[f(x)+k]\sqrt{1+[f'(x)]^2}\,dx$
33. $\frac{8}{3}\pi(17\sqrt{17} - 1)$ **35.** $\frac{\pi}{24}(17\sqrt{17} - 1)$

Exercise Set 6.6 (Page 456)

1. 7.5 ft·lb **3.** $d = 7/4$ **5.** 100 ft·lb **7.** 160 J **9.** 20 lb/ft

Responses to True–False questions may be abridged to save space.

11. False; the work done is the same.
13. True; joules **15.** $47{,}385\pi$ ft·lb **17.** 261,600 J
19. **(a)** 926,640 ft·lb **(b)** hp of motor $= 0.468$ **21.** 75,000 ft·lb
23. 120,000 ft·tons **25.** **(a)** $2{,}400{,}000{,}000/x^2$ lb
(b) $(9.6 \times 10^{10})/(x+4000)^2$ lb **(c)** 2.5344×10^{10} ft·lb
27. $v_f = 100$ m/s
29. **(a)** decrease of 4.5×10^{14} J **(b)** ≈ 0.107 **(c)** ≈ 8.24 bombs

Exercise Set 6.7 (Page 465)

1. **(a)** positive: m_2 is at the fulcrum, so it can be ignored; masses m_1 and m_3 are equidistant from position 5, but $m_1 < m_3$, so the beam will rotate clockwise. **(b)** The fulcrum should be placed $\frac{50}{7}$ units to the right of m_1.
3. $(\frac{1}{2}, \frac{1}{2})$ **5.** $(1, \frac{1}{2})$ **7.** $(\frac{2}{3}, \frac{1}{3})$ **9.** $(\frac{5}{14}, \frac{38}{35})$ **11.** $(\frac{2}{3}, \frac{1}{3})$
13. $(-\frac{1}{2}, 4)$ **15.** $(\frac{1}{2}, \frac{8}{5})$ **17.** $(\frac{9}{20}, \frac{9}{20})$ **19.** $(\frac{49}{48}, \frac{7}{3} - \ln 2)$
23. $\frac{4}{3}$; $(\frac{3}{5}, \frac{3}{8})$ **25.** 3; $(0, \frac{2}{3})$ **27.** 8; $(\frac{\pi}{2}, \frac{\pi}{8})$
29. $\ln 4 - 1$; $\left(\frac{4\ln 4 - 3}{4\ln 4 - 4}, \frac{(\ln 2)^2 + 1 - \ln 4}{\ln 4 - 1}\right)$

Responses to True–False questions may be abridged to save space.

31. True; use symmetry. **33.** True; use symmetry.
35. $\left(\frac{2a}{3}, 0\right)$ **37.** $(\bar{x}, \bar{y}) = \left(0, \frac{(a+2b)c}{3(a+b)}\right)$
41. $2\pi^2 abk$ **43.** $(a/3, b/3)$

Exercise Set 6.8 (Page 472)

1. **(a)** $F = 31{,}200$ lb; $P = 312\ \text{lb/ft}^2$
(b) $F = 2{,}452{,}500$ N; $P = 98.1$ kPa
3. 499.2 lb **5.** 8.175×10^5 N **7.** 1,098,720 N **9.** yes
11. $\rho a^3/\sqrt{2}$ lb

Responses to True–False questions may be abridged to save space.

13. True; this is a consequence of inequalities (4).

15. False; by Equation (7) the force can be arbitrarily large for a fixed volume of water.
17. 61,748 lb **19.** 9.81×10^9 N **21.** **(b)** $80\rho_0$ lb/min

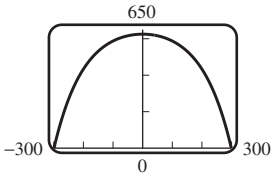
Exercise Set 6.9 (Page 482)

1. **(a)** ≈ 10.0179 **(b)** ≈ 3.7622 **(c)** $15/17 \approx 0.8824$ **(d)** ≈ -1.4436 **(e)** ≈ 1.7627 **(f)** ≈ 0.9730
3. **(a)** $\frac{4}{3}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{312}{313}$ **(d)** $-\frac{63}{16}$
5.

	$\sinh x_0$	$\cosh x_0$	$\tanh x_0$	$\coth x_0$	$\operatorname{sech} x_0$	$\operatorname{csch} x_0$
(a)	2	$\sqrt{5}$	$2/\sqrt{5}$	$\sqrt{5}/2$	$1/\sqrt{5}$	1/2
(b)	3/4	5/4	3/5	5/3	4/5	4/3
(c)	4/3	5/3	4/5	5/4	3/5	3/4

9. $4\cosh(4x-8)$ **11.** $-\frac{1}{x}\operatorname{csch}^2(\ln x)$
13. $\frac{1}{x^2}\operatorname{csch}\left(\frac{1}{x}\right)\coth\left(\frac{1}{x}\right)$ **15.** $\frac{2+5\cosh(5x)\sinh(5x)}{\sqrt{4x+\cosh^2(5x)}}$
17. $x^{5/2}\tanh(\sqrt{x})\operatorname{sech}^2(\sqrt{x})+3x^2\tanh^2(\sqrt{x})$
19. $\frac{1}{\sqrt{9+x^2}}$ **21.** $\frac{1}{(\cosh^{-1}x)\sqrt{x^2-1}}$ **23.** $-\frac{(\tanh^{-1}x)^{-2}}{1-x^2}$
25. $\frac{\sinh x}{|\sinh x|}=\begin{cases}1, & x>0\\ -1, & x<0\end{cases}$ **27.** $-\frac{e^x}{2x\sqrt{1-x}}+e^x\operatorname{sech}^{-1}x$
29. $\frac{1}{7}\sinh^7 x+C$ **31.** $\frac{2}{3}(\tanh x)^{3/2}+C$ **33.** $\ln(\cosh x)+C$
35. $37/375$ **37.** $\frac{1}{3}\sinh^{-1}3x+C$ **39.** $-\operatorname{sech}^{-1}(e^x)+C$
41. $-\operatorname{csch}^{-1}|2x|+C$ **43.** $\frac{1}{2}\ln 3$

Responses to True–False questions may be abridged to save space.
45. True; see Figure 6.9.1 **47.** True; $f(x)=\sinh x$
49. $16/9$ **51.** 5π **53.** $\frac{3}{4}$
55. **(a)** $+\infty$ **(b)** $-\infty$ **(c)** 1 **(d)** -1 **(e)** $+\infty$ **(f)** $+\infty$
63. $|u|<1$: $\tanh^{-1}u+C$; $|u|>1$: $\tanh^{-1}(1/u)+C$
65. **(a)** $\ln 2$ **(b)** $1/2$ **71.** 405.9 ft
73. **(a)**



(b) 1480.2798 ft
(c) ± 283.6249 ft
(d) $82°$

75. **(b)** 14.44 m **(c)** $15\ln 3 \approx 16.48$ m

Chapter 6 Review Exercises (Page 485)

7. **(a)** $\int_a^b (f(x)-g(x))\,dx+\int_b^c (g(x)-f(x))\,dx+\int_c^d (f(x)-g(x))\,dx$ **(b)** $11/4$
9. $4352\pi/105$ **11.** $3/2+\ln 4$ **13.** 9 **15.** $\frac{\pi}{6}\left(65^{3/2}-37^{3/2}\right)$
17. $3\sqrt{3}$ **19.** **(a)** $W=\frac{1}{16}$ J **(b)** 5 m **21.** $\left(\frac{8}{5},0\right)$
23. **(a)** $F=\int_0^1 \rho x 3\,dx$ N **(b)** $F=\int_1^4 \rho(1+x)2x\,dx$ lb/ft^2
(c) $F=\int_{-10}^0 9810|y|2\sqrt{\frac{125}{8}(y+10)}\,dy$ N

Chapter 6 Making Connections (Page 487)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** πA_1 **(b)** $a=\frac{A_1}{2A_2}$ **2.** 1,010,807 ft·lb **3.** $\int_0^a 2\pi r f(r)\,dr$

Exercise Set 7.1 (Page 490)

1. $-2(x-3)^4+C$ **3.** $\frac{1}{2}\tan(x^2)+C$ **5.** $-\frac{1}{3}\ln(2+\cos 3x)+C$
7. $\cosh(e^x)+C$ **9.** $e^{\tan x}+C$ **11.** $-\frac{1}{42}\cos^6 7x+C$
13. $\ln(e^x+\sqrt{e^{2x}+4})+C$ **15.** $2e^{\sqrt{x-1}}+C$ **17.** $2\sinh\sqrt{x}+C$
19. $-\frac{2}{\ln 3}3^{-\sqrt{x}}+C$ **21.** $\frac{1}{2}\coth\frac{2}{x}+C$ **23.** $-\frac{1}{4}\ln\left|\frac{2+e^{-x}}{2-e^{-x}}\right|+C$
25. $\sin^{-1}(e^x)+C$ **27.** $-\frac{1}{2}\cos(x^2)+C$ **29.** $-\frac{1}{\ln 25}5^{-x^2}+C$
31. **(b)** $\ln\left|\tan\frac{x}{2}\right|+C$ **(c)** $\ln\left|\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|+C$
33. **(a)** $\frac{1}{2}\sin^2 x+C$ **(b)** $-\frac{1}{4}\cos 2x+C$

Exercise Set 7.2 (Page 498)

1. $-e^{-2x}\left(\frac{x}{2}+\frac{1}{4}\right)+C$ **3.** $x^2e^x-2xe^x+2e^x+C$
5. $x^2\sin x+2x\cos x-2\sin x+C$ **7.** $-\frac{1}{3}x\cos 3x+\frac{1}{9}\sin 3x+C$
9. $\frac{x^2}{2}\ln x-\frac{x^2}{4}+C$ **11.** $x(\ln x)^2-2x\ln x+2x+C$
13. $x\ln(3x-2)-x-\frac{2}{3}\ln(3x-2)+C$
15. $x\sin^{-1}2x+\frac{1}{2}\sqrt{1-4x^2}+C$
17. $x\tan^{-1}(3x)-\frac{1}{6}\ln(1+9x^2)+C$
19. $(x/2)[\sin(\ln x)-\cos(\ln x)]+C$ **21.** $\frac{1}{2}e^x(\sin x-\cos x)+C$
23. $x\tan x+\ln|\cos x|+C$ **25.** $\frac{1}{2}x^2e^{x^2}-\frac{1}{2}e^{x^2}+C$
27. $\frac{1}{9}(5e^6+1)$ **29.** $(2e^3+1)/9$ **31.** $3\ln 3-2$ **33.** $\frac{5\pi}{6}-\sqrt{3}+1$
35. $-\pi/2$ **37.** $\frac{1}{3}\left(2\sqrt{3}\pi-\frac{\pi}{2}-2+\ln 2\right)$

Responses to True–False questions may be abridged to save space.
39. True; see the subsection "Guidelines for Integration by Parts."
41. False; $\ln x$ isn't a factor of the integrand.
43. $-2\sqrt{x}\cos\sqrt{x}+2\sin\sqrt{x}+C$
47. $(4x^3-6x)\sin 2x-(2x^4-6x^2+3)\cos 2x+C$
49. $-(3x^2+5x+7)e^{-x}+C$
51. $\frac{e^{ax}}{a^2+b^2}(a\sin bx-b\cos bx)+C$ **53.** **(a)** $\frac{1}{2}\sin^2 x+C$
55. **(a)** $A=1$ **(b)** $V=\pi(e-2)$ **57.** $V=2\pi^2$ **59.** π^2-4
61. **(a)** $-\frac{1}{4}\sin^3 x\cos x-\frac{3}{8}\sin x\cos x+\frac{3}{8}x+C$ **(b)** $8/15$
65. **(a)** $\frac{1}{2}\tan^2 x+\ln|\cos(x)|+C$ **(b)** $\frac{1}{3}\sec^2 x\tan x+\frac{2}{3}\tan x+C$
(c) $x^3e^x-3x^2e^x+6xe^x-6e^x+C$
69. $(x+1)\ln(x+1)-x+C$ **71.** $\frac{1}{2}(x^2+1)\tan^{-1}x-\frac{1}{2}x+C$

Exercise Set 7.3 (Page 506)

1. $-\frac{1}{3}\cos^3 x+C$ **3.** $\frac{\theta}{2}-\frac{1}{16}\sin 8\theta+C$
5. $\frac{1}{3a}\cos^3 a\theta-\cos a\theta+C$ **7.** $\frac{1}{2b}\sin^2 bx+C$
9. $\frac{1}{3}\sin^3 t-\frac{1}{5}\sin^5 t+C$ **11.** $\frac{1}{8}x-\frac{1}{32}\sin 4x+C$
13. $-\frac{1}{14}\cos 7x+\frac{1}{2}\cos x+C$ **15.** $-\frac{1}{3}\cos(3x/2)-\cos(x/2)+C$
17. $2/3$ **19.** 0 **21.** $3/8$ **23.** $\frac{1}{2}\tan(2x-1)+C$
25. $\ln|\cos(e^{-x})|+C$ **27.** $\frac{1}{7}\ln|\sec 7x+\tan 7x|+C$
29. $\frac{1}{3}\tan^3 x+C$ **31.** $\frac{1}{16}\sec^4 4x+C$ **33.** $\frac{1}{5}\sec^5 x-\frac{1}{3}\sec^3 x+C$
35. $\frac{1}{4}\sec^3 x\tan x-\frac{5}{8}\sec x\tan x+\frac{3}{8}\ln|\sec x+\tan x|+C$
37. $\frac{1}{3}\sec^3 t+C$ **39.** $\tan x+\frac{1}{3}\tan^3 x+C$
41. $\frac{1}{8}\tan^2 4x+\frac{1}{4}\ln|\cos 4x|+C$ **43.** $\frac{2}{3}\tan^{3/2}x+\frac{2}{7}\tan^{7/2}x+C$
45. $\frac{1}{2}-\frac{\pi}{8}$ **47.** $-\frac{1}{2}+\ln 2$ **49.** $-\frac{1}{5}\csc^5 x+\frac{1}{3}\csc^3 x+C$
51. $-\frac{1}{2}\csc^2 x-\ln|\sin x|+C$

Responses to True–False questions may be abridged to save space.
53. False; use this identity to help evaluate integrals of the form $\int \sin mx\cos nx\,dx$.
55. True; $\int \sin^5 x\cos^8 x\,dx=\int \sin x(1-\cos^2 x)^2\cos^8 x\,dx=-\int(1-u^2)^2u^8\,du=-\int(u^8-2u^{10}+u^{12})\,du$
59. $L=\ln(\sqrt{2}+1)$ **61.** $V=\pi/2$

67. $-\dfrac{1}{\sqrt{a^2+b^2}}\ln\left[\dfrac{\sqrt{a^2+b^2}+a\cos x-b\sin x}{a\sin x+b\cos x}\right]+C$

69. **(a)** $\frac{2}{3}$ **(b)** $3\pi/16$ **(c)** $\frac{8}{15}$ **(d)** $5\pi/32$

▶ Exercise Set 7.4 (Page 513)

1. $\frac{1}{2}\sin^{-1}(x/3)+\frac{1}{2}x\sqrt{9-x^2}+C$ 3. $2\sin^{-1}\left(\dfrac{x}{2}\right)-\dfrac{1}{2}x\sqrt{4-x^2}+C$

5. $\frac{1}{16}\tan^{-1}(x/2)+\dfrac{x}{8(4+x^2)}+C$ 7. $\sqrt{x^2-9}-3\sec^{-1}(x/3)+C$

9. $-(x^2+2)\sqrt{1-x^2}+C$ 11. $\dfrac{\sqrt{9x^2-4}}{4x}+C$ 13. $\dfrac{x}{\sqrt{1-x^2}}+C$

15. $\ln|\sqrt{x^2-16}+x|+C$ 17. $\dfrac{-x}{9\sqrt{4x^2-9}}+C$

19. $\frac{1}{2}\sin^{-1}(e^x)+\frac{1}{2}e^x\sqrt{1-e^{2x}}+C$ 21. $2/3$ 23. $(\sqrt{3}-\sqrt{2})/2$

25. $\dfrac{10\sqrt{3}+18}{243}$

Responses to True–False questions may be abridged to save space.

27. False; use the substitution $x=a\sec\theta$ with $0\le\theta<\pi/2$ $(x\ge a)$ or $\pi/2\le\theta<\pi$ $(x\le -a)$.

29. True; with the restriction $-\pi/2\le\theta\le\pi/2$, this substitution gives $\sqrt{a^2-x^2}=a\cos\theta$ and $dx=a\cos\theta\,d\theta$.

31. $\frac{1}{2}\ln(x^2+9)+C$

33. $L=\sqrt{10}-\sqrt{2}+\ln(\frac{1}{3}(\sqrt{10}-1))-\ln(\sqrt{2}-1)$

35. $S=\dfrac{\pi}{32}\left[18\sqrt{5}-\ln(2+\sqrt{5})\right]$

37. $\dfrac{1}{2}\tan^{-1}\left(\dfrac{x-2}{2}\right)+C$ 39. $\sin^{-1}\left(\dfrac{x-1}{3}\right)+C$

41. $\ln(x-3+\sqrt{(x-3)^2+1})+C$

43. $8\sin^{-1}\left(\dfrac{x+1}{4}\right)+\dfrac{1}{2}(x+1)\sqrt{15-2x-x^2}+C$

45. $\dfrac{1}{\sqrt{14}}\tan^{-1}\left(\sqrt{2/7}(x+1)\right)+C$ 47. $\sin^{-1}\left(\dfrac{2}{3}\right)-\sin^{-1}\left(\dfrac{1}{3}\right)$

49. $u=\sin^2 x$, $\frac{1}{2}\displaystyle\int\sqrt{1-u^2}du$

$=\frac{1}{4}[\sin^2 x\sqrt{1-\sin^4 x}+\sin^{-1}(\sin^2 x)]+C$

51. **(a)** $\sinh^{-1}(x/3)+C$ **(b)** $\ln\left(\dfrac{\sqrt{x^2+9}}{3}+\dfrac{x}{3}\right)+C$

▶ Exercise Set 7.5 (Page 521)

1. $\dfrac{A}{x-3}+\dfrac{B}{x+6}$ 3. $\dfrac{A}{x}+\dfrac{B}{x^2}+\dfrac{C}{x-1}$

5. $\dfrac{A}{x}+\dfrac{B}{x^2}+\dfrac{C}{x^3}+\dfrac{Dx+E}{x^2+2}+\dfrac{Fx+G}{(x^2+2)^2}$ 7. $\dfrac{Ax+B}{x^2-2}+\dfrac{Cx+D}{(x^2-2)^2}$

9. $\dfrac{1}{7}\ln\left|\dfrac{x-5}{x+2}\right|+C$ 11. $9\ln|2x-1|+2\ln|x+5|+C$

13. $\ln\left|\dfrac{x(x+3)^2}{x-3}\right|+C$ 15. $\dfrac{x^2}{2}-3x+2\ln|x+3|+C$

17. $3x+12\ln|x-2|-\dfrac{2}{x-2}+C$

19. $\ln|x^2-3x-10|+C$

21. $x+\dfrac{x^3}{3}+\ln\left|\dfrac{(x-1)^2(x+1)}{x^2}\right|+C$

23. $3\ln|x|-\ln|x-1|-\dfrac{5}{x-1}+C$

25. $\dfrac{2}{x-3}+\ln|x-3|+\ln|x+1|+C$

27. $-\dfrac{1}{x+1}+\dfrac{1}{2(x+1)^2}+C$

29. $-\frac{7}{34}\ln|4x-1|+\frac{6}{17}\ln(x^2+1)+\frac{3}{17}\tan^{-1}x+C$

31. $3\tan^{-1}x+\frac{1}{2}\ln(x^2+3)+C$

33. $\dfrac{x^2}{2}-2x+\dfrac{1}{2}\ln(x^2+1)+C$

Responses to True–False questions may be abridged to save space.

35. True; $\dfrac{2x+3}{x^2}=\dfrac{2x}{x^2}+\dfrac{3}{x^2}=\dfrac{2}{x}+\dfrac{3}{x^2}$.

37. True; partial fractions rewrites proper rational functions $P(x)/Q(x)$ as a sum of terms of the form $\dfrac{A}{(Bx+C)^k}$ and/or $\dfrac{Dx+E}{(Fx^2+Gx+H)^k}$.

39. $\ln\left(\dfrac{1-\sin\theta}{6+\sin\theta}\right)+C$ 41. $e^x-2\tan^{-1}\left(\frac{1}{2}e^x\right)+C$

43. $V=\pi\left(\frac{19}{5}-\frac{9}{4}\ln 5\right)$ 45. $\dfrac{1}{\sqrt{2}}\tan^{-1}\left(\dfrac{x+1}{\sqrt{2}}\right)+\dfrac{1}{x^2+2x+3}+C$

47. $\frac{1}{8}\ln|x-1|-\frac{1}{5}\ln|x-2|+\frac{1}{12}\ln|x-3|-\frac{1}{120}\ln|x+3|+C$

▶ Exercise Set 7.6 (Page 531)

1. Formula (60): $\frac{4}{5}x+\frac{4}{25}\ln|5x-1|+C$

3. Formula (65): $\dfrac{1}{5}\ln\left|\dfrac{x}{5+7x}\right|+C$

5. Formula (102): $\frac{1}{20}(2x-1)(4x+3)^{3/2}+C$

7. Formula (108): $\dfrac{1}{2}\ln\left|\dfrac{\sqrt{4-5x}-2}{\sqrt{4-5x}+2}\right|+C$

9. Formula (69): $\dfrac{1}{6}\ln\left|\dfrac{x+3}{x-3}\right|+C$

11. Formula (73): $\dfrac{x}{2}\sqrt{x^2-2}-\ln|x+\sqrt{x^2-2}|+C$

13. Formula (95): $\dfrac{x}{2}\sqrt{x^2+4}-2\ln(x+\sqrt{x^2+4})+C$

15. Formula (74): $\dfrac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\dfrac{x}{2}+C$

17. Formula (79): $\sqrt{1-x^2}-\ln\left|\dfrac{1+\sqrt{1-x^2}}{x}\right|+C$

19. Formula (38): $-\dfrac{\sin 8x}{16}+\dfrac{1}{4}\sin 2x+C$

21. Formula (50): $\dfrac{x^5}{25}(5\ln x-1)+C$

23. Formula (42): $\dfrac{e^{-2x}}{29}(-2\sin(5x)-5\cos(5x))+C$

25. Formula (62): $\dfrac{1}{2}\displaystyle\int\dfrac{u\,du}{(4-3u)^2}=\dfrac{1}{18}\left[\dfrac{4}{4-3e^{2x}}+\ln|4-3e^{2x}|\right]+C$

27. Formula (68): $\dfrac{2}{3}\displaystyle\int\dfrac{du}{u^2+4}=\dfrac{1}{3}\tan^{-1}\dfrac{3\sqrt{x}}{2}+C$

29. Formula (76): $\dfrac{1}{2}\displaystyle\int\dfrac{du}{\sqrt{u^2-25}}=\dfrac{1}{2}\ln|2x+\sqrt{4x^2-25}|+C$

31. Formula (81): $\dfrac{1}{4}\displaystyle\int\dfrac{u^2}{\sqrt{2-u^2}}du=-\dfrac{1}{4}x^2\sqrt{2-4x^4}$

$+\dfrac{1}{4}\sin^{-1}\left(\sqrt{2}x^2\right)+C$

33. Formula (27): $\displaystyle\int\cos^2 u\,du=\frac{1}{2}\ln x+\frac{1}{4}\sin(2\ln x)+C$

35. Formula (51): $\dfrac{1}{4}\displaystyle\int ue^u\,du=\dfrac{1}{4}(-2x-1)e^{-2x}+C$

37. $u=\sin 3x$, Formula (67): $\dfrac{1}{3}\displaystyle\int\dfrac{du}{u(u+1)^2}$

$=\dfrac{1}{3}\left(\dfrac{1}{\sin 3x+1}+\left|\dfrac{\sin 3x}{\sin 3x+1}\right|\right)+C$

39. $u=4x^2$, Formula (70): $\dfrac{1}{8}\displaystyle\int\dfrac{du}{u^2-1}=\dfrac{1}{16}\ln\left|\dfrac{4x^2-1}{4x^2+1}\right|+C$

41. $u=3x$, Formula (112): $\dfrac{1}{3}\displaystyle\int\sqrt{\tfrac{5}{3}u-u^2}\,du=\dfrac{18x-5}{36}\sqrt{5x-9x^2}$

$+\dfrac{25}{216}\sin^{-1}\left(\dfrac{18x-5}{5}\right)+C$

43. $u=2e^x$, Formula (74): $\dfrac{1}{2}\displaystyle\int\sqrt{3-u^2}\,du=\dfrac{1}{2}e^x\sqrt{3-4e^{2x}}$

$+\dfrac{3}{4}\sin^{-1}\left(\dfrac{2e^x}{\sqrt{3}}\right)+C$

45. $u = 3x$, Formula (44): $\int u \sin u \, du = \sin 3x - 3x \cos 3x + C$

47. $u = -\sqrt{x}$, Formula (51): $2\int ue^u \, du = -2(\sqrt{x}+1)e^{-\sqrt{x}} + C$

49. $x^2 + 6x - 7 = (x+3)^2 - 16$, $u = x + 3$, Formula (70):
$$\int \frac{du}{u^2 - 16} = \frac{1}{8}\ln\left|\frac{x-1}{x+7}\right| + C$$

51. $x^2 - 4x - 5 = (x-2)^2 - 9$, $u = x - 2$, Formula (77):
$$\int \frac{u+2}{\sqrt{9-u^2}}\,du = -\sqrt{5+4x-x^2} + 2\sin^{-1}\left(\frac{x-2}{3}\right) + C$$

53. $u = \sqrt{x-3}$, $\frac{2}{5}(x-3)^{5/2} + 2(x-3)^{3/2} + C$

55. $u = \sqrt{x^3+1}$,
$$\frac{2}{3}\int u^2(u^2-1)\,du = \frac{2}{15}(x^3+1)^{5/2} - \frac{2}{9}(x^3+1)^{3/2} + C$$

57. $u = x^{1/3}$, $\int \frac{3u^2}{u^3 - u}\,du = \frac{3}{2}\ln|x^{2/3} - 1| + C$

59. $u = x^{1/4}$, $4\int \frac{1}{u(1-u)}\,du = 4\ln\frac{x^{1/4}}{|1 - x^{1/4}|} + C$

61. $u = x^{1/6}$,
$$6\int \frac{u^3}{u-1}\,du = 2x^{1/2} + 3x^{1/3} + 6x^{1/6} + 6\ln|x^{1/6} - 1| + C$$

63. $u = \sqrt{1+x^2}$, $\int (u^2 - 1)\,du = \frac{1}{3}(1+x^2)^{3/2} - (1+x^2)^{1/2} + C$

65.
$$\int \frac{1}{1 + \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}} \frac{2}{1+u^2}\,du = \int \frac{1}{u+1}\,du$$
$$= \ln|\tan(x/2) + 1| + C$$

67. $\int \frac{d\theta}{1 - \cos\theta} = \int \frac{1}{u^2}\,du = -\cot(\theta/2) + C$

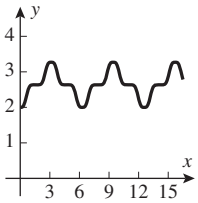
69.
$$\int \frac{1}{\frac{2u}{1+u^2} + \frac{2u}{1+u^2}\cdot\frac{1+u^2}{1-u^2}} \cdot \frac{2}{1+u^2}\,du = \int \frac{1-u^2}{2u}\,du$$
$$= \frac{1}{2}\ln|\tan(x/2)| - \frac{1}{4}\tan^2(x/2) + C$$

71. $x = \frac{6e^3}{1+e^3}$ **73.** $A = 4\sqrt{5} + 18\sin^{-1}\frac{2}{3}$ **75.** $A = \frac{1}{40}\ln 9$

77. $V = \pi(\pi - 2)$ **79.** $V = 2\pi(1 - 4e^{-3})$

81. $L = \sqrt{65} + \frac{1}{8}\ln(8 + \sqrt{65})$ **83.** $S = 2\pi\left[\sqrt{2} + \ln(1+\sqrt{2})\right]$

85.



91. $\frac{1}{31}\cos^{31} x \sin^{31} x + C$

93. $-\frac{1}{9}\ln|1 + x^{-9}| + C$

Exercise Set 7.7 (Page 544)

1. $\int_0^8 \sqrt{x+1}\,dx = \frac{52}{3} \approx 17.3333333$
(a) $M_{10} = 17.34206$; $|E_M| \approx .0087237$
(b) $T_{10} = 17.31575$; $|E_T| \approx .0175868$
(c) $S_{20} = 17.33329$; $|E_S| \approx .00004648$

3. $\int_0^{\pi/2} \cos x\,dx = 1$
(a) $M_{10} = 1.00103$; $|E_M| \approx 0.00102882$
(b) $T_{10} = 0.997943$; $|E_T| \approx 0.00205701$
(c) $S_{20} = 1.00000$; $|E_S| \approx 2.11547 \times 10^{-7}$

5. $\int_1^3 e^{-2x}\,dx = \frac{-1+e^4}{2e^6} \approx 0.0664283$
(a) $M_{10} = 0.0659875$; $|E_M| \approx 0.000440797$
(b) $T_{10} = 0.0673116$; $|E_T| \approx 0.000883357$
(c) $S_{20} = 0.0664289$; $|E_S| \approx 5.87673 \times 10^{-7}$

7. **(a)** $|E_M| \le \frac{9}{3200} = 0.0028125$
(b) $|E_T| \le \frac{9}{1600} = 0.005625$
(c) $|E_S| \le \frac{81}{10{,}240{,}000} \approx 7.91016 \times 10^{-6}$

9. **(a)** $|E_M| \le \frac{\pi^3}{19{,}200} \approx 0.00161491$
(b) $|E_T| \le \frac{\pi^3}{9600} \approx 0.00322982$
(c) $|E_S| \le \frac{\pi^5}{921{,}600{,}000} \approx 3.32053 \times 10^{-7}$

11. **(a)** $|E_M| \le \frac{1}{75e^2} \approx 0.00180447$
(b) $|E_T| \le \frac{2}{75e^2} \approx 0.00360894$
(c) $|E_S| \le \frac{1}{56{,}250e^2} \approx 2.40596 \times 10^{-6}$

13. **(a)** $n = 24$ **(b)** $n = 34$ **(c)** $n = 8$

15. **(a)** $n = 13$ **(b)** $n = 18$ **(c)** $n = 4$

17. **(a)** $n = 43$ **(b)** $n = 61$ **(c)** $n = 8$

Responses to True–False questions may be abridged to save space.

19. False; $S_{50} = \frac{2}{3}M_{25} + \frac{1}{3}T_{25}$

21. False; T_n is the average of L_n and R_n.

23. $g(x) = \frac{1}{24}x^2 - \frac{3}{8}x + \frac{13}{12}$

25. $S_{10} = 1.76414975$; $\int_{-2}^{2} e^{-x^2} \approx 1.76416278$

27. $S_{10} = 14.5496435$; $\int_{-1}^{3} x\sqrt{2+x^3} \approx 14.59935195$

29. $S_{10} = 0.3102678$; $\int_0^1 \sin(x^2)\,dx \approx 0.31026830$

31. **(a)** $M_{10} = 3.14243$; error $E_M \approx -0.000833331$
(b) $T_{10} = 3.13993$; error $E_T \approx 0.00166666$
(c) $S_{20} = 3.14159$; error $E_S \approx 6.20008 \times 10^{-10}$

33. $S_{14} = 0.693147984$, $|E_S| \approx 0.000000803 = 8.03 \times 10^{-7}$

35. $n = 238$ **39.** 1.91009889 **41.** 1071 ft **43.** 37.9 mi **45.** 9.3 L

47. **(a)** $\max|f''(x)| \approx 3.844880$ **(b)** $n = 18$ **(c)** 0.904741

49. **(a)** The maximum value of $|f^{(4)}(x)|$ is approximately 141.09327.
(b) $n = 10$ **(c)** $S_{10} = 0.962257248$

Exercise Set 7.8 (Page 554)

1. **(a)** improper; infinite discontinuity at $x = 4$ **(b)** not improper
(c) improper; infinite discontinuity at $x = 0$
(d) improper; infinite interval of integration
(e) improper; infinite interval of integration and infinite discontinuity at $x = 1$ **(f)** not improper

3. $\frac{1}{3}$ **5.** $\frac{3}{2}\ln 2$ **7.** $\frac{1}{2}$ **9.** $-\frac{1}{4}$ **11.** $\frac{1}{4}$ **13.** divergent **15.** 0

17. divergent **19.** divergent **21.** $\pi/2$ **23.** 1 **25.** divergent

27. $\frac{9}{2}$ **29.** divergent **31.** $\pi/2$

Responses to True–False questions may be abridged to save space.

33. False; the integrand $\frac{1}{x(x-3)}$ is continuous on $[1, 2]$.

35. True; see Theorem 7.8.2 with $p = \frac{4}{3} > 1$.

37. 2 **39.** 2 **41.** $-\frac{1}{2}$

43. **(a)** 2.664069 **(b)** 2.804364 **(c)** 0.219384 **(d)** 0.504067 **45.** 27

47. -1 **49.** $\frac{1}{3}$ **51.** **(a)** $V = \pi/2$ **(b)** $S = \pi[\sqrt{2} + \ln(1+\sqrt{2})]$

53. **(b)** $1/e$ **(c)** It is convergent. **55.** $V = \pi$

59. $\frac{2\pi NI}{kr}\left(1 - \frac{a}{\sqrt{r^2+a^2}}\right)$

61. **(b)** 2.4×10^7 mi·lb **63.** **(a)** $\frac{1}{s^2}$ **(b)** $\frac{2}{s^3}$ **(c)** $\frac{e^{-3s}}{s}$

67. **(a)** 1.047 **71.** 0.621

Chapter 7 Review Exercises (Page 557)

1. $\frac{2}{15}(4+5x)^{3/2}+C$ 3. $-\frac{2}{3}\cos^{3/2}\theta + C$ 5. $\frac{1}{6}\tan^3(x^2)+C$
7. **(a)** $2\sin^{-1}(\sqrt{x/2})+C$; $-2\sin^{-1}(\sqrt{2-x}/\sqrt{2})+C$; $\sin^{-1}(x-1)+C$
9. $-xe^{-x}-e^{-x}+C$ 11. $x\ln(2x+7)-x+\frac{7}{2}\ln(2x+7)+C$
13. $(4x^4-12x^2+6)\sin(2x)+(8x^3-12x)\cos(2x)+C$
15. $\frac{1}{2}\theta-\frac{1}{24}\sin 12\theta + C$ 17. $-\frac{1}{8}\cos 4x+\frac{1}{4}\cos 2x+C$
19. $-\frac{1}{8}\sin^3(2x)\cos 2x-\frac{3}{16}\cos 2x\sin 2x+\frac{3}{8}x+C$
21. $2\sin^{-1}\frac{x}{2}-\frac{1}{2}x\sqrt{4-x^2}+C$ 23. $\ln|x+\sqrt{x^2-1}|+C$
25. $\dfrac{x\sqrt{x^2+9}}{2}-\dfrac{9\ln(|\sqrt{x^2+9}+x|)}{2}+C$ 27. $\dfrac{1}{6}\ln\left|\dfrac{x-1}{x+5}\right|+C$
29. $\frac{1}{2}x^2-2x+6\ln|x+2|+C$
31. $\ln|x+2|+\dfrac{4}{x+2}-\dfrac{2}{(x+2)^2}+C$
35. Formula (40): $-\frac{1}{34}\cos 17x+\frac{1}{6}\cos 3x+C$
37. Formula (113): $\frac{1}{24}(8x^2-2x-3)\sqrt{x-x^2}+\frac{1}{16}\sin^{-1}(2x-1)+C$
39. Formula (28): $\frac{1}{2}\tan 2x - x + C$
41. $\displaystyle\int_1^3 \frac{1}{\sqrt{x+1}} = 4-2\sqrt{2}\approx 1.17157$
(a) $M_{10}=1.17138$; $|E_M|\approx 0.000190169$
(b) $T_{10}=1.17195$; $|E_T|\approx 0.000380588$
(c) $S_{20}=1.17157$; $|E_S|\approx 8.35151\times 10^{-8}$
43. **(a)** $|E_M|\le \dfrac{1}{1600\sqrt{2}}\approx 0.000441942$
(b) $|E_T|\le \dfrac{1}{800\sqrt{2}}\approx 0.000883883$
(c) $|E_S|\le \dfrac{7}{15{,}360{,}000\sqrt{2}}\approx 3.22249\times 10^{-7}$
45. **(a)** $n=22$ **(b)** $n=30$ **(c)** $n=6$ 47. 1 49. 6
51. e^{-1} 53. $a=\pi/2$ 55. $\dfrac{x}{3\sqrt{3+x^2}}+C$ 57. $\frac{5}{12}-\frac{1}{2}\ln 2$
59. $\frac{1}{6}\sin^3 2x-\frac{1}{10}\sin^5 2x+C$
61. $\frac{2}{13}e^{2x}\cos 3x+\frac{3}{13}e^{2x}\sin 3x+C$
63. $-\frac{1}{6}\ln|x-1|+\frac{1}{15}\ln|x+2|+\frac{1}{10}\ln|x-3|+C$
65. $4-\pi$ 67. $\ln\dfrac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1}+C$ 69. $\dfrac{\pi}{12}+\dfrac{\sqrt{3}}{2}-1$
71. $\sqrt{x^2+2x+2}+2\ln(\sqrt{x^2+2x+2}+x+1)+C$ 73. $\dfrac{1}{2(a^2+1)}$

Chapter 7 Making Connections (Page 559)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

3. **(a)** $\Gamma(1)=1$ **(c)** $\Gamma(2)=1, \Gamma(3)=2, \Gamma(4)=6$
5. **(b)** 1.37078 seconds

Exercise Set 8.1 (Page 566)

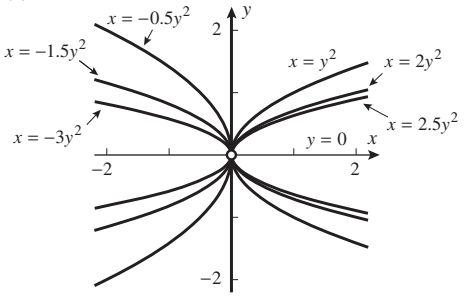
3. **(a)** first order **(b)** second order

Responses to True–False questions may be abridged to save space.

5. False; only first-order derivatives appear.
7. True; it is third order.
15. $y(x)=e^{-2x}+3e^x$ 17. $y(x)=-e^{-2x}+2e^{3x}$
19. $y(x)=-e^{-2x}(\sin 3x+\cos 3x)$ 21. $y(x)=-2x^2+2x+5$
23. $y(x)=2/(3-2x)$ 25. $y(x)=3/x^2$
27. **(a)** $\dfrac{dy}{dt}=ky^2$, $y(0)=y_0$ $(k>0)$
(b) $\dfrac{dy}{dt}=-ky^2$, $y(0)=y_0$ $(k>0)$
29. **(a)** $\dfrac{ds}{dt}=\dfrac{1}{3}s$ **(b)** $\dfrac{d^2s}{dt^2}=2\dfrac{ds}{dt}$ 33. **(b)** $L/2$

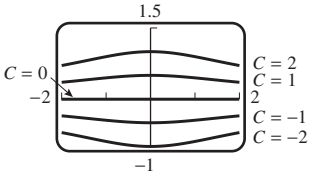
Exercise Set 8.2 (Page 575)

1. $y=Cx$ 3. $y=Ce^{-\sqrt{1+x^2}}-1, C\ne 0$
5. $2\ln|y|+y^2=e^x+C$ 7. $y=\ln(\sec x+C)$
9. $y=\dfrac{1}{1-C(\csc x-\cot x)}, y=0$
11. $y^2+\sin y=x^3+\pi^2$ 13. $y^2-2y=t^2+t-1$
15. **(a)**

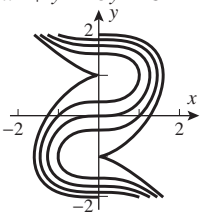


(b) $y^2=x/3$

17. $y=\dfrac{C}{\sqrt{x^2+4}}$

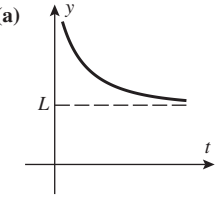


19. $x^3+y^3-3y=C$

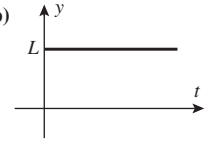


Responses to True–False questions may be abridged to save space.

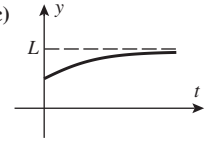
21. True; since $\dfrac{1}{f(y)}\,dy=dx$. 23. True; since $\left(\frac{1}{2}\right)^5 32=1$.
27. $y=\ln\left(\dfrac{x^2}{2}-7\right)$
29. **(a)** $y'(t)=y(t)/50$, $y(0)=20{,}000$ **(b)** $y(t)=20{,}000e^{t/50}$
(c) $50\ln 2\approx 34.66$ hr **(d)** $50\ln(2.25)\approx 40.55$ hr
31. **(a)** $\dfrac{dy}{dt}=-ky$, $k\approx 0.1810$ **(b)** $y=5.0\times 10^7 e^{-0.181t}$
(c) $\approx 219{,}000$ atoms **(d)** 12.72 days
33. $50\ln(50)\approx 196$ days 35. 3.30 days
39. **(b)** 35 years **(c)** 20 years **(d)** 7%
43. **(a)** no **(b)** same, $r\%$ 45. **(b)** $\ln(2)/\ln(1.3)\approx 2.64$ hr
47. **(a)** \$1419.07 **(b)** \$4965.85 **(c)** 9.9 years
51. **(a)**



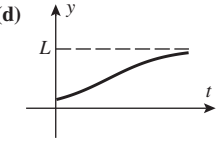
(b)



(c)



(d)

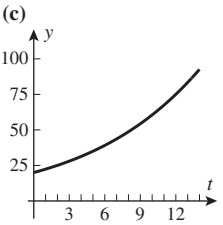


53. $y_0\approx 2$, $L\approx 8$, $k\approx 0.5493$
55. **(a)** $y_0=5$ **(b)** $L=12$ **(c)** $k=1$ **(d)** $t=0.3365$
(e) $\dfrac{dy}{dt}=\dfrac{1}{2}y(12-y)$, $y(0)=5$
57. **(a)** $y=\dfrac{1000}{1+49e^{-0.115t}}$

(b)

t	0	1	2	3	4	5	6	7
$y(t)$	20	22	25	28	31	35	39	44
t	8	9	10	11	12	13	14	
$y(t)$	49	54	61	67	75	83	93	

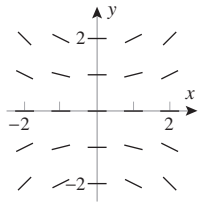
(c)



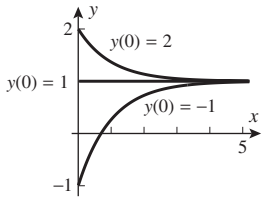
59. **(a)** $T = 21 + 74e^{-kt}$ **(b)** 6.22 min
61. **(a)** $v = c\ln\dfrac{m_0}{m_0 - kt} - gt$ **(b)** 2030 m/s
63. **(a)** $h \approx (2 - 0.003979t)^2$ **(b)** 8.4 min
65. **(a)** $v = 64/(2t+1)$, $x = 32\ln(2t+1)$

Exercise Set 8.3 (Page 584)

1.



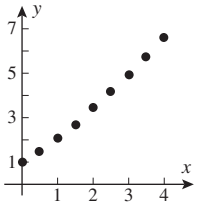
3.



5. $y \to 1$ as $x \to +\infty$

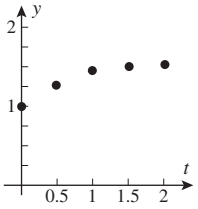
7.

n	0	1	2	3	4	5	6	7	8
x_n	0	0.5	1	1.5	2	2.5	3	3.5	4
y_n	1.00	1.50	2.07	2.71	3.41	4.16	4.96	5.82	6.72



9.

n	0	1	2	3	4
t_n	0.00	0.50	1.00	1.50	2.00
y_n	1.00	1.27	1.42	1.49	1.53



11. 0.63

Responses to True–False questions may be abridged to save space.

13. True; the derivative is positive.
15. True; $y = y_0$ is a solution if y_0 is a root of p.
17. **(b)** $y(1/2) = \sqrt{15}/2$ 19. **(b)** The x-intercept is $\ln 3$.
23. **(a)** $y' = \dfrac{2xy - y^3}{3xy^2 - x^2}$ **(c)** $xy^3 - x^2y = 6$
25. **(b)** $\lim\limits_{n\to+\infty} y_n = \lim\limits_{n\to+\infty}\left(\dfrac{n+1}{n}\right)^n = e$

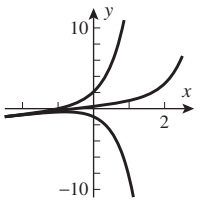
Exercise Set 8.4 (Page 592)

1. $y = \frac{1}{2}e^{-3x} + Ce^{-5x}$ 3. $y = e^{-x}\cos(e^x) + Ce^{-x}$
5. $y = \dfrac{C}{\sqrt{x^2+1}}$ 7. $y = \dfrac{x}{2} + \dfrac{5}{2x}$ 9. $y = 5e^{x^2} - 2$

Responses to True–False questions may be abridged to save space.

11. True; it will approach the concentration of the entering fluid.
13. False; $y = x^2$ is a solution to $dy/dx = 2x$, but $y + y = 2x^2$ is not.

15.



17. $\lim\limits_{x\to+\infty} y = \begin{cases} +\infty & \text{if } y_0 \ge 1/4 \\ -\infty, & \text{if } y_0 < 1/4 \end{cases}$

19. **(a)**

n	0	1	2	3	4	5
x_n	0	0.2	0.4	0.6	0.8	1.0
y_n	1	1.20	1.48	1.86	2.35	2.98

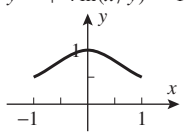
(b) $y = -(x+1) + 2e^x$

x_n	0	0.2	0.4	0.6	0.8	1.0
$y(x_n)$	1	1.24	1.58	2.04	2.65	3.44
Absolute error	0	0.04	0.10	0.19	0.30	0.46
Percentage error	0	3	6	9	11	13

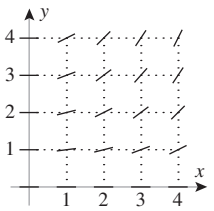
21. **(a)** $240 - 210e^{-t/30}$ oz **(b)** 163 oz 23. 25 lb
27. **(a)** $I(t) = 2 - 2e^{-2t}$ A **(b)** $I(t) \to 2$ A

Chapter 8 Review Exercises (Page 594)

3. $y = \tan(x^3/3 + C)$ 5. $\ln|y| + y^2/2 = e^x + C$ and $y = 0$
7. $y^{-4} + 4\ln(x/y) = 1$
9.

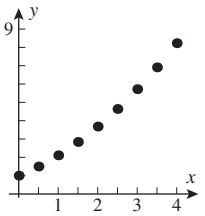


11.



13.

n	0	1	2	3	4	5	6	7	8
x_n	0	0.5	1	1.5	2	2.5	3	3.5	4
y_n	1	1.50	2.11	2.84	3.68	4.64	5.72	6.91	8.23



15. $y(1) \approx 1.00$

n	0	1	2	3	4	5
t_n	0	0.2	0.4	0.6	0.8	1.0
y_n	1.00	1.20	1.26	1.10	0.94	1.00

17. about 2106.5 years 19. $y = \frac{1}{2}e^{-2x} + Ce^{-4x}$
21. $y = -1 + 6e^{x^2/2}$ 23. $y = 2\operatorname{sech} x + \frac{1}{2}(x\operatorname{sech} x + \sinh x)$
25. **(a)** linear **(b)** both **(c)** separable **(d)** neither 27. about 646 oz

Chapter 8 Making Connections (Page 595)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

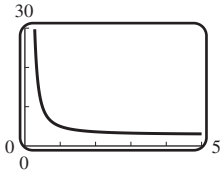
1. **(b)** $y = 2 - e^{-2x}$
3. **(a)** $du/dx = (f(u) - u)/x$ **(b)** $x^2 - 2xy - y^2 = C$

Exercise Set 9.1 (Page 605)

1. (a) $\frac{1}{5^{n-1}}$ (b) $\frac{(-1)^{n-1}}{3^{n-1}}$ (c) $\frac{2n-1}{2n}$ (d) $\frac{n^2}{\pi^{1/(n+1)}}$
3. (a) 2, 0, 2, 0 (b) 1, −1, 1, −1 (c) $2(1+(-1)^n)$; $2+2\cos n\pi$
5. (a) The limit doesn't exist due to repeated oscillation between −1 and 1. (b) 0; 0; 0; 0; 0 (c) yes
7. $\frac{1}{4}, \frac{2}{5}, \frac{3}{6}, \frac{4}{7}, \frac{5}{8}$; converges, $\lim_{n\to+\infty} \frac{n}{n+3} = 1$
9. 3, 3, 3, 3, 3; converges, $\lim_{n\to+\infty} 3 = 3$
11. $\frac{\ln 1}{1}, \frac{\ln 2}{2}, \frac{\ln 3}{3}, \frac{\ln 4}{4}, \frac{\ln 5}{5}$; converges, $\lim_{n\to+\infty} \frac{\ln n}{n} = 0$
13. 0, 2, 0, 2, 0; diverges 15. $-1, \frac{16}{9}, -\frac{54}{28}, \frac{128}{65}, -\frac{250}{126}$; diverges
17. $0, \frac{1}{2}, \frac{10}{18}, \frac{18}{32}, \frac{28}{50}$, converges; $\lim_{n\to+\infty} \frac{1}{2}\left(1-\frac{1}{n}\right)\left(1+\frac{2}{n}\right) = \frac{1}{2}$
19. $e^{-1}, 4e^{-2}, 9e^{-3}, 16e^{-4}, 25e^{-5}$; converges, $\lim_{n\to+\infty} n^2e^{-n} = 0$
21. $2, \left(\frac{5}{3}\right)^2, \left(\frac{6}{4}\right)^3, \left(\frac{7}{5}\right)^4, \left(\frac{8}{6}\right)^5$; converges, $\lim_{n\to+\infty} \left[\frac{n+3}{n+1}\right]^n = e^2$
23. $\left\{\frac{2n-1}{2n}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} \frac{2n-1}{2n} = 1$
25. $\left\{(-1)^{n+1}\frac{1}{3^n}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} (-1)^{n+1}\frac{1}{3^n} = 0$
27. $\left\{(-1)^{n+1}\left(\frac{1}{n}-\frac{1}{n+1}\right)\right\}_{n=1}^{+\infty}$;
converges, $\lim_{n\to+\infty} (-1)^{n+1}\left(\frac{1}{n}-\frac{1}{n+1}\right) = 0$
29. $\{\sqrt{n+1}-\sqrt{n+2}\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} (\sqrt{n+1}-\sqrt{n+2}) = 0$

Responses to True–False questions may be abridged to save space.

31. False; for example, $\{(-1)^{n+1}\}$ diverges with terms that oscillate between 1 and −1.
33. True; a sequence is a function whose domain is a set of integers.
35. The limit is 0.
37. for example, $\{(-1)^n\}_{n=1}^{+\infty}$ and $\{\sin(\pi n/2)+1/n\}_{n=1}^{+\infty}$
39. (a) 1, 2, 1, 4, 1, 6 (b) $a_n = \begin{cases} n, & n \text{ odd} \\ 1/2^n, & n \text{ even} \end{cases}$
(c) $a_n = \begin{cases} 1/n, & n \text{ odd} \\ 1/(n+1); & n \text{ even} \end{cases}$
(d) (a) diverges; (b) diverges; (c) $\lim_{n\to+\infty} a_n = 0$
43. (a) $(0.5)^{2n}$ (c) $\lim_{n\to+\infty} a_n = 0$ (d) $-1 \le a_0 \le 1$
45. (a)



(b) $\lim_{n\to+\infty} (2^n+3^n)^{1/n} = 3$
49. (a) $N = 4$ (b) $N = 10$ (c) $N = 1000$

Exercise Set 9.2 (Page 613)

1. strictly decreasing 3. strictly increasing 5. strictly decreasing
7. strictly increasing 9. strictly decreasing 11. strictly increasing

Responses to True–False questions may be abridged to save space.

13. True; $a_{n+1} - a_n > 0$ for all n is equivalent to $a_1 < a_2 < a_3 < \cdots < a_n < \cdots$.
15. False; for example, $\{(-1)^{n+1}\} = \{1, -1, 1, -1, \ldots\}$ is bounded but diverges.
17. strictly increasing 19. strictly increasing
21. eventually strictly increasing 23. eventually strictly increasing
25. Yes; the limit lies in the interval [1, 2].
27. (a) $\sqrt{2}, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}$ (e) $L = 2$

Exercise Set 9.3 (Page 621)

1. (a) $2, \frac{12}{5}, \frac{62}{25}, \frac{312}{125}; \frac{5}{2}\left(1-\left(\frac{1}{5}\right)^n\right)$; $\lim_{n\to+\infty} s_n = \frac{5}{2}$ (converges)
(b) $\frac{1}{4}, \frac{3}{4}, \frac{7}{4}, \frac{15}{4}; -\frac{1}{4}(1-2^n)$; $\lim_{n\to+\infty} s_n = +\infty$ (diverges)
(c) $\frac{1}{6}, \frac{1}{4}, \frac{3}{10}, \frac{1}{3}; \frac{1}{2} - \frac{1}{n+2}$; $\lim_{n\to+\infty} s_n = \frac{1}{2}$ (converges)
3. $\frac{4}{7}$ 5. $\frac{63}{10}$ 7. $\frac{1}{4}$ 9. $\frac{1}{6}$ 11. diverges 13. $\frac{448}{3}$
15. (a) Exercise 5 (b) Exercise 3 (c) Exercise 7 (d) Exercise 9

Responses to True–False questions may be abridged to save space.

17. False; an infinite series converges if its sequence of *partial sums* converges.
19. True; the sequence of partial sums $\{s_n\}$ for the harmonic series satisfies $s_{2^n} > \frac{n+1}{2}$, so this series diverges.
21. 1 23. $\frac{559}{99}$ 27. 50 m
29. (a) $S_n = -\ln(n+1)$; $\lim_{n\to+\infty} S_n = -\infty$ (diverges)
(b) $S_n = \sum_{k=2}^{n+1}\left[\ln\frac{k-1}{k} - \ln\frac{k}{k+1}\right]$, $\lim_{n\to+\infty} S_n = -\ln 2$
31. (a) converges for $|x| < 1$; $S = \frac{x}{1+x^2}$
(b) converges for $|x| > 2$; $S = \frac{1}{x^2-2x}$
(c) converges for $x > 0$; $S = \frac{1}{e^x-1}$
33. $a_n = \frac{1}{2^{n-1}}a_1 + \frac{1}{2^{n-1}} + \frac{1}{2^{n-2}} + \cdots + \frac{1}{2}$, $\lim_{n\to+\infty} a_n = 1$

Exercise Set 9.4 (Page 629)

1. (a) $\frac{4}{3}$ (b) $-\frac{3}{4}$
3. (a) $p = 4$, converges (b) $p = \frac{1}{2}$, diverges
(c) $p = 1$, diverges (d) $p = \frac{2}{3}$, diverges
5. (a) diverges (b) diverges (c) diverges (d) no information
7. (a) diverges (b) converges
9. diverges 11. diverges 13. diverges 15. diverges 17. diverges
19. converges 21. diverges 23. converges 25. converges for $p > 1$
29. (a) diverges (b) diverges

Responses to True–False questions may be abridged to save space.

31. False; for example, $\sum_{k=0}^{\infty} 2^{-k}$ converges to 2, but $\sum \frac{1}{2^{-k}} = \sum 2^k$ diverges.
33. True; see Theorem 9.4.4.
35. (a) $(\pi^2/2) - (\pi^4/45)$ (b) $(\pi^2/6) - (5/4)$ (c) $\pi^4/90$
37. (d) $\frac{1}{11} < \frac{1}{6}\pi^2 - s_{10} < \frac{1}{10}$
39. (a) $\int_n^{+\infty} \frac{1}{x^4}\,dx = \frac{1}{3n^3}$; apply Exercise 36(b) (b) $n = 6$
(c) $\frac{\pi^4}{90} \approx 1.08238$
41. converges

Exercise Set 9.5 (Page 636)

1. (a) converges (b) diverges 5. converges 7. converges
9. diverges 11. converges 13. inconclusive 15. diverges
17. diverges 19. converges

Responses to True–False questions may be abridged to save space.

21. False; the limit comparison test uses a limit of the quotient of corresponding terms taken from two different sequences.
23. True; use the limit comparison test with the convergent series $\sum(1/k^2)$.
25. converges 27. converges 29. converges 31. diverges
33. converges 35. converges 37. diverges 39. converges
41. diverges 43. converges 45. converges 47. converges

49. $u_k = \dfrac{k!}{1\cdot 3\cdot 5\cdots(2k-1)}$; $\rho = \lim\limits_{k\to+\infty}\dfrac{k+1}{2k+1} = \dfrac{1}{2}$; converges
51. **(a)** converges **(b)** diverges **53.** **(a)** converges

Exercise Set 9.6 (Page 646)

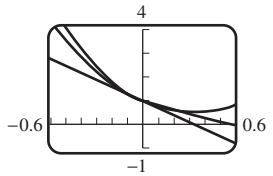
3. diverges **5.** converges **7.** converges absolutely **9.** diverges
11. converges absolutely **13.** conditionally convergent **15.** divergent
17. conditionally convergent **19.** conditionally convergent
21. divergent **23.** conditionally convergent **25.** converges absolutely
27. converges absolutely

Responses to True–False questions may be abridged to save space.

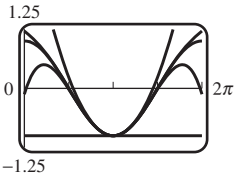
29. True; if a series converges but diverges absolutely, then it converges conditionally.
31. False; an alternating series has terms that alternate between positive and negative.
33. $|\text{error}| < 0.143$ **35.** $|\text{error}| < 0.1$ **37.** $n = 9999$
39. $n = 39{,}999$ **41.** $|\text{error}| < 0.00074$; $s_{10} \approx 0.4995$; $S = 0.5$
43. 0.84 **45.** 0.41 **47.** **(c)** $n = 50$
49. **(a)** If $a_k = \dfrac{(-1)^k}{\sqrt{k}}$, then $\sum a_k$ converges and $\sum a_k^2$ diverges.
If $a_k = \dfrac{(-1)^k}{k}$, then $\sum a_k$ converges and $\sum a_k^2$ also converges.
(b) If $a_k = \dfrac{1}{k}$, then $\sum a_k^2$ converges and $\sum a_k$ diverges. If $a_k = \dfrac{1}{k^2}$, then $\sum a_k^2$ converges and $\sum a_k$ also converges.

Exercise Set 9.7 (Page 657)

1. **(a)** $1 - x + \frac{1}{2}x^2,\ 1 - x$ **(b)** $1 - \frac{1}{2}x^2,\ 1$
3. **(a)** $1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2$ **(b)** 1.04875 **5.** 1.66434809
7. $p_0(x) = 1,\ p_1(x) = 1 - x,\ p_2(x) = 1 - x + \frac{1}{2}x^2,$
$p_3(x) = 1 - x + \dfrac{1}{2}x^2 - \dfrac{1}{3!}x^3,$
$p_4(x) = 1 - x + \dfrac{1}{2}x^2 - \dfrac{1}{3!}x^3 + \dfrac{1}{4!}x^4;\ \sum\limits_{k=0}^{n}\dfrac{(-1)^k}{k!}x^k$
9. $p_0(x) = 1,\ p_1(x) = 1,\ p_2(x) = 1 - \dfrac{\pi^2}{2!}x^2,\ p_3(x) = 1 - \dfrac{\pi^2}{2!}x^2,$
$p_4(x) = 1 - \dfrac{\pi^2}{2!}x^2 + \dfrac{\pi^4}{4!}x^4;\ \sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{(-1)^k\pi^{2k}}{(2k)!}x^{2k}$ (See Exercise 70 of Section 0.2.)
11. $p_0(x) = 0,\ p_1(x) = x,\ p_2(x) = x - \frac{1}{2}x^2,\ p_3(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3,$
$p_4(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4;\ \sum\limits_{k=1}^{n}\dfrac{(-1)^{k+1}}{k}x^k$
13. $p_0(x) = 1,\ p_1(x) = 1,\ p_2(x) = 1 + \dfrac{x^2}{2},$
$p_3(x) = 1 + \dfrac{x^2}{2},\ p_4(x) = 1 + \dfrac{x^2}{2} + \dfrac{x^4}{4!};\ \sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{1}{(2k)!}x^{2k}$ (See Exercise 70 of Section 0.2.)
15. $p_0(x) = 0,\ p_1(x) = 0,\ p_2(x) = x^2,\ p_3(x) = x^2,$
$p_4(x) = x^2 - \frac{1}{6}x^4;\ \sum\limits_{k=0}^{\lfloor n/2\rfloor-1}\dfrac{(-1)^k}{(2k+1)!}x^{2k+2}$ (See Exercise 70 of Section 0.2.)
17. $p_0(x) = e,\ p_1(x) = e + e(x-1),$
$p_2(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2,$
$p_3(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2 + \dfrac{e}{3!}(x-1)^3,$
$p_4(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2 + \dfrac{e}{3!}(x-1)^3 + \dfrac{e}{4!}(x-1)^4;$
$\sum\limits_{k=0}^{n}\dfrac{e}{k!}(x-1)^k$
19. $p_0(x) = -1,\ p_1(x) = -1 - (x+1),$
$p_2(x) = -1 - (x+1) - (x+1)^2,$
$p_3(x) = -1 - (x+1) - (x+1)^2 - (x+1)^3,$
$p_4(x) = -1 - (x+1) - (x+1)^2 - (x+1)^3 - (x+1)^4;$
$\sum\limits_{k=0}^{n}(-1)(x+1)^k$
21. $p_0(x) = p_1(x) = 1,\ p_2(x) = p_3(x) = 1 - \dfrac{\pi^2}{2}\left(x - \dfrac{1}{2}\right)^2,$
$p_4(x) = 1 - \dfrac{\pi^2}{2}\left(x - \dfrac{1}{2}\right)^2 + \dfrac{\pi^4}{4!}\left(x - \dfrac{1}{2}\right)^4;$
$\sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{(-1)^k\pi^{2k}}{(2k)!}\left(x - \dfrac{1}{2}\right)^{2k}$ (See Exercise 70 of Section 0.2.)
23. $p_0(x) = 0,\ p_1(x) = (x-1),\ p_2(x) = (x-1) - \frac{1}{2}(x-1)^2,$
$p_3(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3,$
$p_4(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4;$
$\sum\limits_{k=1}^{n}\dfrac{(-1)^{k-1}}{k}(x-1)^k$
25. **(a)** $p_3(x) = 1 + 2x - x^2 + x^3$
(b) $p_3(x) = 1 + 2(x-1) - (x-1)^2 + (x-1)^3$
27. $p_0(x) = 1,\ p_1(x) = 1 - 3x,$
$p_2(x) = 1 - 3x + \frac{9}{2}x^2,$
$p_3(x) = 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3$

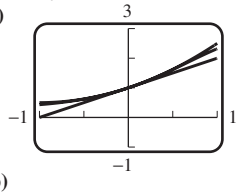


29. $p_0(x) = -1,\ p_2(x) = -1 + \frac{1}{2}(x-\pi)^2,$
$p_4(x) = -1 + \frac{1}{2}(x-\pi)^2 - \frac{1}{24}(x-\pi)^4,$
$p_6(x) = -1 + \frac{1}{2}(x-\pi)^2 - \frac{1}{24}(x-\pi)^4 + \frac{1}{720}(x-\pi)^6$



Responses to True–False questions may be abridged to save space.

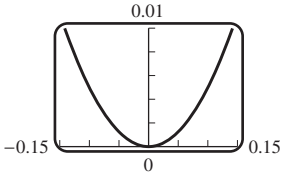
31. True; $y = f(x_0) + f'(x_0)(x - x_0)$ is the first-degree Taylor polynomial for f about $x = x_0$.
33. False; $p_6^{(4)}(x_0) = f^{(4)}(x_0)$ **35.** 1.64870 **37.** IV
39. **(a)**



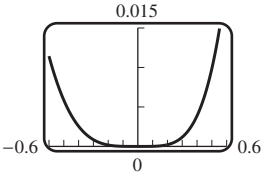
(b)

x	−1.000	−0.750	−0.500	−0.250	0.000	0.250	0.500	0.750	1.000
$f(x)$	0.431	0.506	0.619	0.781	1.000	1.281	1.615	1.977	2.320
$p_1(x)$	0.000	0.250	0.500	0.750	1.000	1.250	1.500	1.750	2.000
$p_2(x)$	0.500	0.531	0.625	0.781	1.000	1.281	1.625	2.031	2.500

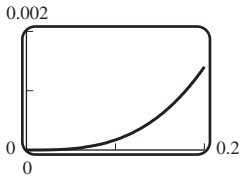
(c) $|e^{\sin x} - (1+x)| < 0.01$ for $-0.14 < x < 0.14$



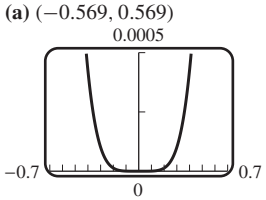
(d) $\left|e^{\sin x} - \left(1 + x + \dfrac{x^2}{2}\right)\right| < 0.01$ for $-0.50 < x < 0.50$



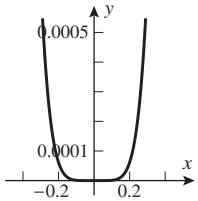
41. **(a)** $[0, 0.137]$ **(b)**



43. **(a)** $(-0.569, 0.569)$



45. $(-0.311, 0.311)$



Exercise Set 9.8 (Page 667)

1. $\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k$ **3.** $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k)!} x^{2k}$ **5.** $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k$

7. $\sum_{k=0}^{\infty} \frac{1}{(2k)!} x^{2k}$ **9.** $\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+2}$ **11.** $\sum_{k=0}^{\infty} \frac{1}{ek!} (x+1)^k$

13. $\sum_{k=0}^{\infty} (-1)(x+1)^k$ **15.** $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k)!} \left(x - \frac{1}{2}\right)^{2k}$

17. $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x-1)^k$ **19.** $-1 < x < 1$; $\frac{1}{1+x}$

21. $1 < x < 3$; $\frac{1}{3-x}$ **23.** **(a)** $-2 < x < 2$ **(b)** $f(0) = 1$; $f(1) = \frac{2}{3}$

Responses to True–False questions may be abridged to save space.

25. True; see Theorem 9.8.2(c).

27. True; the polynomial *is* the Maclaurin series and converges for all x.

29. $R = 1$; $[-1, 1)$ **31.** $R = +\infty$; $(-\infty, +\infty)$ **33.** $R = \frac{1}{6}$; $\left[-\frac{1}{6}, \frac{1}{6}\right]$

35. $R = 1$; $[-1, 1]$ **37.** $R = 1$; $(-1, 1]$ **39.** $R = +\infty$; $(-\infty, +\infty)$

41. $R = 1$; $[-1, 1]$ **43.** $R = 1$; $(-2, 0]$ **45.** $R = \frac{4}{3}$; $\left(-\frac{19}{3}, -\frac{11}{3}\right)$

47. $R = +\infty$; $(-\infty, +\infty)$ **49.** $(-\infty, +\infty)$ **55.** radius $= R$

61. **(a)** $n = 5$; $s_5 \approx 1.1026$ **(b)** $\zeta(3.7) \approx 1.10629$

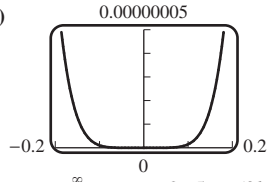
Exercise Set 9.9 (Page 676)

3. 0.069756 **5.** 0.99500 **7.** 0.99619 **9.** 0.52109

11. **(a)** $\sum_{k=1}^{\infty} 2\frac{(1/11)^{2k-1}}{2k-1}$ **(b)** 0.1823

13. **(a)** 0.4635; 0.3218 **(b)** 3.1412 **(c)** no

15. **(a)** error $\le 9 \times 10^{-8}$ **(b)**



17. **(a)** $\sum_{k=0}^{\infty} (-1)^k x^k$ **(b)** $1 + \frac{x}{3} + \sum_{k=2}^{\infty} (-1)^{k-1} \frac{2 \cdot 5 \cdots (3k-4)}{3^k k!} x^k$

(c) $\sum_{k=0}^{\infty} (-1)^k \frac{(k+2)(k+1)}{2} x^k$ **23.** 23.406%

Exercise Set 9.10 (Page 686)

1. **(a)** $1 - x + x^2 - \cdots + (-1)^k x^k + \cdots$; $R = 1$

(b) $1 + x^3 + x^6 + \cdots + x^{3k} + \cdots$; $R = 1$

(c) $1 + 2x + 4x^2 + \cdots + 2^k x^k + \cdots$; $R = \frac{1}{2}$

(d) $\frac{1}{3} + \frac{1}{3^2}x + \frac{1}{3^3}x^2 + \cdots + \frac{1}{3^{k+1}}x^k + \cdots$; $R = 3$

3. **(a)** $(2+x)^{-1/2} = \frac{1}{2^{1/2}} - \frac{1}{2^{5/2}}x + \frac{1 \cdot 3}{2^{9/2} \cdot 2!}x^2 - \frac{1 \cdot 3 \cdot 5}{2^{13/2} \cdot 3!}x^3 + \cdots$

(b) $(1-x^2)^{-2} = 1 + 2x^2 + 3x^4 + 4x^6 + \cdots$

5. **(a)** $3x - \frac{3^3}{3!}x^3 + \frac{3^5}{5!}x^5 - \frac{3^7}{7!}x^7 + \cdots$; $R = +\infty$

(b) $1 - 2x + 2x^2 - \frac{4}{3}x^3 + \cdots$; $R = +\infty$

(c) $1 + x^2 + \frac{1}{2!}x^4 + \frac{1}{3!}x^6 + \cdots$; $R = +\infty$

(d) $x^2 - \frac{\pi^2}{2}x^4 + \frac{\pi^4}{4!}x^6 - \frac{\pi^6}{6!}x^8 + \cdots$; $R = +\infty$

7. **(a)** $x^2 - 4x^3 + 16x^4 - 64x^5 + \cdots$; $R = \frac{1}{4}$

(b) $2x^2 + \frac{2^3}{3!}x^4 + \frac{2^5}{5!}x^6 + \frac{2^7}{7!}x^8 + \cdots$; $R = +\infty$

(c) $x - \frac{3}{2}x^3 + \frac{3}{8}x^5 + \frac{1}{16}x^7 + \cdots$; $R = 1$

9. **(a)** $x^2 - \frac{2^3}{4!}x^4 + \frac{2^5}{6!}x^6 - \frac{2^7}{8!}x^8 + \cdots$

(b) $12x^3 - 6x^6 + 4x^9 - 3x^{12} + \cdots$

11. **(a)** $1 - (x-1) + (x-1)^2 - \cdots + (-1)^k(x-1)^k + \cdots$ **(b)** $(0,2)$

13. **(a)** $x + x^2 + \frac{x^3}{3} - \frac{x^5}{30} + \cdots$ **(b)** $x - \frac{x^3}{24} + \frac{x^4}{24} - \frac{71}{1920}x^5 + \cdots$

15. **(a)** $1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \cdots$ **(b)** $x - 2x^2 + \frac{11}{6}x^3 - x^4 + \cdots$

19. $2 - 4x + 2x^2 - 4x^3 + 2x^4 + \cdots$

25. $[-1, 1]$; $[-1, 1)$ **27.** **(a)** $\sum_{k=0}^{\infty} x^{2k+1}$ **(b)** $f^{(5)}(0) = 5!$, $f^{(6)}(0) = 0$

(c) $f^{(n)}(0) = n!c_n = \begin{cases} n! & \text{if } n \text{ odd} \\ 0 & \text{if } n \text{ even} \end{cases}$

29. **(a)** 1 **(b)** $-\frac{1}{3}$ **31.** 0.9046 **33.** 0.200

35. **(a)** $\sum_{k=0}^{\infty} \frac{x^{5k}}{k!}$; $R = +\infty$ **37.** **(a)** $5/16$ **(b)** $\ln(4/3)$

39. **(a)** $x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7 + \cdots$

(b) $x + \sum_{k=1}^{\infty} (-1)^k \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2^k k!(2k+1)} x^{2k+1}$ **(c)** $R = 1$

41. **(a)** $y(t) = y_0 \sum_{k=0}^{\infty} \frac{(-1)^k (0.000121)^k t^k}{k!}$ **(c)** $0.9998790073y_0$

43. $2\pi\sqrt{\frac{L}{g}}\left(1 + \frac{k^2}{4} + \frac{9k^4}{64}\right)$

Chapter 9 Review Exercises (Page 689)

9. **(a)** true **(b)** sometimes false **(c)** sometimes false
(d) true **(e)** sometimes false **(f)** sometimes false
(g) false **(h)** sometimes false **(i)** true
(j) true **(k)** sometimes false **(l)** sometimes false

11. **(a)** $\left\{\frac{n+2}{(n+1)^2 - n^2}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} \frac{n+2}{(n+1)^2 - n^2} = \frac{1}{2}$

(b) $\left\{(-1)^{n+1}\frac{n}{2n+1}\right\}_{n=1}^{+\infty}$; diverges

15. **(a)** converges **(b)** converges **17.** **(a)** converges **(b)** diverges

19. **(a)** diverges **(b)** converges **21.** $\frac{1}{4 \cdot 5^{99}}$

23. **(a)** 2 **(b)** diverges **(c)** $11/18$ **(d)** $\pi/4$ **25.** $p > 1$

29. **(a)** $p_0(x) = 1$, $p_1(x) = 1 - 6x$, $p_2(x) = 1 - 6x + 5x^2$,
$p_3(x) = 1 - 6x + 5x^2 + 3x^3$, $p_4(x) = 1 - 6x + 5x^2 + 4x^3$

33. **(a)** $e^3 - 1$ **(b)** 0 **(c)** $\cos e$ **(d)** $\frac{1}{3}$

37. **(a)** $x - \frac{2}{3}x^3 + \frac{2}{15}x^5 - \frac{4}{315}x^7$ **(b)** $x - \frac{2}{3}x^3 + \frac{2}{15}x^5 - \frac{4}{315}x^7$

Chapter 9 Making Connections (Page 691)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** $\dfrac{a\sin\theta}{1-\cos\theta}$ **(b)** $a\csc\theta$ **(c)** $a\cot\theta$

2. **(a)** $A = 1,\ B = -2$ **(b)** $s_n = 2 - \dfrac{2^{n+1}}{3^{n+1} - 2^{n+1}}$; 2

4. **(a)** $124.58 < d < 124.77$ **(b)** $1243 < s < 1424$

6. **(b)** $v(t) \approx v_0 - \left(\dfrac{cv_0}{m} + g\right)t + \dfrac{c^2}{2m^2}\left(v_0 + \dfrac{mg}{c}\right)t^2$

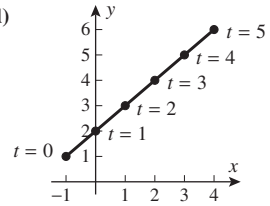
Exercise Set 10.1 (Page 700)

1. **(a)** $y = x + 2\ (-1 \le x \le 4)$

(c)

t	0	1	2	3	4	5
x	−1	0	1	2	3	4
y	1	2	3	4	5	6

(d)



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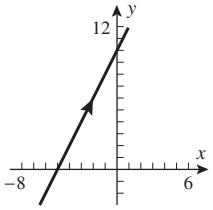

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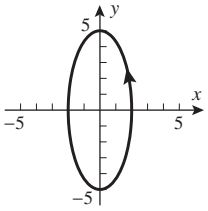

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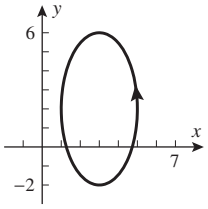

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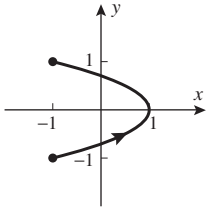

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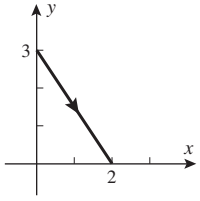
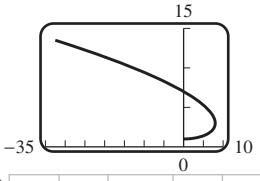


13. $x = 5\cos t,\ y = -5\sin t\ (0 \le t \le 2\pi)$ **15.** $x = 2,\ y = t$

17. $x = t^2,\ y = t\ (-1 \le t \le 1)$

19. **(a)**

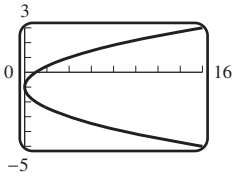


(b)

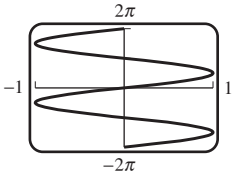
t	0	1	2	3	4	5
x	0	5.5	8	4.5	−8	−32.5
y	1	1.5	3	5.5	9	13.5

(c) $t = 0, 2\sqrt{3}$ **(d)** $0 < t < 2\sqrt{2}$ **(e)** $t = 2$

21. **(a)**



(b)

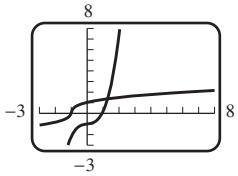


23. **(a)** II **(b)** I **(c)** IV **(d)** III **(e)** VI **(f)** V **25.** **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$

27. **(b)** from (x_0, y_0) to (x_1, y_1)

(c) $x = 3 - 2(t-1),\ y = -1 + 5(t-1)$

29.



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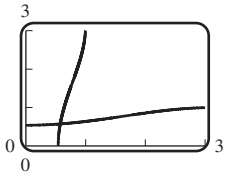
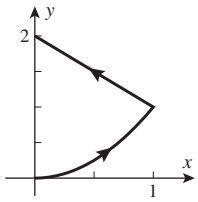


Responses to True–False questions may be abridged to save space.

33. True; $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{12t^3 - 6t^2}{x'(t)}$

35. False; $x = \sin t,\ y = \cos^2 t$ describe only the portion of the parabola $y = 1 - x^2$ with $-1 \le x \le 1$. **37.**



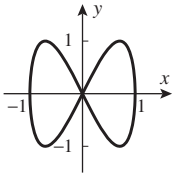
39. **(a)** $x = 4\cos t,\ y = 3\sin t$ **(b)** $x = -1 + 4\cos t,\ y = 2 + 3\sin t$

41. $-4, 4$ **43.** both are positive **45.** $4, 4$ **47.** $\sqrt{2}, -1$

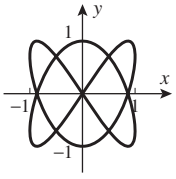
49. $\dfrac{\sqrt{2}(2+\sqrt{2})}{2},\ \dfrac{2}{3-2\sqrt{2}}$ **51.** $y = -e^{-2}x + 2e^{-1}$

53. **(a)** $0, \pi, 2\pi$ **(b)** $\pi/2, 3\pi/2$

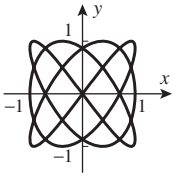
55. **(a)** **(b)** $y = -2x,\ y = 2x$



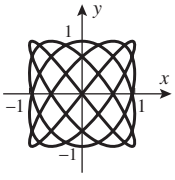
$a = 1,\ b = 2$



$a = 2,\ b = 3$

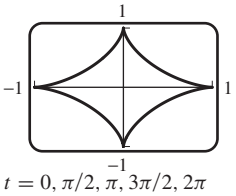


$a = 3,\ b = 4$



$a = 4,\ b = 5$

57. $y = 3x - 27,\ y = -3x + 27$ **59.**



$t = 0, \pi/2, \pi, 3\pi/2, 2\pi$

61. **(a)** $\dfrac{dy}{dx} = \dfrac{3\sin t}{1 - 3\cos t}$ **(b)** $\theta \approx -0.4345$

63. **(a)** ellipses with fixed center, varying axes of symmetry

(b) ellipses with varying center, fixed shape, size, and orientation

(c) circles of radius 1 with centers on line $y = x - 1$

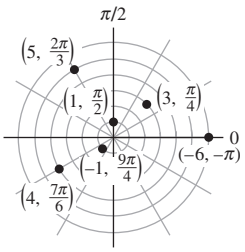
65. $\frac{1}{3}(5\sqrt{5} - 8)$ **67.** 3π **69.** $\frac{\sqrt{10}}{2}(e^2 - e^{-2})$

73. **(b)** $x = \cos t + \cos 2t,\ y = \sin t + \sin 2t$ **(c)** yes

75. $S = \dfrac{205\pi\sqrt{41} - 625}{6}$ **77.** $S = \sqrt{2}\pi$

Exercise Set 10.2 (Page 716)

1.



3. **(a)** $(2\sqrt{3}, 2)$ **(b)** $(-4, 4\sqrt{3})$ **(c)** $(3\sqrt{3}, 3)$ **(d)** $(0, 0)$ **(e)** $(-7\sqrt{3}/2, 7/2)$ **(f)** $(-5, 0)$

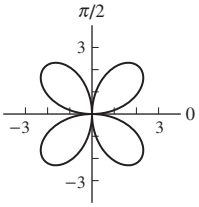
5. **(a)** $(7, \pi), (7, -\pi)$ **(b)** $(4, 11\pi/6), (4, -\pi/6)$ **(c)** $(2, 3\pi/2), (2, -\pi/2)$ **(d)** $(8\sqrt{2}, 5\pi/4), (8\sqrt{2}, -3\pi/4)$ **(e)** $(6, 2\pi/3), (6, -4\pi/3)$ **(f)** $(2\sqrt{2}, \pi/4), (2\sqrt{2}, -7\pi/4)$

7. **(a)** $(5, 0.92730)$ **(b)** $(10, 2.214297)$ **(c)** $(1.27155, 2.47582)$

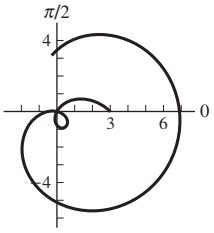
9. **(a)** circle **(b)** line **(c)** circle **(d)** line

11. **(a)** $r = 3\sec\theta$ **(b)** $r = \sqrt{14}$ **(c)** $r = -6\sin\theta$ **(d)** $r^2\cos\theta\sin\theta = 4/9$

13.



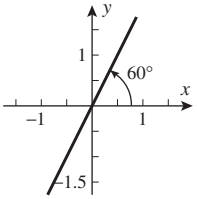
15.



17. **(a)** $r = 5$ **(b)** $r = 6\cos\theta$ **(c)** $r = 1 - \cos\theta$

19. **(a)** $r = 3\sin 2\theta$ **(b)** $r = 3 + 2\sin\theta$ **(c)** $r^2 = 9\cos 2\theta$

21.



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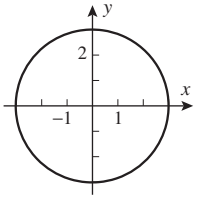
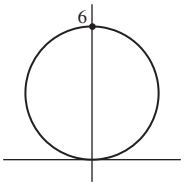
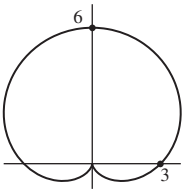


25.



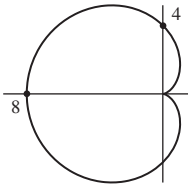
Circle

27.



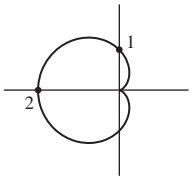
Cardioid

29.



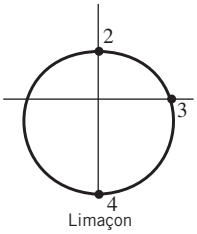
Cardioid

31.



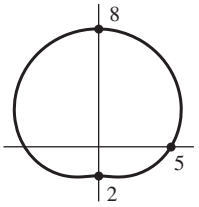
Cardioid

33.



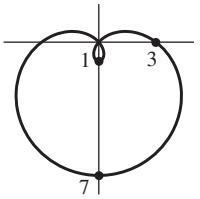
Limaçon

35.



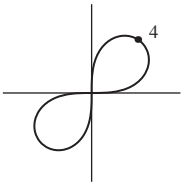
Limaçon

37.



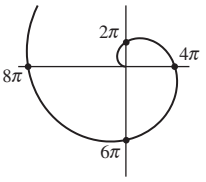
Limaçon

39.



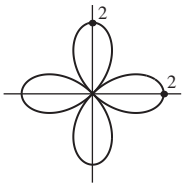
Lemniscate

41.



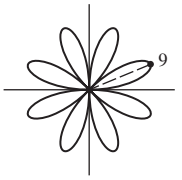
Spiral

43.



Four-petal rose

45.



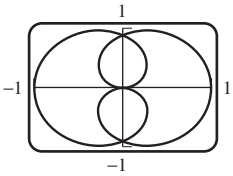
Eight-petal rose

Responses to True–False questions may be abridged to save space.

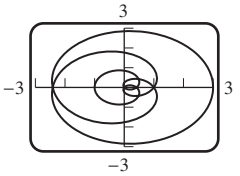
47. True; $\left(-1, \frac{\pi}{3}\right)$ describes the same point as $\left(1, \frac{\pi}{3} + \pi\right)$, which describes the same point as $\left(1, \frac{\pi}{3} + \pi - 2\pi\right) = \left(1, -\frac{2\pi}{3}\right)$.

49. False; $-1 < \sin 2\theta < 0$ for $\pi/2 < \theta < \pi$, so this portion of the graph is in the fourth quadrant.

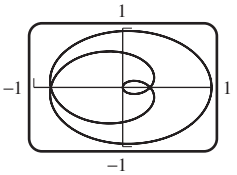
51. $0 \le \theta \le 4\pi$



53. $0 \le \theta \le 8\pi$

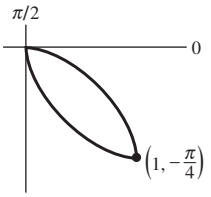


55. $0 \le \theta \le 5\pi$

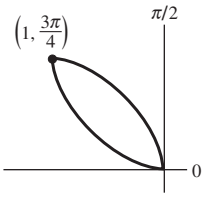


57. **(a)** $-4\pi < \theta < 4\pi$

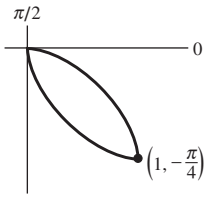
61. **(a)**



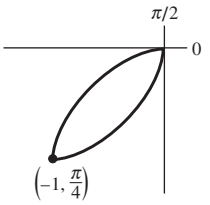
(b)



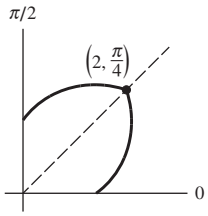
(c)



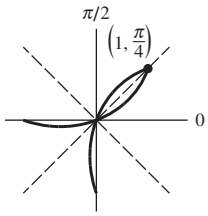
(d)



63. **(a)**



(b)



67. **(a)** $1 + \frac{\sqrt{2}}{2}(\sin\theta - \cos\theta)$ **(b)** $1 - \cos\theta$ **(c)** $1 - \sin\theta$ **(d)** $1 + \frac{\sqrt{2}}{2}(\cos\theta - \sin\theta)$

69. $(3/2, \pi/3)$ **73.** $\sqrt{2}$

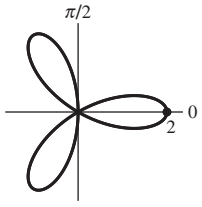
Exercise Set 10.3 (Page 726)

1. $-\sqrt{3}$ **3.** $\dfrac{\tan 3 - 3}{3\tan 3 + 1}$ **5.** $-\sqrt{3}$ **7.** $1, 0, -1$

9. horizontal: $(3a/2, \pi/3)$, $(0, \pi)$, $(3a/2, 5\pi/3)$;
vertical: $(2a, 0)$, $(a/2, 2\pi/3)$, $(a/2, 4\pi/3)$

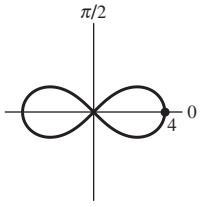
11. $(0, 0)$, $(\sqrt{2}/4, \pi/4)$, $(\sqrt{2}/4, 3\pi/4)$

13.



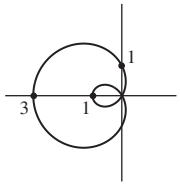
$\theta = \pi/2, \pm\pi/6$

15.



$\theta = \pm\pi/4$

17.



$\theta = \pm\pi/3$

19. $L = 2\pi a$ **21.** $L = 8a$

23. **(b)** ≈ 2.42

(c)

n	2	3	4	5	6	7
L	2.42211	2.22748	2.14461	2.10100	2.07501	2.05816

n	8	9	10	11	12	13	14
L	2.04656	2.03821	2.03199	2.02721	2.02346	2.02046	2.01802

n	15	16	17	18	19	20
L	2.01600	2.01431	2.01288	2.01167	2.01062	2.00971

25. **(a)** $\int_{\pi/2}^{\pi} \frac{1}{2}(1-\cos\theta)^2 d\theta$ **(b)** $\int_0^{\pi/2} 2\cos^2\theta\, d\theta$
(c) $\int_0^{\pi/2} \frac{1}{2}\sin^2 2\theta\, d\theta$ **(d)** $\int_0^{2\pi} \frac{1}{2}\theta^2\, d\theta$
(e) $\int_{-\pi/2}^{\pi/2} \frac{1}{2}(1-\sin\theta)^2\, d\theta$ **(f)** $\int_0^{\pi/4} \cos^2 2\theta\, d\theta$

27. **(a)** $\frac{9}{4}\pi a^2$ **(b)** $\frac{9}{4}\pi a^2$ **29.** 6π **31.** 9π **33.** $\pi - 3\sqrt{3}/2$
35. $\pi/2 - \frac{1}{4}$ **37.** $10\pi/3 - 4\sqrt{3}$ **39.** π **41.** $9\sqrt{3}/2 - 10\pi$
43. $(\pi + 3\sqrt{3})/4$ **45.** $\pi - 2$

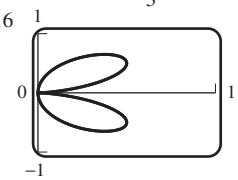
Responses to True–False questions may be abridged to save space.

47. True; apply Theorem 10.3.1: $\cos\frac{\theta}{2}\Big|_{\theta=3\pi} = 0$ and $\frac{dr}{d\theta}\Big|_{\theta=3\pi} = \frac{1}{2} \neq 0$, so the line $\theta = 3\pi$ (the x-axis) is tangent to the curve at the origin.

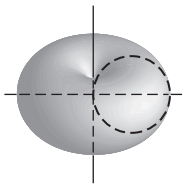
49. False; the area of the sector is $\frac{\theta}{2\pi}\cdot\pi r^2 = \frac{1}{2}\theta r^2$.

51. **(b)** a^2 **(c)** $2\sqrt{3} - \frac{2\pi}{3}$ **53.** $8\pi^3 a^2$

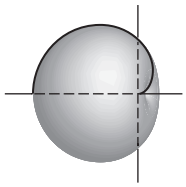
59. $\pi/16$



65. π^2



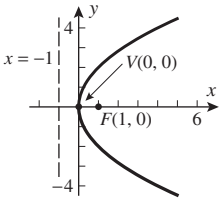
67. $32\pi/5$



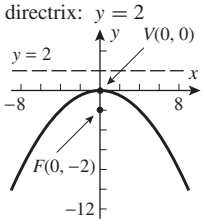
Exercise Set 10.4 (Page 744)

1. **(a)** $x = y^2$ **(b)** $-3y = x^2$ **(c)** $\frac{x^2}{9} + \frac{y^2}{4} = 1$ **(d)** $\frac{x^2}{4} + \frac{y^2}{9} = 1$
(e) $y^2 - x^2 = 1$ **(f)** $\frac{x^2}{4} - \frac{y^2}{4} = 1$

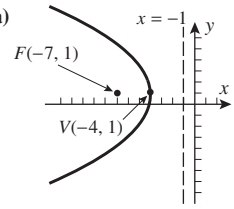
3. **(a)** focus: $(1, 0)$;
vertex: $(0, 0)$;
directrix: $x = -1$



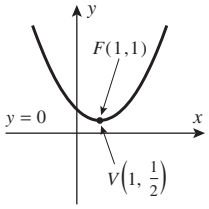
(b) focus: $(0, -2)$;
vertex: $(0, 0)$;
directrix: $y = 2$



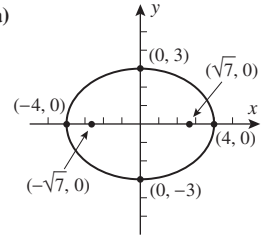
5. **(a)**



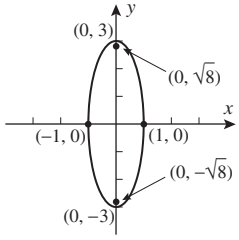
(b)



7. **(a)**

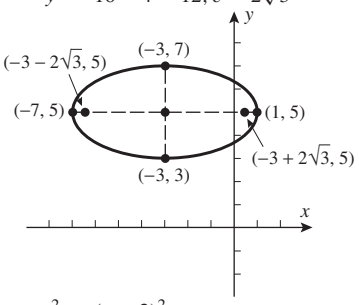


(b)



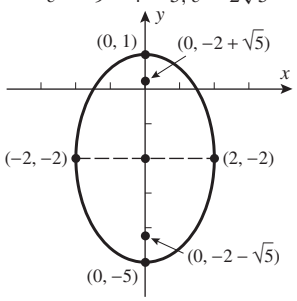
9. **(a)** $\dfrac{(x+3)^2}{16} + \dfrac{(y-5)^2}{4} = 1$

$y^2 = 16 - 4 = 12,\ c = 2\sqrt{3}$

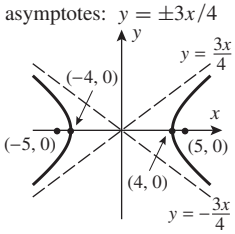


(b) $\dfrac{x^2}{4} + \dfrac{(y+2)^2}{9} = 1$

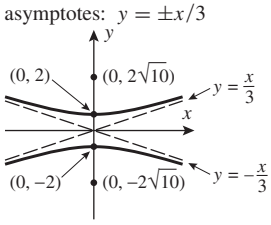
$c^2 = 9 - 4 = 5,\ c = 2\sqrt{5}$



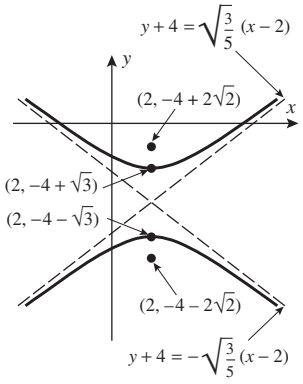
11. **(a)** vertices: $(\pm 4, 0)$;
foci: $(\pm 5, 0)$;
asymptotes: $y = \pm 3x/4$



(b) vertices: $(0, \pm 2)$;
foci: $(0, \pm 2\sqrt{10})$;
asymptotes: $y = \pm x/3$

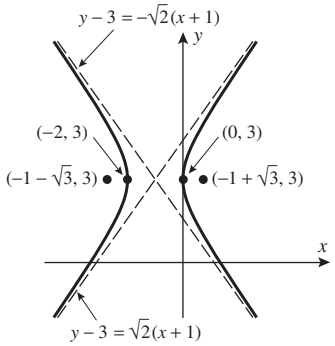


13. **(a)** $c^2 = 3 + 5 = 8,\ c = 2\sqrt{2}$



(b) $\dfrac{(x+1)^2}{1} - \dfrac{(y-3)^2}{2} = 1$

$c^2 = 1 + 2 = 3,\ c = \sqrt{3}$



15. **(a)** $y^2 = 20x$ **(b)** $x^2 = -2y$ **17.** $y^2 = x - 2$

19. **(a)** $\dfrac{x^2}{25} + \dfrac{y^2}{4} = 1$ **(b)** $\dfrac{x^2}{16} + \dfrac{y^2}{25} = 1$

21. **(a)** $\dfrac{x^2}{81/8} + \dfrac{y^2}{36} = 1$ **(b)** $\dfrac{(x+1)^2}{9} + \dfrac{(y-2)^2}{10} = 1$

23. **(a)** $\dfrac{x^2}{4} - \dfrac{y^2}{5} = 1$ **(b)** $\dfrac{y^2}{4} - \dfrac{x^2}{9} = 1$

25. **(a)** $\dfrac{y^2}{9} - \dfrac{x^2}{16} = 1,\ \dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$ **(b)** $\dfrac{x^2}{9/5} - \dfrac{y^2}{36/5} = 1$

Responses to True–False questions may be abridged to save space.

27. False; the distance from the parabola's focus to its directrix is $2p$; see Figure 10.4.6.

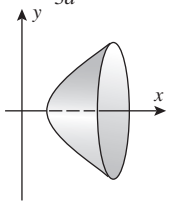
29. False; the description matches a parabola.

31. **(a)** 16 ft **(b)** $8\sqrt{3}$ ft **35.** $\frac{1}{16}$ ft

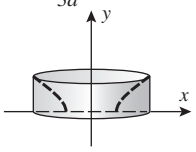
39. $\frac{1}{32}(x-5)^2 + \frac{1}{36}(y-3)^2 = 1$

43. $L = D\sqrt{1+p^2},\ T = \frac{1}{2}pD$ **45.** (64.612, 200)

47. **(a)** $V = \dfrac{\pi b^2}{3a^2}(b^2 - 2a^2)\sqrt{a^2+b^2} + \dfrac{2}{3}ab^2\pi$



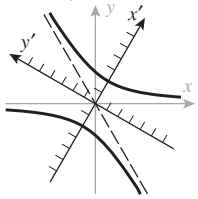
(b) $V = \dfrac{2b^4}{3a}\pi$



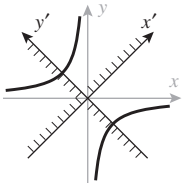
53. **(a)** $(x-2)^2 - 5(y+1)^2 = 8$, hyperbola
(b) $x^2 - 3(y+1)^2 = 0,\ x = \pm\sqrt{3}(y+1)$, two lines
(c) $4(x+2)^2 + 8(y+1)^2 = 8$, ellipse
(d) $3(x+2)^2 + (y+1)^2 = 0$, the point $(-2, -1)$ (degenerate case)
(e) $(x+4)^2 + 2y = 2$, parabola
(f) $5(x+4)^2 + 2y = -14$, parabola

Exercise Set 10.5 (Page 753)

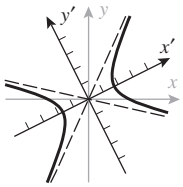
1. (a) $x' = -2 + \sqrt{3}$, $y' = 1 + 2\sqrt{3}$ (b) $3x'^2 - y'^2 = 12$ (c)



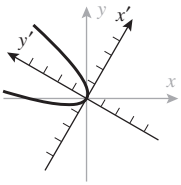
3. $y'^2 - x'^2 = 18$, hyperbola



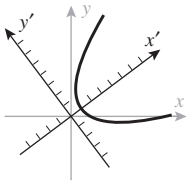
5. $\frac{1}{3}x'^2 - \frac{1}{2}y'^2 = 1$, hyperbola



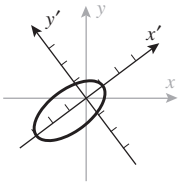
7. $y' = x'^2$, parabola



9. $y'^2 = 4(x' - 1)$, parabola



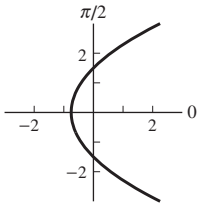
11. $\frac{1}{4}(x' + 1)^2 + y'^2 = 1$, ellipse

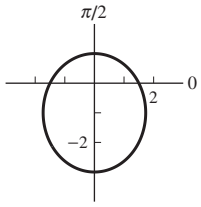


13. $x^2 + xy + y^2 = 5$
19. vertex: $(0, 0)$; focus: $(-1/\sqrt{2}, 1/\sqrt{2})$; directrix: $y = x - \sqrt{2}$
21. vertex: $(8/5, 6/5)$; focus: $(12/5, 9/5)$; directrix: $4x + 3y = 5$
23. foci: $\pm(4\sqrt{7}/5, 3\sqrt{7}/5)$; vertices: $\pm(16/5, 12/5)$; ends: $\pm(-9/5, 12/5)$
25. foci: $(1 - \sqrt{5}/2, -\sqrt{3} + \sqrt{15}/2)$, $(1 + \sqrt{5}/2, -\sqrt{3} - \sqrt{15}/2)$; vertices: $(-1/2, \sqrt{3}/2)$, $(5/2, -5\sqrt{3}/2)$; ends: $(1 + \sqrt{3}, 1 - \sqrt{3})$, $(1 - \sqrt{3}, -1 - \sqrt{3})$
27. foci: $\pm(\sqrt{15}, \sqrt{5})$; vertices: $\pm(2\sqrt{3}, 2)$; asymptotes: $y = \dfrac{5\sqrt{3} \pm 8}{11}x$
29. foci: $\left(-\dfrac{4}{\sqrt{5}} \pm 2\sqrt{\dfrac{13}{5}}, \dfrac{8}{\sqrt{5}} \pm \sqrt{\dfrac{13}{5}}\right)$; vertices: $(2/\sqrt{5}, 11/\sqrt{5})$, $(-2\sqrt{5}, \sqrt{5})$; asymptotes: $y = 7x/4 + 3\sqrt{5}$, $y = -x/8 + 3\sqrt{5}/2$

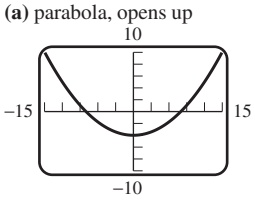
Exercise Set 10.6 (Page 761)

1. (a) $e = 1, d = \frac{3}{2}$ (b) $e = \frac{1}{2}, d = 3$

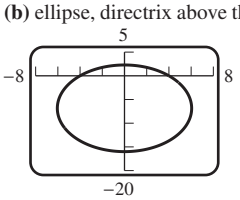




3. (a) parabola, opens up



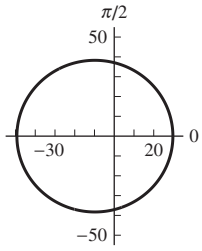
(b) ellipse, directrix above the pole



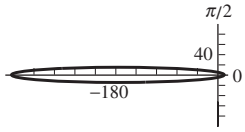
5. (a) $r = \dfrac{12}{4 + 3\cos\theta}$ (b) $r = \dfrac{1}{1 + \cos\theta}$ (c) $r = \dfrac{12}{1 + 2\sin\theta}$
7. (a) $r_0 = 4, r_1 = 12$; $\frac{x^2}{48} + \frac{(y+4)^2}{64} = 1$
(b) $r_0 = \frac{1}{3}, r_1 = 1$; $\frac{9}{4}\left(x - \frac{1}{3}\right)^2 + 3y^2 = 1$
9. (a) $r_0 = 1, r_1 = 3$; $(y - 2)^2 - \dfrac{x^2}{3} = 1$
(b) $r_0 = 1, r_1 = 5$; $\dfrac{(x + 3)^2}{4} - \dfrac{y^2}{5} = 1$
11. (a) $r = \dfrac{15}{2 + \cos\theta}$ (b) $r = \dfrac{64}{25 - 15\sin\theta}$
13. $r = \dfrac{5\sqrt{2} + 5}{1 + \sqrt{2}\cos\theta}$ or $r = \dfrac{5\sqrt{2} - 5}{1 + \sqrt{2}\cos\theta}$

Responses to True–False questions may be abridged to save space.

19. True; the eccentricity e of an ellipse satisfies $0 < e < 1$ (Theorem 10.6.1).
21. False; eccentricity correlates to the "flatness" of an ellipse, which is independent of the distance between its foci.
23. (a) $T \approx 248$ yr
(b) $r_0 \approx 4{,}449{,}675{,}000$ km, $r_1 \approx 7{,}400{,}325{,}000$ km
(c) $r \approx \dfrac{37.05}{1 + 0.249\cos\theta}$ AU (d)



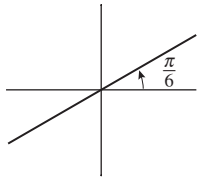
25. (a) $a \approx 178.26$ AU
(b) $r_0 \approx 0.8735$ AU, $r_1 \approx 355.64$ AU
(c) $r \approx \dfrac{1.74}{1 + 0.9951\cos\theta}$ AU (d)



27. 563 km, 4286 km

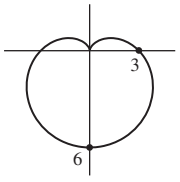
Chapter 10 Review Exercises (Page 763)

1. $x = \sqrt{5}\cos t$, $y = -\sqrt{5}\sin t$ $(0 \le t \le 3\pi/2)$
3. (a) $1/6, 1/6$
5. (a) $t = \pi/2 + n\pi$ for $n = 0, \pm1, \ldots$ (b) $t = n\pi$ for $n = 0, \pm1, \ldots$
7. (a) $(3\sqrt{2}, 3\sqrt{2})$ (b) $(7/\sqrt{2}, -7/\sqrt{2})$ (c) $(4\sqrt{2}, 4\sqrt{2})$ (d) $(7, 0)$ (e) $(0, -2)$ (f) $(0, 0)$
9. (a) $(5, 0.6435)$ (b) $(\sqrt{29}, 5.0929)$ (c) $(1.2716, 0.6658)$
11. (a) parabola (b) hyperbola (c) line (d) circle
13.



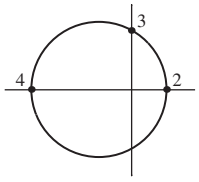
Line

15.



Cardioid

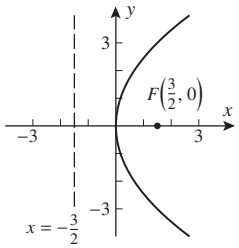
17.



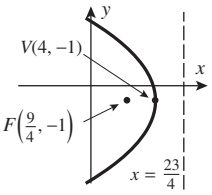
Limaçon

19. (a) $-2, 1/4$ (b) $-3\sqrt{3}/4, 3\sqrt{3}/4$
21. (a) The top is traced from right to left as t goes from 0 to π. The bottom is traced from right to left as t goes from π to 2π, except for the loop, which is traced counterclockwise as t goes from $\pi + \sin^{-1}(1/4)$ to $2\pi - \sin^{-1}(1/4)$. (b) $y = 1$
(c) horizontal: $t = \pi/2, 3\pi/2$; vertical: $t = \pi + \sin^{-1}(1/\sqrt[3]{4})$, $2\pi - \sin^{-1}(1/\sqrt[3]{4})$
(d) $r = 4 + \csc\theta$, $\theta = \pi + \sin^{-1}(1/4)$, $\theta = 2\pi - \sin^{-1}(1/4)$

23. $A = 6\pi$ **25.** $A = \dfrac{5\pi}{12} - \dfrac{\sqrt{3}}{2}$ **27.**

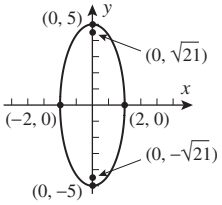


29.



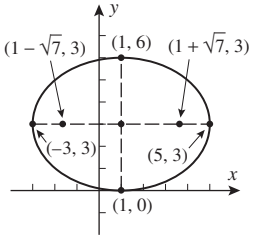
focus: $(9/4, -1)$;
vertex: $(4, -1)$;
directrix: $x = 23/4$

31.

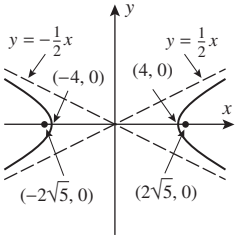


foci: $(0, \pm\sqrt{21})$;
vertices: $(0, \pm 5)$;
ends: $(\pm 2, 0)$

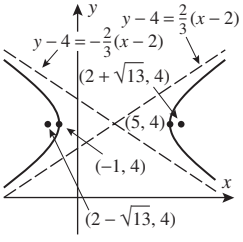
33.



35.



37.



39. $x^2 = -20y$ **41.** $y^2 - x^2 = 4$

43. **(b)** $x = \dfrac{v_0^2}{g}\sin\alpha\cos\alpha$; $y = y_0 + \dfrac{v_0^2\sin^2\alpha}{2g}$

45. $\theta = \pi/4$; $5(y')^2 - (x')^2 = 8$; hyperbola

47. $\theta = \tan^{-1}(1/2)$; $y' = (x')^2$; parabola

49. **(a)** (i) ellipse; (ii) right; (iii) 1 **(b)** (i) hyperbola (ii) left; (iii) 1/3 **(c)** (i) parabola; (ii) above; (iii) 1/3 **(d)** (i) parabola; (ii) above; (iii) 3

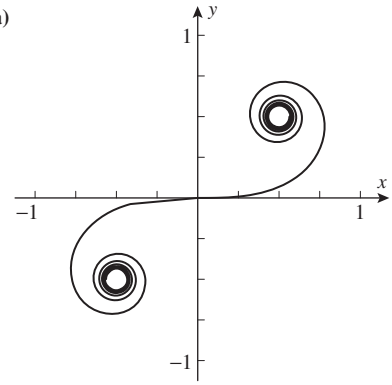
51. **(a)** $\dfrac{(x+3)^2}{25} + \dfrac{(y-2)^2}{9} = 1$ **(b)** $(x+2)^2 = -8y$
(c) $\dfrac{(y-5)^2}{4} - 16(x+1)^2 = 1$

53. 15.86543959

▶ Chapter 10 Making Connections (Page 766)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)**



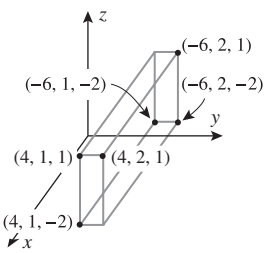
(c) $L = \displaystyle\int_{-1}^{1}\left[\cos^2\left(\frac{\pi t^2}{2}\right) + \sin^2\left(\frac{\pi t^2}{2}\right)\right]dt = 2$

2. **(a)** P: $(b\cos t, b\sin t)$;
Q: $(a\cos t, a\sin t)$;
R: $(a\cos t, b\sin t)$

▶ Exercise Set 11.1 (Page 771)

1. **(a)** (0, 0, 0), (3, 0, 0), (3, 5, 0), (0, 5, 0), (0, 0, 4), (3, 0, 4), (3, 5, 4), (0, 5, 4)
(b) (0, 1, 0), (4, 1, 0), (4, 6, 0), (0, 6, 0), (0, 1, −2), (4, 1, −2), (4, 6, −2), (0, 6, −2)

3. (4, 2, −2), (4, 2, 1), (4, 1, 1), (4, 1, −2), (−6, 1, 1), (−6, 2, 1), (−6, 2, −2), (−6, 1, −2)



5. **(a)** point **(b)** line parallel to the y-axis
(c) plane parallel to the yz-plane

9. radius $\sqrt{59}$, center (2, 1, −5) **11.** **(b)** (2, 1, 6) **(c)** area 147/2

13. **(a)** $(x-1)^2 + y^2 + (z+1)^2 = 9$
(b) $(x+1)^2 + (y-3)^2 + (z-2)^2 = 19$
(c) $\left(x + \frac{1}{2}\right)^2 + (y-2)^2 + (z-2)^2 = \frac{5}{4}$

15. $(x-2)^2 + (y-1)^2 + (z+3)^2 = r^2$;
(a) $r^2 = 9$ **(b)** $r^2 = 1$ **(c)** $r^2 = 4$

Responses to True–False questions may be abridged to save space.

19. False; see Figure 11.1.6.

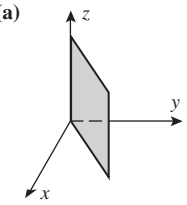
21. True; see Figure 11.1.3.

23. sphere, center (−5, −2, −1), radius 8

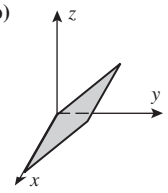
25. sphere; center $\left(\frac{1}{2}, \frac{3}{4}, -\frac{5}{4}\right)$, radius $\dfrac{3\sqrt{6}}{4}$

27. no graph

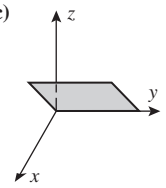
29. **(a)**



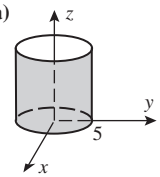
(b)



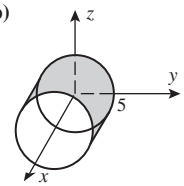
(c)



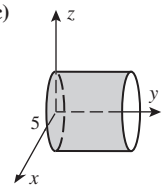
31. **(a)**



(b)

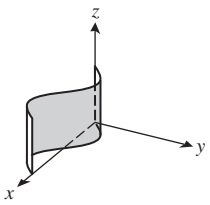


(c)

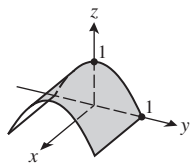


33. **(a)** $-3y + z = 0$ **(b)** $-2x + z = 0$ **(c)** $(x-1)^2 + (y-1)^2 = 4$ **(d)** $(x-1)^2 + (z-1)^2 = 1$

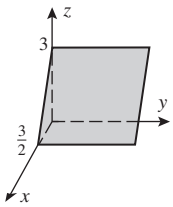
35.



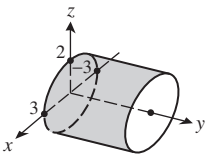
37.



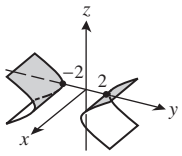
39.



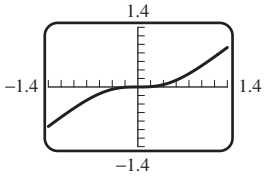
41.

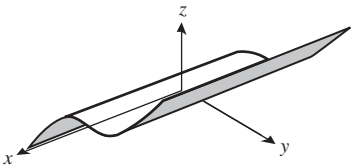


43.



45.





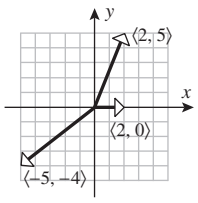
47. largest distance, $4 + \sqrt{6}$; smallest distance, $4 - \sqrt{6}$

49. all points outside the circular cylinder $(y+3)^2 + (z-4)^2 = 36$

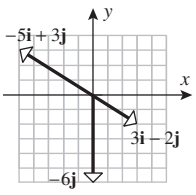
51. $r = (2 - \sqrt{3})R$ **53.** **(b)** $y^2 + z^2 = e^{2x}$

Exercise Set 11.2 (Page 782)

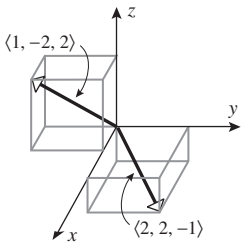
1. **(a-c)**



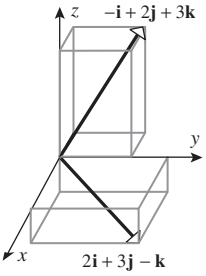
(d-f)



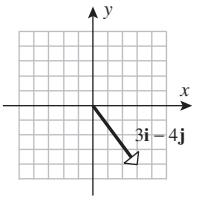
3. **(a,b)**

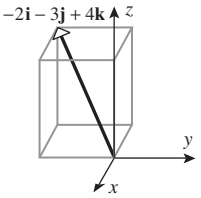


(c,d)



5. **(a)** $\langle 3, -4\rangle$ **(b)** $\langle -2, -3, 4\rangle$





7. **(a)** $\langle -1, 4\rangle$ **(b)** $\langle -7, 2\rangle$ **(c)** $\langle -2, 4, 2\rangle$

9. **(a)** $(5, -1)$ **(b)** $(7, -3, -5)$

11. **(a)** $-\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ **(b)** $21\mathbf{i} + 9\mathbf{j} - 7\mathbf{k}$ **(c)** $-\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$ **(d)** $39\mathbf{i} - 3\mathbf{j} - 6\mathbf{k}$ **(e)** $-2\mathbf{i} - 16\mathbf{j} - 18\mathbf{k}$ **(f)** $-\mathbf{i} + 13\mathbf{j} - 2\mathbf{k}$

13. **(a)** $\sqrt{2}$ **(b)** $2\sqrt{17}$ **(c)** $\sqrt{21}$ **(d)** $\sqrt{14}$

15. **(a)** $2\sqrt{3}$ **(b)** $\sqrt{14} + \sqrt{2}$ **(c)** $2\sqrt{14} + 2\sqrt{2}$ **(d)** $2\sqrt{37}$ **(e)** $(1/\sqrt{6})\mathbf{i} + (1/\sqrt{6})\mathbf{j} - (2/\sqrt{6})\mathbf{k}$ **(f)** 1

Responses to True–False questions may be abridged to save space.

17. False; $\|\mathbf{i} + \mathbf{j}\| = \sqrt{2} \neq 1 + 1 = 2$

19. True; one in the same direction and one in the opposite direction.

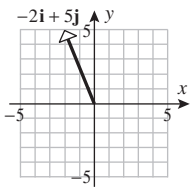
21. **(a)** $(-2/\sqrt{29})\mathbf{i} + (5/\sqrt{29})\mathbf{j}$ **(b)** $(-3\mathbf{i} + 2\mathbf{j} - \mathbf{k})/\sqrt{14}$ **(c)** $(4\mathbf{i} + \mathbf{j} - \mathbf{k})/(3\sqrt{2})$

23. **(a)** $\langle -\frac{3}{2}, 2\rangle$ **(b)** $\dfrac{1}{\sqrt{5}}\langle 7, 0, -6\rangle$

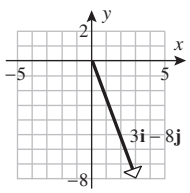
25. **(a)** $\langle 2\sqrt{2}, 2\sqrt{2}\rangle$ **(b)** $\langle 0, 2\rangle$ **(c)** $\langle -5/2, 5\sqrt{3}/2\rangle$ **(d)** $\langle \sqrt{3}/2, 1/2\rangle$

27. $\langle (\sqrt{3} - \sqrt{2})/2, (1 + \sqrt{2})/2\rangle$

29. **(a)** $\langle -2, 5\rangle$



(b) $\langle 3, -8\rangle$



31. $\langle -2, 3\rangle$ **33.** $\mathbf{u} = \frac{5}{7}\mathbf{i} + \frac{2}{7}\mathbf{j} + \frac{1}{7}\mathbf{k}$, $\mathbf{v} = \frac{8}{7}\mathbf{i} - \frac{1}{7}\mathbf{j} - \frac{4}{7}\mathbf{k}$

35. $\sqrt{5}, 3$ **37.** **(a)** $\pm\frac{5}{3}$ **(b)** 2

39. **(a)** $\langle 1/\sqrt{10}, 3/\sqrt{10}\rangle, \langle -1/\sqrt{10}, -3/\sqrt{10}\rangle$ **(b)** $\langle 1/\sqrt{2}, -1/\sqrt{2}\rangle, \langle -1/\sqrt{2}, 1/\sqrt{2}\rangle$ **(c)** $\pm\dfrac{1}{\sqrt{26}}\langle 5, 1\rangle$

41. **(a)** the circle of radius 1 about the origin
(b) the closed disk of radius 1 about the origin
(c) all points outside the closed disk of radius 1 about the origin

43. **(a)** the (hollow) sphere of radius 1 about the origin
(b) the closed ball of radius 1 about the origin
(c) all points outside the closed ball of radius 1 about the origin

45. magnitude $= 30\sqrt{5}$ lb, $\theta \approx 26.57°$

47. magnitude ≈ 207.06 N, $\theta = 45°$

49. magnitude ≈ 94.995 N, $\theta \approx 28.28°$

51. magnitude ≈ 9.165 lb, angle $\approx -70.890°$

53. ≈ 183.02 lb, 224.13 lb

55. **(a)** $c_1 = 1, c_2 = \frac{1}{2}$

Exercise Set 11.3 (Page 792)

1. **(a)** -5; $\cos\theta = -1/\sqrt{5}$ **(b)** -3; $\cos\theta = -3/\sqrt{58}$ **(c)** 0; $\cos\theta = 0$ **(d)** -20; $\cos\theta = -20/(3\sqrt{70})$

3. **(a)** obtuse **(b)** acute **(c)** obtuse **(d)** orthogonal

5. $\sqrt{2}/2, 0, -\sqrt{2}/2, -1, -\sqrt{2}/2, 0, \sqrt{2}/2$

7. **(a)** vertex B **(b)** $82°, 60°, 38°$ **13.** $r = 7/5$

15. **(a)** $\alpha = \beta \approx 55°, \gamma \approx 125°$ **(b)** $\alpha \approx 48°, \beta \approx 132°, \gamma \approx 71°$

19. **(a)** $\approx 35°$ **(b)** $90°$

21. $64°, 41°, 60°$ **23.** $68°, 56°, 42°$

25. **(a)** $\left\langle \frac{2}{3}, \frac{4}{3}, \frac{4}{3} \right\rangle, \left\langle \frac{4}{3}, -\frac{7}{3}, \frac{5}{3} \right\rangle$
(b) $\left\langle \frac{38}{17}, \frac{57}{17}, \frac{38}{17} \right\rangle, \left\langle \frac{30}{17}, -\frac{74}{17}, \frac{81}{17} \right\rangle$

27. **(a)** $\langle -2/5, -1/5 \rangle + \langle -13/5, 26/5 \rangle$ **(b)** $\left\langle 0, -\frac{8}{5}, \frac{4}{5} \right\rangle + \left\langle -2, \frac{13}{5}, \frac{26}{5} \right\rangle$
(c) $\mathbf{v} = \langle 1, 4, 1 \rangle$ is orthogonal to $\mathbf{b}$.

Responses to True–False questions may be abridged to save space.

29. True; $\mathbf{v} + \mathbf{w} = \mathbf{0}$ implies $0 = \mathbf{v} \cdot (\mathbf{v} + \mathbf{w}) = \|\mathbf{v}\|^2 \neq 0$, a contradiction.

31. True; see Equation (12). **33.** $\sqrt{564/29}$ **35.** 210 N

37. $-5\sqrt{3}$ J **45.** **(a)** $40°$ **(b)** $x \approx -0.682328$

Exercise Set 11.4 (Page 803)

1. **(a)** $-\mathbf{j} + \mathbf{k}$ **3.** $\langle 9, 9, 9 \rangle$ **5.** $\langle -3, -8, -4 \rangle$

7. **(a)** $\langle -20, -67, -9 \rangle$ **(b)** $\langle -78, 52, -26 \rangle$
(c) $\langle 0, -56, -392 \rangle$ **(d)** $\langle 0, 56, 392 \rangle$

9. $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$ **11.** $\pm\frac{1}{\sqrt{6}}\langle 2, 1, 1 \rangle$

Responses to True–False questions may be abridged to save space.

13. True; see Theorem 11.4.5(c).

15. False; let $\mathbf{v} = \mathbf{u} = \mathbf{i}$ and let $\mathbf{w} = 2\mathbf{i}$.

17. $\sqrt{30}$ **19.** $\sqrt{339}/2$

21. 80 **23.** -4 **25.** 25.20 **27.** **(a)** yes **(b)** yes **(c)** no

29. **(a)** 9 **(b)** $\sqrt{122}$ **(c)** $\sin^{-1}\left(\frac{9}{14}\right)$

31. **(a)** $2\sqrt{141/29}$ **(b)** $\sqrt{8}$ **33.** $\frac{2}{3}$ **37.** $\theta = \pi/4$

39. **(a)** $10\sqrt{2}$ lb·ft, direction of rotation about P is counterclockwise looking along $\overrightarrow{PQ} \times \mathbf{F} = -10\mathbf{i} + 10\mathbf{k}$ toward its initial point
(b) 10 lb·ft, direction of rotation about P is counterclockwise looking along $-10\mathbf{i}$ toward its initial point
(c) 0 lb·ft, no rotation about P

41. ≈ 36.19 N·m **45.** $-8\mathbf{i} - 20\mathbf{j} + 2\mathbf{k}, -8\mathbf{i} - 8\mathbf{k}$ **49.** 1.887850

Exercise Set 11.5 (Page 810)

1. **(a)** $L_1: x = 1, y = t$, $L_2: x = t, y = 1$, $L_3: x = t, y = t$
(b) $L_1: x = 1, y = 1, z = t$, $L_2: x = t, y = 1, z = 1$, $L_3: x = 1, y = t, z = 1$, $L_4: x = t, y = t, z = t$

3. **(a)** $x = 4 + t, y = -2 + 3t$; line segment: $0 \le t \le 1$
(b) $x = 5 - 3t, y = -2 + 6t, z = 1 + 2t$; line segment: $0 \le t \le 1$

5. **(a)** $x = 2 + t, y = -3 - 5t$ **(b)** $x = t, y = -t, z = 1 + 2t$

7. **(a)** $P(2, -1)$, $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j}$ **(b)** $P(-1, 2, 4)$, $\mathbf{v} = 5\mathbf{i} + 7\mathbf{j} - 8\mathbf{k}$

9. **(a)** $\langle -3, 4 \rangle + t\langle 1, 6 \rangle$; $-3\mathbf{i} + 4\mathbf{j} + t(\mathbf{i} + 6\mathbf{j})$
(b) $\langle 2, -3, 0 \rangle + t\langle -1, 5, 1 \rangle$; $2\mathbf{i} - 3\mathbf{j} + t(-\mathbf{i} + 5\mathbf{j} + \mathbf{k})$

Responses to True–False questions may be abridged to save space.

11. False; the lines could be skew.

13. False; see part (b) of the solution to Example 3.

15. $x = -5 + 2t, y = 2 + 5t$ **17.** $x = 3 + 4t, y = -4 + 3t$

19. $x = -1 + 3t, y = 2 - 4t, z = 4 + 2t$

21. $x = -1 + 2t, y = -t, z = 5 + 3t$

23. **(a)** $x = 7$ **(b)** $y = \frac{7}{3}$ **(c)** $x = \dfrac{-1 \pm \sqrt{85}}{6}, y = \dfrac{43 \mp \sqrt{85}}{18}$

25. $(-2, 8, 0)$; $(-2, 0, -8)$; The line does not intersect the yz-plane.

27. $(0, 4, -2), (4, 0, 6)$ **29.** $(1, -1, 2)$ **33.** The lines are parallel.

35. The points do not lie on the same line.

39. $\langle x, y \rangle = \langle -1, 2 \rangle + t\langle 1, 1 \rangle$

41. the point $1/n$ of the way from $(-2, 0)$ to $(1, 3)$

43. the line segment joining the points $(1, 0)$ and $(-5, 9)$

45. $(5, 2)$ **47.** $2\sqrt{5}$ **49.** distance $= \sqrt{35/6}$

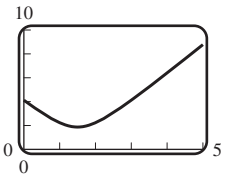
51. **(a)** $x = x_0 + (x_1 - x_0)t, y = y_0 + (y_1 - y_0)t, z = z_0 + (z_1 - z_0)t$
(b) $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$

53. **(b)** $\langle x, y, z \rangle = \langle 1 + 2t, -3 + 4t, 5 + t \rangle$

55. **(b)** $84°$ **(c)** $x = 7 + t, y = -1, z = -2 + t$

57. $x = t, y = 2 + t, z = 1 - t$

59. **(a)** $\sqrt{17}$ cm **(b)**



(d) $\sqrt{14}/2$ cm

Exercise Set 11.6 (Page 819)

1. $x = 3, y = 4, z = 5$ **3.** $2x + 3y + 2z = 26$ **5.** $z = 0$

7. $x - y = 0$ **9.** $y + z = 1$ **11.** $2y - z = 1$

13. **(a)** parallel **(b)** perpendicular **(c)** neither

15. **(a)** parallel **(b)** neither **(c)** perpendicular

17. **(a)** point of intersection is $(3, 3, 3)$ **(b)** no intersection

19. $48°$

Responses to True–False questions may be abridged to save space.

21. True; each will be the negative of the other.

23. True; the direction vector of L must be orthogonal to both normal vectors.

25. $3x - 2y + 7z = 0$ **27.** $4x - 13y + 21z = -14$

29. $x + 2y - 5z = 10$ **31.** $x + 5y + 3z = -6$

33. $x + y + 2z = 9$ **35.** $x = 5 - 2t, y = 5t, z = -2 + 11t$

37. $8x + y + 10z = 28$ **39.** yes

41. $x = -\frac{11}{7} - 23t, y = -\frac{12}{7} + t, z = -7t$

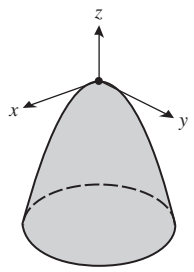
43. $\frac{4}{3}$ **45.** $5/\sqrt{54}$ **47.** $25/\sqrt{126}$

49. $(x - 2)^2 + (y - 1)^2 + (z + 3)^2 = \frac{121}{14}$ **51.** $5/\sqrt{12}$

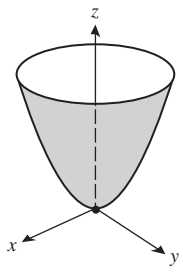
Exercise Set 11.7 (Page 830)

1. **(a)** elliptic paraboloid, $a = 2, b = 5$
(b) hyperbolic paraboloid, $a = 1, b = 5$
(c) hyperboloid of one sheet, $a = b = c = 6$
(d) circular cone, $a = b = 1$ **(e)** elliptic paraboloid, $a = 2, b = 1$
(f) hyperboloid of two sheets, $a = b = c = 1$

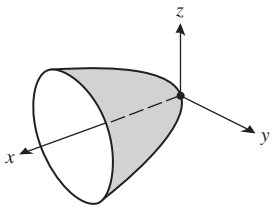
3. **(a)** $-z = x^2 + y^2$, circular paraboloid opening down the negative z-axis



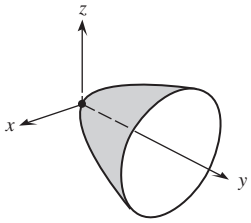
(b) $z = x^2 + y^2$, circular paraboloid, no change
(c) $z = x^2 + y^2$, circular paraboloid, no change
(d) $z = x^2 + y^2$, circular paraboloid, no change



(e) $x = y^2 + z^2$,
circular paraboloid opening along the positive x-axis

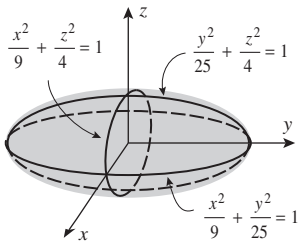


(f) $y = x^2 + z^2$,
circular paraboloid opening along the positive y-axis

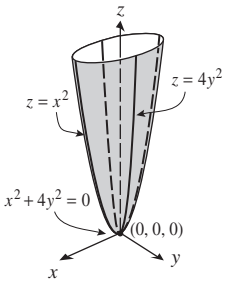


5. **(a)** hyperboloid of one sheet, axis is y-axis
(b) hyperboloid of two sheets separated by yz-plane
(c) elliptic paraboloid opening along the positive x-axis
(d) elliptic cone with x-axis as axis
(e) hyperbolic paraboloid straddling the x-axis
(f) paraboloid opening along the negative y-axis

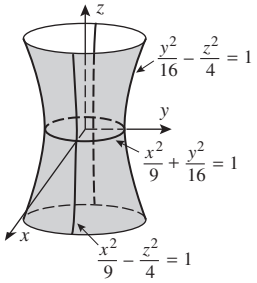
7. **(a)** $x = 0$: $\dfrac{y^2}{25} + \dfrac{z^2}{4} = 1$;
$y = 0$: $\dfrac{x^2}{9} + \dfrac{z^2}{4} = 1$;
$z = 0$: $\dfrac{x^2}{9} + \dfrac{y^2}{25} = 1$



(b) $x = 0$: $z = 4y^2$;
$y = 0$: $z = x^2$;
$z = 0$: $x = y = 0$



(c) $x = 0$: $\dfrac{y^2}{16} - \dfrac{z^2}{4} = 1$;
$y = 0$: $\dfrac{x^2}{9} - \dfrac{z^2}{4} = 1$;
$z = 0$: $\dfrac{x^2}{9} + \dfrac{y^2}{16} = 1$



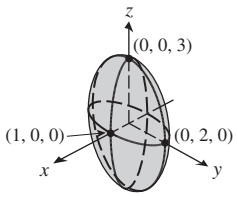
9. **(a)** $4x^2 + 2z^2 = 3$; ellipse **(b)** $y^2 + z^2 = 3$; circle
(c) $y^2 + z^2 = 128$; circle **(d)** $9x^2 - y^2 = 20$; hyperbola
(e) $z = 9x^2 + 16$; parabola **(f)** $9x^2 + 4y^2 = 4$; ellipse

Responses to True–False questions may be abridged to save space.

11. False; "quadric" refers to second powers.

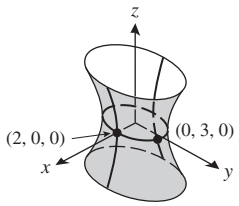
13. False; there need not exist a line such that all cross sections orthogonal to the line are circular.

15.



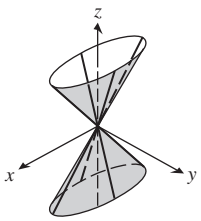
Ellipsoid

17.



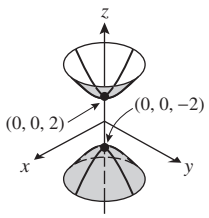
Hyperboloid of one sheet

19.



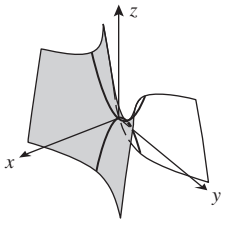
Elliptic cone

21.



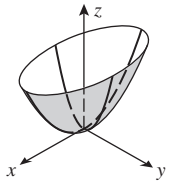
Hyperboloid of two sheets

23.



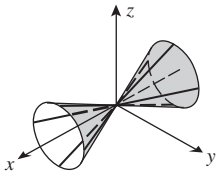
Hyperbolic paraboloid

25.



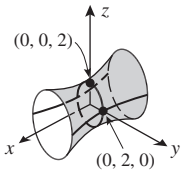
Elliptic paraboloid

27.



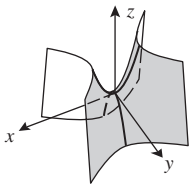
Circular cone

29.



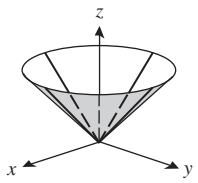
Hyperboloid of one sheet

31.

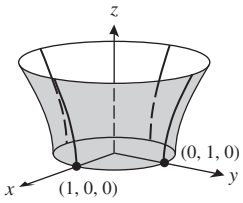


Hyperbolic paraboloid

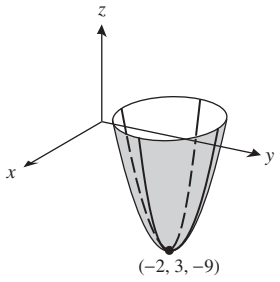
33.



35.

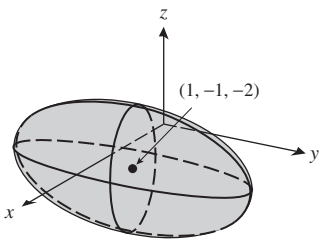


37.



Circular paraboloid

39.



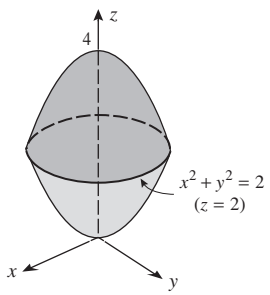
Ellipsoid

41. **(a)** $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ **(b)** 6, 4 **(c)** $(\pm\sqrt{5}, 0, \sqrt{2})$
(d) The focal axis is parallel to the x-axis.

43. **(a)** $\dfrac{y^2}{9} - \dfrac{x^2}{9} = 1$ **(b)** $(0, \pm 3, 9)$ **(c)** $(0, \pm 3\sqrt{2}, 9)$
(d) The focal axis is parallel to the y-axis.

45. **(a)** $z + 9 = y^2$ **(b)** $(3, 0, -9)$ **(c)** $\left(3, 0, -\frac{35}{4}\right)$
(d) The focal axis is parallel to the z-axis.

47. circle of radius $\sqrt{2}$ in the plane $z = 2$, centered at $(0, 0, 2)$



49. $y = 4(x^2 + z^2)$ **51.** $z = (x^2 + y^2)/8$ (circular paraboloid)

▶ Exercise Set 11.8 (Page 837)

1. **(a)** $(4, \pi/6, -4)$ **(b)** $(5\sqrt{2}, 3\pi/4, 6)$
(c) $(2, \pi/2, 0)$ **(d)** $(8, 5\pi/3, 6)$

3. **(a)** $(3, 3\sqrt{3}, 3)$ **(b)** $(-4\sqrt{2}, 4\sqrt{2}, -2)$
(c) $(5, 0, 4)$ **(d)** $(-7, 0, -9)$

5. **(a)** $(4\sqrt{2}, \pi/3, 3\pi/4)$ **(b)** $(2, 7\pi/4, \pi/4)$
(c) $(6, \pi/2, \pi/3)$ **(d)** $(8, 5\pi/6, \pi/2)$

7. **(a)** $(3, \sqrt{3}, 2)$ **(b)** $(7, 0, 0)$
(c) $(0, 0, 1)$ **(d)** $(0, -2, 0)$

9. **(a)** $(4\sqrt{3}, \pi/3, \pi/6)$ **(b)** $(\sqrt{2}, \pi/4, 3\pi/4)$
(c) $(2, 3\pi/4, \pi/2)$ **(d)** $(4\sqrt{3}, 1, 2\pi/3)$

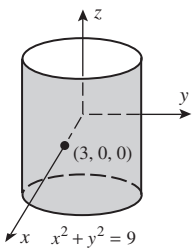
11. **(a)** $(5\sqrt{3}/2, \pi/4, 5/2)$ **(b)** $(0, 5\pi/6, -2)$
(c) $(0, 0, 3)$ **(d)** $(4, \pi/6, 0)$

Responses to True–False questions may be abridged to save space.

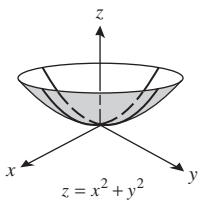
15. True; see Figure 11.8.1*b*.

17. True; see Figures 11.8.3 and 11.8.4.

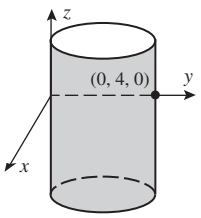
19.



21.

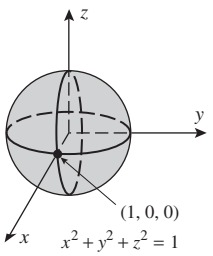


23.



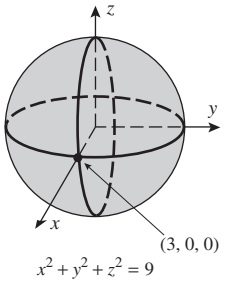
$x^2 + (y - 2)^2 = 4$

25.



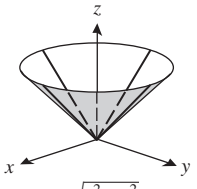
$x^2 + y^2 + z^2 = 1$

27.



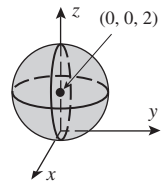
$x^2 + y^2 + z^2 = 9$

29.



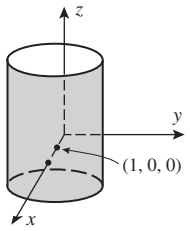
$z = \sqrt{x^2 + y^2}$

31.



$x^2 + y^2 + (z - 2)^2 = 4$

33.



$(x - 1)^2 + y^2 = 1$

35. **(a)** $z = 7$ **(b)** $\rho = 7 \sec\phi$ **37.** **(a)** $z = 3r^2$ **(b)** $\rho = \frac{1}{3}\csc\phi\cot\phi$

39. **(a)** $r = 3$ **(b)** $\rho = 3\csc\phi$ **41.** **(a)** $r^2 + z^2 = 4$ **(b)** $\rho = 4$

43. **(a)** $2r\cos\theta + 3r\sin\theta + 4z = 2$
(b) $2\rho\sin\phi\cos\theta + 3\rho\sin\phi\sin\theta + 4\rho\cos\phi = 2$

45. **(a)** $r^2\cos^2\theta = 16 - z^2$ **(b)** $\rho^2(1 - \sin^2\phi\sin^2\theta) = 16$

47. all points on or above the paraboloid $z = x^2 + y^2$ that are also on or below the plane $z = 4$

49. all points on or between concentric spheres of radii 2 and 5 centered at the origin

51. spherical: $(4000, \pi/6, \pi/6)$; rectangular: $(1000\sqrt{3}, 1000, 2000\sqrt{3})$

53. **(a)** $(10, \pi/2, 1)$ **(b)** $(0, 10, 1)$ **(c)** $(\sqrt{101}, \pi/2, \tan^{-1} 10)$

▶ Chapter 11 Review Exercises (Page 838)

3. **(b)** $-1/2, \pm\sqrt{3}/2$ **(d)** true

5. **(a)** $r^2 = 16$ **(b)** $r^2 = 9$ **(c)** $r^2 = 4$

7. $(8, 5)$

9. **(a)** $-\frac{3}{4}$ **(b)** $\frac{1}{7}$ **(c)** $(48 \pm 25\sqrt{3})/39$ **(d)** $c = \frac{4}{3}$

13. 13 ft·lb **15.** **(a)** $\frac{5}{2}\sqrt{2}$ **(b)** $\frac{5}{3}\sqrt{2}$

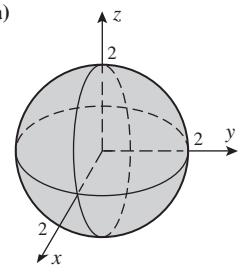
17. **(a)** 29 **(b)** $\dfrac{29}{\sqrt{65}}$ **19.** $x = 5 + t, y = 2 - t, z = 3$

21. $x + 5y - z - 6 = 0$ **23.** $a_1a_2 + b_1b_2 + c_1c_2 = 0$

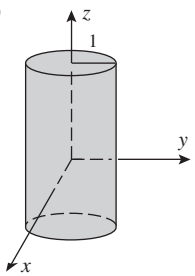
25. **(a)** hyperboloid of one sheet **(b)** sphere **(c)** circular cone

27. **(a)** $z = x^2 - y^2$ **(b)** $xz = 1$

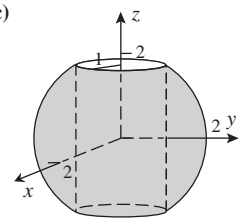
29. **(a)**



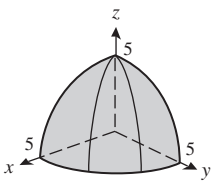
(b)



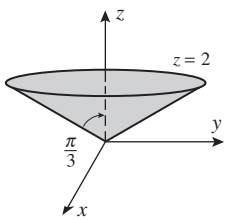
(c)



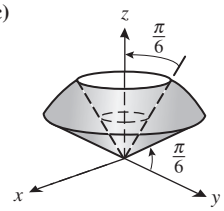
31. **(a)**



(b)



(c)



▶ Chapter 11 Making Connections (Page 840)

Answers are provided in the Student Solutions Manual.

▶ Exercise Set 12.1 (Page 845)

1. $(-\infty, +\infty)$; $\mathbf{r}(\pi) = -4\pi\mathbf{j}$ **3.** $[4, \infty)$; $\mathbf{r}(3) = -\mathbf{i} - \ln 7\mathbf{j} + \mathbf{k}$

5. $\mathbf{r} = 3\cos t\mathbf{i} + (2t + \cos 2t)\mathbf{j}$ **7.** $x = 3t^2$, $y = -2\ln t$

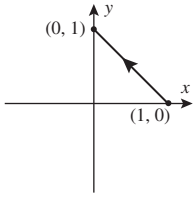
9. the line in 2-space through $(3, 0)$ with direction vector $\mathbf{a} = -\mathbf{i} + 5\mathbf{j}$

11. the line in 3-space through the point $(0, -3, 1)$ and parallel to the vector $2\mathbf{i} - 5\mathbf{k}$

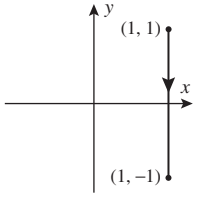
13. an ellipse centered at $(0, 0, 2)$ in the plane $z = 2$

15. **(a)** slope $-\frac{5}{2}$ **(b)** $\left(\frac{5}{2}, 0, \frac{3}{2}\right)$

17. **(a)**

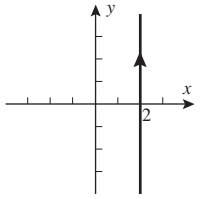


(b)

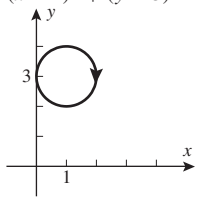


19. $\mathbf{r} = (1 - t)(3\mathbf{i} + 4\mathbf{j})$, $0 \le t \le 1$

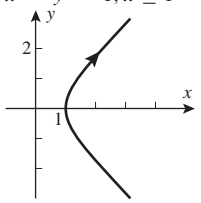
21. $x = 2$



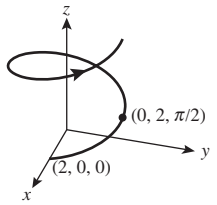
23. $(x - 1)^2 + (y - 3)^2 = 1$



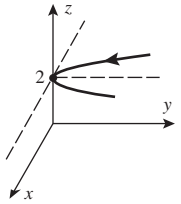
25. $x^2 - y^2 = 1$, $x \ge 1$



27.



29.

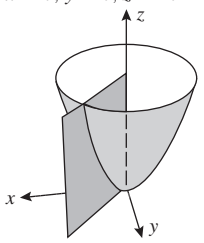


Responses to True–False questions may be abridged to save space.

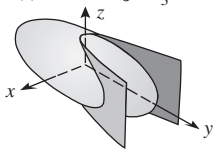
31. True; $\mathbf{r}(t) = (1 - t)\mathbf{r}_0 + t\mathbf{r}_1 (0 \le t \le 1)$ represents the line segment in 3-space that is traced from $\mathbf{r}_0$ to $\mathbf{r}_1$.

33. False; the natural domain of a vector-valued function is the *intersection* of the domains of its component functions.

35. $x = t$, $y = t$, $z = 2t^2$



37. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} \pm \frac{1}{3}\sqrt{81 - 9t^2 - t^4}\mathbf{k}$ **43.** $c = 3/(2\pi)$



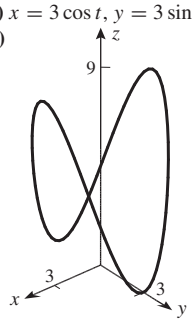
47. **(a)** III, since the curve is a subset of the plane $y = -x$

(b) IV, since only x is periodic in t and y, z increase without bound

(c) II, since all three components are periodic in t

(d) I, since the projection onto the yz-plane is a circle and the curve increases without bound in the x-direction

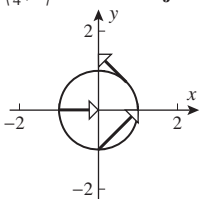
49. **(a)** $x = 3\cos t,\ y = 3\sin t,\ z = 9\cos^2 t$
(b)



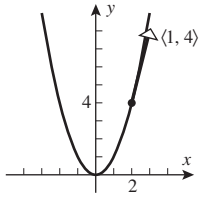
Exercise Set 12.2 (Page 856)

1. $\langle \frac{1}{4}, 0\rangle$ **3.** $10\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ **5.** **(a)** continuous **(b)** not continuous

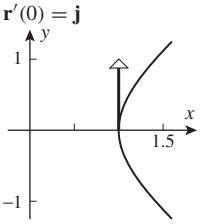
7.



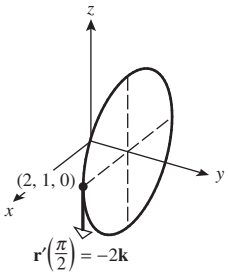
9. $(\cos t)\mathbf{j}$ **11.** $\mathbf{r}'(2) = \langle 1, 4\rangle$



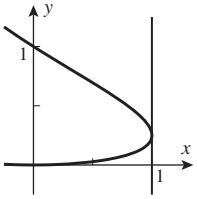
13. $\mathbf{r}'(0) = \mathbf{j}$



15. $\mathbf{r}'(\pi/2) = -2\mathbf{k}$



17.



19. $x = e^2 + 2et,\ y = 1 - \frac{t}{e}$
21. $x = 1 - \sqrt{3}\pi t,\ y = \sqrt{3} + \pi t,\ z = 2 + 6t$
23. $\mathbf{r} = (-\mathbf{i} + 2\mathbf{j}) + t\left(3\mathbf{i} + \frac{3}{4}\mathbf{j}\right)$
25. $\mathbf{r} = (4\mathbf{i} - \mathbf{j}) + t(-4\mathbf{i} + \mathbf{j} + 4\mathbf{k})$
27. **(a)** $\mathbf{i} - \mathbf{j} + \mathbf{k}$ **(b)** $-\mathbf{i} + \mathbf{k}$ **(c)** 0
29. $7t^6;\ 18t^5\mathbf{i} - 10t^4\mathbf{j}$
31. $5t\mathbf{i} + 2t^2\mathbf{j} + \mathbf{C}$

33. $(\sin t)\mathbf{i} - (\cos t)\mathbf{j} + \mathbf{C}$ **35.** $\mathbf{j}$ **37.** $(17\sqrt{17} - 1)/12$
39. $\frac{52}{3}\mathbf{i} + 4\mathbf{j}$

Responses to True–False questions may be abridged to save space.

41. False; for example, $\mathbf{r}(t) = \langle t, |t|\rangle$ is continuous at $t = 0$, but the specified limit doesn't exist at $t = 0$.

43. True; see the definition of $\int_a^b \mathbf{r}(t)\,dt$.

45. $(t^2 + 1)\mathbf{i} + (t^3 - 2)\mathbf{j}$
47. $y(t) = \left(\frac{1}{2}t^2 + 2\right)\mathbf{i} + (e^t - 1)\mathbf{j}$
49. **(a)** $(-2, 4, 6)$ and $(1, 1, -3)$ **(b)** $76°, 71°$ **51.** $68°$

Exercise Set 12.3 (Page 866)

1. smooth **3.** not smooth, $\mathbf{r}'(1) = \mathbf{0}$ **5.** $L = \frac{3}{4}$ **7.** $L = e - e^{-1}$
9. $L = 58$ **11.** $L = 2\pi\sqrt{13}$ **13.** $\mathbf{r}'(\tau) = 5\mathbf{i} + 10(5\tau + 1)\mathbf{j}$
15. $\mathbf{r}'(\tau) = 2\tau e^{\tau^2}\mathbf{i} - 8\tau e^{-\tau^2}\mathbf{j}$

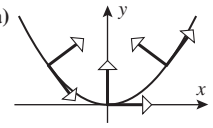
Responses to True–False questions may be abridged to save space.

17. False; $\int_a^b \|\mathbf{r}'(t)\|\,dt$ is a scalar that represents the arc length of the curve in 2-space traced by $\mathbf{r}(t)$ from $t = a$ to $t = b$ (Theorem 12.3.1).

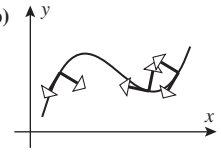
19. False; $\mathbf{r}'$ isn't defined at the point corresponding to the origin.
21. **(a)** $x = \dfrac{s}{\sqrt{2}},\ y = \dfrac{s}{\sqrt{2}}$ **(b)** $x = y = z = \dfrac{s}{\sqrt{3}}$
23. **(a)** $x = 1 + \dfrac{s}{3},\ y = 3 - \dfrac{2s}{3},\ z = 4 + \dfrac{2s}{3}$ **(b)** $\left\langle \frac{28}{3}, -\frac{41}{3}, \frac{62}{3}\right\rangle$
25. $x = 4 + \cos s,\ y = 5 + \sin s,\ 0 \le s \le 2\pi$
27. $x = \frac{1}{3}[(3s + 1)^{2/3} - 1]^{3/2},\ y = \frac{1}{2}[(3s + 1)^{2/3} - 1],\ s \ge 0$
29. $x = \left(\dfrac{s}{\sqrt{2}} + 1\right)\cos\left[\ln\left(\dfrac{s}{\sqrt{2}} + 1\right)\right],$
$y = \left(\dfrac{s}{\sqrt{2}} + 1\right)\sin\left[\ln\left(\dfrac{s}{\sqrt{2}} + 1\right)\right],$ $0 \le s \le \sqrt{2}(e^{\pi/2} - 1)$
33. $x = 2a\cos^{-1}[1 - s/(4a)]$
$-2a(1 - [1 - s/(4a)]^2)^{1/2}(2[1 - s/(4a)]^2 - 1),$
$y = \dfrac{s(8a - s)}{8a}$ for $0 \le s \le 8a$
35. **(a)** $9/2$ **(b)** $9 - 2\sqrt{6}$ **37.** **(a)** $\sqrt{3}(1 - e^{-2})$ **(b)** $4\sqrt{5}$
39. **(a)** $g(\tau) = \pi(\tau)$ **(b)** $g(\tau) = \pi(1 - \tau)$ **41.** 44 in
43. **(a)** $2t + \dfrac{1}{t}$ **(b)** $2t + \dfrac{1}{t}$ **(c)** $15 + 2\ln 2$

Exercise Set 12.4 (Page 872)

1. **(a)**



(b)



5. $\mathbf{T}(1) = \dfrac{2}{\sqrt{5}}\mathbf{i} + \dfrac{1}{\sqrt{5}}\mathbf{j},\ \mathbf{N}(1) = \dfrac{1}{\sqrt{5}}\mathbf{i} - \dfrac{2}{\sqrt{5}}\mathbf{j}$
7. $\mathbf{T}\left(\dfrac{\pi}{3}\right) = -\dfrac{\sqrt{3}}{2}\mathbf{i} + \dfrac{1}{2}\mathbf{j},\ \mathbf{N}\left(\dfrac{\pi}{3}\right) = -\dfrac{1}{2}\mathbf{i} - \dfrac{\sqrt{3}}{2}\mathbf{j}$
9. $\mathbf{T}\left(\dfrac{\pi}{2}\right) = -\dfrac{4}{\sqrt{17}}\mathbf{i} + \dfrac{1}{\sqrt{17}}\mathbf{k},\ \mathbf{N}\left(\dfrac{\pi}{2}\right) = -\mathbf{j}$
11. $\mathbf{T}(0) = \dfrac{1}{\sqrt{3}}\mathbf{i} + \dfrac{1}{\sqrt{3}}\mathbf{j} + \dfrac{1}{\sqrt{3}}\mathbf{k},\ \mathbf{N}(0) = -\dfrac{1}{\sqrt{2}}\mathbf{i} + \dfrac{1}{\sqrt{2}}\mathbf{j}$
13. $x = s,\ y = 1$ **15.** $\mathbf{B} = \frac{4}{5}\cos t\,\mathbf{i} - \frac{4}{5}\sin t\,\mathbf{j} - \frac{3}{5}\mathbf{k}$ **17.** $\mathbf{B} = -\mathbf{k}$
19. $\mathbf{T}\left(\dfrac{\pi}{4}\right) = \dfrac{\sqrt{2}}{2}(-\mathbf{i} + \mathbf{j}),\ \mathbf{N}\left(\dfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2}(\mathbf{i} + \mathbf{j}),$
$\mathbf{B}\left(\dfrac{\pi}{4}\right) = \mathbf{k}$; rectifying: $x + y = \sqrt{2}$; osculating: $z = 1$;
normal: $-x + y = 0$

Responses to True–False questions may be abridged to save space.

21. False; $\mathbf{T}(t)$ points in the direction of increasing parameter but may not be orthogonal to $\mathbf{r}(t)$. For example, if $\mathbf{r}(t) = \langle t, t\rangle$, then $\mathbf{T}(t) = \langle 1/\sqrt{2}, 1/\sqrt{2}\rangle$ is parallel to $\mathbf{r}(t)$.

23. True; $\mathbf{T}(s) = \mathbf{r}'(s)$, the unit tangent vector, and $\mathbf{N}(s) = \dfrac{\mathbf{r}''(s)}{\|\mathbf{r}''(s)\|}$, the unit normal vector, are orthogonal, so $\mathbf{r}'(s)$ and $\mathbf{r}''(s)$ are orthogonal.

Exercise Set 12.5 (Page 879)

1. $\kappa \approx 2$ **3.** **(a)** I is the curvature of II. **(b)** I is the curvature of II.
5. $\dfrac{6}{|t|(4 + 9t^2)^{3/2}}$ **7.** $\dfrac{6e^t}{(4e^{4t} + e^{-2t})^{3/2}}$ **9.** $\frac{5}{26}$ **11.** $\dfrac{1}{2\cosh^2 t}$
13. $\kappa = \frac{6}{13},\ \rho = \frac{13}{6}$ **15.** $\kappa = \dfrac{\sqrt{2}}{3},\ \rho = \dfrac{3\sqrt{2}}{2}$ **17.** $\kappa = \frac{1}{4}$

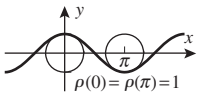
Responses to True–False questions may be abridged to save space.

19. False; see Definition 12.5.1: the curvature of the graph of $\mathbf{r}(s)$ is $\|\mathbf{r}''(s)\|$, the length of $\mathbf{r}''(s)$.

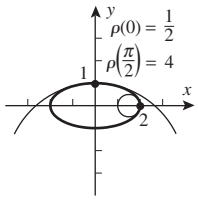
21. True; see Example 1: a circle of radius a has constant curvature $1/a$.

25. 1 **27.** $\dfrac{e^{-2}}{(1 + e^{-4})^{3/2}}$ **29.** $\dfrac{6}{13\sqrt{13}}$ **31.** $\dfrac{16}{17\sqrt{17}}$

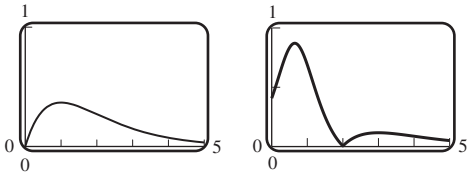
33. **(a)**



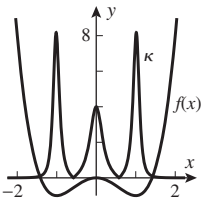
(b)



35.



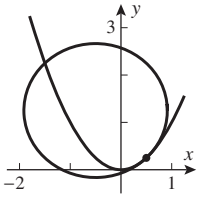
37. **(a)** $\kappa = \dfrac{|12x^2 - 4|}{[1 + (4x^3 - 4x)^2]^{3/2}}$ **(b)**



(c) $\rho = \frac{1}{4}$ for $x = 0$ and $\rho = \frac{1}{8}$ when $x = \pm 1$

41. $\dfrac{3}{4}$ **43.** $\dfrac{18}{27}$ **45.** $\rho = 2|p|$ **47.** $(3, 0), (-3, 0)$

51. **(b)** $\rho = \sqrt{2}$ **(c)**



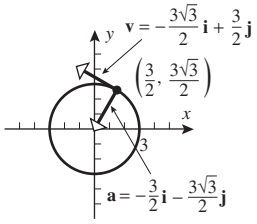
55. $a = \dfrac{1}{2r}$

63. $\tau = \dfrac{2}{(t^2 + 2)^2}$

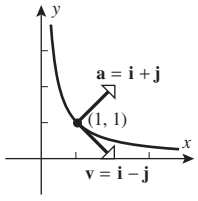
65. $\tau = -\dfrac{\sqrt{2}}{(e^t + e^{-t})^2}$

Exercise Set 12.6 (Page 891)

1. $\mathbf{v}(t) = -3\sin t\,\mathbf{i} + 3\cos t\,\mathbf{j}$
$\mathbf{a}(t) = -3\cos t\,\mathbf{i} - 3\sin t\,\mathbf{j}$
$\|\mathbf{v}(t)\| = 3$



3. $\mathbf{v}(t) = e^t\mathbf{i} - e^{-t}\mathbf{j}$
$\mathbf{a}(t) = e^t\mathbf{i} + e^{-t}\mathbf{j}$
$\|\mathbf{v}(t)\| = \sqrt{e^{2t} + e^{-2t}}$

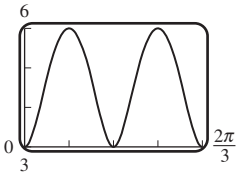


5. $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$, $\|\mathbf{v}\| = \sqrt{2i}$, $\mathbf{a} = \mathbf{j} + 4\mathbf{k}$

7. $\mathbf{v} = -2\mathbf{i} + \mathbf{k}$, $\|\mathbf{v}\| = \sqrt{5}$, $\mathbf{a} = -2\mathbf{j}$

13. minimum speed $3\sqrt{2}$ when $\mathbf{r} = 24\mathbf{i} + 8\mathbf{j}$

15. **(a)**



(b) maximum speed = 6, minimum speed = 3
(d) The maximum speed first occurs when $t = \pi/6$.

17. $\mathbf{v}(t) = (1 - \sin t)\mathbf{i} + (\cos t - 1)\mathbf{j}$;
$\mathbf{r}(t) = (t + \cos t - 1)\mathbf{i} + (\sin t - t + 2)\mathbf{j}$

19. $\mathbf{v}(t) = (1 - \cos t)\mathbf{i} + \sin t\,\mathbf{j} + e^t\mathbf{k}$;
$\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (2 - \cos t)\,\mathbf{j} + (e^t - 1)\mathbf{k}$

21. $9°$ **23.** **(a)** $1.5\mathbf{i} + 4.8\mathbf{j} - 3\mathbf{k}$ **(b)** $\mathbf{r}_0 = -0.7\mathbf{i} - 2.9\mathbf{j} + 4.8\mathbf{k}$

25. $\Delta\mathbf{r} = 12\mathbf{i} + \left(\frac{56}{3}\right)\mathbf{j}$, $s = (40\sqrt{5} - 16\sqrt{2})/3$

27. $\Delta\mathbf{r} = 3\mathbf{i} - \frac{3}{4}\mathbf{j} + \sqrt{2}\ln 4\mathbf{k}$; $s = \frac{15}{4}$

31. **(a)** $a_T = 0$, $a_N = \sqrt{2}$ **(b)** $a_T\mathbf{T} = \mathbf{0}$, $a_N\mathbf{N} = \mathbf{i} + \mathbf{j}$ **(c)** $1/\sqrt{2}$

33. **(a)** $a_T = 2\sqrt{5}$, $a_N = 2\sqrt{5}$ **(b)** $a_T\mathbf{T} = 2\mathbf{i} + 4\mathbf{j}$, $a_N\mathbf{N} = 4\mathbf{i} - 2\mathbf{j}$
(c) $2/\sqrt{5}$

35. **(a)** $a_T = -7/\sqrt{6}$, $a_N = \sqrt{53/6}$
(b) $a_T\mathbf{T} = -\frac{7}{6}(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$, $a_N\mathbf{N} = \frac{13}{6}\mathbf{i} + \frac{19}{3}\mathbf{j} + \frac{7}{6}\mathbf{k}$ **(c)** $\dfrac{\sqrt{53}}{6\sqrt{6}}$

37. $a_T = -3$, $a_N = 2$, $\mathbf{T} = -\mathbf{j}$, $\mathbf{N} = \mathbf{i}$ **39.** -1

41. $a_N = 1.764 \times 10^{11}$ km/s^2

43. $a_N = 32/(1 + 4x^2)^{3/2}$ **45.** $a_N = 0$

Responses to True–False questions may be abridged to save space.

47. True; the velocity and unit tangent vectors have the same direction, so are parallel.

49. False; in this case the velocity and acceleration vectors will be parallel, but they may have opposite direction.

53. ≈ 257.20 N

55. $40\sqrt{3}$ ft **57.** 800 ft/s **59.** $\approx 7°$ or $83°$ **61.** **(c)** ≈ 14.942 ft

63. **(a)** $\rho \approx 176.78$ m **(b)** $\frac{125}{4}$ m

65. **(b)** R is maximum when $\alpha = 45°$, maximum value v_0^2/g

67. **(a)** 3.22 s **(b)** 278.9 ft

69. **(a)** $v_0 \approx 83$ ft/s, $\alpha \approx 8°$ **(b)** 268.76 ft

Exercise Set 12.7 (Page 901)

7. 7.69 km/s **9.** 10.80 km/s

11. **(a)** minimum distance = 220,680 mi, maximum distance = 246,960 mi **(b)** 27.5 days

13. **(a)** 17,224 mi/h **(b)** $e \approx 0.071$, apogee altitude = 819 mi

Chapter 12 Review Exercises (Page 902)

3. the circle of radius 3 in the plane $z = 2$, with center $(0, 0, 2)$

5. a parabola in the plane $x = 4$, vertex at $(4, 0, -1)$, opening upward

11. $x = 1 + t$, $y = -t$, $z = t$ **13.** $-(\cos t)\mathbf{i} + (\sin t)\mathbf{j} + \mathbf{C}$

15. $y(t) = (t^3 + 1)\,\mathbf{i} + (t^2 + 1)\mathbf{j}$ **17.** 15/4

19. $\mathbf{r}(s) = \dfrac{s - 3}{3}\mathbf{i} + \dfrac{12 - 2s}{3}\mathbf{j} + \dfrac{9 + 2s}{3}\mathbf{k}$ **25.** 1/5 **27.** 0

29. **(a)** speed **(b)** distance traveled
(c) distance of the particle from the origin

33. **(a)** $\mathbf{r}(t) = \left(\frac{1}{6}t^4 + t\right)\mathbf{i} + \left(\frac{1}{2}t^2 + 2t\right)\mathbf{j} - \left(\frac{1}{4}\cos 2t + t - \frac{1}{4}\right)\mathbf{k}$
(b) 3.475 **35.** 10.65 km/s **37.** 24.78 ft

Chapter 12 Making Connections (Page 904)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(c)** (i) $\mathbf{N} = \frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j}$ (ii) $\mathbf{N} = -\mathbf{j}$

2. **(b)** (i) $\mathbf{N} = -\sin t\,\mathbf{i} - \cos t\,\mathbf{j}$
(ii) $\mathbf{N} = \dfrac{-(4t + 18t^3)\mathbf{i} + (2 - 18t^4)\mathbf{j} + (6t + 12t^3)\mathbf{k}}{2\sqrt{81t^8 + 117t^6 + 54t^4 + 13t^2 + 1}}$

3. **(c)** $\kappa(s) \to +\infty$, so the spiral winds ever tighter.

4. semicircle: 53.479 ft; quarter-circle: 60.976 ft; point: 64.001 ft

Exercise Set 13.1 (Page 914)

1. **(a)** 5 **(b)** 3 **(c)** 1 **(d)** -1 **(e)** $16a^3 + 1$ **(f)** $a^3b^2 - a^2b^3 + 1$

3. **(a)** $x^2 - y^2 + 3$ **(b)** $3x^3y^5 + 3$ **5.** $x^3e^{x^3(2y-1)}$

7. **(a)** $t^2 + 3t^{10}$ **(b)** 0 **(c)** 4

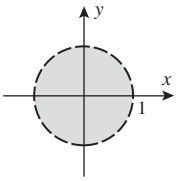
9. **(a)** WCI = 17.2°F **(b)** WCI = 22.6°F

11. **(a)** 59% **(b)** 73.5% **(c)** 60.6%

13. **(a)** 57 **(b)** -9 **(c)** 3 **(d)** $a^6 + 3$ **(e)** $-t^8 + 3$
(f) $(a + b)(a - b)^2b^3 + 3$

15. $(y + 1)e^{2x^2(y+1)z^2}$ **17.** **(a)** $120\sqrt{\pi}$ **(b)** $n(n + 1)/2$

19.



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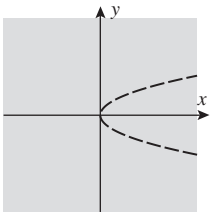


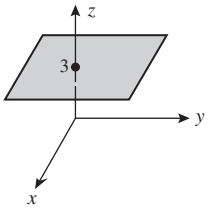
23. **(a)** all points above or on the line $y = -2$ **(b)** all points on or within the sphere $x^2 + y^2 + z^2 = 25$ **(c)** all points in 3-space

Responses to True–False questions may be abridged to save space.

25. True; the interval [0, 1] is the intersection of the domains of $\sin^{-1} t$ and $\sqrt{t}$.

27. False; the natural domain is an infinite solid cylinder.

29.



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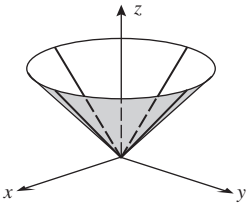

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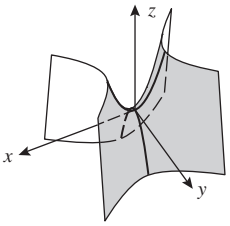

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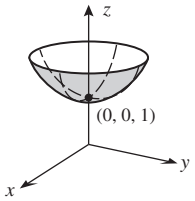

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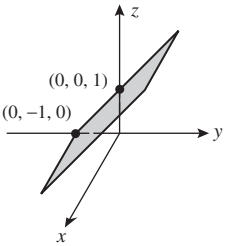
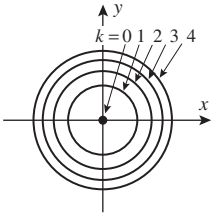


39. **(a)** $1 - x^2 - y^2$
(b) $\sqrt{x^2 + y^2}$
(c) $x^2 + y^2$

41. **(a)** A
(b) B
(c) increase
(d) decrease
(e) increase
(f) decrease

43.



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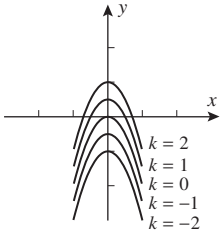

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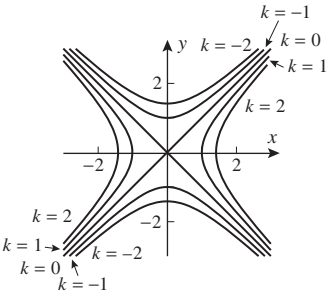

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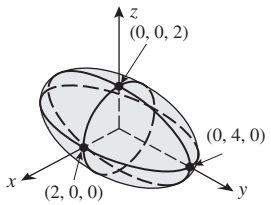

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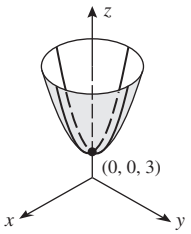


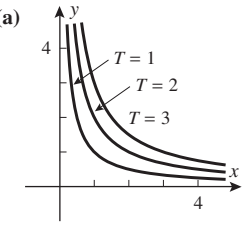
53. concentric spheres, common center at (2, 0, 0)

55. concentric cylinders, common axis the y-axis

57. **(a)** $x^2 - 2x^3 + 3xy = 0$ **(b)** $x^2 - 2x^3 + 3xy = 0$
(c) $x^2 - 2x^3 + 3xy = 14$

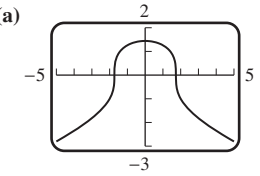
59. **(a)** $x^2 + y^2 - z = 5$ **(b)** $x^2 + y^2 - z = -3$ **(c)** $x^2 + y^2 - z = 0$

61. **(a)**

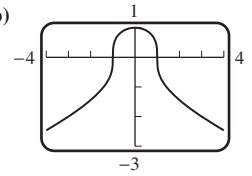


(b) the path $xy = 4$

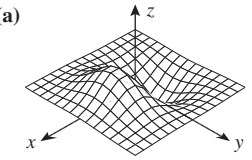
63. **(a)**



(b)



65. **(a)**



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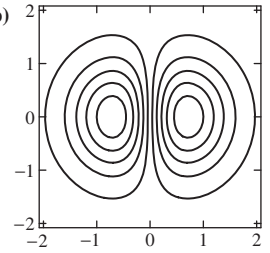


67. **(a)** The graph of g is the graph of f shifted one unit in the negative x-direction.
(b) The graph of g is the graph of f shifted one unit up the z-axis.
(c) The graph of g is the graph of f shifted one unit down the y-axis and then inverted with respect to the plane $z = 0$.

▶ Exercise Set 13.2 (Page 925)

1. 37 **3.** $-\frac{8}{3}$ **5.** 0

7. **(a)** along $x = 0$ limit does not exist
(b) along $x = 0$ limit does not exist

9. $\frac{1}{2}$ **11.** 0 **13.** 0 **15.** limit does not exist **17.** $\frac{48}{5}$ **19.** 0

21. limit does not exist **23.** 0 **25.** 0 **27.** 0

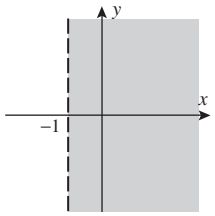
Responses to True–False questions may be abridged to save space.

29. True; by the definition of open set.

31. False; let $f(x, y) = \begin{cases} 1, & x \le 0 \\ -1, & x > 0 \end{cases}$ and let $g(x, y) = -f(x, y)$.

33. **(a)** no **(d)** no; yes **37.** $-\pi/2$ **39.** no

41.



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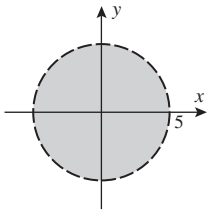

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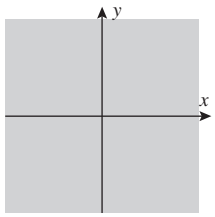

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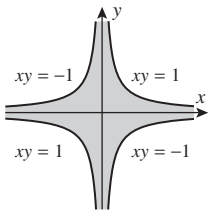


49. all of 3-space

51. all points not on the cylinder $x^2 + z^2 = 2$

Exercise Set 13.3 (Page 936)

1. **(a)** $12x^2y^2$ **(b)** $8x^3y$ **(c)** $12y^2$ **(d)** $12x^2$ **(e)** $8y$ **(f)** $8x^3$ **(g)** 48 **(h)** 16

3. **(a)** $\frac{3}{8}$ **(b)** $\frac{1}{4}$ **5.** **(a)** $-4\cos 11$ **(b)** $2\cos 11$

7. $\partial z/\partial x = -4$; $\partial z/\partial y = \frac{1}{2}$ **9.** **(a)** 4.9 **(b)** 1.2

11. $z = f(x, y)$ has II as its graph, f_x has I as its graph, and f_y has III as its graph.

Responses to True–False questions may be abridged to save space.

13. True; on $y = 2$, $f(x, 2) = c$ is a constant function of x.

15. True; z must be a linear function of x and y.

17. $12xy^3e^{x^2y^3}$, $18x^2y^2e^{x^2y^3}$

19. $x^4/(y^{3/5} + x) + 4x^3\ln(1 + xy^{-3/5})$, $-\frac{3}{5}x^5/(y^{8/5} + xy)$

21. $-\dfrac{y(x^2 - y^2)}{(x^2 + y^2)^2}$, $\dfrac{x(x^2 - y^2)}{(x^2 + y^2)^2}$

23. $(3/2)x^2y(4x - 7)(3x^4y - 7x^3y)^{-1/2}$
$(1/2)x^3(3x - 7)(3x^4y - 7x^3y)^{-1/2}$

25. $\dfrac{y^{-1/2}}{y^2 + x^2}$, $-\dfrac{xy^{-3/2}}{y^2 + x^2} - \dfrac{3}{2}y^{-5/2}\tan^{-1}\left(\dfrac{x}{y}\right)$

27. $-\frac{5}{3}y^2\sec^2 x(y^2\tan x)^{-8/3}$, $-\frac{10}{3}y\tan x(y^2\tan x)^{-8/3}$

29. $-8, -21$ **31.** $1/\sqrt{17}, 8/\sqrt{17}$

33. **(a)** $2xy^4z^3 + y$ **(b)** $4x^2y^3z^3 + x$ **(c)** $3x^2y^4z^2 + 2z$ **(d)** $2y^4z^3 + y$ **(e)** $32z^3 + 1$ **(f)** 438

35. $f_x = 3z/x$, $f_y = z/y$, $f_z = \ln(x^3y\cos z) - z\tan z$

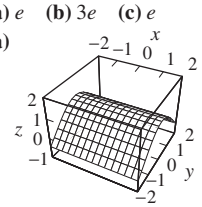
37. $f_x = -y^2z^4/(1 + x^2y^4z^8)$, $f_y = -2xyz^4/(1 + x^2y^4z^8)$, $f_z = -4xy^2z^3/(1 + x^2y^4z^8)$

39. $2yze^z\cos(xz)$, $2e^z\sin(xz)$, $2ye^z(\sin(xz) + 2x\cos(xz))$

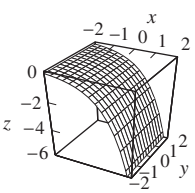
41. $x/\sqrt{x^2 + y^2 + z^2}$, $y/\sqrt{x^2 + y^2 + z^2}$, $z/\sqrt{x^2 + y^2 + z^2}$

43. **(a)** e **(b)** $3e$ **(c)** e

45. **(a)**



(b)



47. 6

49. -2

51. **(a)** $\partial V/\partial r = 2\pi rh$ **(b)** $\partial V/\partial h = \pi r^2$ **(c)** 64π **(d)** 64π

53. **(a)** $\dfrac{1}{4}\dfrac{\text{lb}}{\text{in}^2\cdot\text{K}}$ **(b)** $-2\dfrac{\text{in}^5}{\text{lb}}$

55. **(a)** $\dfrac{\partial V}{\partial l} = 6$ **(b)** $\dfrac{\partial V}{\partial w} = 15$ **(c)** $\dfrac{\partial V}{\partial h} = 10$

59. **(a)** $\pm\sqrt{6}/4$ **61.** $-x/z, -y/z$

63. $-\dfrac{2x + yz^2\cos(xyz)}{xyz\cos(xyz) + \sin(xyz)}$; $-\dfrac{xz^2\cos(xyz)}{xyz\cos(xyz) + \sin(xyz)}$

65. $-x/w, -y/w, -z/w$

67. $-\dfrac{yzw\cos(xyz)}{2w + \sin(xyz)}, -\dfrac{xzw\cos(xyz)}{2w + \sin(xyz)}, -\dfrac{xyw\cos(xyz)}{2w + \cos(xyz)}$

69. $e^{x^2}, -e^{y^2}$

71. $f_x(x, y) = 2xy^3\sin(x^6y^9)$, $f_y(x, y) = 3x^2y^2\sin(x^6y^9)$

73. **(a)** $-\dfrac{\cos y}{4\sqrt{x^3}}$ **(b)** $-\sqrt{x}\cos y$ **(c)** $-\dfrac{1}{2\sqrt{x}}\sin y$ **(d)** $-\dfrac{1}{2\sqrt{x}}\sin y$

75. $-32y^3$ **77.** $-2e^{2x}\sin y$ **79.** $\dfrac{28}{(4x - 7y)^2}$ **81.** $\dfrac{2(x - y)}{(x + y)^3}$

83. **(a)** $\dfrac{\partial^3 f}{\partial x^3}$ **(b)** $\dfrac{\partial^3 f}{\partial y^2\partial x}$ **(c)** $\dfrac{\partial^4 f}{\partial x^2\partial y^2}$ **(d)** $\dfrac{\partial^4 f}{\partial y\partial x^3}$

85. **(a)** $30xy^4 - 4$ **(b)** $60x^2y^3$ **(c)** $60x^3y^2$

87. **(a)** -30 **(b)** -125 **(c)** 150

89. **(a)** $15x^2y^4z^7 + 2y$ **(b)** $35x^3y^4z^6 + 3y^2$ **(c)** $21x^2y^5z^6$ **(d)** $6xy^5z^7$ **(e)** $140x^3y^3z^6 + 6y$ **(f)** $30xy^4z^7$ **(g)** $105x^2y^4z^6$ **(h)** $105x^2y^4z^6$

97. $\dfrac{\partial f}{\partial v} = 8vw^3x^4y^6$, $\dfrac{\partial f}{\partial w} = 12v^2w^2x^4y^6$, $\dfrac{\partial f}{\partial x} = 16v^2w^3x^3y^6$, $\dfrac{\partial f}{\partial y} = 24v^2w^3x^4y^5$

99. $\dfrac{\partial f}{\partial v_1} = \dfrac{2v_1}{v_3^2 + v_4^2}$, $\dfrac{\partial f}{\partial v_2} = \dfrac{-2v_2}{v_3^2 + v_4^2}$, $\dfrac{\partial f}{\partial v_3} = \dfrac{-2v_3(v_1^2 - v_2^2)}{(v_3^2 + v_4^2)^2}$, $\dfrac{\partial f}{\partial v_4} = \dfrac{-2v_4(v_1^2 - v_2^2)}{(v_3^2 + v_4^2)^2}$

101. **(a)** 0 **(b)** 0 **(c)** 0 **(d)** 0 **(e)** $2(1 + yw)e^{yw}\sin z\cos z$ **(f)** $2xw(2 + yw)e^{yw}\sin z\cos z$

103. $-i\sin(x_1 + 2x_2 + \cdots + nx_n)$

105. **(a)** xy-plane, $12x^2 + 6x$ **(b)** $y \neq 0$, $-3x^2/y^2$

107. $f_x(2, -1) = 11$, $f_y(2, -1) = -8$

109. **(b)** does not exist if $y \neq 0$ and $x = -y$

Exercise Set 13.4 (Page 947)

1. 5.05 **3.** 4.14 **9.** $dz = 9\,dx - 2\,dy$ **11.** $dz = 5x^4y^2\,dx + 2x^5y\,dy$

13. $dz = \dfrac{y}{1 + x^2y^2}\,dx + \dfrac{x}{1 + x^2y^2}\,dy$ **15.** $dw = 8\,dx - 5\,dy + 4\,dz$

17. $dw = 3x^2y^2z^2\,dx + 2x^3yz^2\,dy + 2x^3y^2z\,dz$

19. $dw = \dfrac{yz}{1 + x^2y^2z^2}\,dx + \dfrac{xz}{1 + x^2y^2z^2}\,dy + \dfrac{xy}{1 + x^2y^2z^2}\,dz$

21. $df = 0.08$, $\Delta f = 0.0807$ **23.** $df = 0.03$, $\Delta f \approx 0.029412$

25. $df = 0.96$, $\Delta f \approx 0.97929$

Responses to True–False questions may be abridged to save space.

27. False; see the discussion at the beginning of this section.

29. True; see Theorems 13.4.3 and 13.4.4.

31. The increase in the area of the rectangle is given by the sum of the areas of the three small rectangles, and the total differential is given by the sum of the areas of the upper left and lower right rectangles.

33. **(a)** $L = \frac{1}{5} - \frac{4}{125}(x - 4) - \frac{3}{125}(y - 3)$ **(b)** 0.000176603

35. **(a)** $L = 0$ **(b)** 0.0024

37. **(a)** $L = 6 + 6(x - 1) + 3(y - 2) + 2(z - 3)$ **(b)** -0.000481

39. **(a)** $L = e + e(x - 1) - e(y + 1) - e(z + 1)$ **(b)** 0.01554

45. 0.5 **47.** $1, 1, -1, 2$ **49.** $(1, 2)$ **51.** $(1, 1, 1)$ **53.** 7%

55. $r\%$ **57.** 0.3%

59. **(a)** $(r + s)\%$ **(b)** $(r + s)\%$ **(c)** $(2r + 3s)\%$ **(d)** $\left(3r + \dfrac{s}{2}\right)\%$

61. $\approx 39\ \text{ft}^2$

Exercise Set 13.5 (Page 956)

1. $51t^{16}$ **3.** $3t^{-2}\sin(1/t)$ **5.** $-\frac{17}{4}t^{13/4}e^{1-t^{17/4}}$ **7.** $\dfrac{dw}{dt} = 145t^{28}$

9. $-\cos t$ **11.** 11.2688 **13.** 0

17. $24u^2v^2 + 16uv^3 - 2v + 3$, $16u^3v + 24u^2v^2 - 2u + 3$

19. $-\dfrac{2\sin u}{5\sin v}, -\dfrac{2\cos u\cos v}{5\sin^2 v}$ **21.** $e^u, 0$

23. $3r^2\sin\theta\cos^2\theta - 4r^3\sin^3\theta\cos\theta,$
$-2r^3\sin^2\theta\cos\theta + r^4\sin^4\theta + r^3\cos^3\theta - 3r^4\sin^2\theta\cos^2\theta$

25. $\dfrac{x^2+y^2}{4x^2y}, \dfrac{-(x^2+y^2)}{4xy^2}$ **27.** $\dfrac{\partial z}{\partial r} = \dfrac{2r\sin^2\theta}{r^2\sin^2\theta+1}, \dfrac{\partial z}{\partial\theta} = \dfrac{2r^2\sin\theta\cos\theta}{r^2\sin^2\theta+1}$

29. $\dfrac{dw}{d\rho} = 2\rho(4\sin^2\phi+\cos^2\phi), \dfrac{\partial w}{\partial\phi} = 6\rho^2\sin\phi\cos\phi, \dfrac{dw}{d\theta} = 0$

31. $-\pi$ **33.** $\sqrt{3}e^{\sqrt{3}}, (2-4\sqrt{3})e^{\sqrt{3}}$

Responses to True–False questions may be abridged to save space.

35. False; the symbols ∂z and ∂x have no individual meaning.

37. False; consider $z = xy, x = t, y = t$.

39. $-\dfrac{2xy^3}{3x^2y^2+\cos y}$

41. $-\dfrac{ye^{xy}}{xe^{xy}+ye^y+e^y}$ **43.** $\dfrac{2x+yz}{6yz-xy}, \dfrac{xz-3z^2}{6yz-xy}$

45. $\dfrac{ye^x}{15\cos 3z+4}, \dfrac{e^x}{15\cos 3z+4}$

59. $\dfrac{\partial w}{\partial\rho} = (\sin\phi\cos\theta)\dfrac{\partial w}{\partial x} + (\sin\phi\sin\theta)\dfrac{\partial w}{\partial y} + (\cos\phi)\dfrac{\partial w}{\partial z},$
$\dfrac{\partial w}{\partial\phi} = (\rho\cos\phi\cos\theta)\dfrac{\partial w}{\partial x} + (\rho\cos\phi\sin\theta)\dfrac{\partial w}{\partial y} - (\rho\sin\phi)\dfrac{\partial w}{\partial z},$
$\dfrac{\partial w}{\partial\theta} = -(\rho\sin\phi\sin\theta)\dfrac{\partial w}{\partial x} + (\rho\sin\phi\cos\theta)\dfrac{\partial w}{\partial y}$

63. **(a)** $\dfrac{dw}{dt} = \displaystyle\sum_{i=1}^{4}\dfrac{\partial w}{\partial x_i}\dfrac{dx_i}{dt}$ **(b)** $\dfrac{\partial w}{\partial v_j} = \displaystyle\sum_{i=1}^{4}\dfrac{\partial w}{\partial x_i}\dfrac{\partial x_i}{\partial v_j}, j = 1, 2, 3$

Exercise Set 13.6 (Page 968)

1. $\frac{81}{4}\sqrt{2}$ **3.** 0 **5.** $-\frac{704}{3}$ **7.** $-\frac{410}{949}$ **9.** 0 **11.** $-\frac{25}{\sqrt{2}}$

13. $\sqrt{2}/2$ **15.** $72/\sqrt{14}$ **17.** $-8/63$ **19.** $1/2+\sqrt{3}/8$ **21.** $2\sqrt{2}$

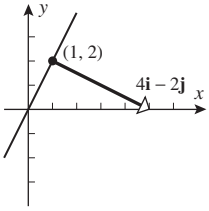
23. $1/\sqrt{5}$ **25.** $-\frac{3}{2}e$ **27.** $3/\sqrt{11}$ **29.** **(a)** 5 **(b)** 10 **(c)** $-5\sqrt{5}$

31. III **33.** $4\mathbf{i}-10\mathbf{j}$

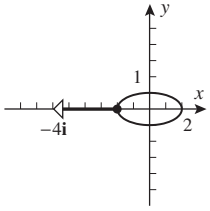
35. $\nabla w = \dfrac{x}{x^2+y^2+z^2}\mathbf{i} + \dfrac{y}{x^2+y^2+z^2}\mathbf{j} + \dfrac{z}{x^2+y^2+z^2}\mathbf{k}$

37. $-540\mathbf{i}-216\mathbf{j}$ **39.** $5(\mathbf{i}+\mathbf{j}+\mathbf{k})$

41.



43.



45. $\pm(-4\mathbf{i}+\mathbf{j})/\sqrt{17}$ **47.** $\mathbf{u} = (3\mathbf{i}-\mathbf{j})/\sqrt{10}, \|\nabla f(-1,1)\| = 4\sqrt{10}$

49. $\mathbf{u} = (4\mathbf{i}+3\mathbf{j})/5, \|\nabla f(4,-3)\| = 1$ **51.** $\dfrac{1}{\sqrt{2}}(\mathbf{i}-\mathbf{j}), 3\sqrt{2}$

53. $\dfrac{1}{\sqrt{2}}(-\mathbf{i}+\mathbf{j}), \dfrac{1}{\sqrt{2}}$

55. $\mathbf{u} = (-\mathbf{i}+3\mathbf{j})/\sqrt{10}, -\|\nabla f(-1,3)\| = -2\sqrt{10}$

57. $\mathbf{u} = (3\mathbf{i}-\mathbf{j})/\sqrt{10}, -\|\nabla f(\pi/6,\pi/4)\| = -\sqrt{5}$

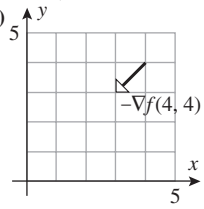
59. $(\mathbf{i}-11\mathbf{j}+12\mathbf{k})/\sqrt{266}, -\sqrt{266}$

Responses to True–False questions may be abridged to save space.

61. False; they are equal. **63.** False; let $\mathbf{u} = \mathbf{i}$ and let $f(x, y) = y$.

65. $8/\sqrt{29}$

67. **(a)** $\approx 1/\sqrt{2}$
(b)



69. $16x^2+y^2 = 16$

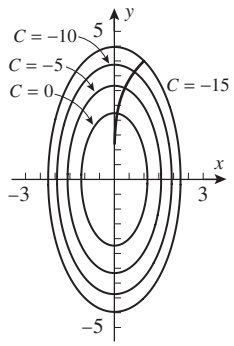
71. $36/\sqrt{17}$

73. **(a)** $2e^{-\pi/2}\mathbf{i}$

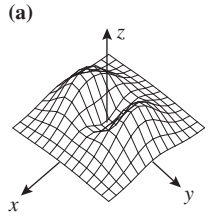
75. $-\frac{5}{3}(2\mathbf{i}-\mathbf{j}-2\mathbf{k})$

81. $x(t) = e^{-8t}, y(t) = 4e^{-2t}$

83.



85. **(a)**



(c) $\nabla f = [2x - 2x(x^2+3y^2)]e^{-(x^2+y^2)}\mathbf{i} + [6y - 2y(x^2+3y^2)]e^{-(x^2+y^2)}\mathbf{j}$
(d) $x = y = 0$ or $x = 0, y = \pm 1$ or $x = \pm 1, y = 0$

Exercise Set 13.7 (Page 975)

1. **(a)** $x+y+2z = 6$ **(b)** $x = 2+t, y = 2+t, z = 1+2t$
(c) 35.26°

3. tangent plane: $3x+4z = 25$;
normal line: $x = 3-(3t/4), y = 0, z = 4-t$

5. tangent plane: $48x-14y-z = 66$;
normal line: $x = 1+48t, y = -2-14t, z = 10-t$

7. tangent plane: $x-y-ez = -e$;
normal line: $x = 1+\frac{1}{e}t, y = 1-\frac{1}{e}t, z = 1-t$

9. tangent plane: $3y-z = -3$;
normal line: $x = \pi/6, y = 3t, z = 3-t$

11. **(a)** all points on the x-axis or y-axis **(b)** $(0, -2, -4)$

13. $(-1, -2, -3)$ **15.** **(a)** $(-2, 1, 5), (0, 3, 9)$ **(b)** $\dfrac{4}{3\sqrt{14}}, \dfrac{4}{\sqrt{222}}$

Responses to True–False questions may be abridged to save space.

17. False; they need only be parallel.

19. True; see Formula (15) of Section 13.4.

21. $\pm\dfrac{1}{\sqrt{1218}}(\mathbf{i}-16\mathbf{j}-31\mathbf{k})$ **25.** $(1, 2/3, 2/3), (-1, -2/3, -2/3)$

27. $x = 1+8t, y = -1+5t, z = 2+6t$

29. $x = 3+4t, y = -3-4t, z = 4-3t$

Exercise Set 13.8 (Page 985)

1. **(a)** minimum at $(2, -2)$, no maxima
(b) maximum at $(0, 0)$, no minima **(c)** no maxima or minima

3. minimum at $(3, -1)$, no maxima **5.** relative minimum at $(0, 0)$

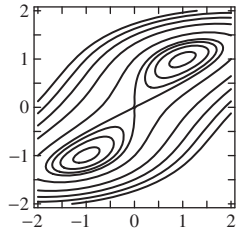
7. relative minimum at $(0, 0)$; saddle points at $(\pm 2, 1)$

9. saddle point at $(0, -2)$ **11.** relative minimum at $(4, -2)$

13. relative minima at $(-2, -2)$ and $(2, 2)$ **15.** saddle point at $(-1, 0)$

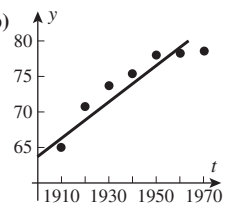
17. no critical points **19.** relative maximum at $(-2, 0)$

21. saddle point at $(0, 0)$; relative minima at $(1, 1)$ and $(-1, -1)$

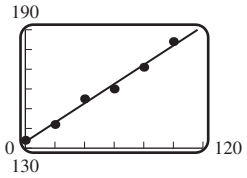


Responses to True–False questions may be abridged to save space.

23. False; let $f(x, y) = y$.
25. True; this follows from Theorem 13.8.6.
27. **(b)** relative minimum at $(0, 0)$
31. absolute maximum 0,
absolute minimum -12
33. absolute maximum 3,
absolute minimum -1
35. absolute maximum $\frac{33}{4}$,
absolute minimum $-\frac{1}{4}$
37. 12, 12, 12
39. maximum at $(1, 2, 2)$
41. $2a/\sqrt{3}, 2a/\sqrt{3}, 2a/\sqrt{3}$
43. length and width 2 ft, height 4 ft
45. **(a)** $x = 0$: minimum -3, maximum 0;
$x = 1$: minimum 3, maximum $13/3$;
$y = 0$: minimum 0, maximum 4;
$y = 1$: minimum -3, maximum 3
(b) $y = x$: minimum 0, maximum 3;
$y = 1 - x$: maximum 4, minimum -3
(c) minimum -3, maximum $13/3$
47. length and width $\sqrt[3]{2V}$, height $\sqrt[3]{2V}/2$ **51.** $y = \frac{3}{4}x + \frac{19}{12}$
53. $y = 0.5x + 0.8$
55. **(a)** $y = 63.73 + 0.2565t$ **(b)**
(c) about 84 years

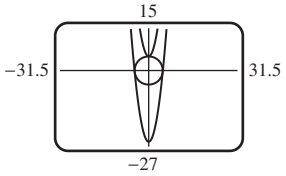


57. **(a)** $P = \dfrac{2798}{21} + \dfrac{171}{350}T$ **(b)**
(c) $T \approx -272.7096°\text{C}$



Exercise Set 13.9 (Page 996)

1. **(a)** 4 **3.** **(a)**

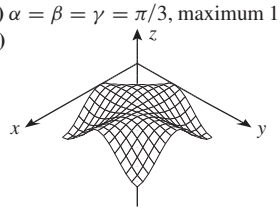


(c) maximum $\frac{101}{4}$,
minimum -5

5. maximum $4\sqrt{2}$ at $(-2\sqrt{2}, -2)$ and $(2\sqrt{2}, 2)$,
minimum $-4\sqrt{2}$ at $(-2\sqrt{2}, 2)$ and $(2\sqrt{2}, -2)$
7. maximum $\sqrt{2}$ at $(1/\sqrt{2}, 0)$, minimum $-\sqrt{2}$ at $(-1/\sqrt{2}, 0)$
9. maximum 6 at $\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$, minimum -6 at $\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$
11. maximum is $1/(3\sqrt{3})$ at $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$,
$(1/\sqrt{3}, -1/\sqrt{3}, -1/\sqrt{3})$, $(-1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3})$, and
$(-1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3})$; minimum is $-1/(3\sqrt{3})$ at
$(1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3})$, $(1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3})$,
$(-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$, and $(-1/\sqrt{3}, -1/\sqrt{3}, -1/\sqrt{3})$

Responses to True–False questions may be abridged to save space.

13. False; a Lagrange multiplier is a scalar.
15. False; we must solve three equations in three unknowns.
17. $\left(\frac{2}{5}, -\frac{4}{5}\right)$ **19.** $\left(\frac{1}{3}, \frac{2}{3}, \frac{1}{3}\right)$
21. $(3, 6)$ is closest and $(-3, -6)$ is farthest **23.** $5(\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$
25. 9, 9, 9 **27.** $(\pm\sqrt{5}, 0, 0)$ **29.** length and width 2 ft, height 4 ft
33. **(a)** $\alpha = \beta = \gamma = \pi/3$, maximum $1/8$
(b)



Chapter 13 Review Exercises (Page 997)

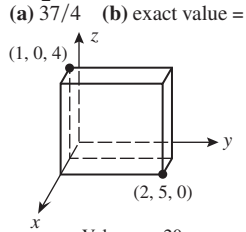
1. **(a)** xy **(b)** $e^{rs}\ln(r + s)$
5. **(a)** not defined on line $y = x$ **(b)** not continuous
9. **(a)** 10 Pa/min **(b)** $500/3$ Pa/min
15. df (the differential of f) is an approximation for Δf (the change in f)
17. $dV = -0.2\ \text{m}^3$; $\Delta V = -0.1973333\ \text{m}^3$ **19.** 2
21. $\dfrac{-f_y^2 f_{xx} + 2f_x f_y f_{xy} - f_x^2 f_{yz}}{f_y^3}$ **25.** $\dfrac{7}{2} + \dfrac{4}{5}\ln 2$ **27.** $-7/\sqrt{5}$
29. $(0, 0, 2)$, $(1, 1, 1)$, $(-1, -1, 1)$ **31.** $\left(-\frac{1}{2}, -1, \frac{25}{4}\right)$
33. relative minimum at $(15, -8)$
35. saddle point at $(0, 0)$, relative minimum at $(1, 1)$
37. absolute maximum of 4 at $(\pm 1, \pm 2)$, absolute minimum of 0 at
$(\pm\sqrt{2}, 0)$ and $(0, \pm 2\sqrt{2})$
39. $I_1 : I_2 : I_3 = \dfrac{1}{R_1} : \dfrac{1}{R_2} : \dfrac{1}{R_3}$
41. **(a)** $\partial P/\partial L = c\alpha L^{\alpha-1}K^{\beta}$, $\partial P/\partial K = c\beta L^{\alpha}K^{\beta-1}$

Chapter 13 Making Connections (Page 999)

Answers are provided in the Student Solutions Manual.

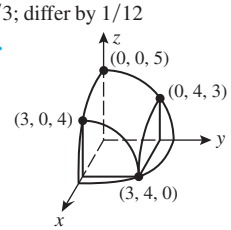
Exercise Set 14.1 (Page 1007)

1. 11 **3.** 16 **5.** 3 **7.** 4 **9.** $2 - \ln 3$
11. $\dfrac{1 - \ln 2}{2}$ **13.** 0 **15.** $\frac{1}{3}$
17. **(a)** $37/4$ **(b)** exact value $= 28/3$; differ by $1/12$
19.



Volume = 20

21.



Responses to True–False questions may be abridged to save space.

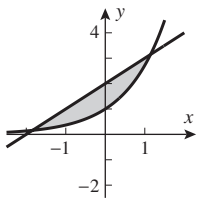
23. False; ΔA_k is the area of such a rectangular region.
25. False; $\displaystyle\iint_R f(x, y)\,dA = \int_1^5 \int_2^4 f(x, y)\,dy\,dx$.
29. $\frac{63}{2}$ **31.** 16 **33.** $\dfrac{1}{3\pi}$ **35.** $1 - \dfrac{2}{\pi}$ **37.** $\frac{14}{3}$ °C **39.** 1.381737122
41. first integral equals $\frac{1}{2}$, second equals $-\frac{1}{2}$; no

Exercise Set 14.2 (Page 1015)

1. $\frac{2}{15}$ **3.** 18 **5.** $\dfrac{\pi}{2}$ **7.** $\frac{4}{3}$
9. **(a)** $\displaystyle\int_0^2 \int_0^{x^2} f(x, y)\,dy\,dx$ **(b)** $\displaystyle\int_0^4 \int_{\sqrt{y}}^2 f(x, y)\,dx\,dy$
11. **(a)** $\displaystyle\int_1^2 \int_{-2x+5}^3 f(x, y)\,dy\,dx + \int_2^4 \int_1^3 f(x, y)\,dy\,dx +$
$\displaystyle\int_4^5 \int_{2x-7}^3 f(x, y)\,dy\,dx$ **(b)** $\displaystyle\int_1^3 \int_{(5-y)/2}^{(y+7)/2} f(x, y)\,dx\,dy$
13. **(a)** $\frac{16}{3}$ **(b)** 38 **15.** 576 **17.** 0 **19.** $\dfrac{\sqrt{26} - 1}{2}$ **21.** $\frac{50}{3}$

23. $-\frac{2}{15}$ **25.** $\dfrac{1-\cos 8}{3}$

27. **(a)**



(b) $(-1.8414, 0.1586)$, $(1.1462, 3.1462)$
(c) -0.4044
(d) -0.4044

29. $\sqrt{2}-1$ **31.** 32

Responses to True–False questions may be abridged to save space.

33. False; $\int_0^1\int_{x^2}^{2x} f(x,y)\,dy\,dx$ integrates $f(x,y)$ over the region between the graphs of $y=x^2$ and $y=2x$ for $0\le x\le 1$ and results in a number, but $\int_{x^2}^{2x}\int_0^1 f(x,y)\,dx\,dy$ produces an expression involving x.

35. False; although R is symmetric across the x-axis, the integrand may not be.

37. 12 **39.** 64π **41.** 170 **43.** 18π **45.** 8π

47. $\int_0^{\sqrt{2}}\int_{y^2}^{2} f(x,y)\,dx\,dy$ **49.** $\int_1^{e^3}\int_{\ln x}^{3} f(x,y)\,dy\,dx$

51. $\int_0^{\pi/2}\int_0^{\sin x} f(x,y)\,dy\,dx$ **53.** $\dfrac{1-e^{-16}}{8}$ **55.** $\dfrac{e^8-1}{3}$

57. **(a)** 0 **(b)** $\tan 1$ **59.** 0 **61.** $\dfrac{\pi}{4}-\dfrac{\ln 2}{2}$ **63.** $\frac{2}{3}$°C **65.** 0.676089

▶ Exercise Set 14.3 (Page 1024)

1. $\frac{\sqrt{2}}{24}$ **3.** $\frac{4}{9}a^3$ **5.** $\frac{1}{12}$ **7.** $\dfrac{3\pi}{2}$ **9.** $\dfrac{7\pi}{16}$ **11.** $\int_{\pi/6}^{5\pi/6}\int_2^{4\sin\theta} f(r,\theta)r\,dr\,d\theta$

13. $8\int_0^{\pi/2}\int_1^3 r\sqrt{9-r^2}\,dr\,d\theta$ **15.** $2\int_0^{\pi/2}\int_0^{\cos\theta}(1-r^2)r\,dr\,d\theta$

17. $\dfrac{64\sqrt{2}}{3}\pi$ **19.** $\dfrac{5\pi}{32}$ **21.** $\dfrac{27\pi}{16}$ **23.** $(1-e^{-1})\pi$ **25.** $\dfrac{\pi}{8}\ln 10$ **27.** 2π

29. $\frac{16}{9}$ **31.** $\dfrac{\pi}{2}\left(1-\dfrac{1}{\sqrt{1+a^2}}\right)$ **33.** $\dfrac{\pi}{4}(\sqrt{5}-1)$

Responses to True–False questions may be abridged to save space.

35. True; the disk is given in polar coordinates by $0\le r\le 2, 0\le\theta\le 2\pi$.

37. False; the integrand is missing a factor of r:

$$\iint_R f(r,\theta)\,dA=\int_0^{\pi/2}\int_1^2 f(r,\theta)r\,dr\,d\theta.$$

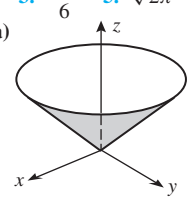
39. $\pi a^2 h$ **41.** $\dfrac{1}{5}+\dfrac{\pi}{2}$

43. **(a)** $\dfrac{4}{3}\pi a^2 c$ **(b)** $\approx 1.0831682\times 10^{21}\ \text{m}^3$ **45.** $2a^2$

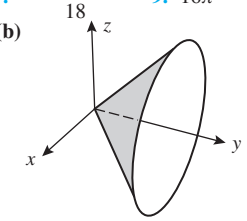
▶ Exercise Set 14.4 (Page 1036)

1. 3π **3.** $\dfrac{\sqrt{7}}{6}$ **5.** $\sqrt{2}\pi$ **7.** $\dfrac{(5\sqrt{5}-1)\pi}{18}$ **9.** 16π

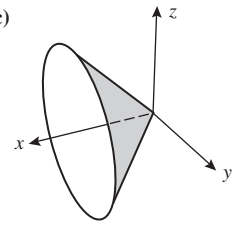
11. **(a)**



(b)



(c)



13. **(a)** $x=u, y=v, z=3+\frac{3}{2}u-2v$ **(b)** $x=u, y=v, z=u^2$

15. **(a)** $x=\sqrt{7}\cos u, y=\sqrt{7}\sin u, z=v; 0\le u\le 2\pi, 0\le v\le 1$
(b) $x=2\cos u, y=v, z=2\sin u; 0\le u\le 2\pi, 1\le v\le 5$

17. $x=u, y=\sin u\cos v, z=\sin u\sin v$

19. $x=r\cos\theta, y=r\sin\theta, z=\dfrac{1}{3+r^2}$

21. $x=r\cos\theta, y=r\sin\theta, z=2r^3\cos\theta\sin^2\theta$

23. $x=r\cos\theta, y=r\sin\theta, z=\sqrt{9-r^2}; r\le 2\sqrt{2}$

25. $x=\dfrac{1}{2}\rho\cos\theta, y=\dfrac{1}{2}\rho\sin\theta, z=\dfrac{\sqrt{3}}{2}\rho$ **27.** $z=x-y$; a plane

29. $(x/3)^2+(y/2)^2=1; 2\le z\le 4$; part of an elliptic cylinder

31. $(x/2)^2+(y/5)^2=z^2; 0\le z\le 1$; part of an elliptic cone

33. **(a)** $x=r\cos\theta, y=r\sin\theta, z=r, 0\le r\le 2$;
$x=u, y=v, z=\sqrt{u^2+v^2}, 0\le u^2+v^2\le 4$

35. **(a)** $0\le u\le 3, 0\le v\le\pi$ **(b)** $0\le u\le 4, -\pi/2\le v\le\pi/2$

37. **(a)** $0\le\phi\le\pi/2, 0\le\theta\le 2\pi$ **(b)** $0\le\phi\le\pi, 0\le\theta\le\pi$

39. $2x+4y-z=5$ **41.** $z=0$ **43.** $x-y+\dfrac{\sqrt{2}}{2}z=\dfrac{\pi\sqrt{2}}{8}$

45. $\dfrac{(17\sqrt{17}-5\sqrt{5})\pi}{6}$

Responses to True–False questions may be abridged to save space.

47. False; the surface area is $S=\iint_R\sqrt{[f_x(x,y)]^2+[f_y(x,y)]^2+1}\,dA$.

49. True; see the discussion preceding Definition 14.4.1.

51. $4\pi a^2$ **55.** $4\pi^2 ab$ **57.** 9.099

59. $(x/a)^2+(y/b)^2+(z/c)^2=1$; ellipsoid

61. $(x/a)^2+(y/b)^2-(z/c)^2=-1$; hyperboloid of two sheets

▶ Exercise Set 14.5 (Page 1045)

1. 16 **3.** $\frac{1}{24}$ **5.** $\frac{81}{5}$ **7.** $-\frac{32}{15}$ **9.** $\pi(\pi-3)/8$ **11.** $\frac{27}{32}$ **13.** 4.712

15. 4 **17.** $\frac{256}{15}$

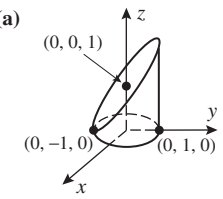
19. **(a)** $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{4x^2+y^2}^{4-3y^2} f(x,y,z)\,dz\,dy\,dx$

(b) $\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4x^2+y^2}^{4-3y^2} f(x,y,z)\,dz\,dx\,dy$

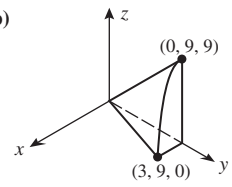
21. $4\int_0^1\int_0^{\sqrt{1-x^2}}\int_{4x^2+y^2}^{4-3y^2} dz\,dy\,dx$

23. $2\int_{-3}^{3}\int_0^{\frac{1}{3}\sqrt{9-x^2}}\int_0^{x+3} dz\,dy\,dx$

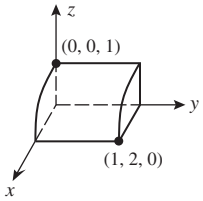
25. **(a)**



(b)



(c)



Responses to True–False questions may be abridged to save space.

27. True; apply Fubini's Theorem (Theorem 14.5.1).

29. False;

$$\iiint_G f(x, y, z)\, dV = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} f(x, y, z)\, dz\, dy\, dx.$$

33. $\frac{3}{4}$ **35.** 3.291

37. **(a)** $\int_0^a \int_0^{b(1-x/a)} \int_0^{c(1-x/a-y/z)} dz\, dy\, dx$ is one example.

39. **(a)** $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^6 f(x, y, z)\, dz\, dy\, dx$

(b) $\int_0^9 \int_0^{3-\sqrt{x}} \int_y^{3-\sqrt{x}} f(x, y, z)\, dz\, dy\, dx$

(c) $\int_0^2 \int_0^{4-x^2} \int_y^{8-y} f(x, y, z)\, dz\, dy\, dx$

Exercise Set 14.6 (Page 1056)

1. -2π **3.** $\dfrac{\pi}{8}$

5. The region is bounded by the xy-plane and the upper half of a sphere of radius 1 centered at the origin; $f(r, \theta, z) = z$.

7. The region is the portion of the first octant inside a sphere of radius 1 centered at the origin; $f(\rho, \theta, \phi) = \rho \cos \phi$.

9. 128π **11.** $\frac{152}{3}\pi + \frac{80}{3}\pi\sqrt{5}$ **13.** $\dfrac{64\pi}{3}(2 - \sqrt{3})$ **15.** $\dfrac{11\pi a^3}{3}$

17. $\dfrac{\pi a^6}{48}$ **19.** $\dfrac{32(2\sqrt{2} - 1)\pi}{15}$

Responses to True–False questions may be abridged to save space.

21. False; the factor r^2 should be r [Formula (6)]:

$$\iiint_G f(x, y, z)\, dV = \iiint_{\substack{\text{appropriate}\\ \text{limits}}} f(r\cos\theta, r\sin\theta, z) r\, dz\, dr\, d\theta.$$

23. True; G is the spherical wedge bounded by the spheres $\rho = 1$ and $\rho = 3$, the half-planes $\theta = 0$ and $\theta = 2\pi$, and above the cone $\phi = \pi/4$, so

$$(\text{volume of } G) = \iiint_G dV = \int_0^{\pi/4} \int_0^{2\pi} \int_1^3 \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi.$$

25. **(a)** $\frac{5}{2}(-8 + 3\ln 3)\ln(\sqrt{5} - 2)$ **(b)** $f(x, y, z) = \dfrac{y^3}{x^3\sqrt{1+z^2}}$; G is the cylindrical wedge $1 \le r \le 4$, $\dfrac{\pi}{6} \le \theta \le \dfrac{\pi}{3}$, $-2 \le z \le 2$

27. $\dfrac{4\pi a^3}{3}$ **29.** $\dfrac{2\pi}{3}(\sqrt{2} - 1)$

Exercise Set 14.7 (Page 1068)

1. -20 **3.** $\cos(u - v)$ **5.** $x = \frac{6}{17}u + \frac{5}{17}v$, $y = -\frac{1}{17}u + \frac{2}{17}v$; $\frac{1}{17}$

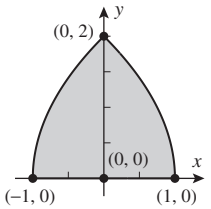
7. $x = \dfrac{\sqrt{u+v}}{\sqrt{2}}$, $y = \dfrac{\sqrt{v-u}}{\sqrt{2}}$; $\dfrac{1}{4\sqrt{v^2 - u^2}}$ **9.** 7 **11.** $-\dfrac{1}{v}$

Responses to True–False questions may be abridged to save space.

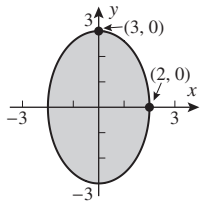
13. False; $|\partial(x, y)/\partial(u, v)| = \|\partial\mathbf{r}/\partial u \times \partial\mathbf{r}/\partial v\|$; evaluating this at (u_0, v_0) gives the area of the indicated parallelogram.

15. False; $\partial(x, y)/\partial(r, \theta) = r$.

17.



19.



21. $\frac{3}{2}\ln 3$ **23.** $1 - \frac{1}{2}\sin 2$ **25.** $\frac{800}{3}\pi$ **27.** $\dfrac{\pi}{24}(1 - \cos 1)$ **29.** $\frac{192}{5}\pi$

31. $u = \begin{cases} \cot^{-1}(x/y), & y \ne 0 \\ 0, & y = 0 \text{ and } x > 0 \\ \pi, & y = 0 \text{ and } x < 0 \end{cases}$

$v = \sqrt{x^2 + y^2}$; other answers possible

33. $u = (3/7)x - (2/7)y$, $v = (-1/7)x + (3/7)y$; other answers possible

35. $\frac{1}{4}\ln\frac{5}{2}$

37. $\dfrac{1}{2}\left[\ln(\sqrt{2} + 1) - \dfrac{\pi}{4}\right]$ **39.** $\frac{7}{50}$ **41.** $2\ln 2$ **45.** $21/8$

Exercise Set 14.8 (Page 1077)

1. $M = \frac{9}{10}$, center of gravity $\left(\frac{130}{189}, \frac{13}{27}\right)$

3. $M = a^4/8$, center of gravity $(8a/15, 8a/15)$

5. $\left(\frac{1}{2}, \frac{1}{2}\right)$ **7.** $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Responses to True–False questions may be abridged to save space.

9. True; recall this from Section 6.7.

11. False; the center of gravity of the lamina is $(\bar{x}, \bar{y}) = (M_y/M, M_x/M)$, where M_y and M_x are the lamina's first moments about the y- and x-axes, respectively, and M is the mass of the lamina.

15. $\left(\dfrac{128}{105\pi}, \dfrac{128}{105\pi}\right)$ **17.** $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ **19.** $\left(\frac{1}{2}, 0, \frac{12}{5}\right)$

21. $(3a/8, 3a/8, 3a/8)$

23. $M = b^4/2$, center of gravity $(b/3, b/2, b/2)$

25. $M = \frac{1}{3}$, center of gravity $\left(0, \frac{16}{35}, \frac{1}{2}\right)$ **27.** **(a)** $\left(\frac{5}{8}, \frac{5}{8}\right)$ **(b)** $\left(\frac{2}{3}, \frac{1}{2}\right)$

29. (1.177406, 0.353554, 0.231557)

31. $\dfrac{27\pi}{4}$ **33.** $\pi k a^4$ **35.** $\left(0, 0, \frac{7}{16\sqrt{2} - 14}\right)$ **37.** $(3a/8, 3a/8, 3a/8)$

39. $4\pi(2 - \sqrt{2})$ **41.** $(0, 0, 8/15)$ **43.** $(0, 45/28, 0)$

47. $\frac{1}{2}\delta\pi a^4 h$ **49.** $\frac{1}{2}\delta\pi h(a_1^4 - a_1^4)$ **53.** $2\pi^2 abk$ **55.** $(a/3, b/3)$

Chapter 14 Review Exercises (Page 1080)

3. **(a)** $\iint_R dA$ **(b)** $\iiint_G dV$ **(c)** $\iint_R \sqrt{1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2}\, dA$

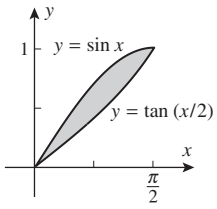
5. $\int_0^1 \int_{1-\sqrt{1-y^2}}^{1+\sqrt{1-y^2}} f(x, y)\, dx\, dy$

7. **(a)** $a = 2, b = 1, c = 1, d = 2$ or $a = 1, b = 2, c = 2, d = 1$ **(b)** 3

9. $-\dfrac{3}{2\sqrt{2}\pi}$

11. $\int_0^1 \int_{2y}^2 e^x e^y\, dx\, dy$

13.



15. $\frac{1}{3}(1 - \cos 64)$

17. a^2

19. $\frac{3}{2}$

21. 32π

23. **(a)** $\int_0^{2\pi} \int_0^{\pi/3} \int_0^a \rho^4 \sin^3\phi\, d\rho\, d\phi\, d\theta$

(b) $\int_0^{2\pi} \int_0^{\sqrt{3}a/2} \int_{r/\sqrt{3}}^{\sqrt{a^2-r^2}} r^3\, dz\, dr\, d\theta$

(c) $\int_{-\sqrt{3}a/2}^{\sqrt{3}a/2} \int_{-\sqrt{(3a^2/4)-x^2}}^{\sqrt{(3a^2/4)-x^2}} \int_{\sqrt{x^2+y^2}/\sqrt{3}}^{\sqrt{a^2-x^2-y^2}} (x^2 + y^2)\, dz\, dy\, dx$

25. $\dfrac{\pi a^3}{9}$ **27.** $\dfrac{13}{6}\sqrt{2} \approx 3.0641$ **29.** $2x + 6y - z = 10$

33. **(a)** $\dfrac{1}{2(u+w)}$ **(b)** $\frac{1}{2}(7\ln 7 - \ln 84,375)$ **35.** $\left(\frac{8}{5}, 0\right)$
37. $(0, 0, h/4)$

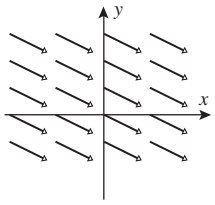
▶ Chapter 14 Making Connections (Page 1082)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

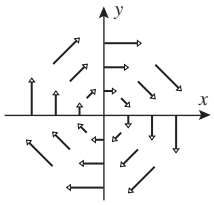
1. **(b)** $\dfrac{\pi}{4}$ **3.** **(a)** 1.173108605 **(b)** 1.173108605
4. **(a)** the sphere $0 \le x^2 + y^2 + z^2 \le 1$ **(b)** 4.934802202 **(c)** $\pi^2/2$
5. **(b)** 4.4506 **6.** $\frac{4}{35}\pi a^3$

▶ Exercise Set 15.1 (Page 1092)

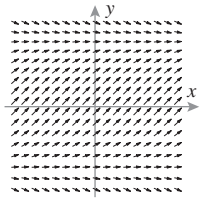
1. **(a)** III **(b)** IV **3.** **(a)** true **(b)** true **(c)** true
5.



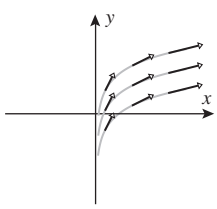
7.



9.



Responses to True–False questions may be abridged to save space.
11. False; the vector field has a nonzero **k**-component.
13. True; this is the curl of **F**.
15. **(a)** all x, y **(b)** all x, y **17.** div $\mathbf{F} = 2x + 2yz$, curl $\mathbf{F} = z^2\mathbf{i}$
19. div $\mathbf{F} = 0$, curl $\mathbf{F} = (40x^2z^4 - 16xy^3)\mathbf{i} + (14y^3z + 4y^4)\mathbf{j} - (16xz^5 + 21y^2z^2)\mathbf{k}$
21. div $\mathbf{F} = \dfrac{2}{\sqrt{x^2+y^2+z^2}}$, curl $\mathbf{F} = 0$ **23.** $5x$ **25.** 0
27. $\mathbf{i} + z\mathbf{j} + y\mathbf{k}$
39. $\nabla\cdot(k\mathbf{F}) = k\nabla\cdot\mathbf{F}$, $\nabla\cdot(\mathbf{F}+\mathbf{G}) = \nabla\cdot\mathbf{F} + \nabla\cdot\mathbf{G}$, $\nabla\cdot(\phi\mathbf{F}) = \phi\nabla\cdot\mathbf{F} + \nabla\phi\cdot\mathbf{F}$, $\nabla\cdot(\nabla\times\mathbf{F}) = 0$ **47.** **(b)** $x^2 + y^2 = K$
49. $\dfrac{dy}{dx} = \dfrac{1}{x}$, $y = \ln x + K$



▶ Exercise Set 15.2 (Page 1108)

1. **(a)** 1 **(b)** 0 **3.** 16
7. **(a)** $\frac{331}{108}\sqrt{10} - \frac{1}{36}\ln(3+\sqrt{10}) - \frac{4}{27}$ **(b)** 4 **(c)** $\frac{17}{2}$
9. **(a)** 4 **(b)** 4 **(c)** 4 **(d)** 4
Responses to True–False questions may be abridged to save space.
11. False; line integrals of functions are independent of the orientation of the curve. **13.** True; this is Equation (26).
15. $2\sqrt{5} - 2$ **17.** $\frac{46}{63}$ **19.** $3 - 2\pi$ **21.** 3
23. $-1 - (\pi/4)$ **25.** $1 - e^2$
27. **(a)** $\dfrac{63\sqrt{17}}{64} + \dfrac{1}{4}\ln(4+\sqrt{17}) - \dfrac{1}{8}\ln\dfrac{\sqrt{17}+1}{\sqrt{17}-1} - \dfrac{1}{4}\ln(\sqrt{2}+1) + \dfrac{1}{8}\ln\dfrac{\sqrt{2}+1}{\sqrt{2}-1}$ **(b)** $1/2 - \pi/4$
29. **(a)** -1 **(b)** -2 **31.** $\frac{5}{2}$ **33.** 0 **35.** $1 - e^{-1}$ **37.** $6\sqrt{3}$
39. $5k\tan^{-1}3$ **41.** $\frac{19}{35}$ **43.** $\frac{27}{28}$ **45.** $\frac{3}{4}$ **47.** $\dfrac{17\sqrt{17}-1}{4}$
49. **(b)** $S = \displaystyle\int_C z(t)\,dt$ **(c)** 4π **51.** $\lambda = -12$

▶ Exercise Set 15.3 (Page 1120)

1. conservative, $\phi = x^2 + y^2 + K$ **3.** not conservative
5. conservative, $\phi = x\cos y + y\sin x + K$
9. -8 **11.** $27e^2$ **13.** 32 **15.** $W = -8$ **17.** $W = e^2 - e^{-1}$
Responses to True–False questions may be abridged to save space.
19. False; the integral must be 0 for *all* closed curves C.
21. True; if $\nabla\phi$ is constant, then ϕ must be a linear function.
23. $\ln 2 - 1$ **25.** ≈ -0.307 **27.** no **33.** $h(x) = Ce^x$
35. **(a)** $W = -\dfrac{1}{\sqrt{14}} + \dfrac{1}{\sqrt{16}}$ **(b)** $W = -\dfrac{1}{\sqrt{14}} + \dfrac{1}{\sqrt{6}}$ **(c)** $W = 0$

▶ Exercise Set 15.4 (Page 1127)

1. 0 **3.** -6 **5.** π^2 **7.** 24π **9.** -4 **11.** -1 **13.** 0
Responses to True–False questions may be abridged to save space.
15. False; Green's Theorem applies to closed curves.
17. True; the integral is the area of the region bounded by C.
19. **(a)** ≈ -3.550999378 **(b)** ≈ -0.269616482 **21.** $\frac{3}{8}a^2\pi$ **23.** $\frac{1}{2}abt_0$
27. Formula (1) of Section 6.1 **29.** $\frac{128}{3}$ **31.** $-3\pi a^2$ **33.** $\left(\frac{8}{15}, \frac{8}{21}\right)$
35. $\left(0, \dfrac{4a}{3\pi}\right)$ **37.** the circle $x^4 + y^4 = 1$ **39.** 69

▶ Exercise Set 15.5 (Page 1136)

1. $40\pi\sqrt{2}$ **3.** $\dfrac{\pi}{4}$ **5.** $-2\sqrt{2}$ **7.** 9
Responses to True–False questions may be abridged to save space.
9. True; this follows from the definition.
11. False; the integral is the total mass of the lamina.
13. **(b)** $2\pi\left[1 - \sqrt{1-r^2} + \dfrac{r^2}{2}\right] \to 3\pi$ as $r \to 1^-$
(c) $\mathbf{r}(\phi,\theta) = \sin\phi\cos\theta\mathbf{i} + \sin\phi\sin\theta\mathbf{j} + \cos\phi\mathbf{k}$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/2$;
$$\iint (1+z)\,dS = \int_0^{2\pi}\int_0^{\pi/2}(1+\cos\phi)\sin\phi\,d\phi\,d\theta = 3\pi$$
17. **(c)** $4\pi/3$
19. **(a)** $\dfrac{\sqrt{29}}{16}\displaystyle\int_0^6\int_0^{(12-2x)/3} xy(12-2x-3y)\,dy\,dx$
(b) $\dfrac{\sqrt{29}}{4}\displaystyle\int_0^3\int_0^{(12-4z)/3} yz(12-3y-4z)\,dy\,dz$
(c) $\dfrac{\sqrt{29}}{9}\displaystyle\int_0^3\int_0^{6-2z} xz(12-2x-4z)\,dx\,dz$
21. $\dfrac{18\sqrt{29}}{5}$
23. $\displaystyle\int_0^4\int_1^2 y^3z\sqrt{4y^2+1}\,dy\,dz$; $\dfrac{1}{2}\displaystyle\int_0^4\int_1^4 xz\sqrt{1+4x}\,dx\,dz$
25. $\dfrac{391\sqrt{17}}{15} - \dfrac{5\sqrt{5}}{3}$ **27.** $\frac{5}{3}\pi\delta_0$ **29.** $\frac{1}{3}(37\sqrt{37}-1)$ **31.** $M = \delta_0 S$
33. $(0, 0, 149/65)$ **35.** $\dfrac{363\sqrt{10}}{5}$ **37.** $\dfrac{\pi}{2}$ **39.** 57.895751

▶ Exercise Set 15.6 (Page 1146)

1. **(a)** zero **(b)** zero **(c)** positive **(d)** negative **(e)** zero **(f)** zero
3. **(a)** positive **(b)** zero **(c)** positive **(d)** zero **(e)** positive **(f)** zero
5. **(a)** $n = -\cos v\mathbf{i} - \sin v\mathbf{j}$ **(b)** inward **7.** 32π **9.** $\dfrac{52\pi}{3}$ **11.** 0
13. 21π **15.** $\frac{4}{9}$ **17.** **(a)** 8 **(b)** 24 **(c)** 0

Responses to True–False questions may be abridged to save space.

19. False; the Möbius strip has no orientation.

21. False; the net volume can be zero because as much fluid passes through the surface in the negative direction as in the positive direction.

23. $\frac{-52\pi}{3}$ **25.** **(a)** $0\ \text{m}^3/\text{s}$ **(b)** $0\ \text{kg/s}$ **27.** **(b)** $32/3$

29. **(a)** $4\pi a^{k+3}$ **(b)** $k = -3$ **31.** $a = 2, 3$

Exercise Set 15.7 (Page 1157)

1. 3 **3.** $\frac{4\pi}{3}$

Responses to True–False questions may be abridged to save space.

5. False; it equates a surface integral and a triple integral.

7. True; see subsection entitled Sources and Sinks.

9. 18 **11.** $4\pi a^2$ **13.** 352π **15.** $\frac{192\pi}{5}$ **17.** 8π

19. $\frac{4608}{35}$ **21.** 189π **33.** no sources or sinks

35. sources at all points except the origin, no sinks **37.** $\frac{7\pi}{4}$

Exercise Set 15.8 (Page 1164)

1. $\frac{3}{2}$ **3.** 0 **5.** 6π **7.** 16π **9.** 0 **11.** $2\pi a^2$

Responses to True–False questions may be abridged to save space.

13. True; see Theorem 15.8.1. **15.** False; the circulation is $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$.

17. **(a)** $\frac{3}{2}$ **(b)** -1 **(c)** $-\frac{1}{\sqrt{2}}\mathbf{j} - \frac{1}{\sqrt{2}}\mathbf{k}$ **23.** $-\frac{5\pi}{4}$

Chapter 15 Review Exercises (Page 1166)

3. $\dfrac{1-x}{\sqrt{(1-x)^2+(3-y)^2}}\mathbf{i} + \dfrac{3-y}{\sqrt{(1-x)^2+(3-y)^2}}\mathbf{j}$ **5.** $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

7. **(a)** $\displaystyle\int_a^b \left[f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt} \right] dt$

(b) $\displaystyle\int_a^b f(x(t), y(t))\sqrt{x'(t)^2 + y'(t)^2}\,dt$

11. 0 **13.** -7 **17.** **(a)** $h(x) = Cx^{-3/2}$ **(b)** $g(y) = C/y^3$

21. $A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta r^2\,d\theta$

23. $\displaystyle\iint_R f(x(u,v), y(u,v), z(u,v))\|r_u \times r_v\|\,du\,dv$ **25.** yes **27.** 8π

31. -8π **35.** **(a)** conservative **(b)** not conservative

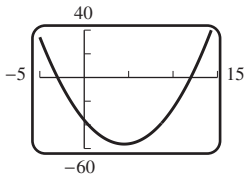
Chapter 15 Making Connections (Page 1168)

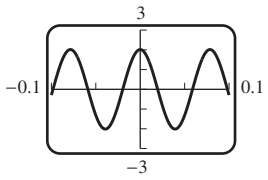
Answers are provided in the Student Solutions Manual.

Appendix A (Page A11)

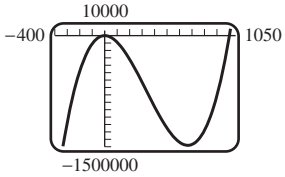
1. (e) **3.** (b), (c) **5.** $[-3, 3] \times [0, 5]$

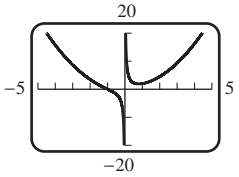
7. $[-5, 14] \times [-60, 40]$ **9.** $[-0.1, 0.1] \times [-3, 3]$





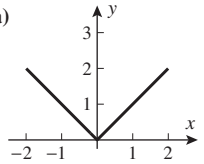
11. $[-400, 1050] \times [-1500000, 10000]$ **13.** $[-2, 2] \times [-20, 20]$



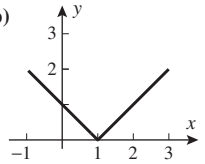


17. **(a)** $f(x) = \sqrt{16 - x^2}$ **(b)** $f(x) = -\sqrt{16 - x^2}$ **(e)** no

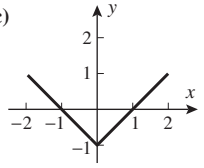
19. **(a)**



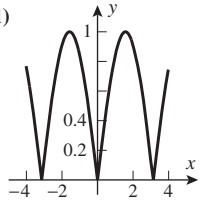
(b)



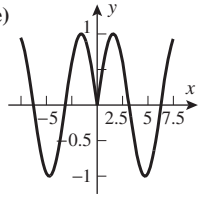
(c)



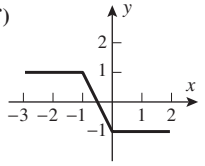
(d)



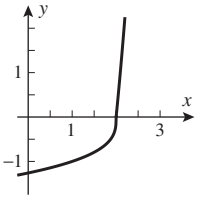
(e)



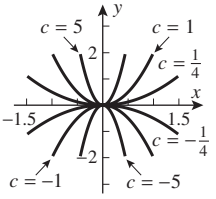
(f)



21.

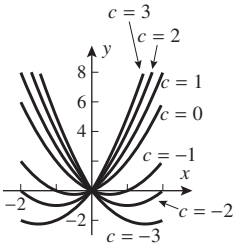


25. **(a)**



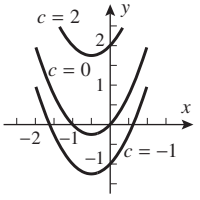
The graph is stretched in the vertical direction, and reflected across the x-axis if $c < 0$.

(b)



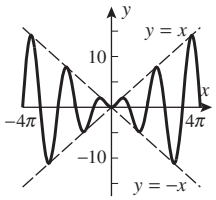
The graph is translated so its vertex is on the parabola $y = -x^2$.

(c)

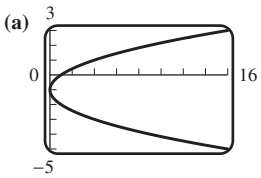


The graph is translated vertically.

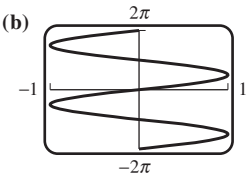
27.



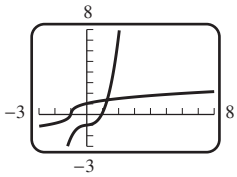
29. **(a)**



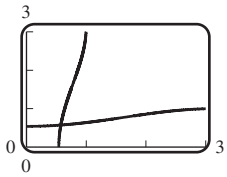
(b)



31.



33.



35. **(a)** $x = 4\cos t,\ y = 3\sin t$ **(b)** $x = -1 + 4\cos t,\ y = 2 + 3\sin t$

▶ Appendix B (Page A23)

1. **(a)** $\frac{5}{12}\pi$ **(b)** $\frac{13}{6}\pi$ **(c)** $\frac{1}{9}\pi$ **(d)** $\frac{23}{30}\pi$

3. **(a)** $12°$ **(b)** $(270/\pi)°$ **(c)** $288°$ **(d)** $540°$

5.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$\sqrt{21}/5$	$2/5$	$\sqrt{21}/2$	$5/\sqrt{21}$	$5/2$	$2/\sqrt{21}$
(b)	$3/4$	$\sqrt{7}/4$	$3/\sqrt{7}$	$4/3$	$4/\sqrt{7}$	$\sqrt{7}/3$
(c)	$3/\sqrt{10}$	$1/\sqrt{10}$	3	$\sqrt{10}/3$	$\sqrt{10}$	$1/3$

7. $\sin\theta = 3/\sqrt{10}, \cos\theta = 1/\sqrt{10}$ **9.** $\tan\theta = \sqrt{21}/2, \csc\theta = 5/\sqrt{21}$

11. 1.8

13.

	θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$225°$	$-1/\sqrt{2}$	$-1/\sqrt{2}$	1	$-\sqrt{2}$	$-\sqrt{2}$	1
(b)	$-210°$	$1/2$	$-\sqrt{3}/2$	$-1/\sqrt{3}$	2	$-2/\sqrt{3}$	$-\sqrt{3}$
(c)	$5\pi/3$	$-\sqrt{3}/2$	$1/2$	$-\sqrt{3}$	$-2/\sqrt{3}$	2	$-1/\sqrt{3}$
(d)	$-3\pi/2$	1	0	—	1	—	0

15.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$4/5$	$3/5$	$4/3$	$5/4$	$5/3$	$3/4$
(b)	$-4/5$	$3/5$	$-4/3$	$-5/4$	$5/3$	$-3/4$
(c)	$1/2$	$-\sqrt{3}/2$	$-1/\sqrt{3}$	2	$-2/\sqrt{3}$	$-\sqrt{3}$
(d)	$-1/2$	$\sqrt{3}/2$	$-1/\sqrt{3}$	-2	$2/\sqrt{3}$	$-\sqrt{3}$
(e)	$1/\sqrt{2}$	$1/\sqrt{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
(f)	$1/\sqrt{2}$	$-1/\sqrt{2}$	-1	$\sqrt{2}$	$-\sqrt{2}$	-1

17. **(a)** 1.2679 **(b)** 3.5753

19.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$a/3$	$\sqrt{9-a^2}/3$	$a/\sqrt{9-a^2}$	$3/a$	$3/\sqrt{9-a^2}$	$\sqrt{9-a^2}/a$
(b)	$a/\sqrt{a^2+25}$	$5/\sqrt{a^2+25}$	$a/5$	$\sqrt{a^2+25}/a$	$\sqrt{a^2+25}/5$	$5/a$
(c)	$\sqrt{a^2-1}/a$	$1/a$	$\sqrt{a^2-1}$	$a/\sqrt{a^2-1}$	a	$1/\sqrt{a^2-1}$

21. **(a)** $3\pi/4 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/3 \pm 2n\pi$ and $5\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$

23. **(a)** $\pi/6 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $4\pi/3 \pm 2n\pi$ and $5\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$

25. **(a)** $3\pi/4 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/6 \pm n\pi, n = 0, 1, 2, \ldots$

27. **(a)** $\pi/3 \pm 2n\pi$ and $2\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/6 \pm 2n\pi$ and $11\pi/6 \pm 2n\pi, n = 0, 1, 2, \ldots$

29. $\sin\theta = 2/5, \cos\theta = -\sqrt{21}/5, \tan\theta = -2/\sqrt{21}$,
$\csc\theta = 5/2, \sec\theta = -5/\sqrt{21}, \cot\theta = -\sqrt{21}/2$

31. **(a)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$ **(b)** $\theta = \pi/2 \pm n\pi, n = 0, 1, 2, \ldots$
(c) $\theta = \pm n\pi, n = 0, 1, 2, \ldots$ **(d)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$
(e) $\theta = \pi/2 \pm n\pi, n = 0, 1, 2, \ldots$ **(f)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$

33. **(a)** $2\pi/3$ cm **(b)** $10\pi/3$ cm **35.** $\frac{2}{5}$

37. **(a)** $\dfrac{2\pi - \theta}{2\pi}R$ **(b)** $\dfrac{\sqrt{4\pi\theta - \theta^2}}{2\pi}R$ **39.** $\frac{21}{4}\sqrt{3}$ **41.** 9.2 ft

43. $h = d(\tan\beta - \tan\alpha)$ **45.** **(a)** $4\sqrt{5}/9$ **(b)** $-\frac{1}{9}$

47. $\sin 3\theta = 3\sin\theta\cos^2\theta - \sin^3\theta, \cos 3\theta = \cos^3\theta - 3\sin^2\theta\cos\theta$

61. **(a)** $\cos\theta$ **(b)** $-\sin\theta$ **(c)** $-\cos\theta$ **(d)** $\sin\theta$

69. **(a)** $153°$ **(b)** $45°$ **(c)** $117°$ **(d)** $89°$ **71.** **(a)** $60°$ **(b)** $117°$

▶ Appendix C (Page A32)

1. **(a)** $q(x) = x^2 + 4x + 2, r(x) = -11x + 6$
(b) $q(x) = 2x^2 + 4, r(x) = 9$
(c) $q(x) = x^3 - x^2 + 2x - 2, r(x) = 2x + 1$

3. **(a)** $q(x) = 3x^2 + 6x + 8, r(x) = 15$
(b) $q(x) = x^3 - 5x^2 + 20x - 100, r(x) = 504$
(c) $q(x) = x^4 + x^3 + x^2 + x + 1, r(x) = 0$

5.

x	0	1	−3	7
$p(x)$	−4	−3	101	5001

7. **(a)** $q(x) = x^2 + 6x + 13, r = 20$ **(b)** $q(x) = x^2 + 3x - 2, r = -4$

9. **(a)** $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12, \pm 24$
(b) $\pm 1, \pm 2, \pm 5, \pm 10, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{5}{3}, \pm\frac{10}{3}$ **(c)** $\pm 1, \pm 17$

11. $(x+1)(x-1)(x-2)$ **13.** $(x+3)^3(x+1)$

15. $(x+3)(x+2)(x+1)^2(x-3)$ **17.** -3 **19.** $-2, -\frac{2}{3}, -1 \pm \sqrt{3}$

21. $-2, 2, 3$ **23.** $2, 5$ **25.** 7 cm

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